A HANDBOOK FOR QUANT FINANCE

STOCHASTIC CALCULUS AND DERIVATIVE PRICING.

First Edition

PREFACE

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CHAPITRE 1

FOUNDATIONS OF PROBABILITY AND MEASURE

Probability theory provides a mathematical framework for studying random phenomena. Many processes in nature, science, engineering, and finance involve outcomes that cannot be predicted with certainty. Examples include the outcome of a coin toss, the arrival time of customers in a queue, or the evolution of asset prices in financial markets. Probability theory allows us to model such situations and to reason quantitatively about uncertainty.

The modern formulation of probability theory is based on **measure theory**. In measure theory, the goal is to assign a non-negative number to certain subsets of a given set. This number is called a **measure** and represents a generalized notion of size, extending familiar ideas such as length, area, or volume. A fundamental property of a measure is **countable additivity**: if a collection of sets is disjoint, the measure of their union is the sum of their measures.

For mathematical reasons, it is not possible in general to assign a measure to every subset of a set. Instead, one restricts attention to a special collection of subsets called a σ-field . A set equipped with a σ-field is called a **measurable space**, and measures are defined only on these measurable sets.

Probability theory can be viewed as a special case of measure theory, where we are assigning a measure that represents our belief of occurrence of events. This measure is called a **Probability measure** and it has the additional property that the total measure of the entire space is equal to one. The triplet $(\Omega, \mathcal{A}, \mathbb{P})$ is called a **probability space** where :

- The set Ω is the **sample space**. It contains all possible outcomes of the experiment. Once an outcome $\omega \in \Omega$ is fixed, all random quantities associated with the experiment are completely determined.
- The collection $\mathcal{A}$ is a σ-field of subsets of Ω, called **events**. These are the sets to which probabilities can be assigned. From the point of view of measure theory, they are precisely the measurable sets. An event $A \in \mathcal{A}$ represents the occurrence of a given property of the outcome ω.
- The function $\mathbb{P} : \mathcal{A} \to [0, 1]$ is a **probability measure**. For each event $A \in \mathcal{A}$, the value $\mathbb{P}(A)$ represents the probability that the event A occurs.

This axiomatic formulation of probability theory was introduced by **Andrey Kolmogorov in 1933**. The Kolmogorov axioms provide a simple and powerful foundation for probability and allow many results to be developed in a unified and rigorous way.

Although probability theory relies on measure theory, the goals of the two subjects are different. Measure theory studies general notions of size and integration, while probability theory focuses on modeling and analyzing randomness. In this chapter, we mainly develop probability theory while referring to measure-theoretic concepts whenever they are needed.

A prior knowledge of measure theory can be helpful but is not strictly required. The reader is assumed to have seen basic notions such as σ-fields , measurable functions, and measures. Whenever these ideas are used, we briefly recall their meaning and explain their role in probability.

The aim of this chapter is to present the fundamental concepts of probability theory in a clear and structured way, combining mathematical rigor with intuitive interpretations based on random experiments.

Important remark. In many situations, the explicit form of the probability measure $\mathbb{P}$ will not be specified. Instead, we focus on the properties of measurable functions defined on $(\Omega, \mathcal{A})$, which are called *random variables.*

Terminology. As in measure theory, sets of measure zero play an important role in probability theory. A property depending on $\omega \in \Omega$ is said to hold *almost surely* (a.s.) if it holds for all ω in an event of probability one. Thus, *almost surely* has the same meaning as "almost everywhere" in measure theory.

Notation. For a random variable $\mathbf{X}$, the notation $\mathbb{P}(\mathbf{X} \in A)$ means :

$$\mathbb{P}(\{\omega \in \Omega : \mathbf{X}(\omega) \in A\}).$$

1.1 General Definitions

In this section, we introduce the fundamental objects of probability theory. We begin with the definition of a probability space, which provides the mathematical framework for modeling random experiments. We then define random variables, which allow us to describe and analyze numerical quantities generated by randomness.

These definitions form the basis of all subsequent results in this chapter. They are stated in a general and rigorous setting, with close connections to measure theory whenever necessary.

1.1.1 Probability space

Definition 1.1.1: Probability space

A **probability space** is a triple $(\Omega, \mathcal{A}, \mathbb{P})$, where:

- Ω is a non-empty set called the **sample space**,
- $\mathcal{A}$ is a σ-field of subsets of Ω, and
- $\mathbb{P} : \mathcal{A} \to [0, 1]$ is a **probability measure**.

We remind the reader that a σ-field $\mathcal{A}$ is a collection of subsets of Ω such that:

- $\Omega \in \mathcal{A}$,
- if $A \in \mathcal{A}$ then $A^c = \Omega \setminus A \in \mathcal{A}$,
- if $A_1, A_2, \ldots \in \mathcal{A}$ then $\cup_{i=1}^{\infty} A_i \in \mathcal{A}$.

Definition 1.1.2: Probability measure

A **probability measure** $\mathbb{P}$ on a σ-field $\mathcal{A}$ of a set Ω is a function $\mathbb{P} : \mathcal{A} \to [0, 1]$ satisfying:

- $\mathbb{P}(\Omega) = 1$
- For any countable sequence of disjoint sets $A_1, A_2, \ldots \in \mathcal{A}$,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

Proposition 1.1.3

We have the following results :

1. If $A \subset B$ then :

$$\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$$

2. If $A, B \in \mathcal{A}$ then :

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

3. If $A_n \in \mathcal{A}$ and $A_n \subset A_{n+1}$ for every $n \in \mathbb{N}$ then :

$$\mathbb{P}(\bigcup_{n\in\mathbb{N}} A_n) = \lim_{n\to\infty} \mathbb{P}(A_n)$$

4. If $A_n \in \mathcal{A}$ and $A_{n+1} \subset A_n$ for every $n \in \mathbb{N}$ then :

$$\mathbb{P}(\bigcap_{n\in\mathbb{N}} A_n) = \lim_{n\to\infty} \mathbb{P}(A_n)$$

5. If $A_n \in \mathcal{A}$ for every $n \in \mathbb{N}$ then :

$$\mathbb{P}(\bigcup_{n\in\mathbb{N}} A_n) \leq \sum_{n\in\mathbb{N}} \mathbb{P}(A_n)$$

Proof for Proposition.

Left exercise for the reader. ∎

1.1.2 Monotone class

The axioms defining a σ-field are often stronger than those required in many arguments. In practice, one frequently begins with a simpler collection of sets, such as intervals or rectangles, and seeks to extend properties established on this collection to the σ-field it generates. This motivates the study of weaker systems whose closure properties are sufficient for such extension arguments. Among the most important are monotone classes (also called λ-systems or Dynkin systems). In this section, we focus on monotone classes, which are obtained by replacing the requirement of closure under arbitrary countable unions in the definition of a σ-field (Definition 1.1.1) with the weaker requirement of closure under countable disjoint unions. We conclude by proving Dynkin's Monotone Class Theorem (also known as the π–λ theorem), a fundamental result that allows properties established on a π-system to be extended to the σ-field it generates.

Definition 1.1.4: Monotone class

Let Ω be a set and let $\mathcal{M} \subset \mathcal{P}(\Omega)$. The collection $\mathcal{M}$ is called a *monotone class* if:

1. $\Omega \in \mathcal{M}$;
2. if $A, B \in \mathcal{M}$ and $A \subset B$, then $B \setminus A \in \mathcal{M}$;
3. if $(A_n)_{n\in\mathbb{N}}$ is an increasing sequence in $\mathcal{M}$, i.e.

$$A_1 \subset A_2 \subset \cdots ,$$

then

$$\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{M}.$$

Every σ-algebra is a monotone class. The converse is false in general. However, a monotone class is a σ-algebra if and only if it is closed under finite intersections. Indeed, closure under finite intersections yields closure under complements and finite unions, while the monotone class property then extends finite unions to countable unions, giving the defining properties of a σ-algebra.

It is therefore useful to introduce the notion of a π-system, namely a collection of sets closed under finite intersections. With this terminology, a σ-algebra may be viewed as the combination of a π-system and a λ-system (equivalently, a monotone class), that is,

$$\boxed{\sigma\text{-algebra} = \pi\text{-system} + \lambda\text{-system}.}$$

This observation underlies many uniqueness arguments in measure theory, most notably the π-λ theorem.

Arbitrary intersections of monotone classes are again monotone classes. Given a collection $\mathcal{C} \subset \mathcal{P}(\Omega)$, we define the *monotone class generated by* $\mathcal{C}$, denoted $\mathcal{M}(\mathcal{C})$, as the intersection of all monotone classes containing $\mathcal{C}$.

Theorem 1.1.5: Monotone Class Theorem

Let $\mathcal{C} \subset \mathcal{P}(\Omega)$ be a collection of sets that is closed under finite intersections. Then

$$\mathcal{M}(\mathcal{C}) = \sigma(\mathcal{C}).$$

Consequently, if $\mathcal{M}$ is any monotone class such that $\mathcal{C} \subset \mathcal{M}$, then

$$\sigma(\mathcal{C}) \subset \mathcal{M}.$$

Proof for Theorem.

It is enough to prove the first assertion. Since any σ-field is a monotone class, it is clear that $\mathcal{M}(C) \subset \sigma(\mathcal{C})$. To prove the reverse inclusion, it is enough to verify that $\mathcal{M}(\mathcal{C})$ is a σ-field . As explained above, it suffices to verify that $\mathcal{M}(\mathcal{C})$ is closed under finite intersections.

For every $A \in \mathcal{P}(\Omega)$, set

$$\mathcal{M}_A = \{B \in \mathcal{M}(\mathcal{C}) : A \cap B \in \mathcal{M}(\mathcal{C})\}.$$

Fix $A \in \mathcal{C}$. Since $\mathcal{C}$ is closed under finite intersections, we have $\mathcal{C} \subset \mathcal{M}_A$. Let us verify that $\mathcal{M}_A$ is a monotone class:

- $\Omega \in \mathcal{M}_A$ is trivial.
- If $B, B' \in \mathcal{M}_A$ and $B \subset B'$, then $A \cap (B' \setminus B) = (A \cap B') \setminus (A \cap B) \in \mathcal{M}(\mathcal{C})$ and thus $B' \setminus B \in \mathcal{M}_A$.

- If $B_n \in \mathcal{M}_A$ for every $n \geq 0$ and the sequence $(B_n)_{n\geq 0}$ is increasing, we have $A \cap (\cup_{n\in\mathbb{N}} B_n) = \cup_{n\in\mathbb{N}}(A \cap B_n) \in \mathcal{M}(\mathcal{C})$ and therefore $\cup_{n\in\mathbb{N}} B_n \in \mathcal{M}_A$.

Since $\mathcal{M}_A$ is a monotone class that contains $\mathcal{C}$, $\mathcal{M}_A$ also contains $\mathcal{M}(\mathcal{C})$. We have thus obtained that

$$\forall A \in \mathcal{C},\ \forall B \in \mathcal{M}(\mathcal{C}),\ A \cap B \in \mathcal{M}(\mathcal{C}).$$

This is not yet the desired result, but we can use the same idea one more time. Precisely, we now fix $A \in \mathcal{M}(\mathcal{C})$. According to the first part of the proof, $\mathcal{C} \subset \mathcal{M}_A$. By exactly the same arguments as in the first part of the proof, we get that $\mathcal{M}_A$ is a monotone class. It follows that $\mathcal{M}(\mathcal{C}) \subset \mathcal{M}_A$, which shows that $\mathcal{M}(\mathcal{C})$ is closed under finite intersections, and completes the proof. ■

Although the Monotone Class Theorem is easy to state, its true importance lies in the proof strategy that it provides. In practice, one rarely works directly with the monotone class $\mathcal{M}(\mathcal{C})$. Instead, one defines the collection of all sets for which a desired property holds, proves that this collection forms a monotone class containing a suitable generating algebra, and then invokes the Monotone Class Theorem to extend the property to the entire generated σ-field. This four-step argument appears repeatedly throughout probability theory and measure theory. For future reference, the general scheme is summarized in Figure 1.

Step 1. Choose a generating algebra $\mathcal{C}$.

Verify that the desired property $P(A)$ holds for every $A \in \mathcal{C}$.

Step 2. Define the good class

$$\mathcal{G} = \{A \in \sigma(\mathcal{C}) : P(A) \text{ holds}\}.$$

Show that $\mathcal{C} \subseteq \mathcal{G}$.

Step 3. Show that $\mathcal{G}$ is a monotone class.

Verify that the property is preserved under increasing unions and decreasing intersections.

Step 4. Apply the Monotone Class Theorem

$$\sigma(\mathcal{C}) \subseteq \mathcal{G}.$$

Hence, the property $P(A)$ holds for every measurable set $A \in \sigma(\mathcal{C})$.

FIGURE 1 – The standard proof strategy for applying the Monotone Class Theorem.

Here are some examples :

Example : Uniqueness of probability measures
Suppose $\mathbb{P}$ and $\mathbb{Q}$ are probability measures on $(\Omega, \mathcal{A})$ satisfying

$$\mathbb{P}(A) = \mathbb{Q}(A), \qquad A \in \mathcal{C},$$

where $\mathcal{C}$ is an algebra generating $\mathcal{A}$. To prove that $\mathbb{P} = \mathbb{Q}$, proceed as follows.

1. Define the collection of *good* sets:

$$\mathcal{G} = \{A \in \mathcal{A} : \mathbb{P}(A) = \mathbb{Q}(A)\}.$$

2. By assumption,

$$\mathcal{C} \subseteq \mathcal{G}.$$

3. Show that $\mathcal{G}$ is a monotone class. This follows from the continuity of finite measures from below and above.
4. Since $\mathcal{C}$ is an algebra generating $\mathcal{A}$, the Monotone Class Theorem yields

$$\mathcal{A} = \sigma(\mathcal{C}) \subseteq \mathcal{G}.$$

Hence,

$$\mathbb{P}(A) = \mathbb{Q}(A), \qquad A \in \mathcal{A},$$

so that $\mathbb{P} = \mathbb{Q}$.

■

Example : Extension of an integral identity
Suppose $f, g \geq 0$ are measurable functions satisfying

$$\int_A f \, d\mu = \int_A g \, d\mu, \qquad A \in \mathcal{C},$$

where $\mathcal{C}$ is an algebra generating $\mathcal{A}$. To extend this identity to all measurable sets, proceed as follows.

1. Define

$$\mathcal{G} = \left\{ A \in \mathcal{A} : \int_A f \, d\mu = \int_A g \, d\mu \right\}.$$

2. Clearly,

$$\mathcal{C} \subseteq \mathcal{G}.$$

3. Show that $\mathcal{G}$ is a monotone class. This follows from the Monotone Convergence Theorem.
4. The Monotone Class Theorem implies

$$\mathcal{A} = \sigma(\mathcal{C}) \subseteq \mathcal{G},$$

so the identity holds for every measurable set.

■

Example : Equality of localized expectations
Let X and Y be integrable random variables such that

$$\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A], \qquad A \in \mathcal{C},$$

where $\mathcal{C}$ is an algebra generating $\mathcal{A}$. To prove that the equality holds for every measurable set, proceed as follows.

1. Define
$$\mathcal{G} = \{A \in \mathcal{A} : \mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A]\}.$$
2. By hypothesis,
$$\mathcal{C} \subseteq \mathcal{G}.$$
3. Verify that $\mathcal{G}$ is a monotone class. This follows from the Dominated Convergence Theorem.
4. Applying the Monotone Class Theorem gives
$$\mathcal{A} = \sigma(\mathcal{C}) \subseteq \mathcal{G},$$
and therefore
$$\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A], \qquad A \in \mathcal{A}.$$

1.1.3 Random variables

Definition 1.1.6: Random variable

A function from Ω to $\mathbb{R}^n$ is called a random variable (in measure theory is called measurable function), meaning an application $\mathbf{X}$ such that:

$$\mathbf{X}^{-1}(B) \in \mathcal{A}, \text{ for each set } B \in \mathcal{B}(\mathbb{R}^n)$$

where $\mathcal{B}(\mathbb{R}^n)$ denotes the Borel [a] σ-field .

a. The Borel σ-field on $\mathbb{R}^n$ is generated by the open subsets of $\mathbb{R}^n$. The Lebesgue measure on $\mathbb{R}^n$ can be first defined on rectangles of the form $[a_1, b_1] \times \cdots \times [a_n, b_n]$ by $(b_1 - a_1) \cdots (b_n - a_n)$, and then uniquely extended to a measure on the Borel σ-field . For more details, see the chapter [3, chap. 3].

Based on this definition, we can define a probability measure in the probabilistic space $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \mathbb{P}_\mathbf{X})$, where the probability measure (also called the distribution of $\mathbf{X}$) $\mathbb{P}_\mathbf{X}$ is defined as follows:

$$\mathbb{P}_\mathbf{X}(B) = \mathbb{P}(\mathbf{X}^{-1}(B)) = \mathbb{P}(\{\mathbf{X} \in B\}), \text{ for each set } B \in \mathcal{B}(\mathbb{R}^n)$$

We call $\mathbb{P}_\mathbf{X}$ the law (or the distribution) of the random variable $\mathbf{X}$. In order to characterize the distribution of a random variable, we use cumulative distribution functions and probability density functions.

Definition 1.1.7: Cumulative distribution function

Let $\mathbf{X}$ be a random variable with value space $\mathbb{R}^n$. For each $\mathbf{x} \in \mathbb{R}^n$, we define the **cumulative distribution function** as follows:

$$F_{\mathbf{X}}(\mathbf{x}) = \mathbb{P}(\mathbf{X} \leq \mathbf{x}) = \mathbb{P}(X_1 \leq x_1, \cdots, X_n \leq x_n)$$

Definition 1.1.8: Probability density function

Let $\mathbf{X}$ be a random variable with value space $\mathbb{R}^n$. For each $\mathbf{x} \in \mathbb{R}^n$, we define the **probability density function** as follows:

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{\partial^n}{\partial \mathbf{x}} F_{\mathbf{X}}(\mathbf{x}) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F_{\mathbf{X}}(\mathbf{x})$$

if it exists on a non-negligible set of $\mathbb{R}^n$. This function is a density for the probability measure $\mathbb{P}_{\mathbf{X}}$ such that:

$$\mathbb{P}_{\mathbf{X}}(B) = \int_B d\mathbb{P}_{\mathbf{X}} = \int_B f_{\mathbf{X}}(\mathbf{x}) d\lambda(\mathbf{x})$$

Where λ is the Lebesgue measure on $\mathcal{B}(\mathbb{R}^n)$

Finally, we have the following theorem, which ensures that a constructed variable is indeed a random variable.

Theorem 1.1.9: Operations on random variables

1. Let X_1, X_2 be two random variables on $\mathbb{R}$, and $f : \mathbb{R}^2 \mapsto \mathbb{R}$ a measurable function (w.r.t Lebesgue measure and Borel σ-field), then $(X_1, X_2), \quad f(X_1, X_2)$ are random variables.
2. Let (X_n) be a sequence of random variables that map into $\bar{\mathbb{R}}$ then :

$$\sup_{n \in \mathbb{N}} X_n, \quad \inf_{n \in \mathbb{N}} X_n, \quad \limsup_{n \to \infty} X_n, \quad \liminf_{n \to \infty} X_n$$

   are also random variables. In particular, if the sequence (X_n) converges pointwise, its limit is a random variable.
3. Let (X_n) be a sequence of random variables that map into $\bar{\mathbb{R}}$ then : The set A of all $x \in \Omega$ such that $X_n(\omega)$ converges in $\mathbb{R}$ as $n \to \infty$ is measurable ($A \in \mathcal{A}$). Furthermore, the function $h : \Omega \longrightarrow \mathbb{R}$ defined by

$$Y(\omega) := \begin{cases} \lim_{n \to \infty} X_n(\omega) & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A, \end{cases}$$

is a random variable.

Proof for Theorem.

The proofs could be found in [3, sec. 1.3]. ■

Finally we have this two lemmas, that would be used frequently in most of probability-propositions proofs.

Lemma 1.1.10: Approximation by Simple Random Variables

Let X be a non-negative random variable with values in $\mathbb{R}$. Then there is a sequence (φ_n) of simple random variables with $\varphi_{n+1} \geq \varphi_n$ such that $X = \lim \varphi_n$ at each point of Ω. Where a simple random variable is a random variable having the form :

$$X = \sum_{i=1}^{m} c_i \mathbb{1}_{E_i}$$

For some integer m and real positive values c_i.

Proof for Lemma

The proof is left exercise for reader with the hint to use :

$$E_{n,k} = \{x : k2^{-n} \leq X(\omega) < (k+1)2^{-n}\}, \text{ and the sequence } \varphi_n = 2^{-n} \sum_{k=0}^{2^{2n}} k \mathbb{1}_{E_{n,k}}.$$

■

Lemma 1.1.11: Doob-Dynkin Lemma

If $\mathbf{Y}$ with values in $\mathbb{R}^m$ is $\sigma(\mathbf{X})$-measurable where $\mathbf{X}$ is a random variable with values in $\mathbb{R}^n$, there is a measurable function $f : \mathbb{R}^n \to \mathbb{R}^m$ such that $\mathbf{Y} = f(\mathbf{X})$.

Proof for Lemma

We'll proof the theorem for real valued random variables (with values in $\mathbb{R}$).

1. **Step 1: Indicator Functions** First, consider the case where $Y = \mathbb{1}_A$ for some $A \in \sigma(X)$. Since $A \in \sigma(X)$, by definition, there exists a Borel set $B \in \mathcal{B}(\mathbb{R})$ such that $A = X^{-1}(B)$. Define the function $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \mathbb{1}_B(x).$$

Then, for any $\omega \in \Omega$,

$$Y(\omega) = \mathbb{1}_A(\omega) = \mathbb{1}_{\mathbf{X}^{-1}(B)}(\omega) = \mathbb{1}_B(X(\omega)) = f(X(\omega)).$$

Hence, $Y = f(X)$ for this case.

2. **Step 2: Simple Functions** Now, consider Y as a simple function, i.e., $Y =$

$\sum_{i=1}^n c_i \mathbb{1}_{A_i}$ where $c_i \in \mathbb{R}$ and $A_i \in \sigma(X)$. For each $A_i \in \sigma(X)$, there exists a Borel set $B_i \in \mathcal{B}(\mathbb{R})$ such that $A_i = X^{-1}(B_i)$. Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \sum_{i=1}^n c_i \mathbb{1}_{B_i}(x).$$

Then, for any $\omega \in \Omega$,

$$Y(\omega) = \sum_{i=1}^n c_i \mathbb{1}_{A_i}(\omega) = \sum_{i=1}^n c_i \mathbb{1}_{\mathbf{X}^{-1}(B_i)}(\omega) = \sum_{i=1}^n c_i \mathbb{1}_{B_i}(X(\omega)) = f(X(\omega)).$$

Thus, $Y = f(X)$ for this case.

3. **Step 3: Positive Measurable Functions** Consider Y as a positive $\sigma(X)$-measurable function. There exists a sequence of simple functions Y_n such that $Y_n \nearrow Y$. (use of the lemma 1.1.10) From Step 2, for each n, there exists a Borel-measurable function $f_n : \mathbb{R} \to \mathbb{R}$ such that

$$Y_n = f_n(X).$$

Since $Y_n \nearrow Y$, we have $f_n(X) \to Y$ pointwise almost everywhere. Define $f(x) = \lim_{n\to\infty} f_n(x)$. Since each f_n is Borel-measurable, and the pointwise limit of Borel-measurable functions is Borel-measurable, f is Borel-measurable. Therefore, $Y = f(X)$.

4. **Step 4: General Measurable Functions** Finally, consider Y as an arbitrary $\sigma(X)$-measurable function. We can decompose Y into its positive and negative parts:

$$Y = Y^+ - Y^-,$$

where $Y^+ = \max(Y, 0)$ and $Y^- = -\min(Y, 0)$. Both Y^+ and Y^- are positive $\sigma(X)$-measurable functions. By Step 3, there exist Borel-measurable functions f^+ and f^- such that

$$Y^+ = f^+(X) \quad \text{and} \quad Y^- = f^-(X).$$

Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = f^+(x) - f^-(x).$$

Then,

$$Y = Y^+ - Y^- = f^+(X) - f^-(X) = f(X).$$

Thus, $Y = f(X)$ for some Borel-measurable function f.

■

A large number of proofs in probability theory follow the same general strategy. The idea is to first establish a result for very simple random variables and then extend it step by step to more general cases. This approach reflects the way many objects in probability theory are constructed from simpler ones.

More precisely, proofs often proceed through the following steps:

Step 1: Indicator Functions

Prove the result for random variables of the form

$$Y = \mathbb{1}_A, \qquad A \in \mathcal{A}.$$

Step 2: Simple Random Variables

Extend the result by linearity to finite linear combinations of indicator functions,

$$Y = \sum_{i=1}^{n} a_i \mathbb{1}_{A_i}.$$

Step 3: Positive Measurable Random Variables

Approximate by an increasing sequence of simple functions

$$Y_n \uparrow Y,$$

and use the approximation lemma.

Step 4: General Measurable Random Variables

Write

$$Y = Y^+ - Y^-,$$

apply Step 3 to Y^+ and Y^- separately, and combine the results.

Conclusion

The desired statement now holds for all measurable random variables.

FIGURE 2 – The standard four-step proof strategy used throughout probability theory.

In many situations, some of these steps may be omitted when they follow directly from previous arguments. Nevertheless, this four-step method provides a common structure that appears repeatedly in probability theory.

1.2 Expectation and Moments of Random Variables

1.2.1 Expectation

Definition 1.2.1: Expectation

Let $\mathbf{X}$ be a random variable with value space $\mathbb{R}^n$. The **expectation** or **expected value** of a random variable $\mathbf{X}$ with respect to a probability measure $\mathbb{P}$ is defined as

$$\mathbb{E}[\mathbf{X}] = \int_{\Omega} \mathbf{X}\, d\mathbb{P} = \int_{\mathbb{R}^n} \mathbf{x}\, d\mathbb{P}_{\mathbf{X}} = \int_{\mathbb{R}^n} \mathbf{x} f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x}$$

The integral here is a Lebesgue integral [a] with respect to the probability measure. We require the integral to exist, and the last equality holds only if a density exists.

a. The Lebesgue integral is the generalisation of integration on the measurable spaces, for the rigorous definition check the chapter [3, chap. 2]

We recall some properties of expectations as follow : Let X and Y be random variables, and let a and b be constants. The main properties of expectations are:

1. **Probability representation:**

$$\mathbb{P}[X \in A] = \mathbb{E}(\mathbb{1}_{X \in A})$$

2. **Linearity:**

$$\mathbb{E}[a\mathbf{X} + b\mathbf{Y}] = a\mathbb{E}[\mathbf{X}] + b\mathbb{E}[\mathbf{Y}]$$

3. **Expectation of a Constant:**

$$\mathbb{E}[a] = a$$

4. **Non-negativity:** If $\mathbf{X} \geq 0$ almost surely, then

$$\mathbb{E}[\mathbf{X}] \geq 0$$

5. **Finiteness:**

$$\mathbb{E}[X] < \infty, \Longrightarrow X < \infty \quad \mathbb{P}\text{-a.s.}$$

6. **Zero Expectation:** If X is a nonnegative random variable and

$$\mathbb{E}[X] = 0 \iff X = 0 \quad \mathbb{P}\text{-a.s.}$$

7. **Equality of Expectations:** If Y is another nonnegative random variable and

$$X = Y \quad \mathbb{P}\text{-a.s.}$$

then

$$\mathbb{E}[X] = \mathbb{E}[Y].$$

8. **Law of the Unconscious Statistician:** If g is a measurable function and $\mathbf{X}$ is a random variable, then
$$\mathbb{E}[g(\mathbf{X})] = \int g(\mathbf{X}) d\mathbb{P}$$
9. **Fatou's Lemma:** If (X_n) is a sequence of positive real random variables, then
$$\mathbb{E}[\liminf_{n\to\infty} X_n] \leq \liminf_{n\to\infty} \mathbb{E}[X_n]$$
10. **Monotone Convergence Theorem:** If (X_n) is a sequence of positive real random variables that converges to X almost surely and the sequence is a.s. increasing, then
$$\mathbb{E}[\lim_{n\to\infty} X_n] = \lim_{n\to\infty} \mathbb{E}[X_n]$$
11. **Dominated Convergence Theorem:** If (X_n) is a sequence of real random variables that converges to X almost surely and is dominated by an integrable random variable Y (i.e., $|X_n| \leq Y$ almost surely for all n), then
$$\mathbb{E}[\lim_{n\to\infty} X_n] = \lim_{n\to\infty} \mathbb{E}[X_n]$$

The last three theorems are fundamental results from measure theory and Lebesgue integration. We do not present their proofs here, since most applications in probability only require knowing when they can be applied. They are the principal tools for exchanging limits and expectations.

A good rule of thumb is to proceed in the following order whenever you encounter an expression of the form
$$\mathbb{E}\left[\lim_{n\to\infty} X_n\right].$$

1. Check whether X_n is increasing and $X_n \to X$ almost surely[1]. If so, apply the **Monotone Convergence Theorem**.
2. Otherwise, check whether there exists an integrable random variable Y satisfying
$$|X_n| \leq Y \quad \text{a.s. for every } n.$$
If so, apply the **Dominated Convergence Theorem**.
3. If neither of the above hypotheses is satisfied, but the random variables are nonnegative, try using **Fatou's Lemma**. Although it only provides an inequality,
$$\mathbb{E}[\liminf X_n] \leq \liminf \mathbb{E}[X_n],$$
it is often sufficient to obtain bounds or prove convergence results.

Another tool from real analysis is the Stone-Weierstrass theorem which is generally paired with the fact that :

$\mathbb{E}[f(X)] = \mathbb{E}[f(Y)]$ for every **bounded continuous** function $f : \mathbb{R} \to \mathbb{R}$. Then $\mathbb{P}_X = \mathbb{P}_Y$.

1. this is saying that limite of X_n is X on a set A of $\omega \in A$ with $\mathbb{P}(A) = 1$

Theorem 1.2.2: Stone-Weierstrass theorem

Let $K \subseteq \mathbb{R}$ be a compact set (for example, $K = [a, b]$). Let

$$A \subseteq C(K, \mathbb{C})$$

be a subalgebra, where $C(K, \mathbb{C})$ denotes the space of all continuous complex-valued functions on K, equipped with the uniform norm

$$\|f\|_\infty = \sup_{x \in K} |f(x)|.$$

Assume that:

1. A is a subalgebra of $C(K, \mathbb{C})$; that is, whenever $f, g \in A$ and $\alpha, \beta \in \mathbb{C}$,

$$\alpha f + \beta g \in A, \qquad fg \in A.$$

2. The constant function 1 belongs to A.
3. A separates points of K; that is, for every distinct $x, y \in K$, there exists $f \in A$ such that

$$f(x) \neq f(y).$$

Then A is dense in $C(K, \mathbb{C})$ with respect to the uniform norm. Equivalently,

$$\overline{A}^{\|\cdot\|_\infty} = C(K, \mathbb{C}).$$

That is, for every $f \in C(K, \mathbb{C})$ and every $\varepsilon > 0$, there exists $g \in A$ such that

$$\sup_{x \in K} |f(x) - g(x)| < \varepsilon.$$

Before completing the section we highly invit the reader to consult the section of $\mathcal{L}^p$ spaces 1.3

1.2.2 Variance

Definition 1.2.3: Variance – Covariance – Correlation

Let $X \in L^2_{\mathbb{R}}(\Omega, \mathcal{A}, \mathbb{P})$ be a real-valued random variable.

1. **Variance.** The **variance** of the random variable X measures how much X varies around its mean. It is defined by

$$\mathrm{Var}(X) = \mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big].$$

The **standard deviation** of X is defined as

$$\sigma(X) = \sqrt{\mathrm{Var}(X)}.$$

2. **Covariance.** Let X and Y be two random variables in $L^2(\Omega, \mathcal{A}, \mathbb{P})$. The **covariance** of X and Y measures how the two variables vary together and is defined by

$$\mathrm{Cov}(X,Y) = \mathbb{E}\Big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\Big] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

3. **Correlation.** The **correlation** between X and Y is a normalized version of the covariance and is defined by

$$\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma(X)\sigma(Y)}.$$

It takes values between -1 and 1.

4. **Covariance matrix.** Let $\mathbf{X} = (X_1, \ldots, X_n)$ be a random vector in $\mathbb{R}^n$ with $\mathbf{X} \in L^2_{\mathbb{R}^n}(\Omega, \mathcal{A}, \mathbb{P})$. The **covariance matrix** of $\mathbf{X}$ is defined by

$$\mathrm{Var}(\mathbf{X}) = \mathbb{E}\Big[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T\Big] = \Big(\mathrm{Cov}(X_i, X_j)\Big)_{1 \le i,j \le n}.$$

This matrix belongs to $M_n(\mathbb{R})$.

Theorem 1.2.4: Markov Inequalities

Markov's Inequality If X is a nonnegative random variable and $a > 0$,

$$\mathbb{P}(X \ge a) \le \frac{1}{a}\mathbb{E}[X]$$

Bienaymé-Chebyshev Inequality If $X \in L^2(\Omega, \mathcal{A}, \mathbb{P})$ and $a > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \ge a) \le \frac{1}{a^2}\,\mathrm{var}(X)$$

Proof for Theorem.

A classical proof, left exercise for the reader.
Hint : Use $\mathbb{E}[\mathbb{1}_{X \ge a}] \le \mathbb{E}[X/a]$ for Markov's inequality, and apply Markov's inequality to the nonnegative random variable $(X - \mathbb{E}[X])^2$ for Chebyshev's inequality. ■

1.2.3 Characteristic Function

We start with a basic definition. We use the notation $x \cdot y$ for the standard Euclidean scalar product in $\mathbb{R}^d$.

Definition 1.2.5: Characteristic function

Let X be a random variable with values in $\mathbb{R}^d$. The characteristic function of X is the function $\Phi_X : \mathbb{R}^d \longrightarrow \mathbb{C}$ defined by

$$\Phi_X(\xi) := \mathbb{E}[\exp(i\xi \cdot X)], \qquad \xi \in \mathbb{R}^d.$$

The definition of expectation 1.2.1 allows us to write

$$\Phi_X(\xi) = \int_{\mathbb{R}^d} e^{i\xi\cdot x}\, d\mathbb{P}_X(x)$$

and we can also view Φ_X as the Fourier transform of the law $\mathbb{P}_X$ of X. For this reason, we sometimes write $\Phi_X(\xi) = \widehat{\mathbb{P}_X}(\xi)$. It's easy to notice (in the case $d = 1$), the dominated convergence theorem ensures that the function Φ_X is (bounded and) continuous on $\mathbb{R}^d$.

Our first goal is to show that the characteristic function determines the law $\mathbb{P}_X$ of X (as the name suggests!). This is equivalent to showing that the Fourier transform acting on probability measures on $\mathbb{R}^d$ is injective. We begin with an important calculation for a special case of the Gaussian distribution. We will study this case in more detail in Section 6, where we will derive the following formula:

$$\Phi_X(\xi) = \exp\left(im\xi - \frac{\sigma^2\xi^2}{2}\right), \quad \xi \in \mathbb{R}$$

Theorem 1.2.6: Characteristic function determines the law

The characteristic function of a random variable X with values in $\mathbb{R}^d$ determines the law of this random variable. In other words, the Fourier transform acting on probability measures on $\mathbb{R}^d$ is injective.

Proof for Theorem.

Let us consider the case $d = 1$. For every $\sigma > 0$, let g_σ be the density of the Gaussian $\mathcal{N}(0, \sigma^2)$ distribution:

$$g_\sigma(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}.$$

Let μ be a probability measure on $\mathbb{R}$, and set

$$\begin{cases} f_\sigma(x) & = \int_{\mathbb{R}} g_\sigma(x-y)\,\mu(dy) \overset{\text{(def)}}{=} g_\sigma * \mu(x), \\ d\mu_\sigma(x) & = f_\sigma(x)\,dx. \end{cases}$$

Let $C_b(\mathbb{R})$ denote the space of all bounded continuous functions $\varphi : \mathbb{R} \longrightarrow \mathbb{R}$, and recall that a probability measure ν on $\mathbb{R}$ is characterized by the values of $\int \varphi(x)\,d\nu(x)$ for $\varphi \in C_b(\mathbb{R})$. The result of the theorem (when $d = 1$) will follow if we are able to prove that:

1. μ_σ is determined by $\widehat{\mu}$.
2. For every $\varphi \in C_b(\mathbb{R})$, $\int \varphi(x)\mu_\sigma(dx) \longrightarrow \int \varphi(x)\mu(dx)$ as $\sigma \to 0$.

In order to establish (1), we use the formula of the characteristic function for gaussian distribution to write, for every $x \in \mathbb{R}$,

$$\sigma\sqrt{2\pi}\, g_\sigma(x) = \exp\left(-\frac{x^2}{2\sigma^2}\right) = \int_{\mathbb{R}} e^{i\xi x} g_{1/\sigma}(\xi)\, d\xi.$$

It follows that

$$\begin{aligned} f_\sigma(x) &= \int_{\mathbb{R}} g_\sigma(x-y)\, \mu(dy) \\ &= (\sigma\sqrt{2\pi})^{-1} \int_{\mathbb{R}} \left(\int_{\mathbb{R}} e^{i\xi(x-y)} g_{1/\sigma}(\xi)\, d\xi\right) \mu(dy) \\ &= (\sigma\sqrt{2\pi})^{-1} \int_{\mathbb{R}} e^{i\xi x} g_{1/\sigma}(\xi) \left(\int_{\mathbb{R}} e^{-i\xi y}\mu(dy)\right) d\xi \\ &= (\sigma\sqrt{2\pi})^{-1} \int_{\mathbb{R}} e^{i\xi x} g_{1/\sigma}(\xi)\, \widehat{\mu}(-\xi)\, d\xi. \end{aligned}$$

In the penultimate equality, we used the Fubini-Lebesgue theorem 2.1.4. The application of this theorem is easily justified since μ is a probability measure and the function $g_{1/\sigma}$ is integrable with respect to Lebesgue measure. The formula of the last display shows that f_σ (and hence μ_σ) is determined by $\widehat{\mu}$. As for (2), we start by writing, for every $\varphi \in C_b(\mathbb{R})$,

$$\begin{aligned} \int \varphi(x)\mu_\sigma(dx) &= \int \varphi(x) \left(\int g_\sigma(y-x)\mu(dy)\right) dx \\ &= \int \left(\int g_\sigma(y-x)\varphi(x)\, dx\right) \mu(dy) \\ &= \int g_\sigma * \varphi(y)\, \mu(dy). \end{aligned}$$

where we have again applied the Fubini-Lebesgue theorem, with the same justification as above, also the last equality use the convolution that we will see in 2.1. Then, we note that the properties

$$\begin{cases} \int g_\sigma(x)\, dx = 1, \\ \lim_{\sigma\to 0} \int_{\{|x|>\varepsilon\}} g_\sigma(x)\, dx = 0, \quad \forall \varepsilon > 0. \end{cases}$$

imply that, for every $y \in \mathbb{R}$,

$$\lim_{\sigma\to 0} g_\sigma * \varphi(y) = \varphi(y)$$

thanks to Proposition **??** (1) that we will see in the section measure product and independence 2.1. By dominated convergence, using the fact that $|g_\sigma * \varphi| \le \sup\{|\varphi(x)| : x \in \mathbb{R}\}$, we get

$$\lim_{\sigma\to 0} \int \varphi(x)\mu_\sigma(dx) = \lim_{\sigma\to 0} \int g_\sigma * \varphi(y)\mu(dy) = \int \varphi(x)\mu(dx),$$

which completes the proof of (2) and of the case $d = 1$ of the theorem.
The proof in the general case is similar. We now use the auxiliary functions

$$g_\sigma^{(d)}(x_1, \dots, x_d) = \prod_{j=1}^{d} g_\sigma(x_j)$$

noting that, for $\xi = (\xi_1, \dots, \xi_d) \in \mathbb{R}^d$,

$$\int_{\mathbb{R}^d} g_\sigma^{(d)}(x) e^{i\xi\cdot x}\, dx = \prod_{j=1}^{d} \int_{\mathbb{R}} g_\sigma(x_j) e^{i\xi_j x_j}\, dx_j = (2\pi/\sigma^2)^{d/2} g_{1/\sigma}^{(d)}(\xi).$$

The arguments of the case $d = 1$ then apply with obvious changes. ■

Proposition 1.2.7

Let $X = (X_1, \dots, X_d)$ be a random variable with values in $\mathbb{R}^d$. Assume that $\mathbb{E}[(X_j)^2] < \infty$ for every $j \in \{1, \dots, d\}$. Then, Φ_X is twice continuously differentiable and

$$\Phi_X(\xi) = 1 + i \sum_{j=1}^{d} \xi_j \mathbb{E}[X_j] - \frac{1}{2} \sum_{j=1}^{d} \sum_{k=1}^{d} \xi_j \xi_k \mathbb{E}[X_j X_k] + o(|\xi|^2)$$

when $\xi = (\xi_1, \dots, \xi_d)$ tends to 0.

Proof for Proposition.

By differentiating under the integral sign (easy application of dominance convergence theorem), we get, for every $1 \le j \le d$,

$$\frac{\partial \Phi_X}{\partial \xi_j}(\xi) = i\, \mathbb{E}[X_j e^{i\xi\cdot X}].$$

The justification is easy since $|iX_j e^{i\xi\cdot X}| = |X_j|$ and $X_j \in L^2(\Omega, \mathcal{A}, \mathbb{P}) \subset L^1(\Omega, \mathcal{A}, \mathbb{P})$. Similarly, for every $j, k \in \{1, \dots, d\}$, the fact that $X_j X_k \in L^1(\Omega, \mathcal{A}, \mathbb{P})$ allows us to differentiate once more and to get

$$\frac{\partial^2 \Phi_X}{\partial \xi_j \partial \xi_k}(\xi) = -\mathbb{E}[X_j X_k e^{i\xi\cdot X}].$$

Moreover, the dominated convergence theorem ensures that the quantity $\mathbb{E}[X_j X_k e^{i\xi\cdot X}]$ is a continuous function of ξ. We have thus proved that Φ_X is twice continuously differentiable.

Finally, the last assertion of the proposition is just Taylor's expansion of Φ_X at order two at the origin. ■

1.2.4 Laplace transform

For a random variable with values in $\mathbb{R}_+$, one often uses the Laplace transform instead of the characteristic function.

Definition 1.2.8: Laplace transform

Let X be a nonnegative random variable. The Laplace transform of (the law of) X is the function L_X defined on $\mathbb{R}_+$ by

$$L_X(\lambda) := \mathbb{E}[e^{-\lambda X}] = \int_{\mathbb{R}_+} e^{-\lambda x}\, \mathbb{P}_X(dx).$$

The dominated convergence theorem shows that L_X is continuous on $[0, \infty)$. An application of the theorem of differentiation under the integral [which is just easy result of convergence theorem] sign shows that L_X is infinitely differentiable on $(0, \infty)$. The function L_X is also convex (as a consequence of the fact that the functions $\lambda \mapsto e^{-\lambda x}$ are convex, for every $x \geq 0$) and thus has a (possibly infinite) right derivative at 0, which is given by

$$L'_X(0) = -\lim_{\lambda \downarrow 0} \mathbb{E}\left[\frac{1 - e^{-\lambda X}}{\lambda}\right] = -\mathbb{E}[X]$$

where the second equality follows from the monotone convergence theorem.

Theorem 1.2.9: Laplace transform determines the law

If X is a nonnegative random variable, the Laplace transform of X determines the law $\mathbb{P}_X$ of X.

Proof for Theorem.

To prove this theorem, the reader may follow the steps below:

1. Show that equality of the Laplace transforms implies equality of expectations for the exponential functions $x \mapsto e^{-\lambda x}$, and hence for all their finite linear combinations.
2. Apply the Stone–Weierstrass theorem to uniformly approximate every function in $C([0, \infty])$ by elements of the linear span of the exponential functions.
3. Pass to the limit using the uniform convergence obtained in the previous step to extend the equality of expectations to all continuous functions.
4. Conclude that the induced probability measures coincide by the measure-determining property of continuous functions with compact support.

■

Finally, in the case of integer-valued random variables, one often uses generating functions.

Definition 1.2.10: Generating function

Let X be a random variable with values in $\mathbb{N}$. The generating function of X is the function $g_X : [0,1] \to [0,1]$ defined by

$$g_X(r) := \mathbb{E}[r^X] = \sum_{n=0}^{\infty} \mathbb{P}(X = n)\, r^n.$$

Note that $g_X(0) = \mathbb{P}(X = 0)$ and $g_X(1) = 1$. The function g_X is continuous on $[0,1]$. Since the radius of convergence of the series in the definition is at least 1, the function g_X is infinitely differentiable on $[0,1)$. Moreover, it is obvious that g_X determines $\mathbb{P}_X$, since the quantities $\mathbb{P}(X = n)$ appear in the Taylor expansion of g_X at the origin.

The derivative of g_X is

$$g'_X(r) = \sum_{n=1}^{\infty} n\, \mathbb{P}(X = n)\, r^{n-1}$$

for $r \in [0,1)$. It follows that g_X has a (possibly infinite) left-derivative at $r = 1$, and

$$g'_X(1) = \sum_{n=1}^{\infty} n\, \mathbb{P}(X = n) = \mathbb{E}[X].$$

More generally, for every integer $p \geq 1$, if $g_X^{(p)}$ denotes the p-th derivative of g_X,

$$\lim_{r \uparrow 1} g_X^{(p)}(r) = \mathbb{E}[X(X-1)\cdots(X-p+1)]$$

which shows how to recover all moments of X from the knowledge of its generating function.

1.3 Lp Spaces

This section is mainly devoted to the study of the Lebesgue space $\mathcal{L}^p$ of random variables whose p-th power of the absolute value is having expectaiton (integrable) on a given probability measure. The fundamental inequalities of Hölder, Minkowski and Jensen provide essential ingredients for this study. We investigate the Banach space structure of $\mathcal{L}^p$, and in the special case $p = 2$, the Hilbert space structure of $\mathcal{L}^2$, which has important applications in probability theory.

Density theorems showing that any function of $\mathcal{L}^p$ can be well approximated by *smoother* functions play an important role in many analytic developments. As an application of the Hilbert space structure of L^2 , we establish the Radon-Nikodym theorem, which allows one to decompose an arbitrary measure as the sum of a measure having a density with respect to the reference measure and a singular measure. The Radon-Nikodym theorem will be a crucial ingredient of the theory of conditional expectations.

1.3.1 Definition of the space

Throughout this section, we consider a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. For a real $p \geq 1$, we let $\mathcal{L}^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ denote the space of all random variables $\mathbf{X} : \Omega \mapsto \mathbb{R}^d$ such that:

$$\int |\mathbf{X}|^p \, \mathrm{d}\mathbb{P} < \infty$$

We also introduce the space $\mathcal{L}^\infty_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ of all random variables $\mathbf{X} : \Omega \to \mathbb{R}$ such that there exists a constant $C \in \mathbb{R}_+$with

$$|\mathbf{X}| \leq C, \mathbb{P} \text{ a.e.}$$

For every $p \in [1, \infty]$, we define an equivalence relation on the set of p-integrable random variables by setting :

$$\mathbf{X} \equiv \mathbf{Y} \quad \text{if and only if} \quad \mathbf{X} = \mathbf{Y}, \ \mathbb{P}\text{-a.s.}$$

We then consider the quotient space

$$L^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P}) = \mathcal{L}^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P}) / \equiv,$$

An element of $L^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ is thus an equivalence class of random variables that are equal almost surely. In fact, $\langle X \rangle_c \in \mathcal{L}^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ is defined as :

$$\langle X \rangle_c = \Big\{ Y \in \mathcal{L}^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P}) \quad \text{such that } \mathbf{X} = \mathbf{Y} \quad \mathbb{P} - a.s. \Big\}$$

In fact we can define operation

In what follows, we will almost always abuse notation and identify an element of $L^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ with one of its representatives: when we speak of a random variable in $L^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$, we mean that a particular representative has been chosen, and all subsequent considerations do not depend on this choice.

We will often write $L^p(\Omega, \mathcal{A}, \mathbb{P})$, $L^p(\mathbb{P})$, $L^p_{\mathbb{R}^d}$ or simply L^p, instead of $L^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ when there is no risk of confusion. Notice that the space L^0 is the space of all random variables, the space L^1 is precisely the space of integrable random variables, and that the space L^2 is the space of square-integrable random variables, which is of particular importance in probability theory.

For every random variable $\mathbf{X}$ on $(\Omega, \mathcal{A}, \mathbb{P})$ and every $p \in [1, \infty)$, we set :

$$\|\mathbf{X}\|_p = \left(\mathbb{E}[|\mathbf{X}|^p]\right)^{1/p},$$

with the convention $\infty^{1/p} = \infty$, and

$$\|\mathbf{X}\|_\infty = \inf\{C \in [0, \infty] : |\mathbf{X}| \leq C, \ \mathbb{P}\text{-a.s.}\}.$$

These definitions are such that $|\mathbf{X}| \leq \|\mathbf{X}\|_\infty$ $\mathbb{P}$-a.s., and $\|\mathbf{X}\|_\infty$ is the smallest number in $[0, \infty]$ with this property. We observe that, if X and Y are two random variables such that $X = Y$ $\mathbb{P}$-a.s., then $\|\mathbf{X}\|_p = \|\mathbf{Y}\|_p$. Hence $\|\mathbf{X}\|_p$ is well defined for $\mathbf{X} \in L^p(\Omega, \mathcal{A}, \mathbb{P})$.

It is worth noting that when dealing with random variables taking values in $\mathbb{R}^d$, the absolute value can be replaced by the Euclidean norm, indeed for a for a vector $|\mathbf{x}|$ the euclidian norm is $\sqrt{x_1^1 + ... + x_d^2}$.

1.3.2 Principale inequalities

Definition 1.3.1: Conjugate exponents

Let $p, q \in [1, \infty]$. We say that p and q are *conjugate exponents* if

$$\frac{1}{p} + \frac{1}{q} = 1.$$

In particular, $p = 1$ and $q = \infty$

Theorem 1.3.2: Holder Inequality

Let p and q be conjugate exponents. Then, if $\mathbf{X}$ and $\mathbf{Y}$ are two random variables on $(\Omega, \mathcal{A}, \mathbb{P})$,

$$\mathbb{E}[|\mathbf{XY}|] \leq \|\mathbf{X}\|_p \, \|\mathbf{Y}\|_q.$$

In particular, $\mathbf{XY} \in L^1(\Omega, \mathcal{A}, \mathbb{P})$ whenever $\mathbf{X} \in L^p(\Omega, \mathcal{A}, \mathbb{P})$ and $\mathbf{Y} \in L^q(\Omega, \mathcal{A}, \mathbb{P})$.

Proof for Theorem.

If $\|\mathbf{X}\|_p = 0$, then $\mathbf{X} = 0$ $\mathbb{P}$-a.s., which implies $\mathbb{E}[|\mathbf{XY}|] = 0$, and the inequality is trivial. We may therefore assume that $\|\mathbf{X}\|_p > 0$ and $\|\mathbf{Y}\|_q > 0$. Without loss of generality, we may also assume that $\mathbf{X} \in L^p(\Omega, \mathcal{A}, \mathbb{P})$ and $\mathbf{Y} \in L^q(\Omega, \mathcal{A}, \mathbb{P})$ (otherwise $\|\mathbf{X}\|_p \|\mathbf{Y}\|_q = \infty$ and there is nothing to prove).

The case $p = 1$ and $q = \infty$ is immediate, since $|\mathbf{XY}| \leq \|\mathbf{Y}\|_\infty |\mathbf{X}|$ $\mathbb{P}$-a.s., which implies :

$$\mathbb{E}[|\mathbf{XY}|] \leq \|\mathbf{Y}\|_\infty \, \mathbb{E}[|\mathbf{X}|] = \|Y\|_\infty \, \|\mathbf{X}\|_1.$$

In what follows, we assume that $1 < p < \infty$ (and thus $1 < q < \infty$). Let $\alpha \in (0, 1)$. Then, for every $x \in \mathbb{R}_+$,

$$x^\alpha - \alpha x \leq 1 - \alpha.$$

Indeed, define $\varphi_\alpha(x) = x^\alpha - \alpha x$ for $x \geq 0$. For $x > 0$,

$$\varphi'_\alpha(x) = \alpha(x^{\alpha-1} - 1),$$

so that $\varphi'_\alpha(x) > 0$ for $x \in (0, 1)$ and $\varphi'_\alpha(x) < 0$ for $x \in (1, \infty)$. Hence φ_α attains its maximum at $x = 1$, which yields the inequality above.

Applying this inequality to $x = u/v$, where $u \geq 0$ and $v > 0$, we obtain

$$u^\alpha v^{1-\alpha} \leq \alpha u + (1 - \alpha)v,$$

and the inequality still holds when $v = 0$. We now take $\alpha = 1/p$ (so that $1 - \alpha = 1/q$) and set :

$$u = \frac{|\mathbf{X}(\omega)|^p}{\|\mathbf{X}\|_p^p}, \qquad v = \frac{|\mathbf{Y}(\omega)|^q}{\|\mathbf{Y}\|_q^q}.$$

This yields

$$\frac{|\mathbf{X}(\omega)Y(\omega)|}{\|\mathbf{X}\|_p\|\mathbf{Y}\|_q} \leq \frac{1}{p}\frac{|\mathbf{X}(\omega)|^p}{\|\mathbf{X}\|_p^p} + \frac{1}{q}\frac{|Y(\omega)|^q}{\|\mathbf{Y}\|_q^q}, \quad \mathbb{P}\text{-a.s.}$$

Taking expectations on both sides gives

$$\frac{1}{\|\mathbf{X}\|_p\|\mathbf{Y}\|_q}\,\mathbb{E}[|\mathbf{X}\mathbf{Y}|] \leq \frac{1}{p} + \frac{1}{q} = 1,$$

which completes the proof. ■

Theorem 1.3.3: Cauchy-Schwartz Inequality

If $\mathbf{X}$ and $\mathbf{Y}$ are two random variables on $(\Omega, \mathcal{A}, \mathbb{P})$, then

$$\mathbb{E}[|\mathbf{X}\mathbf{Y}|] \leq \left(\mathbb{E}\left[|\mathbf{X}|^2\right]\right)^{1/2} \left(\mathbb{E}\left[|\mathbf{Y}|^2\right]\right)^{1/2} = \|\mathbf{X}\|_2\, \|\mathbf{Y}\|_2.$$

Proof for Theorem.

Just a special case of Holder Inequality with $p = 2$ ■

Proposition 1.3.4

Suppose that $\mathbb{P}$ is a probability measure and let $p > 1$. Then, for any random variable $\mathbf{X} \in L^p(\Omega, \mathcal{A}, \mathbb{P})$,

$$\|\mathbf{X}\|_1 \leq \|\mathbf{X}\|_p.$$

Consequently, $L^p(\Omega, \mathcal{A}, \mathbb{P}) \subset L^1(\Omega, \mathcal{A}, \mathbb{P})$ for every $p \in (1, \infty]$. More generally, for any $1 \leq r < r' \leq \infty$ and any random variable $\mathbf{X} \in L^{r'}(\Omega, \mathcal{A}, \mathbb{P})$,

$$\|\mathbf{X}\|_r \leq \|\mathbf{X}\|_{r'},$$

and thus $L^{r'}(\Omega, \mathcal{A}, \mathbb{P}) \subset L^r(\Omega, \mathcal{A}, \mathbb{P})$.

Proof for Proposition.

The inequality $\|\mathbf{X}\|_1 \leq \|\mathbf{X}\|_p$ follows directly from Hölder's inequality by taking $\mathbf{Y} = 1$. For the second bound in the corollary, it suffices to replace $\mathbf{X}$ by $|\mathbf{X}|^r$ and to take $p = r'/r$. ■

Theorem 1.3.5: Jensen Inequality

Suppose that $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{A})$, and let $\varphi : \mathbb{R} \to \mathbb{R}_+$ be a convex function. Then, for every integrable real-valued random variable $X \in L^1_{\mathbb{R}}(\Omega, \mathcal{A}, \mathbb{P})$,

$$\mathbb{E}[\varphi(X)] \geq \varphi(\mathbb{E}[X]) .$$

remark : The expectation $\mathbb{E}[\varphi(X)]$ is well defined, since $\varphi(X)$ is a nonnegative measurable random variable.

Proof for Theorem.

Set

$$\mathcal{E}_\varphi = \{(a,b) \in \mathbb{R}^2 : \forall x \in \mathbb{R},\ \varphi(x) \geq ax + b\}.$$

Elementary properties of convex functions show that

$$\forall x \in \mathbb{R}, \quad \varphi(x) = \sup_{(a,b)\in\mathcal{E}_\varphi} (ax+b).$$

Since $\varphi(X) \geq aX + b$ almost surely for every $(a,b) \in \mathcal{E}_\varphi$, we obtain

$$\mathbb{E}[\varphi(X)] \geq \sup_{(a,b)\in\mathcal{E}_\varphi} \mathbb{E}[aX+b] = \sup_{(a,b)\in\mathcal{E}_\varphi} \Big(a\,\mathbb{E}[X] + b\Big) = \varphi(\mathbb{E}[X]) .$$

■

Theorem 1.3.6: Minkowski's Inequality

Let $p \in [1, \infty]$, and let $\mathbf{X}, \mathbf{Y} \in L^p(\Omega, \mathcal{A}, \mathbb{P})$. Then :

$$\mathbf{X} + \mathbf{Y} \in L^p(\Omega, \mathcal{A}, \mathbb{P}) \quad \text{and} \quad \|\mathbf{X} + \mathbf{Y}\|_p \leq \|\mathbf{X}\|_p + \|\mathbf{Y}\|_p.$$

Proof for Theorem.

The cases $p = 1$ and $p = \infty$ are immediate, using the inequality $|\mathbf{X} + \mathbf{Y}| \leq |\mathbf{X}| + |\mathbf{Y}|$. We therefore assume that $1 < p < \infty$.

Writing

$$|\mathbf{X}+\mathbf{Y}|^p \leq (|\mathbf{X}| + |\mathbf{Y}|)^p \leq \Big(2\max(|\mathbf{X}|, |\mathbf{Y}|)\Big)^p \leq 2^p\Big(|\mathbf{X}|^p + |\mathbf{Y}|^p\Big),$$

we see that $\mathbb{E}[|\mathbf{X}+\mathbf{Y}|^p] < \infty$, and hence $\mathbf{X}+\mathbf{Y} \in L^p$.

Next, observe that

$$|\mathbf{X}+\mathbf{Y}|^p = |\mathbf{X}+\mathbf{Y}| \cdot |\mathbf{X}+\mathbf{Y}|^{p-1} \leq |\mathbf{X}|\,|\mathbf{X}+\mathbf{Y}|^{p-1} + |\mathbf{Y}|\,|\mathbf{X}+\mathbf{Y}|^{p-1}.$$

Taking expectations yields

$$\mathbb{E}[|\mathbf{X}+\mathbf{Y}|^p] \leq \mathbb{E}\Big[|\mathbf{X}|\,|\mathbf{X}+\mathbf{Y}|^{p-1}\Big] + \mathbb{E}\Big[|\mathbf{Y}|\,|\mathbf{X}+\mathbf{Y}|^{p-1}\Big] .$$

Applying Hölder's inequality with conjugate exponents p and $q = p/(p-1)$, we obtain

$$\mathbb{E}[|\mathbf{X}+\mathbf{Y}|^p] \leq \|\mathbf{X}\|_p \left(\mathbb{E}[|\mathbf{X}+\mathbf{Y}|^p]\right)^{(p-1)/p} + \|\mathbf{Y}\|_p \left(\mathbb{E}[|\mathbf{X}+\mathbf{Y}|^p]\right)^{(p-1)/p}.$$

If $\mathbb{E}[|\mathbf{X}+\mathbf{Y}|^p] = 0$, the inequality is trivial. Otherwise, dividing both sides by $(\mathbb{E}[|\mathbf{X}+\mathbf{Y}|^p])^{(p-1)/p}$ yields

$$\|\mathbf{X}+\mathbf{Y}\|_p \leq \|\mathbf{X}\|_p + \|\mathbf{Y}\|_p,$$

which completes the proof. ■

1.3.3 Topological structure of Lp-space

Proposition 1.3.7

The formula

$$d(X,Y) = \mathbb{E}\Big[\,|X-Y| \wedge 1\,\Big]$$

defines a distance on $L^0_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$, and the space $L^0_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ is complete for the distance d.

Proof for Proposition.

Let's prove that the space $L^0_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ is complete for the distance d. So let $(X_n)_{n\in\mathbb{N}}$ be a Cauchy sequence for the distance d. We can then find a subsequence $Y_k = X_{n_k}$, $k \in \mathbb{N}$, such that for every $k \geq 1$,

$$d(Y_k, Y_{k+1}) \leq 2^{-k}.$$

Then,

$$\mathbb{E}\left[\sum_{k=1}^{\infty}\Big(|Y_{k+1}-Y_k| \wedge 1\Big)\right] = \sum_{k=1}^{\infty} d(Y_k, Y_{k+1}) < \infty,$$

which implies that $\sum_{k=1}^{\infty}(|Y_{k+1}-Y_k| \wedge 1) < \infty$ almost surely, and therefore also $\sum_{k=1}^{\infty}|Y_{k+1}-Y_k| < \infty$ almost surely (there are almost surely only finitely many values of k such that $|Y_{k+1}-Y_k| \geq 1$). We then define a random variable in $L^0_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ by setting

$$X = Y_1 + \sum_{k=1}^{\infty}(Y_{k+1}-Y_k),$$

on the event where the series converges absolutely, and $X = 0$ on the complementary event, which has zero probability. By construction, the sequence $(Y_k)_{k\in\mathbb{N}}$ converges almost surely to X, and this implies

$$d(Y_k, X) = \mathbb{E}\Big[\,|Y_k - X| \wedge 1\,\Big] \longrightarrow 0 \quad \text{as } k \to \infty,$$

by an application of the dominated convergence theorem. Hence the sequence $(Y_k)_{k\in\mathbb{N}}$ converges to X for the distance d, and since the sequence $(X_n)_{n\in\mathbb{N}}$ is Cauchy and has a convergent subsequence, it must also converge. ■

Theorem 1.3.8: Riesz Theorem

For every $p \in [1, \infty]$, the space $L^p(\Omega, \mathcal{A}, \mathbb{P})$ equipped with the norm $\mathbf{X} \mapsto \|\mathbf{X}\|_p$ is a Banach space, that is, a complete normed linear space.

Proof for Theorem.

Can be found in [3, p. 68] ■

Theorem 1.3.9: The Hilbert Space L2

The space $L^2_{\mathbb{R}}(\Omega, \mathcal{A}, \mathbb{P})$ equipped with the scalar product

$$\langle X, Y \rangle = \int XY \, d\mathbb{P} = \mathbb{E}(XY)$$

is a real Hilbert space.

Proof for Theorem.

The Cauchy-Schwarz inequality shows that, if $X, Y \in L^2(\Omega, \mathcal{A}, \mathbb{P})$, then XY is integrable and hence $\langle X, Y \rangle$ is well defined. It is then clear that the mapping $(X, Y) \mapsto \langle X, Y \rangle$ defines an inner product on $L^2(\Omega, \mathcal{A}, \mathbb{P})$, and that $\langle X, X \rangle = \|X\|_2^2$. Finally, the fact that $\left(L^2(\Omega, \mathcal{A}, \mathbb{P}), \|\cdot\|_2\right)$ is complete is a special case of the Riesz theorem 5.2.1. ■

Classical results from the theory of Hilbert spaces [see appendix 5] can therefore be applied to $L^2(\Omega, \mathcal{A}, \mathbb{P})$. In particular, if $\Phi : L^2_{\mathbb{R}}(\Omega, \mathcal{A}, \mathbb{P}) \to \mathbb{R}$ is a continuous linear functional, then there exists a unique element $Y \in L^2_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ such that

$$\Phi(X) = \langle X, Y \rangle \quad \text{for every } X \in L^2_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P}),$$

where $\langle X, Y \rangle = \mathbb{E}[XY]$; .

1.3.4 The Radon-Nikodym Theorem

Definition 1.3.10: Measures continuity

Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on $(\Omega, \mathcal{A})$. We say that:

1. $\mathbb{Q}$ is *absolutely continuous* with respect to $\mathbb{P}$ (notation $\mathbb{Q} \ll \mathbb{P}$) if

$$\forall A \in \mathcal{A}, \quad \mathbb{P}(A) = 0 \Rightarrow \mathbb{Q}(A) = 0.$$

2. $\mathbb{Q}$ is *singular* with respect to $\mathbb{P}$ (notation $\mathbb{Q} \perp \mathbb{P}$) if there exists $\Omega_0 \in \mathcal{A}$ such that

$$\mathbb{P}(\Omega_0) = 0 \quad \text{and} \quad \mathbb{Q}(\Omega_0^c) = 0.$$

3. $\mathbb{P}$ and $\mathbb{Q}$ are said to be *equivalent* (notation $\mathbb{P} \sim \mathbb{Q}$) if

$$\mathbb{P} \ll \mathbb{Q} \quad \text{and} \quad \mathbb{Q} \ll \mathbb{P}.$$

Now we present the theorem of Radon-Nikodym :

Theorem 1.3.11: Radon-Nikodym Theorem

Suppose that $\mathbb{P}$ is a σ-finite measure [a] on $(\Omega, \mathcal{A})$, and let $\mathbb{Q}$ be another σ-finite measure on $(\Omega, \mathcal{A})$. Then there exists a unique pair $(\mathbb{Q}_a, \mathbb{Q}_s)$ of σ-finite measures on $(\Omega, \mathcal{A})$ such that:

1. $\mathbb{Q} = \mathbb{Q}_a + \mathbb{Q}_s$,
2. $\mathbb{Q}_a \ll \mathbb{P}$ and $\mathbb{Q}_s \perp \mathbb{P}$.

Moreover, there exists a unique [b] nonnegative measurable function $g : \Omega \to \mathbb{R}_+$ such that

$$d\mathbb{Q}_a = g \cdot d\mathbb{P},$$

meaning that

$$\forall A \in \mathcal{A}, \qquad \mathbb{Q}_a(A) = \int_A g \, d\mathbb{P}.$$

a. A measure $\mathbb{P}$ on $(\Omega, \mathcal{A})$ is called *finite* if $\mathbb{P}(\Omega) < \infty$. It is called *σ-finite* if there exists a sequence of measurable sets $(A_n)_{n\geq 1}$ such that $\Omega = \bigcup_{n=1}^{\infty} A_n$ and $\mathbb{P}(A_n) < \infty$ for all n.

b. Uniqueness is in the sense that if $\tilde{g}$ is another measurable function satisfying the same properties, then $\tilde{g} = g$ $\mathbb{P}$-a.s.

Proof for Theorem.

Could be found in [3, p. 76] proven in the cadre of measure theory. ∎

Remark: A useful formula one can use in *change of measures* in Radon-Nikodym calculations is this :

$$d\mathbb{Q} = g \cdot d\mathbb{P} \qquad \textit{then} \qquad \int_A f d\mathbb{Q} = \int_A f \cdot g \cdot d\mathbb{P}$$

Theorem 1.3.12: Radon-Nikodym Theorem in Probability

Suppose that $\mathbb{P}$ and $\mathbb{Q}$ are two probability measures such that $\mathbb{Q} \ll \mathbb{P}$ then :
There exists a unique [a] nonnegative random variable $X : \Omega \to \mathbb{R}_+$ such that

$$d\mathbb{Q} = X \cdot d\mathbb{P},$$

Furthermore, if $\mathbb{P} \ll \mathbb{Q}$ then $1/X$ satisfies :

$$d\mathbb{P} = \frac{1}{X} \cdot d\mathbb{Q},$$

a. Uniqueness is in the sense that if $\tilde{X}$ is another random variable satisfying the same properties, then $\tilde{X} = X$ $\mathbb{P}$-a.s.

Proposition 1.3.13

Let $\mathbb{P}$ be a σ-finite measure on $(\Omega, \mathcal{A})$, and let $\mathbb{Q}$ be a finite measure on $(\Omega, \mathcal{A})$. The following conditions are equivalent:

1. $\mathbb{Q} \ll \mathbb{P}$;
2. For every $\varepsilon > 0$, there exists $\delta > 0$ such that, for every $A \in \mathcal{A}$,

$$\mathbb{P}(A) < \delta \implies \mathbb{Q}(A) < \varepsilon.$$

Proof for Proposition.

The implication (ii) $\Rightarrow$ (i) is immediate. Conversely, suppose that $\mathbb{Q} \ll \mathbb{P}$. By the Radon-Nikodym theorem 1.3.11, there exists a nonnegative measurable function $g : \Omega \to \mathbb{R}_+$ such that

$$d\mathbb{Q} = g \cdot d\mathbb{P}.$$

Note that

$$\int_\Omega g \, d\mathbb{P} = \mathbb{Q}(\Omega) < \infty.$$

The dominated convergence theorem then yields

$$\int_\Omega \mathbb{1}_{\{g>n\}} \, g \, d\mathbb{P} \xrightarrow[n\to\infty]{} 0.$$

Given $\varepsilon > 0$, we may therefore choose $M \in \mathbb{N}$ large enough such that

$$\int_\Omega \mathbb{1}_{\{g>M\}} \, g \, d\mathbb{P} < \frac{\varepsilon}{2}.$$

Setting $\delta = \varepsilon/(2M)$, we obtain, for every $A \in \mathcal{A}$ such that $\mathbb{P}(A) < \delta$,

$$\mathbb{Q}(A) = \int_A g \, d\mathbb{P} \le \int_\Omega \mathbb{1}_{\{g>M\}} \, g \, d\mathbb{P} + \int_{A\cap\{g\le M\}} g \, d\mathbb{P} < \frac{\varepsilon}{2} + M \, \mathbb{P}(A) \le \varepsilon.$$

This proves (i) $\Rightarrow$ (ii) and completes the proof. ∎

CHAPITRE 2

INDEPENDENCE, CONDITIONING AND ASYMPTOTIC THEORY

2.1 Measure product and Independence

The notion of independence is one of the first genuinely new concepts introduced in probability theory, in the sense that it does not arise as a straightforward adaptation of a corresponding notion from measure theory. While the independence of two events or two random variables admits a natural intuitive interpretation, the most fundamental and structurally relevant notion is that of independence between two or more σ-fields .

A central result shows that two random variables are independent if and only if the joint distribution of the pair coincides with the product of their marginal distributions. Combined with Fubini's theorem, which we present first, this characterization yields several equivalent and practical formulations of independence.

We then establish the classical Borel-Cantelli lemma and illustrate its strength through an application to the dyadic expansion of a randomly chosen real number, leading to several striking probabilistic properties.

2.1.1 Probabilities measures product

Let $(\Omega_1, \mathcal{A}_1)$ and $(\Omega_2, \mathcal{A}_2)$ be two probabilistic spaces. The product σ-field $\mathcal{A}_1 \otimes \mathcal{A}_2$ is defined as the σ-field on $\Omega_1 \times \Omega_2$ given by

$$\mathcal{A}_1 \otimes \mathcal{A}_2 = \sigma\Big(A \times B : A \in \mathcal{A}_1,\ B \in \mathcal{A}_2\Big).$$

Sets of the form $A \times B$, with $A \in \mathcal{A}_1$ and $B \in \mathcal{A}_2$, are called *measurable rectangles*.

remark : The definition of the product σ-field extends naturally to a finite collection of probabilistic spaces $(\Omega_1, \mathcal{A}_1), \ldots, (\Omega_n, \mathcal{A}_n)$ by setting

$$\mathcal{A}_1 \otimes \cdots \otimes \mathcal{A}_n = \sigma(\mathcal{A}_1 \times \cdots \times \mathcal{A}_n).$$

The expected associativity properties hold; for instance, if $n = 3$,

$$(\mathcal{A}_1 \otimes \mathcal{A}_2) \otimes \mathcal{A}_3 = \mathcal{A}_1 \otimes (\mathcal{A}_2 \otimes \mathcal{A}_3) = \mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \mathcal{A}_3.$$

Let $C \subset \Omega_1 \times \Omega_2$ and let $\omega_1 \in \Omega_1$. We define the *section* of C at ω_1 by

$$C_{\omega_1} = \{\omega_2 \in \Omega_2 : (\omega_1, \omega_2) \in C\}.$$

Similarly, for $\omega_2 \in \Omega_2$, define

$$C^{\omega_2} = \{\omega_1 \in \Omega_1 : (\omega_1, \omega_2) \in C\}.$$

If X is a random variable on $\Omega_1 \times \Omega_2$ and $\omega_1 \in \Omega_1$, we denote by X_{ω_1} the random variable on Ω_2 defined by

$$X_{\omega_1}(\omega_2) = X(\omega_1, \omega_2).$$

Similarly, for $\omega_2 \in \Omega_2$, we define $X^{\omega_2}(\omega_1) = X(\omega_1, \omega_2)$.

Proposition 2.1.1

1. If $C \in \mathcal{A}_1 \otimes \mathcal{A}_2$, then for every $\omega_1 \in \Omega_1$, the section C_{ω_1} belongs to $\mathcal{A}_2$, and for every $\omega_2 \in \Omega_2$, the section C^{ω_2} belongs to $\mathcal{A}_1$.
2. Let X be a random variable on $\Omega_1 \times \Omega_2$ with respect to $\mathcal{A}_1 \otimes \mathcal{A}_2$. Then, for every $\omega_1 \in \Omega_1$, X_{ω_1} is $\mathcal{A}_2$-measurable, and for every $\omega_2 \in \Omega_2$, X^{ω_2} is $\mathcal{A}_1$-measurable.

Proof for Proposition.

1. Fix $\omega_1 \in \Omega_1$ and Consider the class

$$\mathcal{G} = \{C \in \mathcal{A}_1 \otimes \mathcal{A}_2 : C_{\omega_1} \in \mathcal{A}_2\}.$$

Let $\mathcal{R} = \{A \times B : \ A \in \mathcal{A}_1, \ B \in \mathcal{A}_2\}$. One checks that:

- $\mathcal{R}$ is an algebra generating $\mathcal{A}_1 \otimes \mathcal{A}_2$;
- $\mathcal{R} \subset \mathcal{G}$ (if $C = A \times B$, we have $C_{\omega_1} = B$ or $C_{\omega_1} = \emptyset$ according as $\omega_1 \in A$ or $\omega_1 \notin A$) ;
- $\mathcal{G}$ is a monotone class.

Hence, by the Monotone Class Theorem,

$$\mathcal{A}_1 \otimes \mathcal{A}_2 = \sigma(\mathcal{R}) \subset \mathcal{G},$$

proving that $C_{\omega_1} \in \mathcal{A}_2$ for every $C \in \mathcal{A}_1 \otimes \mathcal{A}_2$. The property $C^{\omega_2} \in \mathcal{A}_1$ is derived in a similar manner.

2. For every measurable subset D of G,

$$\begin{aligned} X_{\omega_1}^{-1}(D) &= \{\omega_2 \in \Omega_2 : X_{\omega_1}(\omega_2) \in D\} \\ &= \{\omega_2 \in \Omega_2 : (\omega_1, \omega_2) \in X^{-1}(D)\} \\ &= (X^{-1}(D))_{\omega_1}. \end{aligned}$$

which belongs to $\mathcal{A}_2$ by part (1).

■

Theorem 2.1.2: Existence of product measure

Let $\mathbb{P}_1$ and $\mathbb{P}_2$ be two probability measures on $(\Omega_1, \mathcal{A}_1)$ and $(\Omega_2, \mathcal{A}_2)$, respectively.

1. There exists a unique measure $\mathbb{P}_1 \otimes \mathbb{P}_2$ on $(\Omega_1 \times \Omega_2, \mathcal{A}_1 \otimes \mathcal{A}_2)$ such that

$$(\mathbb{P}_1 \otimes \mathbb{P}_2)(A \times B) = \mathbb{P}_1(A)\,\mathbb{P}_2(B), \qquad A \in \mathcal{A}_1,\ B \in \mathcal{A}_2,$$

2. For every $C \in \mathcal{A}_1 \otimes \mathcal{A}_2$, the mappings

$$\omega_1 \mapsto \mathbb{P}_2(C_{\omega_1}) \quad \text{and} \quad \omega_2 \mapsto \mathbb{P}_1(C^{\omega_2})$$

are measurable (ie random variables), and

$$(\mathbb{P}_1 \otimes \mathbb{P}_2)(C) = \int_{\Omega_1} \mathbb{P}_2(C_{\omega_1})\, d\mathbb{P}_1(\omega_1) = \int_{\Omega_2} \mathbb{P}_1(C^{\omega_2})\, d\mathbb{P}_2(\omega_2).$$

Proof for Theorem.

1. **Uniqueness.** Left as an exercise. Apply the Monotone Class Theorem (Theorem 1.1.5) to

$$\mathcal{C} = \{D \in \mathcal{A}_1 \otimes \mathcal{A}_2 :\ p(D) = q(D)\},$$

where p and q are probability measures satisfying

$$p(A \times B) = q(A \times B) = \mathbb{P}_1(A)\mathbb{P}_2(B), \qquad A \in \mathcal{A}_1,\ B \in \mathcal{A}_2.$$

(See Figure 1 for the general proof strategy.)

Existence. Motivated by the last identity stated in the theorem, define

$$\mathbb{P}(C) := \int_{\Omega_1} \mathbb{P}_2(C_{\omega_1})\, d\mathbb{P}_1(\omega_1), \qquad C \in \mathcal{A}_1 \otimes \mathcal{A}_2.$$

We verify that this mapping is well defined and defines a probability measure satisfying

$$\mathbb{P}(A \times B) = \mathbb{P}_1(A)\mathbb{P}_2(B), \qquad A \in \mathcal{A}_1,\ B \in \mathcal{A}_2.$$

- *Well-definedness.* By Proposition 2.1.1, the section C_{ω_1} belongs to $\mathcal{A}_2$, so

$\mathbb{P}_2(C_{\omega_1})$ is well defined. It remains to prove that

$$\phi_C : \omega_1 \longmapsto \mathbb{P}_2(C_{\omega_1})$$

is $\mathcal{A}_1$-measurable.
Consider the class

$$\mathcal{G} = \{C \in \mathcal{A}_1 \otimes \mathcal{A}_2 : \phi_C \text{ is } \mathcal{A}_1\text{-measurable}\}.$$

ANd use the Monotone Class Theorem to prove :

$$\mathcal{A}_1 \otimes \mathcal{A}_2 = \sigma(\mathcal{R}) \subset \mathcal{G},$$

proving that ϕ_C is measurable for every $C \in \mathcal{A}_1 \otimes \mathcal{A}_2$.

- $\mathbb{P}$ *is a measure.* Let $(C_n)_{n\geq 1}$ be pairwise disjoint subsets of $\mathcal{A}_1 \otimes \mathcal{A}_2$. Since the sections $(C_n)_{\omega_1}$ are pairwise disjoint for every ω_1,

$$\begin{aligned} \mathbb{P}\left(\bigcup_{n=1}^{\infty} C_n\right) &= \int_{\Omega_1} \mathbb{P}_2\left(\left(\bigcup_{n=1}^{\infty} C_n\right)_{\omega_1}\right) d\mathbb{P}_1(\omega_1) \\ &= \int_{\Omega_1} \sum_{n=1}^{\infty} \mathbb{P}_2((C_n)_{\omega_1})\, d\mathbb{P}_1(\omega_1) \\ &= \sum_{n=1}^{\infty} \int_{\Omega_1} \mathbb{P}_2((C_n)_{\omega_1})\, d\mathbb{P}_1(\omega_1) \\ &= \sum_{n=1}^{\infty} \mathbb{P}(C_n), \end{aligned}$$

where the third equality follows from the Monotone Convergence Theorem. Thus $\mathbb{P}$ is countably additive.
Finally, for every $A \in \mathcal{A}_1$ and $B \in \mathcal{A}_2$,

$$\mathbb{P}(A \times B) = \int_{\Omega_1} \mathbb{P}_2((A \times B)_{\omega_1})\, d\mathbb{P}_1(\omega_1) = \mathbb{P}_1(A)\mathbb{P}_2(B).$$

Therefore, $\mathbb{P}$ is a probability measure satisfying the required property.

The final identity of the theorem is established similarly. Indeed, the corresponding integrand is well defined by the same measurability argument used above. ■

Remarks:

1. Suppose now that we consider an arbitrary finite number of σ-finite measures $\mathbb{P}_1, \ldots, \mathbb{P}_n$, defined on $(E_1, \mathcal{A}_1), \ldots, (E_n, \mathcal{A}_n)$ respectively. We can then define the product measure

$$\mathbb{P}_1 \otimes \cdots \otimes \mathbb{P}_n$$

on

$$(E_1 \times \cdots \times E_n,\ \mathcal{A}_1 \otimes \cdots \otimes \mathcal{A}_n)$$

by setting

$$\mathbb{P}_1 \otimes \cdots \otimes \mathbb{P}_n = \mathbb{P}_1 \otimes (\mathbb{P}_2 \otimes (\cdots \otimes \mathbb{P}_n)).$$

The way we insert parentheses is in fact unimportant since $\mathbb{P}_1 \otimes \cdots \otimes \mathbb{P}_n$ is characterized by its values on (measurable) rectangles,

$$\mathbb{P}_1 \otimes \cdots \otimes \mathbb{P}_n(A_1 \times \cdots \times A_n) = \mathbb{P}_1(A_1) \cdots \mathbb{P}_n(A_n).$$

Theorem 2.1.3: Fubini-Tonelli Theorem

Let $(\Omega_1, \mathcal{A}_1)$ and $(\Omega_2, \mathcal{A}_2)$ be probabilistic spaces, and equip $\Omega_1 \times \Omega_2$ with the product σ-field $\mathcal{A}_1 \otimes \mathcal{A}_2$ and Let $\mathbb{P}_1$ and $\mathbb{P}_2$ be two probability measures on $(\Omega_1, \mathcal{A}_1)$ and $(\Omega_2, \mathcal{A}_2)$, respectively, and let $X : \Omega_1 \times \Omega_2 \to [0, \infty]$ be a positive random variable.

1. The functions

$$\Omega_1 \ni \omega_1 \longmapsto \int_{\Omega_2} X(\omega_1, \omega_2)\, d\mathbb{P}_2(\omega_2), \qquad \Omega_2 \ni \omega_2 \longmapsto \int_{\Omega_1} X(\omega_1, \omega_2)\, d\mathbb{P}_1(\omega_1),$$

   with values in $[0, \infty]$, are $\mathcal{A}_1$-measurable and $\mathcal{A}_2$-measurable, respectively (ie. they are random variables).

2. We have :

$$\begin{aligned}\int_{\Omega_1 \times \Omega_2} X\, d(\mathbb{P}_1 \otimes \mathbb{P}_2) &= \int_{\Omega_1} \left(\int_{\Omega_2} X(\omega_1, \omega_2)\, d\mathbb{P}_2(\omega_2) \right) d\mathbb{P}_1(\omega_1) \\ &= \int_{\Omega_2} \left(\int_{\Omega_1} X(\omega_1, \omega_2)\, d\mathbb{P}_1(\omega_1) \right) d\mathbb{P}_2(\omega_2).\end{aligned}$$

   Or more simply :

$$\mathbb{E}_{\mathbb{P}_1 \otimes \mathbb{P}_2}[X] = \mathbb{E}_{\mathbb{P}_1}\Big[\mathbb{E}_{\mathbb{P}_2}\big[X(\omega_1, \cdot)\big]\Big] = \mathbb{E}_{\mathbb{P}_2}\Big[\mathbb{E}_{\mathbb{P}_1}\big[X(\cdot, \omega_2)\big]\Big].$$

Proof for Theorem.

We give the proof for the special case $X = \mathbb{1}_C$ for some $C \in \mathcal{A}_1 \otimes \mathcal{A}_2$. The general case can be deduced from this special case by applying the monotone convergence theorem to the sequence of random variables $X_n \xrightarrow[n\to\infty]{} X$.
In fact, let $X = \mathbb{1}_C$ for some $C \in \mathcal{A}_1 \otimes \mathcal{A}_2$ the functions in the theorem are reduced to :

$$\Omega_1 \ni \omega_1 \longmapsto \int_{\Omega_2} \mathbb{1}_{C_{\omega_1}}(\omega_2), d\mathbb{P}_2(\omega_2), \qquad \Omega_2 \ni \omega_2 \longmapsto \int_{\Omega_1} \mathbb{1}_{C^{\omega_2}}(\omega_1)\, d\mathbb{P}_1(\omega_1),$$

Which are exactly :

$$\omega_1 \mapsto \mathbb{P}_2(C_{\omega_1}) \quad \text{and} \quad \omega_2 \mapsto \mathbb{P}_1(C^{\omega_2})$$

By the second point of Theorem 2.1.2, these functions are measurable, and we have

$$(\mathbb{P}_1 \otimes \mathbb{P}_2)(C) = \int_{\Omega_1} \mathbb{P}_2(C_{\omega_1})\, d\mathbb{P}_1(\omega_1) = \int_{\Omega_2} \mathbb{P}_1(C^{\omega_2})\, d\mathbb{P}_2(\omega_2).$$

Which is :

$$\mathbb{E}_{\mathbb{P}_1\otimes\mathbb{P}_2}[X] = \mathbb{E}_{\mathbb{P}_1}\Big[\mathbb{E}_{\mathbb{P}_2}\big[X(\omega_1,\cdot)\big]\Big] = \mathbb{E}_{\mathbb{P}_2}\Big[\mathbb{E}_{\mathbb{P}_1}\big[X(\cdot,\omega_2)\big]\Big].$$

■

Theorem 2.1.4: Fubini-Lebesgue Theorem

Let $X \in L^1(\Omega_1 \times \Omega_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \mathbb{P}_1 \otimes \mathbb{P}_2)$. Then:

1. For $\mathbb{P}_1$-almost every $\omega_1 \in \Omega_1$, the random variable $\omega_2 \mapsto X(\omega_1,\omega_2)$ belongs to $L^1(\Omega_2, \mathcal{A}_2, \mathbb{P}_2)$, and for $\mathbb{P}_2$-almost every $\omega_2 \in \Omega_2$, the random variable $\omega_1 \mapsto X(\omega_1,\omega_2)$ belongs to $L^1(\Omega_1, \mathcal{A}_1, \mathbb{P}_1)$.
2. The functions

$$\omega_1 \longmapsto \int_{\Omega_2} X(\omega_1,\omega_2)\, d\mathbb{P}_2(\omega_2), \qquad \omega_2 \longmapsto \int_{\Omega_1} X(\omega_1,\omega_2)\, d\mathbb{P}_1(\omega_1),$$

   belong to $L^1(\Omega_1, \mathcal{A}_1, \mathbb{P}_1)$ and $L^1(\Omega_2, \mathcal{A}_2, \mathbb{P}_2)$, respectively.
3. We have :

$$\begin{aligned}\int_{\Omega_1\times\Omega_2} X\, d(\mathbb{P}_1\otimes\mathbb{P}_2) &= \int_{\Omega_1}\Big(\int_{\Omega_2} X(\omega_1,\omega_2)\, d\mathbb{P}_2(\omega_2)\Big)\, d\mathbb{P}_1(\omega_1)\\ &= \int_{\Omega_2}\Big(\int_{\Omega_1} X(\omega_1,\omega_2)\, d\mathbb{P}_1(\omega_1)\Big)\, d\mathbb{P}_2(\omega_2).\end{aligned}$$

   Or more simply :

$$\mathbb{E}_{\mathbb{P}_1\otimes\mathbb{P}_2}[X] = \mathbb{E}_{\mathbb{P}_1}\Big[\mathbb{E}_{\mathbb{P}_2}\big[X(\omega_1,\cdot)\big]\Big] = \mathbb{E}_{\mathbb{P}_2}\Big[\mathbb{E}_{\mathbb{P}_1}\big[X(\cdot,\omega_2)\big]\Big].$$

Proof for Theorem.

1. By theorem 2.1.3 applied to $|X|$, we have

$$\int_{\Omega_1}\Big(\int_{\Omega_2} |X(\omega_1,\omega_2)|\, d\mathbb{P}_2(\omega_2)\Big)\, d\mathbb{P}_1(\omega_1) = \int |X|\, d\mathbb{P}_1\otimes\mathbb{P}_2 < \infty.$$

   Thanks to propertry (5) following definition of expectation 1.2.1, this implies that

$$\int_{\Omega_2} |X(\omega_1,\omega_2)|\, d\mathbb{P}_2(\omega_2) < \infty \quad \text{for } \mathbb{P}_1\text{-a.e. } \omega_1.$$

   Thus, the function $\omega_1 \mapsto X(\omega_1,\omega_2)$ (which is measurable by proposition 2.1.1) belongs to $L^1(\Omega_2, \mathcal{A}_2, \mathbb{P}_2)$ except on the $\mathcal{A}_1$-measurable set

$$N := \left\{\omega_1 \in \Omega_1 : \int_{\Omega_2} |X(\omega_1,\omega_2)|\, d\mathbb{P}_2(\omega_2) = \infty\right\},$$

   which has zero $\mathbb{P}_1$-measure. The same argument applies to the function $\omega_2 \mapsto X(\omega_1,\omega_2)$.
2. If $\omega_1 \in N^c$, $\int_{\Omega_2} X(\omega_1,\omega_2)\, d\mathbb{P}_2(\omega_2)$ is well defined since the function is only defined

(thanks to (1)) outside a measurable set of values of ω_1 of zero $\mathbb{P}_1$-measure. We can always agree that that $\int_{\Omega_2} X(\omega_1, \omega_2)\, d\mathbb{P}_2(\omega_2) = 0$ if $\omega_1 \in N$. Then the function

$$\begin{aligned}\omega_1 \mapsto \int_{\Omega_2} X(\omega_1, \omega_2)\, d\mathbb{P}_2(\omega_2) &= \mathbb{1}_{N^c}(\omega_1) \int_{\Omega_2} X^+(\omega_1, \omega_2)\, d\mathbb{P}_2(\omega_2) \\ &\quad - \mathbb{1}_{N^c}(\omega_1) \int_{\Omega_2} X^-(\omega_1, \omega_2)\, d\mathbb{P}_2(\omega_2).\end{aligned}$$

is $\mathcal{A}_1$-measurable by 2.1.3 (1), and we have also, thanks to part (2) of the same theorem,

$$\begin{aligned}\int_{\Omega_1} \left| \int_{\Omega_2} X(\omega_1, \omega_2)\, d\mathbb{P}_2(\omega_2) \right| d\mathbb{P}_1(\omega_1) &\leq \int_{\Omega_1} \left(\int_{\Omega_2} |X(\omega_1, \omega_2)|\, d\mathbb{P}_2(\omega_2) \right) d\mathbb{P}_1(\omega_1) \\ &= \int |X|\, d\mathbb{P}_1 \otimes \mathbb{P}_2 \\ &< \infty.\end{aligned}$$

This proves that the function $\omega_1 \mapsto \int_{\Omega_2} X(\omega_1, \omega_2)\, d\mathbb{P}_2(\omega_2)$ is in $L^1(\Omega_1, \mathcal{A}_1, \mathbb{P}_1)$. The same argument applies to the function $\omega_2 \mapsto \int_{\Omega_1} X(\omega_1, \omega_2)\, d\mathbb{P}_1(\omega_1)$.

3. Theorem 2.1.3 gives the two equalities

$$\begin{aligned}\int_{\Omega_1} \left(\int_{\Omega_2} X^+(\omega_1, \omega_2)\, d\mathbb{P}_2(\omega_2) \right) d\mathbb{P}_1(\omega_1) &= \int_{\Omega_1 \times \Omega_2} X^+\, d\mathbb{P}_1 \otimes \mathbb{P}_2, \\ \int_{\Omega_1} \left(\int_{\Omega_2} X^-(\omega_1, \omega_2)\, d\mathbb{P}_2(\omega_2) \right) d\mathbb{P}_1(\omega_1) &= \int_{\Omega_1 \times \Omega_2} X^-\, d\mathbb{P}_1 \otimes \mathbb{P}_2.\end{aligned}$$

We can replace the integral over Ω_1 by an integral over N^c in the left-hand sides of these formulas. Then we just have to subtract the second equality from the first one (using part (2)), in order to arrive at the first equality in (3), and the second equality is derived in a similar manner.

■

2.1.2 Convolution

Throughout this section, we consider the measure space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$. Let f and g be two real measurable functions on $\mathbb{R}^d$. For $x \in \mathbb{R}^d$, the convolution

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x - y) g(y)\, dy$$

is well defined provided

$$\int_{\mathbb{R}^d} |f(x - y) g(y)|\, dy < \infty.$$

In that case, the fact that Lebesgue measure on $\mathbb{R}^d$ is invariant under translations and under the symmetry $y \mapsto -y$ shows that $(g * f)(x)$ is also well defined and $(g * f)(x) = (f * g)(x)$.

Proposition 2.1.5

Let $f \in L^1(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$ and $g \in L^p(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$, where $p \in [1, \infty)$. Then, for λ a.e. $x \in \mathbb{R}^d$ [a], the convolution $(f * g)(x)$ is well defined. Moreover $f * g \in L^p(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$ and :

$$\|f * g\|_p \leq \|f\|_1 \|g\|_p$$

a. for all x except a set of null measure

Proof for Proposition.

Again, we may agree that, on the set of zero Lebesgue measure where $f * g$ is not well defined, we take $f * g(x) = 0$.

We leave aside the case $p = \infty$, which is easier (and where stronger results are proved in Proposition 2.1.6 below). We assume that $\|f\|_1 > 0$, since otherwise the results are trivial. Since the measure with density $\|f\|_1^{-1}|f(x)|$ with respect to λ is a probability measure, Jensen's inequality gives, for every $x \in \mathbb{R}^d$,

$$\begin{aligned}
\left(\int_{\mathbb{R}^d} |g(x-y)||f(y)|\,dy\right)^p &= \|f\|_1^p \left(\int_{\mathbb{R}^d} |g(x-y)| \frac{|f(y)|}{\|f\|_1} dy\right)^p \\
&\leq \|f\|_1^{p-1} \int_{\mathbb{R}^d} |g(x-y)|^p |f(y)|\,dy.
\end{aligned}$$

Using Fubini-Tonelli Theorem 2.1.3, we have then :

$$\begin{aligned}
\int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |g(x-y)||f(y)|\,dy\right)^p dx &\leq \|f\|_1^{p-1} \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |g(x-y)|^p |f(y)|\,dy\right) dx \\
&= \|f\|_1^{p-1} \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |g(y)|^p |f(x-y)|\,dy\right) dx \\
&= \|f\|_1^{p-1} \int_{\mathbb{R}^d} |g(y)|^p \left(\int_{\mathbb{R}^d} |f(x-y)| dx\right) dy \\
&= \|f\|_1^p \|g\|_p^p < \infty,
\end{aligned}$$

which shows that

$$\int_{\mathbb{R}^d} |f(x-t)||g(t)| dt < \infty, \qquad \lambda(dx) \text{ a.e.}$$

and gives the first assertion. The second one also follows since the previous calculation gives

$$\int_{\mathbb{R}^d} |f * g(x)|^p dx \leq \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |g(x-y)||f(y)|\,dy\right)^p dx \leq \|f\|_1^p \|g\|_p^p.$$

■

The next proposition gives another important case where the convolution $f * g$ is well defined and enjoys nice continuity properties.

Proposition 2.1.6

Let $p \in [1, \infty)$ and $q \in (1, \infty)$ be conjugate exponents $\left(\frac{1}{p} + \frac{1}{q} = 1\right)$. Let $f \in L^p(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$ and $g \in L^q(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$. Then, the convolution $f * g(x)$ is well defined and $|f * g(x)| \leq \|f\|_p \|g\|_q$, for every $x \in \mathbb{R}^d$. Furthermore, the function $f * g$ is uniformly continuous on $\mathbb{R}^d$.

Proof for Proposition.

By the Hölder inequality,

$$\int_{\mathbb{R}^d} |f(x-y)g(y)|dy \leq \left(\int |f(x-y)|^p dy\right)^{1/p} \|g\|_q = \|f\|_p \|g\|_q.$$

This gives the first assertion and also proves that $f * g$ is bounded above by $\|f\|_p\|g\|_q$. To get the property of uniform continuity, we use the Lemma 2.1.7 it is easy to complete the proof of the proposition. Indeed, for every $x, x' \in \mathbb{R}^d$,

$$\begin{aligned}
|f * g(x) - f * g(x')| &\leq \int |f(x-y) - f(x'-y)||g(y)|dy \\
&\leq \|g\|_q \left(\int |f(x-y) - f(x'-y)|^p dy\right)^{1/p} \\
&= \|g\|_q \|f \circ \sigma_{-x} - f \circ \sigma_{-x'}\|_p
\end{aligned}$$

and we use Lemma 2.1.7 to say that $\|f \circ \sigma_{-x} - f \circ \sigma_{-x'}\|_p$ tends to 0 when $x - x' \to 0$.

Lemma 2.1.7: Continuity of translation

Set $\sigma_x(y) = y - x$ for every $x, y \in \mathbb{R}^d$. Let $p \in [1, \infty)$ and $f \in L^p(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$. Then the mapping $x \mapsto f \circ \sigma_x$ is uniformly continuous from $\mathbb{R}^d$ into $L^p(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$.

Proof for Lemma

The proof of this lemma is omitted, but could be found in the book [3, lem 5.7]

Approximations of the Dirac Measure

Definition 2.1.8: Approximation of the Dirac Measure

Let us say that a sequence $(\varphi_n)_{n\in\mathbb{N}}$ of nonnegative continuous functions on $\mathbb{R}^d$ is an approximation of the Dirac measure δ_0 if:

1. For every $n \in \mathbb{N}$,

$$\int_{\mathbb{R}^d} \varphi_n(x)\, dx = 1.$$

2. For every $\delta > 0$,
$$\lim_{n\to\infty} \int_{\{|x|>\delta\}} \varphi_n(x)\, dx = 0.$$

It is easy to construct approximations of δ_0. If φ is any nonnegative continuous function on $\mathbb{R}^d$ such that $\int \varphi(x)\, dx = 1$, we can set $\varphi_n(x) = n^d\varphi(nx)$ for every $x \in \mathbb{R}^d$ and every $n \in \mathbb{N}$. Note that, if we start from a function φ with compact support, all functions φ_n will vanish outside the same closed ball.

Proposition 2.1.9

Let $(\varphi_n)_{n\in\mathbb{N}}$ be an approximation of the Dirac measure δ_0.

1. For any bounded continuous function $f : \mathbb{R}^d \longrightarrow \mathbb{R}$, the sequence $\varphi_n * f$ converges to f as $n \to \infty$, uniformly on every compact subset of $\mathbb{R}^d$.
2. Let $f \in L^p(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$, where $p \in [1, \infty)$. Then $\varphi_n * f \to f$ in L^p.

Proof for Proposition.

1. First note that $\varphi_n * f$ is well defined since φ_n is integrable and f is bounded. Then, we have, for every $\delta > 0$,
$$\begin{aligned}\varphi_n * f(x) - f(x) &= \int_{|y|\le\delta} (f(x-y) - f(x))\varphi_n(y)\, dy \\ &\quad + \int_{|y|>\delta} (f(x-y) - f(x))\varphi_n(y)\, dy. \qquad \text{(i)}\end{aligned}$$
Fix $\varepsilon > 0$ and a compact subset H of $\mathbb{R}^d$. Since f is uniformly continuous on any compact subset of $\mathbb{R}^d$, we can choose $\delta > 0$ such that $|f(x-y) - f(x)| \le \varepsilon$ for every $x \in H$ and $y \in \mathbb{R}^d$ such that $|y| \le \delta$. Recalling (1) in Definition 2.1.8, we get that the first term in the right-hand side of (i) is smaller than ε in absolute value, for every $x \in H$ and $n \in \mathbb{N}$. Furthermore, (2) in Definition 2.1.8 and the boundedness of f show that the second term in the right-hand side of (i) tends to 0 when $n \to \infty$, uniformly when x varies in $\mathbb{R}^d$. It follows that we have $|\varphi_n * f(x) - f(x)| < 2\varepsilon$ for every $x \in H$, as soon as n is sufficiently large.
2. The fact that $\varphi_n * f$ is well defined a.e. and belongs to L^p follows from Proposition 2.1.5 since φ_n is integrable. We then observe that
$$\begin{aligned}\int |\varphi_n * f(x) - f(x)|^p dx &\le \int \left(\int \varphi_n(y)|f(x-y) - f(x)|\, dy\right)^p dx \\ &\le \int \left(\int \varphi_n(y)|f(x-y) - f(x)|^p dy\right) dx \\ &= \int \left(\int \varphi_n(y)|f(x-y) - f(x)|^p dx\right) dy \\ &= \int \varphi_n(y) \left(\int |f(x-y) - f(x)|^p dx\right) dy\end{aligned}$$
where the second inequality is a consequence of Jensen's inequality (note that

$\varphi_n(y)dy$ is a probability measure), and the next equality uses Fubini-Tonelli Theorem 2.1.3. For every $y \in \mathbb{R}^d$, set

$$h(y) = \int |f(x-y) - f(x)|^p dx = \|f \circ \sigma_y - f\|_p^p,$$

with the notation of Lemma 2.1.7. The function h is bounded and $h(y) \to 0$ as $y \to 0$ by Lemma 2.1.7. It then immediately follows from (1) and (2) in Definition 2.1.8 that

$$\lim_{n\to\infty} \int \varphi_n(y)h(y)\,dy = 0.$$

This completes the proof.

■

Application In dimension $d = 1$, we can take

$$\varphi_n(x) = c_n(1-x^2)^n \mathbb{1}_{\{|x|\leq 1\}}$$

where the constant c_n is chosen so that $\int \varphi_n(x)dx = 1$. Then let $[a, b]$ be a compact subinterval of $(0, 1)$, and let f be a continuous function on $[a, b]$. It is easy to extend f to a continuous function on $\mathbb{R}$ that vanishes outside $[0, 1]$ (for instance, take f to be affine on both intervals $[0, a]$ and $[b, 1]$). Then the preceding proposition shows that

$$\varphi_n * f(x) = c_n \int (1-(x-y)^2)^n \mathbb{1}_{\{|x-y|\leq 1\}} f(y)\,dy \longrightarrow f(x)$$

uniformly on $[a, b]$. If $x \in [a, b]$, we can remove the indicator function $\mathbb{1}_{\{|x-y|\leq 1\}}$ in the last display, since the conditions $x \in [a, b]$ and $|x-y| > 1$ force $f(y) = 0$. It now follows that f is a uniform limit of polynomials on the interval $[a, b]$ (this is a special case of the Stone-Weierstrass theorem 1.2.2).

2.1.3 Independent Events

Definition 2.1.10: Independence of events

1. Two events A and B are said to be *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B).$$

2. Let $A_1, \ldots, A_n$ be events. We say that the events $A_1, \ldots, A_n$ are *independent* if, for every nonempty subset $\{i_1, \ldots, i_p\} \subset \{1, \ldots, n\}$, we have

$$\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_p}) = \mathbb{P}(A_{i_1}) \cdots \mathbb{P}(A_{i_p}).$$

remark: In particular, for $n \geq 3$, pairwise independence does not imply independence.

Proposition 2.1.11

The events $A_1, \dots, A_n$ are independent if and only if

$$\mathbb{P}(B_1 \cap \cdots \cap B_n) = \mathbb{P}(B_1) \cdots \mathbb{P}(B_n)$$

for every choice of events $B_i \in \sigma(A_i) = \{\varnothing, A_i, A_i^c, \Omega\}$, $i = 1, \dots, n$.

Proof for Proposition.

Left exercise for the reader. ■

2.1.4 Independence for σ-fields and Random Variables

We say that $\mathcal{G}$ is a *sub-σ-field* of $\mathcal{A}$ if $\mathcal{G}$ is a σ-field and $\mathcal{G} \subset \mathcal{A}$. Roughly speaking, a sub-σ-field represents partial information in the probability space, namely the information obtained by knowing which events in $\mathcal{G}$ have occurred. For instance, if X is a random variable, the σ-field $\sigma(X)$ corresponds to the information provided by the value of X.

This suggests that the most general notion of independence is that of independent sub-σ-fields : informally, two sub-σ-fields $\mathcal{G}$ and $\mathcal{G}'$ are independent if knowing which events of $\mathcal{G}$ have occurred gives no information about the occurrence of events in $\mathcal{G}'$, and conversely. We make this precise in the following definition.

Definition 2.1.12: Independence of [finite] σ-fields

1. Let $\mathcal{G}_1, \dots, \mathcal{G}_n$ be sub-σ-fields of $\mathcal{A}$. We say that $\mathcal{G}_1, \dots, \mathcal{G}_n$ are *independent* if, for every choice of events

$$A_1 \in \mathcal{G}_1,\ A_2 \in \mathcal{G}_2,\ \dots,\ A_n \in \mathcal{G}_n,$$

we have

$$\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\, \mathbb{P}(A_2) \cdots \mathbb{P}(A_n).$$

2. Let $X_1, \dots, X_n$ be random variables taking values in the measurable spaces $(E_1, \mathcal{E}_1), \dots, (E_n, \mathcal{E}_n)$, respectively. We say that the random variables $X_1, \dots, X_n$ are *independent* if the σ-fields $\sigma(X_1), \dots, \sigma(X_n)$ are independent.
3. Equivalently, $X_1, \dots, X_n$ are independent if and only if, for every choice of sets $F_1 \in \mathcal{E}_1, \dots, F_n \in \mathcal{E}_n$,

$$\mathbb{P}\Big(X_1 \in F_1, \dots, X_n \in F_n\Big) = \mathbb{P}(X_1 \in F_1) \cdots \mathbb{P}(X_n \in F_n).$$

Firstly, we introduce a suffisant condition for the independence of σ-fields .

Theorem 2.1.13: Independence Extension Theorem

Let $\mathcal{G}_1, \ldots, \mathcal{G}_n$ be sub-σ-fields of $\mathcal{A}$. For each $i \in \{1, \ldots, n\}$, let $\mathcal{C}_i \subset \mathcal{G}_i$ be a class of sets such that:

- $\mathcal{C}_i$ is closed under finite intersections,
- $\Omega \in \mathcal{C}_i$,
- $\sigma(\mathcal{C}_i) = \mathcal{G}_i$.

Assume that

$$\forall C_1 \in \mathcal{C}_1, \ldots, C_n \in \mathcal{C}_n, \qquad \mathbb{P}(C_1 \cap \cdots \cap C_n) = \prod_{i=1}^{n} \mathbb{P}(C_i).$$

Then the σ-fields $\mathcal{G}_1, \ldots, \mathcal{G}_n$ are independent.

Proof for Theorem.

We first fix $C_2 \in \mathcal{C}_2, \ldots, C_n \in \mathcal{C}_n$ and define

$$\mathcal{M}_1 = \left\{ B_1 \in \mathcal{G}_1 : \mathbb{P}(B_1 \cap C_2 \cap \cdots \cap C_n) = \mathbb{P}(B_1) \prod_{i=2}^{n} \mathbb{P}(C_i) \right\}.$$

By assumption, $\mathcal{C}_1 \subset \mathcal{M}_1$. It is straightforward to check that $\mathcal{M}_1$ is a monotone class. By the monotone class theorem,

$$\mathcal{M}_1 \supset \sigma(\mathcal{C}_1) = \mathcal{G}_1.$$

Hence, for all $B_1 \in \mathcal{G}_1$ and all $C_2 \in \mathcal{C}_2, \ldots, C_n \in \mathcal{C}_n$,

$$\mathbb{P}(B_1 \cap C_2 \cap \cdots \cap C_n) = \mathbb{P}(B_1) \prod_{i=2}^{n} \mathbb{P}(C_i).$$

We now fix $B_1 \in \mathcal{G}_1$, $C_3 \in \mathcal{C}_3, \ldots, C_n \in \mathcal{C}_n$ and define

$$\mathcal{M}_2 = \left\{ B_2 \in \mathcal{G}_2 : \mathbb{P}(B_1 \cap B_2 \cap C_3 \cap \cdots \cap C_n) = \mathbb{P}(B_1)\mathbb{P}(B_2) \prod_{i=3}^{n} \mathbb{P}(C_i) \right\}.$$

Again, $\mathcal{C}_2 \subset \mathcal{M}_2$, and $\mathcal{M}_2$ is a monotone class, so $\mathcal{M}_2 = \mathcal{G}_2$.

Proceeding inductively, we obtain that, for all $B_1 \in \mathcal{G}_1, \ldots, B_n \in \mathcal{G}_n$,

$$\mathbb{P}(B_1 \cap \cdots \cap B_n) = \prod_{i=1}^{n} \mathbb{P}(B_i).$$

This is precisely the definition of independence of the σ-fields $\mathcal{G}_1, \ldots, \mathcal{G}_n$. ■

The next theorem provides a fundamental characterization of the independence of n random variables.

Theorem 2.1.14: Independence Characterization

Let $X_1, \ldots, X_n$ be random variables taking values in $\mathbb{R}$, the following statements are equivalent:

1. The random variables $X_1, \ldots, X_n$ are independent.
2. the law of the random vector $(X_1, \ldots, X_n)$ is the product of the respective laws of $X_1, \ldots, X_n$, that is,
$$\mathbb{P}_{(X_1,\ldots,X_n)} = \mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}.$$
3. For every $a_1, \ldots, a_n \in \mathbb{R}$,
$$\mathbb{P}(X_1 \leq a_1, \ldots, X_n \leq a_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \leq a_i).$$
4. For every choice of continuous functions $f_1, \ldots, f_n : \mathbb{R} \to \mathbb{R}_+$ with compact support,
$$\mathbb{E}\left[\prod_{i=1}^{n} f_i(X_i)\right] = \prod_{i=1}^{n} \mathbb{E}[f_i(X_i)].$$
5. The characteristic function of the random vector $X = (X_1, \ldots, X_n)$ factorizes as
$$\Phi_X(\xi_1, \ldots, \xi_n) = \prod_{i=1}^{n} \Phi_{X_i}(\xi_i), \qquad (\xi_1, \ldots, \xi_n) \in \mathbb{R}^n.$$

Proof for Theorem.

- (1) $\Longleftrightarrow$ (2) :
Let $F_i \in \mathcal{E}_i$, for every $i \in \{1, \ldots, n\}$. On one hand,
$$\mathbb{P}_{(X_1,\ldots,X_n)}(F_1 \times \cdots \times F_n) = \mathbb{P}\Big((X_1 \in F_1) \cap \cdots \cap (X_n \in F_n)\Big).$$
On the other hand,
$$\mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}(F_1 \times \cdots \times F_n) = \prod_{i=1}^{n} \mathbb{P}_{X_i}(F_i) = \prod_{i=1}^{n} \mathbb{P}(X_i \in F_i).$$
Comparing with the definition of independence, we see that $X_1, \ldots, X_n$ are independent if and only if the two probability measures $\mathbb{P}_{(X_1,\ldots,X_n)}$ and $\mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}$ take the same values on boxes $F_1 \times \cdots \times F_n$. This is equivalent to saying that :
$$\mathbb{P}_{(X_1,\ldots,X_n)} = \mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n},$$
since we know that a probability measure on a product space is characterized by its values on boxes (theorem 2.1.2) we conclude the result.
- (2) $\Longleftrightarrow$ (3) :
Just by taking $F_i =]-\infty, a_i]$ we can prove the (2) $\Longleftarrow$ (3), conversely, Just

by taking $C_i = \Big\{]-\infty, a] \;\Big|\; a \in \mathbb{R} \Big\}$, and apply the theorem **??** we obtain the independence (1) and therefore (2).

- (1) $\Longleftrightarrow$ (4) :
  We have (4) $\Longrightarrow$ (3) $\Longrightarrow$ (1) by taking $f_i(x) = \mathbb{1}_{x \le a_i}$. The other implication is imediate from the next proposition.
- (2) $\Longleftrightarrow$ (5) :
  To show this equivalence, we note that the Fourier transform of the product measure $\mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}$ is

$$\begin{aligned}(\xi_1, \ldots, \xi_n) \longmapsto \int \exp\left(i \sum_{j=1}^{n} \xi_j x_j \right) \, \mathbb{P}_{X_1}(dx_1) \cdots \mathbb{P}_{X_n}(dx_n) \\ = \prod_{j=1}^{n} \int e^{i\xi_j x_j} \, \mathbb{P}_{X_j}(dx_j) \\ = \prod_{j=1}^{n} \Phi_{X_j}(\xi_j).\end{aligned}$$

using the Fubini theorem in the first equality. Hence the property (5) is equivalent to the fact that the Fourier transform of $\mathbb{P}_{(X_1,\ldots,X_n)}$ coincides with the Fourier transform of $\mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}$. By pervious property, this holds if and only if

$$\mathbb{P}_{(X_1,\ldots,X_n)} = \mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n},$$

■

From this theorem follow this interesting results :

Proposition 2.1.15

Let $X_1, \ldots, X_n$ be independent real-valued random variables.

1. Let $f_1, \ldots, f_n$ be non negative measurable functions we have :

$$\mathbb{E}\left[\prod_{i=1}^{n} f_i(X_i) \right] = \prod_{i=1}^{n} \mathbb{E}[f_i(X_i)] \,,$$

2. Let $f_1, \ldots, f_n$ be measurable functions such that

$$\mathbb{E}[|f_i(X_i)|] < \infty \quad \text{for every } i \in \{1, \ldots, n\}.$$

   Then

$$\mathbb{E}\left[\prod_{i=1}^{n} f_i(X_i) \right] = \prod_{i=1}^{n} \mathbb{E}[f_i(X_i)],$$

   and the expectation on the left-hand side is well defined. The same statement holds for complex-valued functions f_i under the same integrability assumption.

3. As a special case of (1), if $X_1, \ldots, X_n$ are independent random variables in

$\mathcal{L}^1$, then $X_1 \cdots X_n \in \mathcal{L}^1$ and

$$\mathbb{E}[X_1 \cdots X_n] = \prod_{i=1}^{n} \mathbb{E}[X_i].$$

In general, the product of random variables in $\mathcal{L}^1$ need not belong to $\mathcal{L}^1$ without independence.

4. If X_1 and X_2 are independent real-valued random variables in L^2, then

$$\operatorname{cov}(X_1, X_2) = 0.$$

5. Suppose that, for every $i \in \{1, \ldots, n\}$, the random variable X_i admits a density p_i with respect to Lebesgue measure and that $X_1, \ldots, X_n$ are independent. Then the random vector $(X_1, \ldots, X_n)$ admits a density p given by

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p_i(x_i).$$

6. Conversely, suppose that $(X_1, \ldots, X_n)$ admits a density p of the form

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} q_i(x_i),$$

where $q_1, \ldots, q_n$ are nonnegative measurable functions on $\mathbb{R}$. Then $X_1, \ldots, X_n$ are independent, and for every $i \in \{1, \ldots, n\}$, X_i admits a density p_i of the form $p_i = C_i q_i$, where $C_i > 0$ is a normalization constant.

Proof for Proposition.

1. Using the Fubini-Tonelli theorem 2.1.3 for the case of two independent variables :

$$\begin{aligned}
\mathbb{E}\Big(f_1(X_1) f_2(X_2)\Big) &= \int_{\mathbb{R}^2} f_1(x_1) f_2(x_2), d\mathbb{P}_{(X_1,X_2)}(x_1, x_2) \\
&= \int_{\mathbb{R}^2} f_1(x_1) f_2(x_2), d(\mathbb{P}_{X_1} \otimes \mathbb{P}_{X_2})(x_1, x_2) \\
&= \int_{\mathbb{R}} \int_{\mathbb{R}} f_1(x_1) f_2(x_2), d\mathbb{P}_{X_2}(x_2), d\mathbb{P}_{X_1}(x_1) \\
&= \int_{\mathbb{R}} f_1(x_1), d\mathbb{P}_{X_1}(x_1) \int_{\mathbb{R}} f_2(x_2), d\mathbb{P}_{X_2}(x_2) \\
&= \mathbb{E}\Big(f_1(X_1)\Big) \times \mathbb{E}\Big(f_2(X_2)\Big).
\end{aligned}$$

The general case can be achived by a simple reccurence.

2. Similar to (1), but using the Fubini-Lebesgue theorem 2.1.4 and ensuring the $f_i(X_i) \in \mathcal{L}^1$ which is the condition :

$$\mathbb{E}[|f_i(X_i)|] < \infty \quad \text{for every } i \in \{1, \ldots, n\}.$$

3. Clear conclusion from (1).
4. This follows from the preceding remarks since :

$$\mathrm{Cov}(X_1, X_2) = \mathbb{E}(X_1 X_2) - \mathbb{E}(X_1)\mathbb{E}(X_2) = 0$$

5. This is an immediate consequence of the caracterisation of independece $\mathbb{P}_{(X_1,\dots,X_n)} = \mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}$. Indeed, by the Fubini theorem, under the assumption

$$\mathbb{P}_{X_i}(dx_i) = p_i(x_i)\, dx_i \quad \text{for } i \in \{1, \dots, n\},$$

we have

$$\mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}(dx_1 \cdots dx_n) = \left(\prod_{i=1}^n p_i(x_i)\right) dx_1 \cdots dx_n.$$

6. We first observe, again thanks to the Fubini theorem, that

$$\prod_{i=1}^n \left(\int q_i(x)\, dx\right) = \int_{\mathbb{R}^n} p(x_1, \dots, x_n)\, dx_1 \cdots dx_n = 1.$$

In particular, the quantity

$$K_i = \int q_i(x)\, dx$$

is positive and finite for every $i \in \{1, \dots, n\}$. Then, by Properties of density function 1.1.8, the density of X_i is

$$\begin{aligned} p_i(x_i) &= \int_{\mathbb{R}^{n-1}} p(x_1, \dots, x_n)\, dx_1 \cdots dx_{i-1}\, dx_{i+1} \cdots dx_n \\ &= \left(\prod_{j \neq i} K_j\right) q_i(x_i) = \frac{1}{K_i}\, q_i(x_i). \end{aligned}$$

Thus $p_i = C_i q_i$ with $C_i = 1/K_i$. This also allows us to rewrite the density of $(X_1, \dots, X_n)$ as

$$p(x_1, \dots, x_n) = \prod_{i=1}^n q_i(x_i) = \prod_{i=1}^n p_i(x_i),$$

and therefore

$$\mathbb{P}_{(X_1,\dots,X_n)} = \mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n},$$

which shows that $X_1, \dots, X_n$ are independent.

■

Definition 2.1.16: Independence of [infinite] σ-fields

Let $(\mathcal{G}_i)_{i\in I}$ be an arbitrary collection of sub-σ-fields of $\mathcal{A}$. We say that the σ-fields $\mathcal{G}_i$, $i \in I$, are *independent* if, for every finite subset $\{i_1, \dots, i_p\} \subset I$, the σ-fields $\mathcal{G}_{i_1}, \dots, \mathcal{G}_{i_p}$ are independent.
If $(X_i)_{i\in I}$ is an arbitrary collection of random variables, we say that the random

variables X_i, $i \in I$, are *independent* if the σ-fields $\sigma(X_i)$, $i \in I$, are independent. Similarly, for an arbitrary collection $(A_i)_{i\in I}$ of events, we say that the events A_i, $i \in I$, are independent if the σ-fields $\sigma(A_i)$, $i \in I$, are independent.

Proposition 2.1.17

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent random variables. Then, for every $p \in \mathbb{N}$, the σ-fields

$$\mathcal{G}_1 = \sigma(X_1, \dots, X_p), \qquad \mathcal{G}_2 = \sigma(X_{p+1}, X_{p+2}, \dots)$$

are independent.

Proof for Proposition.

We apply the suffisant condition of σ-fields independence given in **??**. Set

$$\mathcal{C}_1 = \sigma(X_1, \dots, X_p) = \mathcal{G}_1$$

and

$$\mathcal{C}_2 = \bigcup_{k=p+1}^{\infty} \sigma(X_{p+1}, \dots, X_k) \subset \mathcal{G}_2.$$

The class $\mathcal{C}_2$ is closed under finite intersections, contains Ω, and satisfies $\sigma(\mathcal{C}_2) = \mathcal{G}_2$.

By independence of the random variables $(X_n)_{n\in\mathbb{N}}$, and by the grouping-by-blocks principle, we have for every $C_1 \in \mathcal{C}_1$ and $C_2 \in \mathcal{C}_2$,

$$\mathbb{P}(C_1 \cap C_2) = \mathbb{P}(C_1)\mathbb{P}(C_2).$$

Therefore, the assumptions of theorem **??** are satisfied, and it follows that the σ-fields $\mathcal{G}_1$ and $\mathcal{G}_2$ are independent. ∎

2.1.5 Sums of Independent Random Variables

Sums of independent random variables play a central role in probability theory. Their distributional and analytic properties are conveniently described using the notion of convolution of probability measures. Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on $\mathbb{R}^d$. The *convolution* $\mathbb{P} * \mathbb{Q}$ is defined as the pushforward of the product measure $\mathbb{P} \otimes \mathbb{Q}$ under the mapping $(x, y) \mapsto x + y$. Equivalently, for every nonnegative measurable function φ on $\mathbb{R}^d$,

$$\int_{\mathbb{R}^d} \varphi(z)\,(\mathbb{P} * \mathbb{Q})(dz) = \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} \varphi(x + y)\, d\mathbb{P}(x)\, d\mathbb{Q}(y).$$

If $\mathbb{P}$ and $\mathbb{Q}$ admit densities f and g with respect to Lebesgue measure, then $\mathbb{P} * \mathbb{Q}$ admits the density $f * g$ given by

$$(f * g)(z) = \int_{\mathbb{R}^d} f(x)\, g(z - x)\, dx.$$

Proposition 2.1.18

Let X and Y be two independent random variables with values in $\mathbb{R}^d$.

1. The law of $X+Y$ is the convolution of the laws of X and Y:

$$\mathbb{P}_{X+Y} = \mathbb{P}_X * \mathbb{P}_Y.$$

   In particular, if X and Y admit densities p_X and p_Y, then $X+Y$ admits the density $p_X * p_Y$.

2. The characteristic function of $X+Y$ satisfies

$$\Phi_{X+Y}(\xi) = \Phi_X(\xi)\,\Phi_Y(\xi), \qquad \xi \in \mathbb{R}^d.$$

3. If $\mathbb{E}[\|X\|^2] < \infty$ and $\mathbb{E}[\|Y\|^2] < \infty$, then the covariance matrices satisfy

$$K_{X+Y} = K_X + K_Y.$$

   In particular, if $d = 1$ and $X, Y \in L^2$, then

$$\mathrm{var}(X+Y) = \mathrm{var}(X) + \mathrm{var}(Y).$$

Proof for Proposition.

1. We know that $\mathbb{P}_{(X,Y)} = \mathbb{P}_X \otimes \mathbb{P}_Y$, and thus, for every nonnegative measurable function φ on $\mathbb{R}^d$,

$$\mathbb{E}[\varphi(X+Y)] = \int\!\!\int \varphi(x+y)\,\mathbb{P}_X(dx)\,\mathbb{P}_Y(dy) = \int \varphi(z)\,(\mathbb{P}_X * \mathbb{P}_Y)(dz),$$

   by the definition of $\mathbb{P}_X * \mathbb{P}_Y$. The case with density follows from the remarks before the proposition.

2. This follows from the equality

$$\mathbb{P}_{X+Y} = \mathbb{P}_X * \mathbb{P}_Y$$

   and the last observation before the proposition.

3. If $X = (X_1, \dots, X_d)$ and $Y = (Y_1, \dots, Y_d)$, proposition 2.1.15 implies that

$$\mathrm{cov}(X_i, Y_j) = 0 \quad \text{for every } i, j \in \{1, \dots, d\}.$$

   Consequently, using bilinearity,

$$\mathrm{cov}(X_i + Y_i,\, X_j + Y_j) = \mathrm{cov}(X_i, X_j) + \mathrm{cov}(Y_i, Y_j),$$

   which gives $K_{X+Y} = K_X + K_Y$.

■

2.1.6 The Borel–Cantelli Lemma

Let $(A_n)_{n\in\mathbb{N}}$ be a sequence of events. Recall that

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k.$$

The event $\limsup_{n\to\infty} A_n$ consists of those $\omega \in \Omega$ that belong to infinitely many of the events A_n. Note that It's easy to check that :

$$\mathbb{P}\Big(\limsup_{n\to\infty} A_n\Big) \geq \limsup_{n\to\infty} \mathbb{P}(A_n).$$

Theorem 2.1.19: Borel–Cantelli Lemma

Let $(A_n)_{n\in\mathbb{N}}$ be a sequence of events.

1. If $\sum_{n\in\mathbb{N}} \mathbb{P}(A_n) < \infty$, then

$$\mathbb{P}\Big(\limsup_{n\to\infty} A_n\Big) = 0.$$

Equivalently,

$$\Big\{n \in \mathbb{N} :\ \omega \in A_n\Big\} \quad \text{is finite almost surely.}$$

2. If $\sum_{n\in\mathbb{N}} \mathbb{P}(A_n) = \infty$ and the events $(A_n)_{n\in\mathbb{N}}$ are independent, then

$$\mathbb{P}\Big(\limsup_{n\to\infty} A_n\Big) = 1.$$

Equivalently,

$$\Big\{n \in \mathbb{N} :\ \omega \in A_n\Big\} \quad \text{is infinite almost surely.}$$

Proof for Theorem.

1. If $\sum_{n\in\mathbb{N}} \mathbb{P}(A_n) < \infty$, then

$$\mathbb{E}\left[\sum_{n\in\mathbb{N}} \mathbb{1}_{A_n}\right] = \sum_{n\in\mathbb{N}} \mathbb{P}(A_n) < \infty.$$

It follows that $\sum_{n\in\mathbb{N}} \mathbb{1}_{A_n} < \infty$ almost surely, which means that ω belongs to only finitely many of the events A_n. Hence,

$$\mathbb{P}\Big(\limsup_{n\to\infty} A_n\Big) = 0.$$

2. Assume now that $\sum_{n\in\mathbb{N}} \mathbb{P}(A_n) = \infty$ and that the events A_n are independent. Fix

$n_0 \in \mathbb{N}$. For $n \geq n_0$, we have

$$\mathbb{P}\left(\bigcap_{k=n_0}^{n} A_k^c\right) = \prod_{k=n_0}^{n} \mathbb{P}(A_k^c) = \prod_{k=n_0}^{n} \Big(1 - \mathbb{P}(A_k)\Big).$$

Since the series $\sum_{k\in\mathbb{N}} \mathbb{P}(A_k)$ diverges, the right-hand side converges to 0 as $n \to \infty$. Consequently, $\mathbb{P}\Big(\bigcap_{k=n_0}^{\infty} A_k^c\Big) = 0$. As this holds for every $n_0 \in \mathbb{N}$, we obtain

$$\mathbb{P}\left(\bigcup_{n_0=1}^{\infty} \bigcap_{k=n_0}^{\infty} A_k^c\right) = 0.$$

Taking complements yields

$$\mathbb{P}\left(\bigcap_{n_0=1}^{\infty} \bigcup_{k=n_0}^{\infty} A_k\right) = 1,$$

that is, $\mathbb{P}\Big(\limsup_{n\to\infty} A_n\Big) = 1.$

■

The Borel–Cantelli Lemma is a powerful tool for proving (or disproving) that a certain property holds eventually almost surely, meaning that from an index onward the property is true almost surely. It achieves this by reducing the problem to studying whether a series of probabilities converges or diverges. The standard application of the lemma is summarized in Figure 3.

Step 1. Identify the bad event.

Define $A_n = \{\text{the desired property fails at time } n\}$.

Step 2. Estimate the probabilities.

Find an upper bound for $P(A_n)$ Usually use Markov, Chebyshev, Hoeffding, Chernoff, or another inequality.

Step 3. Check the series.

Determine whether $\sum_{n=1}^{\infty} P(A_n)$ converges or diverges.

Step 4. Apply Borel–Cantelli.

If $\sum P(A_n) < \infty$, then $P(A_n \text{ i.o.}) = 0$.
so eventually the desired property always holds.

FIGURE 3 – Standard strategy for applying the First Borel–Cantelli Lemma.

A more general version of the second lemma of Borel–Cantelli is the zero-one law of kolmogrov that stated as follow :

Theorem 2.1.20: Kolmogorov's Zero–One Law

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $(\mathcal{F}_n)_{n\geq 1}$ be a sequence of independent sub-σ-algebras of $\mathcal{F}$. For each $n \geq 1$, define the tail σ-algebra from time n onward by

$$\mathcal{B}_n := \sigma\left(\bigcup_{k=n}^{\infty} \mathcal{F}_k\right).$$

The *tail* (or *terminal*) σ-algebra associated with $(\mathcal{F}_n)_{n\geq 1}$ is defined by

$$\mathcal{B}_\infty := \bigcap_{n=1}^{\infty} \mathcal{B}_n.$$

Then the tail σ-algebra is $\mathbb{P}$-trivial. That is,

$$\forall A \in \mathcal{B}_\infty, \qquad \mathbb{P}(A) \in \{0, 1\}.$$

Let $(X_n)_{n\geq 1}$ be a sequence of independent random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. We can state the zero-one law as follow : consider the associated tail σ-algebra is

$$\mathcal{B}_\infty = \bigcap_{n=1}^{\infty} \sigma(X_n, X_{n+1}, \ldots).$$

For every event $A \in \mathcal{B}_\infty$,

$$\mathbb{P}(A) \in \{0, 1\}.$$

2.2 Conditioning

2.2.1 Definition of conditional expectaiton

Discret case

As in the previous chapters, we consider a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. If $B \in \mathcal{A}$ is an event of positive probability, we can define a new probability measure on $(\Omega, \mathcal{A})$ by setting, for every $A \in \mathcal{A}$,

$$\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

This probability measure $A \mapsto \mathbb{P}(A \mid B)$ is called **the conditional probability given** B. Similarly, for every nonnegative random variable X, or for $X \in L^1(\Omega, \mathcal{A}, \mathbb{P})$, the **conditional expectation of** X **given** B is defined by

$$\mathbb{E}[X \mid B] := \frac{\mathbb{E}[X \mathbb{1}_B]}{\mathbb{P}(B)}.$$

One verifies that $\mathbb{E}[X \mid B]$ is also the expected value of X under $\mathbb{P}(\cdot \mid B)$. We can interpret $\mathbb{E}[X \mid B]$ as the mean value of X knowing that the event B occurs.

Let us now define the conditional expectation knowing a discrete random variable. We consider a random variable Y taking values in a countable space E equipped as usual with the σ-field of all its subsets. Let

$$E' = \{y \in E : \mathbb{P}(Y = y) > 0\}.$$

If $X \in L^1(\Omega, \mathcal{A}, \mathbb{P})$, we may consider, for every $y \in E'$,

$$\mathbb{E}[X \mid Y = y] = \frac{\mathbb{E}[X\mathbb{1}_{\{Y=y\}}]}{\mathbb{P}(Y = y)},$$

as a special case of the conditional expectation $\mathbb{E}[X \mid B]$ when $\mathbb{P}(B) > 0$.

The conditional expectation of X knowing Y is the real random variable

$$\mathbb{E}[X \mid Y] = \varphi(Y),$$

where the function $\varphi : E \to \mathbb{R}$ is given by

$$\varphi(y) = \begin{cases} \mathbb{E}[X \mid Y = y], & \text{if } y \in E', \\ 0, & \text{if } y \in E \setminus E'. \end{cases}$$

General case

Now we turn to the definition of conditional expectation with respect to a sub-σ-field . For that purpose we need the next theorem that provides the existance and uniqueness of conditional expectation.

Theorem 2.2.1: Existance of Conditional Expectation

Let $\mathcal{B}$ be a sub-σ-field of $\mathcal{A}$, and let $X \in L^1(\Omega, \mathcal{A}, \mathbb{P})$. There exists a unique element of $L^1(\Omega, \mathcal{B}, \mathbb{P})$, which is denoted by $\mathbb{E}[X \mid \mathcal{B}]$, such that

$$\forall B \in \mathcal{B}, \quad \mathbb{E}[X\mathbb{1}_B] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]\mathbb{1}_B]. \tag{i}$$

We have more generally, for every bounded $\mathcal{B}$-measurable real random variable Z,

$$\mathbb{E}[XZ] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]Z]. \tag{ii}$$

If $X \geq 0$, we have $\mathbb{E}[X \mid \mathcal{B}] \geq 0$ a.s.

Proof for Theorem.

Let us start by proving uniqueness. Let X' and X'' be two random variables in $L^1(\Omega, \mathcal{B}, \mathbb{P})$ such that

$$\forall B \in \mathcal{B}, \quad \mathbb{E}[X'\mathbb{1}_B] = \mathbb{E}[X\mathbb{1}_B] = \mathbb{E}[X''\mathbb{1}_B].$$

Taking $B = \{X' > X''\}$ (which is in $\mathcal{B}$ since both X' and X'' are $\mathcal{B}$-measurable), we get

$$\mathbb{E}[(X' - X'')\mathbb{1}_{\{X'>X''\}}] = 0,$$

which implies that $X' \leq X''$ a.s., and we have similarly $X' \geq X''$ a.s. Thus $X' = X''$ a.s., which means that X' and X'' are equal as elements of $L^1(\Omega, \mathcal{B}, \mathbb{P})$.

Let us now turn to existence. We first assume that $X \geq 0$ and $\mathbb{E}(X) \neq 0$, then we let $\mathbb{Q}$ be the finite measure on $(\Omega, \mathcal{B})$ defined by

$$\forall B \in \mathcal{B}, \quad \mathbb{Q}(B) := \frac{1}{\mathbb{E}(X)}\mathbb{E}[X\mathbb{1}_B].$$

Let us emphasize that we define $\mathbb{Q}(B)$ only for $B \in \mathcal{B}$. We may also view $\mathbb{P}$ as a probability measure on $(\Omega, \mathcal{B})$, by restricting the mapping $B \mapsto \mathbb{P}(B)$ to $B \in \mathcal{B}$, and it is immediate that $\mathbb{Q} \ll \mathbb{P}$. The Radon–Nikodym theorem 1.3.11 applied to the probability measures $\mathbb{P}$ and $\mathbb{Q}$ on the measurable space $(\Omega, \mathcal{B})$ yields the existence of a nonnegative $\mathcal{B}$-measurable random variable $\tilde{X}$ such that

$$\forall B \in \mathcal{B}, \quad \mathbb{E}[X\mathbb{1}_B] = \mathbb{Q}(B) = \mathbb{E}[\tilde{X}\mathbb{1}_B].$$

Taking $B = \Omega$, we have $\mathbb{E}[\tilde{X}] = \mathbb{E}[X] < \infty$, and thus $\tilde{X} \in L^1(\Omega, \mathcal{B}, \mathbb{P})$. The random variable $\mathbb{E}[X \mid \mathcal{B}] = \tilde{X}$ satisfies (1). (for the case of $\mathbb{E}(X) = 0$ we have $X = 0$ a.s. and we can take $\tilde{X} = 0$)

When X is of arbitrary sign, we just have to take

$$\mathbb{E}[X \mid \mathcal{B}] = \mathbb{E}[X^+ \mid \mathcal{B}] - \mathbb{E}[X^- \mid \mathcal{B}],$$

and it is clear that (1) also holds in that case.

Finally, to see that (1) implies (2), we rely on the usual measure-theoretic arguments. (2) follows from (1) when Z is a simple random variable (taking only finitely many values), and in the general case lemma 1.1.10 allows us to write Z as the pointwise limit of a sequence $(Z_n)_{n\in\mathbb{N}}$ of simple $\mathcal{B}$-measurable random variables that are uniformly bounded by the same constant K (such that $|Z| \leq K$) and the dominated convergence theorem yields the desired result. ■

The crucial point is the fact that $\mathbb{E}[X \mid \mathcal{B}]$ is $\mathcal{B}$-measurable. Either of properties (1) and (2) characterizes the conditional expectation $\mathbb{E}[X \mid \mathcal{B}]$ among random variables of $L^1(\Omega, \mathcal{B}, \mathbb{P})$. In what follows, we will refer to (1) or (2) as the *characteristic property* of $\mathbb{E}[X \mid \mathcal{B}]$.

In the special case where $\mathcal{B} = \sigma(Y)$ is generated by a random variable Y, we will write indifferently

$$\mathbb{E}[X \mid \mathcal{B}] = \mathbb{E}[X \mid \sigma(Y)] = \mathbb{E}[X \mid Y].$$

We now turn to the definition of $\mathbb{E}[X \mid \mathcal{B}]$ for a nonnegative random variable X which allows X to take the value $+\infty$.

Theorem 2.2.2: Existence of Conditional Expectation for nonnegative random variables

Let X be a random variable with values in $[0,\infty]$. There exists a $\mathcal{B}$-measurable random variable with values in $[0,\infty]$, which is denoted by $\mathbb{E}[X \mid \mathcal{B}]$ and is such that, for every nonnegative $\mathcal{B}$-measurable random variable Z,

$$\mathbb{E}[XZ] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]Z]. \tag{iii}$$

Furthermore $\mathbb{E}[X \mid \mathcal{B}]$ is unique up to a $\mathcal{B}$-measurable set of probability zero.

Proof for Theorem.

We define $\mathbb{E}[X \mid \mathcal{B}]$ by setting

$$\mathbb{E}[X \mid \mathcal{B}] := \lim_{n\uparrow\infty} \mathbb{E}[X \wedge n \mid \mathcal{B}] \quad \text{a.s.}$$

This definition makes sense because, for every $n \in \mathbb{N}$, $X \wedge n$ is bounded hence integrable and thus $\mathbb{E}[X \wedge n \mid \mathcal{B}]$ is well defined by the previous section. Furthermore, the fact that the sequence $(\mathbb{E}[X \wedge n \mid \mathcal{B}])_{n\in\mathbb{N}}$ is (a.s.) increasing follows from the fact that the conditional expectation of a.s. non-negative random variable is also an a.s. non-negative random variable. Then, if Z is nonnegative and $\mathcal{B}$-measurable, the monotone convergence theorem implies that

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]Z] = \lim_{n\to\infty} \mathbb{E}[\mathbb{E}[X \wedge n \mid \mathcal{B}]Z\mathbb{1}_{\{Z<\infty\}}] = \lim_{n\to\infty} \mathbb{E}[(X \wedge n)Z\mathbb{1}_{\{Z<\infty\}}] = \mathbb{E}[XZ].$$

It remains to establish uniqueness. Let X' and X'' be two $\mathcal{B}$-measurable random variables with values in $[0,\infty]$, such that

$$\mathbb{E}[X'Z] = \mathbb{E}[X''Z]$$

for every nonnegative $\mathcal{B}$-measurable random variable Z. Let us fix two nonnegative rationals $a < b$, and take

$$Z = \mathbb{1}_{\{X' \le a < b \le X''\}}.$$

It follows that

$$a\,\mathbb{P}(X' \le a < b \le X'') \le \mathbb{E}[\mathbb{1}_{\{X' \le a < b \le X''\}}X'] = \mathbb{E}[\mathbb{1}_{\{X' \le a < b \le X''\}}X''] \ge b\,\mathbb{P}(X' \le a < b \le X''),$$

which is only possible if $\mathbb{P}(X' \le a < b \le X'') = 0$. Hence,

$$\mathbb{P}\left(\bigcup_{\substack{a,b\in\mathbb{Q}_+ \\ a<b}} \{X' \le a < b \le X''\}\right) = 0,$$

which implies $X' \ge X''$ a.s., and interchanging the roles of X' and X'' also gives

$X'' \geq X'$ a.s. ■

Remark: The uniqueness part of the theorem means that we should view $\mathbb{E}[X \mid \mathcal{B}]$ as an equivalence class of $\mathcal{B}$-measurable random variables that are equal a.s. But of course it will be more convenient to speak about "the random variable" $\mathbb{E}[X \mid \mathcal{B}]$.

In the case where X is also integrable, the definition of the preceding theorem is consistent with that of Theorem 2.2.1. Indeed (3) with $Z = 1$ shows that $\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]] = \mathbb{E}[X]$, and in particular $\mathbb{E}[X \mid \mathcal{B}] < \infty$ a.s. so that we can take a representative with values in $[0, \infty)$ and $\mathbb{E}[X \mid \mathcal{B}] \in L^1$. We then just have to note that (1) (in the theorem we are citing) is the special case of (3) where $Z = \mathbb{1}_B$, $B \in \mathcal{B}$.

Similarly as in the case of integrable variables, we will refer to (3) (or to its variant when $Z = \mathbb{1}_B$, $B \in \mathcal{B}$) as the characteristic property of $\mathbb{E}[X \mid \mathcal{B}]$.

2.2.2 Properties of conditional Expectations

We now list the properties of conditional expectaitons :

Proposition 2.2.3

1. For a $X, Y \in L^1(\Omega, \mathcal{A}, \mathbb{P}))$ and $\alpha, \beta \in \mathbb{R}$ we have :
$$\mathbb{E}(\alpha X + \beta Y \mid \mathcal{B}) = \alpha \mathbb{E}(X \mid \mathcal{B}) + \beta \mathbb{E}(Y \mid \mathcal{B})$$
2. For a X, Y non negative random variables and $\alpha, \beta \in \mathbb{R}^+$ we have :
$$\mathbb{E}(\alpha X + \beta Y \mid \mathcal{B}) = \alpha \mathbb{E}(X \mid \mathcal{B}) + \beta \mathbb{E}(Y \mid \mathcal{B})$$
3. If $X \in L^1(\Omega, \mathcal{A}, \mathbb{P}))$ (or if X is just non negative) and X is $\mathcal{B}$-measurable, then :
$$\mathbb{E}[X \mid \mathcal{B}] = X$$
4. If $X \in L^1(\Omega, \mathcal{A}, \mathbb{P}))$ (or if X is just non negative), then :
$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]] = \mathbb{E}[X]$$
5. If $X, X' \in L^1(\Omega, \mathcal{A}, \mathbb{P}))$ and $X \geq X'$, then :
$$\mathbb{E}[X \mid \mathcal{B}] \geq \mathbb{E}[X' \mid \mathcal{B}] \quad \text{a.s}$$
6. If $X \in L^1(\Omega, \mathcal{A}, \mathbb{P}))$, then :
$$|\mathbb{E}[X \mid \mathcal{B}]| \leq \mathbb{E}[|X| \mid \mathcal{B}] \quad \text{a.s.}$$
and, consequently,
$$\mathbb{E}[|\mathbb{E}[X \mid \mathcal{B}]|] \leq \mathbb{E}[|X|].$$
Therefore the mapping $X \mapsto \mathbb{E}[X \mid \mathcal{B}]$ is a contraction of $L^1(\Omega, \mathcal{A}, \mathbb{P}))$.

Proof for Proposition.

The proofs for these properties are left exercise for the reader. ■

Now the properties allowing intereaction between limits and conditional expectations :

Proposition 2.2.4

1. If $(X_n)_{n\in\mathbb{N}}$ is an increasing sequence of nonnegative random variables, and $X = \lim \uparrow X_n$, then

$$\mathbb{E}[X \mid \mathcal{B}] = \lim_{n\to\infty} \uparrow \mathbb{E}[X_n \mid \mathcal{B}], \quad \text{a.s.}$$

   As consequence : if $(Y_n)_{n\in\mathbb{N}}$ is a sequence of nonnegative random variables, we have

$$\mathbb{E}\left[\sum_{n\in\mathbb{N}} Y_n \,\middle|\, \mathcal{B}\right] = \sum_{n\in\mathbb{N}} \mathbb{E}[Y_n \mid \mathcal{B}].$$

2. If $(X_n)_{n\in\mathbb{N}}$ is any sequence of nonnegative random variables, then

$$\mathbb{E}[\liminf X_n \mid \mathcal{B}] \leq \liminf \mathbb{E}[X_n \mid \mathcal{B}], \quad \text{a.s.}$$

3. Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of integrable random variables that converges a.s. to X. Assume that there exists a nonnegative random variable Z such that $|X_n| \leq Z$ a.s. for every $n \in \mathbb{N}$, and $\mathbb{E}[Z] < \infty$. Then,

$$\mathbb{E}[X \mid \mathcal{B}] = \lim_{n\to\infty} \mathbb{E}[X_n \mid \mathcal{B}], \quad \text{a.s.}$$

   The convergence is also in $\mathcal{L}^1$ that is :

$$\mathbb{E}\left[\big|\mathbb{E}[X \mid \mathcal{B}] - \mathbb{E}[X_n \mid \mathcal{B}]\big|\right] \longrightarrow 0 \quad \textit{as } n \to +\infty$$

Proof for Proposition.

1. It follows from last property that we have $\mathbb{E}[X \mid \mathcal{B}] \geq \mathbb{E}[Y \mid \mathcal{B}]$ if $X \geq Y \geq 0$. Under the assumptions of (1), we can therefore set

$$X' = \lim \uparrow \mathbb{E}[X_n \mid \mathcal{B}],$$

   which is a $\mathcal{B}$-measurable random variable with values in $[0, \infty)$. Then, for every nonnegative $\mathcal{B}$-measurable random variable Z, the monotone convergence theorem gives

$$\mathbb{E}[ZX'] = \lim \uparrow \mathbb{E}[Z\mathbb{E}[X_n \mid \mathcal{B}]] = \lim \uparrow \mathbb{E}[ZX_n] = \mathbb{E}[ZX],$$

   which implies $X' = \mathbb{E}[X \mid \mathcal{B}]$ thanks to the characteristic property (3) from theorem 2.2.1. The second assertion in (1) follows by applying the first one to $X_n = Y_1 + \cdots + Y_n$.

2. Using (1), we have

$$\mathbb{E}[\liminf X_n \mid \mathcal{B}] = \mathbb{E}\left[\lim_{k\uparrow\infty}\left(\inf_{n\geq k} X_n\right)\middle|\,\mathcal{B}\right] = \lim_{k\uparrow\infty}\mathbb{E}\left[\inf_{n\geq k} X_n\,\middle|\,\mathcal{B}\right].$$

3. It suffices to apply (2) twice:

$$\mathbb{E}[Z - X \mid \mathcal{B}] = \mathbb{E}[\liminf(Z - X_n) \mid \mathcal{B}] \leq \mathbb{E}[Z \mid \mathcal{B}] - \limsup \mathbb{E}[X_n \mid \mathcal{B}],$$

$$\mathbb{E}[Z + X \mid \mathcal{B}] = \mathbb{E}[\liminf(Z + X_n) \mid \mathcal{B}] \leq \mathbb{E}[Z \mid \mathcal{B}] + \liminf \mathbb{E}[X_n \mid \mathcal{B}],$$

which leads to

$$\mathbb{E}[X \mid \mathcal{B}] \leq \liminf \mathbb{E}[X_n \mid \mathcal{B}] \leq \limsup \mathbb{E}[X_n \mid \mathcal{B}] \leq \mathbb{E}[X \mid \mathcal{B}],$$

giving the desired almost sure convergence. The convergence in L^1 is now a consequence of the dominated convergence theorem, since we have

$$|\mathbb{E}[X_n \mid \mathcal{B}]| \leq \mathbb{E}[|X_n| \mid \mathcal{B}] \leq \mathbb{E}[Z \mid \mathcal{B}]$$

and $\mathbb{E}[\mathbb{E}[Z \mid \mathcal{B}]] = \mathbb{E}[Z] < \infty$.

■

Theorem 2.2.5: Jensen Inequality conditional Expectation

If $f : \mathbb{R} \to \mathbb{R}_+$ is convex, and if $X \in L^1$, then

$$\mathbb{E}[f(X) \mid \mathcal{B}] \geq f(\mathbb{E}[X \mid \mathcal{B}]).$$

Proof for Theorem.

Set

$$E_f = \{(a,b) \in \mathbb{R}^2 : \forall x \in \mathbb{R},\ f(x) \geq ax + b\}.$$

Then,

$$\forall x \in \mathbb{R}, \quad f(x) = \sup_{(a,b)\in E_f}(ax+b) = \sup_{(a,b)\in E_f\cap\mathbb{Q}^2}(ax+b).$$

We can take advantage of the fact that $\mathbb{Q}^2$ is countable to discard a countable collection of sets of probability zero and to get that, a.s.,

$$\mathbb{E}[f(X) \mid \mathcal{B}] = \mathbb{E}\left[\sup_{(a,b)\in E_f\cap\mathbb{Q}^2}(aX+b)\,\middle|\,\mathcal{B}\right] \geq \sup_{(a,b)\in E_f\cap\mathbb{Q}^2}\mathbb{E}[aX+b \mid \mathcal{B}] = f(\mathbb{E}[X \mid \mathcal{B}]).$$

■

As a consequence of Jensen's inequality, we obtain that, for every $p \geq 1$, the mapping $X \mapsto \mathbb{E}[X \mid \mathcal{B}]$ maps $L^p(\Omega, \mathcal{A}, \mathbb{P})$ into itself, and is a contraction of $L^p(\Omega, \mathcal{A}, \mathbb{P})$. Indeed, for every $X \in L^p(\Omega, \mathcal{A}, \mathbb{P})$, we have using property (6) in Proposition 2.2.3 ,

$$\mathbb{E}[|\mathbb{E}[X \mid \mathcal{B}]|^p] \leq \mathbb{E}[\mathbb{E}[|X| \mid \mathcal{B}]^p] \leq \mathbb{E}[\mathbb{E}[|X|^p \mid \mathcal{B}]] = \mathbb{E}[|X|^p].$$

This contraction property also holds for $p = \infty$.

Theorem 2.2.6: The orthogonal projection property

If $X \in L^2(\Omega, \mathcal{A}, \mathbb{P})$, then $\mathbb{E}[X \mid \mathcal{B}]$ is the orthogonal projection of X on $L^2(\Omega, \mathcal{B}, \mathbb{P})$.

Proof for Theorem.

Jensen's inequality shows that $\mathbb{E}[X \mid \mathcal{B}]^2 \leq \mathbb{E}[X^2 \mid \mathcal{B}]$, a.s. This implies that

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]^2] \leq \mathbb{E}[\mathbb{E}[X^2 \mid \mathcal{B}]] = \mathbb{E}[X^2] < \infty,$$

and thus the random variable $\mathbb{E}[X \mid \mathcal{B}]$ belongs to $L^2(\Omega, \mathcal{B}, \mathbb{P})$.

On the other hand, for every bounded $\mathcal{B}$-measurable random variable Z,

$$\mathbb{E}[Z(X - \mathbb{E}[X \mid \mathcal{B}])] = \mathbb{E}[ZX] - \mathbb{E}[Z\mathbb{E}[X \mid \mathcal{B}]] = 0,$$

by the characteristic property (2) in Theorem 2.2.1 . Hence $X - \mathbb{E}[X \mid \mathcal{B}]$ is orthogonal to the space of bounded $\mathcal{B}$-measurable random variables, and the latter space is dense in $L^2(\Omega, \mathcal{B}, \mathbb{P})$ (for instance by Lemma 1.1.10). It follows that $X - \mathbb{E}[X \mid \mathcal{B}]$ is orthogonal to $L^2(\Omega, \mathcal{B}, \mathbb{P})$, which gives the desired result. ■

Theorem 2.2.7: Pull-out property of conditional expectation

Let X and Y be real random variables and assume that Y is $\mathcal{B}$-measurable. Then

$$\mathbb{E}[YX \mid \mathcal{B}] = Y\,\mathbb{E}[X \mid \mathcal{B}],$$

provided X and Y are both nonnegative, or X and YX are both integrable.

Proof for Theorem.

Suppose that $X \geq 0$ and $Y \geq 0$. Then, for every nonnegative $\mathcal{B}$-measurable random variable Z, we have

$$\mathbb{E}[Z(Y\mathbb{E}[X \mid \mathcal{B}])] = \mathbb{E}[(ZY)\mathbb{E}[X \mid \mathcal{B}]] = \mathbb{E}[ZYX],$$

using *characteristic property* (3) from theorem 2.2.1 and the fact that ZY is $\mathcal{B}$-measurable. Since $Y\mathbb{E}[X \mid \mathcal{B}]$ is a nonnegative $\mathcal{B}$-measurable random variable, the characteristic property from theorem 2.2.1 implies that

$$Y\mathbb{E}[X \mid \mathcal{B}] = \mathbb{E}[YX \mid \mathcal{B}].$$

In the case when X and YX are integrable, we get the desired result by decomposing

$$X = X^+ - X^-, \qquad Y = Y^+ - Y^-.$$

■

Theorem 2.2.8: Nested σ-Fields

Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be two sub-σ-fields of $\mathcal{A}$ such that $\mathcal{B}_1 \subset \mathcal{B}_2$. Then, for every nonnegative (or integrable) random variable X, we have

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}_2] \mid \mathcal{B}_1] = \mathbb{E}[X \mid \mathcal{B}_1].$$

Proof for Theorem.

Consider the case where $X \geq 0$. Let Z be a nonnegative $\mathcal{B}_1$-measurable random variable. Then, since Z is also $\mathcal{B}_2$-measurable,

$$\mathbb{E}[Z\,\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}_2] \mid \mathcal{B}_1]] = \mathbb{E}[Z\,\mathbb{E}[X \mid \mathcal{B}_2]] = \mathbb{E}[ZX].$$

Hence $\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}_2] \mid \mathcal{B}_1]$ satisfies the characteristic property of $\mathbb{E}[X \mid \mathcal{B}_1]$, and the desired result follows. ■

Remark: We have also $\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}_1] \mid \mathcal{B}_2] = \mathbb{E}[X \mid \mathcal{B}_1]$ under the same assumptions, but this is trivial since $\mathbb{E}[X \mid \mathcal{B}_1]$ is $\mathcal{B}_2$-measurable.

Theorem 2.2.9: Independence and conditional expectaiton

Two sub-σ-fields $\mathcal{B}_1$ and $\mathcal{B}_2$ of $\mathcal{A}$ are independent if and only if, for every $B \in \mathcal{B}_2$, we have

$$\mathbb{E}[\mathbb{1}_B \mid \mathcal{B}_1] = \mathbb{P}(B).$$

Furthermore, if $\mathcal{B}_1$ and $\mathcal{B}_2$ are independent, we have also $\mathbb{E}[X \mid \mathcal{B}_1] = \mathbb{E}[X]$ for every nonnegative $\mathcal{B}_2$-measurable random variable X and for every $X \in L^1(\Omega, \mathcal{B}_2, \mathbb{P})$.

Proof for Theorem.

Left exercise for the reader. ■

Theorem 2.2.10: Factorization under independence

Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be two measurable spaces, and let X and Y be random variables taking values in E and F respectively. Assume that X is independent of $\mathcal{B}$ and that Y is $\mathcal{B}$-measurable. Then, for every $\mathcal{E} \otimes \mathcal{F}$-measurable function $g : E \times F \to \mathbb{R}_+$,

$$\mathbb{E}[g(X, Y) \mid \mathcal{B}] = \int g(x, Y)\,\mathbb{P}_X(dx),$$

where we recall that $\mathbb{P}_X$ denotes the law of X. The right-hand side is the composition of the random variable Y with the mapping $\Psi : F \to \mathbb{R}_+$ defined by

$$\Psi(y) = \int g(x, y)\, \mathbb{P}_X(dx).$$

Proof for Theorem.

We have to prove that, for any nonnegative $\mathcal{B}$-measurable random variable Z, we have

$$\mathbb{E}[g(X, Y)Z] = \mathbb{E}[\Psi(Y)Z].$$

Write $\mathbb{P}_{X,Y,Z}$ for the law of the triple (X, Y, Z), which is a probability measure on $E \times F \times \mathbb{R}_+$. Since X is independent of $\mathcal{B}$, X is independent of the pair (Y, Z), and therefore

$$\mathbb{P}_{X,Y,Z} = \mathbb{P}_X \otimes \mathbb{P}_{Y,Z}.$$

Then, using the Fubini theorem, we have

$$\begin{aligned}
\mathbb{E}[g(X, Y)Z] &= \int g(x, y)z\, \mathbb{P}_{X,Y,Z}(dx\, dy\, dz) \\
&= \int g(x, y)z\, \mathbb{P}_X(dx)\, \mathbb{P}_{Y,Z}(dy\, dz) \\
&= \int_{F\times\mathbb{R}_+} z \left(\int_E g(x, y)\, \mathbb{P}_X(dx)\right) \mathbb{P}_{Y,Z}(dy\, dz) \\
&= \int_{F\times\mathbb{R}_+} z\, \Psi(y)\, \mathbb{P}_{Y,Z}(dy\, dz) \\
&= \mathbb{E}[\Psi(Y)Z].
\end{aligned}$$

which was the desired result. ■

Remark: The mapping Ψ is measurable thanks to the Fubini theorem. The theorem can be explained informally as follows. If we condition with respect to the σ-field $\mathcal{B}$, the random variable Y, which is $\mathcal{B}$-measurable, behaves like a constant, but on the other hand $\mathcal{B}$ gives no information on the random variable X. Thus the best approximation of $g(X, Y)$ knowing $\mathcal{B}$ is obtained by integrating $g(\cdot, Y)$ with respect to the law of X.

Theorem 2.2.11: Invariance under independent information

Let Z be a random variable in L^1, and let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two sub-σ-fields of $\mathcal{A}$. Assume that $\mathcal{H}_2$ is independent of $\sigma(Z) \vee \mathcal{H}_1$. Then,

$$\mathbb{E}[Z \mid \mathcal{H}_1 \vee \mathcal{H}_2] = \mathbb{E}[Z \mid \mathcal{H}_1].$$

Proof for Theorem.

It suffices to prove that the equality

$$\mathbb{E}[\mathbb{1}_A Z] = \mathbb{E}[\mathbb{1}_A \mathbb{E}[Z \mid \mathcal{H}_1]]$$

holds. ■

2.2.3 Transition Probabilities and Conditional Distributions

Definition 2.2.12: Transition probability

Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be two measurable spaces. A transition probability (or transition kernel) from E into F is a mapping

$$\nu : E \times \mathcal{F} \longrightarrow [0, 1]$$

which satisfies the following two properties:

1. for every $x \in E$, $A \mapsto \nu(x, A)$ is a probability measure on $(F, \mathcal{F})$;
2. for every $A \in \mathcal{F}$, the function $x \mapsto \nu(x, A)$ is $\mathcal{E}$-measurable.

At an intuitive level, each time one fixes a "starting point" $x \in E$, the probability measure $\nu(x, \cdot)$ gives a way to choose an "arrival point" $y \in F$ in a random manner (but with a law depending on the starting point x). This notion plays a fundamental role in the theory of Markov chains.

Proposition 2.2.13

Let ν be a transition probability from E into F.

1. If h is a nonnegative (resp. bounded) measurable function on $(F, \mathcal{F})$, then

$$\varphi(x) = \int h(y)\, \nu(x, dy), \qquad x \in E,$$

is a nonnegative (resp. bounded) measurable function on E.

2. If γ is a probability measure on $(E, \mathcal{E})$, then

$$\mu(A) = \int \gamma(dx)\, \nu(x, A), \qquad A \in \mathcal{F},$$

is a probability measure on $(F, \mathcal{F})$.

Proof for Proposition.

We omit the easy proof of the proposition. When treating the nonnegative case of (1), we first consider simple functions, and then use an increasing passage to the limit. ∎

We now come to the relation between the notion of a transition probability and conditional expectations.

Definition 2.2.14: The conditional distribution

Let X and Y be two random variables taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$ respectively. Any transition probability ν from E into F such that, for every nonnegative measurable function h on F,

$$\mathbb{E}[h(Y) \mid X] = \int h(y)\,\nu(X, dy), \qquad \text{a.s.},$$

is called a conditional distribution of Y knowing X.

Remark The random variable $\int h(y)\,\nu(X, dy)$ is the composition of X with the function

$$x \mapsto \int h(y)\,\nu(x, dy),$$

which is measurable by last Proposition. In particular, this random variable is a function of X, as $\mathbb{E}[h(Y) \mid X]$ should be.

By definition, if ν is a conditional distribution of Y knowing X, we have, for every $A \in \mathcal{F}$,

$$\mathbb{P}(Y \in A \mid X) = \nu(X, A), \qquad \text{a.s.}$$

It is tempting to replace this equality of random variables by an equality of real numbers,

$$\mathbb{P}(Y \in A \mid X = x) = \nu(x, A),$$

for every $x \in E$. Although it gives the intuition behind the notion of conditional distribution, this last equality has no mathematical meaning in general, because one has typically $\mathbb{P}(X = x) = 0$ for every $x \in E$, which makes it impossible to define the conditional probability given the event $\{X = x\}$. The only rigorous formulation is thus the first equality $\mathbb{P}(Y \in A \mid X) = \nu(X, A)$.

Let us briefly discuss the uniqueness of the conditional distribution of Y knowing X. If ν and ν' are two such conditional distributions, we have, for every $A \in \mathcal{F}$,

$$\nu(X, A) = \mathbb{P}(Y \in A \mid X) = \nu'(X, A), \qquad \text{a.s.}$$

which is equivalent to saying that, for every $A \in \mathcal{F}$,

$$\nu(x, A) = \nu'(x, A), \qquad \mathbb{P}_X(dx)\text{-a.s.}$$

Suppose that the measurable space $(F, \mathcal{F})$ is such that any probability measure on $(F, \mathcal{F})$ is characterized by its values on a (given) countable collection of measurable sets (this property holds for $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, by considering the collection of all rectangles with rational coordinates). Then, we conclude from the preceding display that

$$\nu(x, \cdot) = \nu'(x, \cdot), \qquad \mathbb{P}_X(dx)\text{-a.s.}$$

So uniqueness holds in this sense (and clearly we cannot expect more). By abuse of language, we will nonetheless speak about *the conditional distribution* of Y knowing X.

Let us now consider the problem of the existence of conditional distributions.

Theorem 2.2.15: Existence of conditional distribution

Let X and Y be two random variables taking values in $(E,\mathcal{E})$ and $(F,\mathcal{F})$, respectively. Suppose that $(F,\mathcal{F})$ is a complete separable metric space equipped with its Borel σ-field. Then, there exists a conditional distribution of Y knowing X.

Proof for Theorem.

We will not prove this theorem, see Theorem 6.3 in [1] for the proof of a more general statement. ■

2.2.4 Evaluation of Conditional Expectation

Discret Case

If X is a discrete variable (E is countable), and Y any random variable. then we have the conditional distibution of Y given X, denoted by $\nu(x,A)$, defined by :

$$\nu(x,A) := \begin{cases} \frac{1}{\mathbb{P}(X=x)}\mathbb{P}(Y\in A, X=x) & \text{if } x\in E' := \{y\in E : \mathbb{P}(X=y)>0\}, \\ \delta_{y_0}(A) & \text{if } x\notin E', \end{cases}$$

where y_0 is an arbitrary fixed point of F. Theorefore we can conclude that :

$$\mathbb{E}[h(Y)\mid X] = \varphi(X)$$

where

$$\varphi(x) = \int h(y)\,\nu(x,dy) = \begin{cases} \frac{1}{\mathbb{P}(X=x)}\mathbb{E}(h(Y)\mathbb{1}_{\{X=x\}}) & \text{if } x\in E', \\ y_0 & \text{if } x\notin E', \end{cases}$$

Where y_0 is an arbitrary fixed point of F.

Continuous Case [Random Variables with a Density]

Suppose that X and Y take values in $\mathbb{R}^m$ and in $\mathbb{R}^n$ respectively, and that the pair (X,Y) has density $p(x,y)$, $(x,y)\in\mathbb{R}^m\times\mathbb{R}^n$. The density of X is then

$$q(x) = \int_{\mathbb{R}^n} p(x,y)\,dy,$$

where we take $q(x)=0$ if the integral is infinite. We may define the conditional distribution of Y knowing X by :

$$\nu(x,A) = \begin{cases} \frac{1}{q(x)}\int_A p(x,y)\,dy & \text{if } q(x)>0, \\ \delta_0(A) & \text{if } q(x)=0. \end{cases}$$

Theorefore we can conclude that :

$$\mathbb{E}[h(Y) \mid X] = \varphi(X)$$

where

$$\varphi(x) = \int h(y)\,\nu(x,dy) = \begin{cases} \dfrac{1}{q(y)} \displaystyle\int_{\mathbb{R}^m} h(x)\,p(x,y)\,dx & \text{if } q(y) > 0, \\ h(0) & \text{if } q(y) = 0. \end{cases}$$

2.3 Convergence of random variables

The study of convergence of random variables is fundamental in probability theory, particularly in the analysis of asymptotic behaviors and limit theorems. Different notions of convergence allow us to formalize how a sequence of random variables approximates another random variable under various probabilistic criteria. These modes of convergence differ in strength and applicability, and understanding their relationships is essential for results such as the law of large numbers and the central limit theorem.

Throughout this section, we consider a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. Let $(\mathbf{X}_n)_{n\in\mathbb{N}}$ be a sequence of random variables that mapsto $\mathbb{R}^d$ and let $\mathbf{X}$ be another random variable taking values in $\mathbb{R}^d$. We introduce below the principal notions of convergence that will be used in the sequel.

Definition 2.3.1: Almost sure convergence

The sequence $(\mathbf{X}_n)$ is said to converge almost surely to $\mathbf{X}$, denoted by

$$\mathbf{X}_n \xrightarrow[n\to\infty]{a.s.} \mathbf{X},$$

if

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n\to\infty} \mathbf{X}_n(\omega) = \mathbf{X}(\omega)\right\}\right) = 1.$$

The *Borel–Cantelli Lemma* 2.1.19 is a powerful tool for proving (or disproving) almost sure convergence. The main idea is to convert the convergence problem into the study of a sequence of *bad* events as discribed in Figure 3 and define the bad events as follow :

$$A_n = \{|X_n - X| > \varepsilon\},$$

Definition 2.3.2: Convergence in probability

We say that the sequence $(\mathbf{X}_n)_{n\in\mathbb{N}}$ converges in probability to $\mathbf{X}$, and write

$$\mathbf{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mathbf{X},$$

if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\Big(|\mathbf{X}_n - \mathbf{X}| > \varepsilon\Big) = 0.$$

Proposition 2.3.3

We remind that d is the distance on $L^0(\Omega, \mathcal{A}, \mathbb{P})$ defined by :

$$d(\mathbf{X}, \mathbf{Y}) = \mathbb{E}[|\mathbf{X} - \mathbf{Y}| \wedge 1]$$

We have :

$$\mathbf{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mathbf{X} \Longleftrightarrow d(\mathbf{X}_n, \mathbf{X}) \xrightarrow[n\to\infty]{} 0$$

Proof for Proposition.

Left exercice for the reader.
Hint: use the fact that the space $L^0_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ is complete for the distance d by proposition 1.3.7. ■

Definition 2.3.4: Convergence in Lp

Let $p \in [1, \infty)$ and assume that

$$\mathbb{E}[|\mathbf{X}|^p] < \infty \quad \text{and} \quad \mathbb{E}[|\mathbf{X}_n|^p] < \infty \quad \text{for all } n \in \mathbb{N}.$$

The sequence $(\mathbf{X}_n)$ is said to converge to $\mathbf{X}$ in L^p, written

$$\mathbf{X}_n \xrightarrow[n\to\infty]{\mathcal{L}^p} \mathbf{X},$$

if

$$\lim_{n\to\infty} \mathbb{E}\Big[|\mathbf{X}_n - \mathbf{X}|^p\Big] = 0.$$

This is equivalent to saying that each component sequence of $(\mathbf{X}_n)_{n\in\mathbb{N}}$ converges to the corresponding component of $\mathbf{X}$ in the Banach space $L^p(\Omega, \mathcal{A}, \mathbb{P})$. One could also consider the convergence in L^∞, but this convergence is very rarely used in probability theory and we will not discuss it.

A relevant remark is that domination transforms a pointwise convergence problem into an L^p-convergence considerably more straightforward.

2.3.1 Relationship between modes of convergences

The different notions of convergence introduced in this chapter are related by a number of fundamental implications. Among them, convergence in probability occupies an intermediate position: it is weaker than both almost sure convergence and L^p-convergence, yet stronger than convergence in distribution. Consequently, every almost sure convergence or L^p-convergence yields convergence in probability, while convergence in probability itself implies convergence in distribution.

The latter notion is introduced and studied in the following sections. Briefly, convergence in distribution describes the convergence of the laws $(\mathbb{P}_{X_n})_{n\in\mathbb{N}}$ toward the law $\mathbb{P}_X$.

The three direct implications are summarized by the following chain:

- almost sure convergence imply probability
- L^p-convergence imply probability
- convergence in probability imply convergence in distribution

The converse implications are, in general, false. Throughout this section, we establish the direct implications above and investigate the conditions under which some of the converse implications may nevertheless hold.

Proposition 2.3.5

If

$$X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$$

Then :

$$X_n \xrightarrow[n\to\infty]{\mathcal{D}} X$$

Proof for Proposition.

Suppose first that X_n converges a.s. to X. Then, for every $\varphi \in C_b(\mathbb{R}^d)$, $\varphi(X_n)$ converges a.s. to $\varphi(X)$ and the dominated convergence theorem gives $\mathbb{E}[\varphi(X_n)] \to \mathbb{E}[\varphi(X)]$.

If X_n only converges in probability to X, then we know from Proposition 2.3.7 that there is a subsequence of $(X_n)_{n\in\mathbb{N}}$ that converges a.s. to X, and thus, for every $\varphi \in C_b(\mathbb{R}^d)$, $\mathbb{E}[\varphi(X_n)]$ converges to $\mathbb{E}[\varphi(X)]$ along this subsequence. But then, the same argument shows that, from any subsequence of $(X_n)_{n\in\mathbb{N}}$, we can extract a subsequence along which $\mathbb{E}[\varphi(X_n)]$ converges to $\mathbb{E}[\varphi(X)]$. This is only possible if $\mathbb{E}[\varphi(X_n)] \to \mathbb{E}[\varphi(X)]$ as $n \to \infty$. ■

Remark: The converse to the proposition is (of course) false, because, as we already mentioned, the convergence in distribution of X_n to X does not determine the limiting variable X.

Proposition 2.3.6

If :

$$\mathbf{X}_n \xrightarrow[n\to\infty]{a.s.} \mathbf{X}, \quad \text{or} \quad \mathbf{X}_n \xrightarrow[n\to\infty]{\mathcal{L}^p} \mathbf{X}.$$

Then :

$$\mathbf{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mathbf{X}.$$

Proof for Proposition.

1. If $\mathbf{X}_n$ converges to $\mathbf{X}$ in almost surely, then define :

$$Z_n = \mathbb{1}_{\{||\mathbf{X}_n - \mathbf{X}|| > \varepsilon\}}$$

Since $||\mathbf{X}_n - \mathbf{X}|| \xrightarrow[n\to\infty]{} 0$ then Z_n converge almost surely to 0, then apply the Dominated Convergence Theorem to conclude that :

$$\mathbb{P}(||\mathbf{X}_n - \mathbf{X}|| > \varepsilon) = \mathbb{E}[\mathbb{1}_{\{||\mathbf{X}_n - \mathbf{X}|| > \varepsilon\}}] = \mathbb{E}[Z_n] \xrightarrow[n\to\infty]{} 0$$

And therefore the convergence in probability.

2. If $\mathbf{X}_n$ converges to $\mathbf{X}$ in L^p, then by Markov inequality 1.2.4 we conclude the convergence in probability as follow :

$$P(||X_n - X|| > \varepsilon) = P(||X_n - X||^p > \varepsilon^p) \leq \frac{E|X_n - X|^p}{\varepsilon^p} \xrightarrow[n\to\infty]{} 0$$

■

Although convergence in probability is weaker than almost sure convergence, it has a key compactness property: every sequence converging in probability admits a subsequence that converges almost surely to the same limit.

Proposition 2.3.7

If :

$$\mathbf{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mathbf{X}.$$

Then :

$$\exists, (n_k) \uparrow \infty \quad \text{such that} \quad \mathbf{X}_{n_k} \xrightarrow[k\to\infty]{a.s.} \mathbf{X}$$

Proof for Proposition.

Choose the subsequence so that

$$P(|X_{n_k} - X| > 2^{-k}) < 2^{-k}.$$

Then

$$\sum_k P(|X_{n_k} - X| > 2^{-k}) < \infty.$$

Apply the first Borel-Cantelli lemma. Eventually,

$$|X_{n_k} - X| \leq 2^{-k},$$

hence almost sure convergence. ■

The previous result only yields almost sure convergence along a subsequence and cannot, in general, be strengthened to full almost sure convergence. One may then ask whether convergence in probability can be upgraded to (L^p)-convergence. This is false without extra assumptions. A key sufficient condition is a uniform (L^r) bound with $(r > p)$, which ensures (L^p)-convergence via uniform integrability.

Proposition 2.3.8

If :

$$\begin{cases} \mathbf{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mathbf{X}. \\ \exists r \in (1,\infty) \quad \text{such :} \quad (\mathbb{E}[|\mathbf{X}_n|^r])_{n\in\mathbb{N}} \quad \text{is bounded.} \end{cases}$$

Then :

$$\begin{cases} \mathbb{E}[|\mathbf{X}|^r] < \infty \\ \forall p \in [1,r), \quad \mathbf{X}_n \xrightarrow[n\to\infty]{\mathcal{L}^p} \mathbf{X}. \end{cases}$$

Proof for Proposition.

By assumption, there is a constant C such that $\mathbb{E}[|\mathbf{X}_n|^r] \leq C$ for every n. By the previous proposition we can find a subsequence $(\mathbf{X}_{n_k})$ that converges almost surely to $\mathbf{X}$. By Fatou's lemma,

$$\begin{aligned} \mathbb{E}[|\mathbf{X}|^r] &= \mathbb{E}\Big[\liminf_{k\to\infty} |\mathbf{X}_{n_k}|^r\Big] \\ &\leq \liminf_{k\to\infty} \mathbb{E}[|\mathbf{X}_{n_k}|^r] \\ &\leq C. \end{aligned}$$

Then, for every $p \in [1,r)$ and every $\varepsilon > 0$,

$$\begin{aligned} \mathbb{E}[|\mathbf{X}_n - \mathbf{X}|^p] &= \mathbb{E}\Big[|\mathbf{X}_n - \mathbf{X}|^p \mathbb{1}_{\{|\mathbf{X}_n - \mathbf{X}| \leq \varepsilon\}}\Big] + \mathbb{E}\Big[|\mathbf{X}_n - \mathbf{X}|^p \mathbb{1}_{\{|\mathbf{X}_n - \mathbf{X}| > \varepsilon\}}\Big] \\ &\leq \varepsilon^p + \mathbb{E}[|\mathbf{X}_n - \mathbf{X}|^r]^{p/r} \mathbb{P}(|\mathbf{X}_n - \mathbf{X}| > \varepsilon)^{1-p/r} \\ &\leq \varepsilon^p + (2C)^{p/r} \mathbb{P}(|\mathbf{X}_n - \mathbf{X}| > \varepsilon)^{1-p/r}. \end{aligned}$$

Since $\mathbf{X}_n$ converges in probability to $\mathbf{X}$,

$$\limsup_{n\to\infty} \mathbb{E}[|\mathbf{X}_n - \mathbf{X}|^p] \leq \varepsilon^p.$$

Letting $\varepsilon \to 0$ yields $\mathbb{E}[|\mathbf{X}_n - \mathbf{X}|^p] \to 0$. ■

Another important criterion for strengthening convergence in probability is provided by Scheffé's lemma. Instead of requiring higher-order moment bounds, it assumes the convergence of the first moments. Under the additional assumptions of non-negativity and convergence of expectations, convergence in probability is sufficient to conclude convergence in (L^1).

Lemma 2.3.9: Scheffé's lemma

If :

$$\begin{cases} \mathbf{X}_n \geq 0 \quad \text{and} \quad \mathbf{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mathbf{X}. \\ \forall n \in \mathbb{N} : \quad \mathbf{X}_n \in L^1 \quad \text{and} \quad \mathbf{X} \in L^1 \\ \mathbb{E}[\mathbf{X}_n] \xrightarrow[n\to\infty]{} \mathbb{E}[\mathbf{X}] \end{cases}$$

Then :

$$\mathbf{X}_n \xrightarrow[n\to\infty]{\mathcal{L}^1} \mathbf{X}.$$

Proof for Lemma

We write

$$\mathbb{E}[|\mathbf{X}_n - \mathbf{X}|] = \mathbb{E}[\mathbf{X}_n - \mathbf{X}] + 2\mathbb{E}[(\mathbf{X}_n - \mathbf{X})^-].$$

We have $\mathbb{E}[\mathbf{X}_n - \mathbf{X}] \to 0$, and the bound $(\mathbf{X}_n - \mathbf{X})^- \leq \mathbf{X}$ allows us to apply the dominated convergence theorem to obtain $\mathbb{E}[(\mathbf{X}_n - \mathbf{X})^-] \to 0$ (using that almost sure convergence in the dominated convergence theorem can be replaced by convergence in probability). ■

A more general criterion for upgrading convergence in probability to (L^1)-convergence is provided by **Vitali's theorem**, which replaces moment assumptions by the weaker notion of uniform integrability. Since the proof of this result relies on techniques from martingale theory, it will be deferred to the next chapter. We therefore conclude this section by introducing the notion of uniform integrability.

A family $(X_i)_{i\in I}$ of random variables in $L^1(\Omega, \mathcal{A}, \mathbb{P})$ is said to be *uniformly integrable (u.i.)* if

$$\lim *a \to +\infty \sup_{i\in I} \mathbb{E}! \left[|X_i| \mathbb{1}_{|X_i|>a}\right] = 0.$$

Theorem 2.3.10: Vitali convergence theorem

If

$$\begin{cases} \mathbf{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mathbf{X}. \\ \mathbf{X}_n \in L^1 \quad \text{for every } n \in \mathbb{N}. \\ \text{the sequence } (\mathbf{X}_n)_{n\in\mathbb{N}} \text{ is u.i.} \end{cases}$$

Then the following assertions are equivalent :

$$\mathbf{X}_n \xrightarrow[n\to\infty]{L^1} \mathbf{X}$$

Proof for Theorem.

It would be shown in the section of uniform integrability 3.3.4 (next chapter). ■

It's also relevant to see that if there is $Z \in L^1$ such $|\mathbf{X}_n| \leq Z$ a.s. for every $n \in \mathbb{N}$, then the sequence $(\mathbf{X}_n)_{n\in\mathbb{N}}$ is u.i., which gives a simpler condition to prove L^1-convergence. In fact, we have :

$$\mathbb{E}[|\mathbf{X}_n \mathbb{1}_{|\mathbf{X}_n \geq a}] \leq \mathbb{E}[Z\mathbb{1}_{Z\geq a}]$$

Finaly, taking supremum and limits :

$$\lim_{a\to\infty} \sup_{n\in\mathbb{N}} \mathbb{E}[|\mathbf{X}_n|\mathbb{1}_{|\mathbf{X}_n \geq a}] \leq \lim_{a\to\infty} \mathbb{E}[Z\mathbb{1}_{Z\geq a}] = 0 \quad \text{Because :} \quad Z\mathbb{1}_{Z\geq a} \xrightarrow[n\to\infty]{a.s.} 0$$

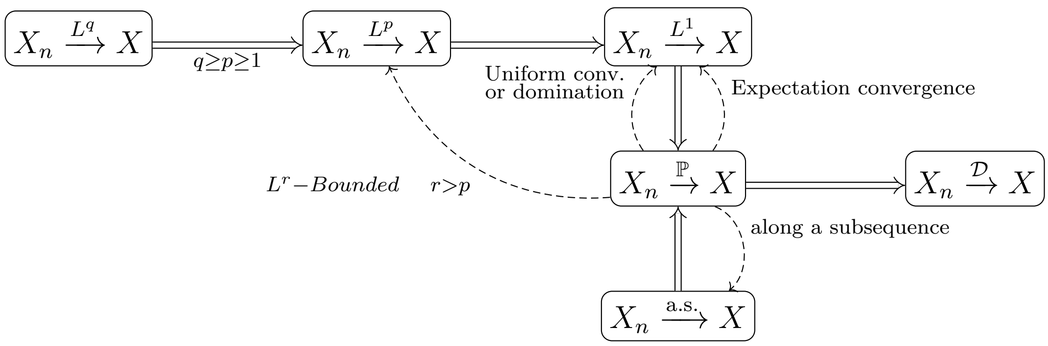

FIGURE 4 – Relations between the different convergences

Figure 4 illustrates the relations between the different types of convergence that we have introduced. The dashed lines give partial converses (under the stated conditions) that correspond to results stated in Propositions 2.3.6, 2.3.7, 2.3.9, 2.3.8, and to the dominated convergence theorem.

2.3.2 Convergence in Distribution

Recall that $C_b(\mathbb{R}^d)$ denote the set of all bounded continuous functions from $\mathbb{R}^d$ into $\mathbb{R}$. We equip $C_b(\mathbb{R}^d)$ with the supremum norm

$$\|\varphi\| = \sup_{x\in\mathbb{R}^d} |\varphi(x)|.$$

Also recall the notation $C_c(\mathbb{R}^d)$ for the space of all real continuous functions with compact support on $\mathbb{R}^d$. We let $M_1(\mathbb{R}^d)$ denote the set of all probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$.

Definition 2.3.11: Convergence in distribution

A sequence $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ in $M_1(\mathbb{R}^d)$ converges weakly to $\mathbb{Q} \in M_1(\mathbb{R}^d)$ if

$$\forall \varphi \in C_b(\mathbb{R}^d), \qquad \int \varphi \, d\mathbb{Q}_n \xrightarrow[n\to\infty]{} \int \varphi \, d\mathbb{Q}.$$

A sequence $(X_n)_{n\in\mathbb{N}}$ of random variables with values in $\mathbb{R}^d$ converges in distribution to a random variable X with values in $\mathbb{R}^d$ if the sequence $(P_{X_n})_{n\in\mathbb{N}}$ converges weakly to P_X.

By Properties of Expectation 1.2.1, this is equivalent to saying that

$$\forall \varphi \in C_b(\mathbb{R}^d), \qquad \mathbb{E}[\varphi(X_n)] \xrightarrow[n\to\infty]{} \mathbb{E}[\varphi(X)].$$

We will write $\mathbb{Q}_n \xrightarrow[n\to\infty]{(w)} \mathbb{Q}$, resp. $X_n \xrightarrow[n\to\infty]{\mathcal{D}} X$, if the sequence $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ converges weakly to $\mathbb{Q}$, resp. if $(X_n)_{n\in\mathbb{N}}$ converges in distribution to X.

Remark

- There is some abuse of terminology in saying that the sequence $(X_n)_{n\in\mathbb{N}}$ converges in distribution to X, because the limiting random variable X is not determined uniquely (only its distribution P_X is determined). For this reason, we will sometimes write that a sequence $(X_n)_{n\in\mathbb{N}}$ of random variables converges in distribution to a probability measure $\mathbb{Q}$ — one should of course understand that the laws P_{X_n} converge weakly to $\mathbb{Q}$. We may also note that the convergence in distribution makes sense even if the random variables X_n, $n \in \mathbb{N}$ are defined on different probability spaces (in the present book, we will however always assume that they are defined on the same probability space). This makes the convergence in distribution very different from the other types of convergence studied in this section.
- Although the definition of weak convergence is formulated in terms of all bounded continuous functions, such a family is unnecessarily large in practice. It is therefore natural to ask whether convergence only needs to be verified on a smaller collection of test functions. The next proposition shows that this is indeed the case.

Proposition 2.3.12

Let $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ and $\mathbb{Q}$ be probability measures on $\mathbb{R}^d$. Let H be a subset of $C_b(\mathbb{R}^d)$ whose closure (with respect to the supremum norm) contains $C_c(\mathbb{R}^d)$. The following properties are equivalent.

1. The sequence $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ converges weakly to $\mathbb{Q}$.
2. We have
$$\forall \varphi \in C_c(\mathbb{R}^d), \qquad \int \varphi \, d\mathbb{Q}_n \xrightarrow[n\to\infty]{} \int \varphi \, d\mathbb{Q}.$$
3. We have
$$\forall \varphi \in H, \qquad \int \varphi \, d\mathbb{Q}_n \xrightarrow[n\to\infty]{} \int \varphi \, d\mathbb{Q}.$$

Proof for Proposition.

It is obvious that (1)⇒(2) and (1)⇒(3). Then suppose that (2) holds. Let $\varphi \in C_b(\mathbb{R}^d)$ and let $(f_k)_{k\in\mathbb{N}}$ be a sequence in $C_c(\mathbb{R}^d)$ such that $0 \le f_k \le 1$ for every $k \in \mathbb{N}$, and $f_k \uparrow 1$ as $k \to \infty$. Then, for every $k \in \mathbb{N}$, $\varphi f_k \in C_c(\mathbb{R}^d)$ and thus

$$\int \varphi f_k \, d\mathbb{Q}_n \xrightarrow[n\to\infty]{} \int \varphi f_k \, d\mathbb{Q}. \tag{i}$$

On the other hand,

$$\left|\int \varphi\, d\mathbb{Q}_n - \int \varphi f_k\, d\mathbb{Q}_n\right| \le \|\varphi\| \int (1-f_k)\, d\mathbb{Q}_n = \|\varphi\| \left(1 - \int f_k\, d\mathbb{Q}_n\right),$$

and similarly

$$\left|\int \varphi\, d\mathbb{Q} - \int \varphi f_k\, d\mathbb{Q}\right| \le \|\varphi\| \left(1 - \int f_k\, d\mathbb{Q}\right).$$

Hence, using (i), we get for every $k \in \mathbb{N}$,

$$\begin{aligned}\limsup_{n\to\infty} \left|\int \varphi\, d\mathbb{Q}_n - \int \varphi\, d\mathbb{Q}\right| &\le \limsup_{n\to\infty} \left|\int \varphi\, d\mathbb{Q}_n - \int \varphi f_k\, d\mathbb{Q}_n\right| + \left|\int \varphi\, d\mathbb{Q} - \int \varphi f_k\, d\mathbb{Q}\right| \\ &\le \|\varphi\| \left(\limsup_{n\to\infty}\left(1 - \int f_k\, d\mathbb{Q}_n\right) + \left(1 - \int f_k\, d\mathbb{Q}\right)\right) \\ &= 2\|\varphi\| \left(1 - \int f_k\, d\mathbb{Q}\right).\end{aligned}$$

Now we just have to let k tend to ∞ (noting that $\int f_k\, d\mathbb{Q} \uparrow 1$, by monotone convergence) and we find that $\int \varphi\, d\mathbb{Q}_n$ converges to $\int \varphi\, d\mathbb{Q}$, so that (1) holds.

We complete the proof by verifying that (3)⇒(2). So assume that (3) holds. If $\varphi \in C_c(\mathbb{R}^d)$, then, for every $k \in \mathbb{N}$, we can find a function $\varphi_k \in H$ such that $\|\varphi - \varphi_k\| \le 1/k$, and it follows that

$$\begin{aligned}&\limsup_{n\to\infty} \left|\int \varphi\ \mathrm{d}\mathbb{Q}_n - \int \varphi \mathrm{d}\mathbb{Q}\right| \\ &\le \limsup_{n\to\infty} \left(\left|\int (\varphi - \varphi_k)\, \mathrm{d}\mathbb{Q}_n\right| + \left|\int \varphi_k\ \mathrm{d}\mathbb{Q}_n - \int \varphi_k\ \mathrm{d}\mathbb{Q}\right| + \left|\int (\varphi_k - \varphi)\, \mathrm{d}\mathbb{Q}\right|\right) \\ &\le \frac{2}{k}\end{aligned}$$

Since this holds for every k, we get that

$$\int \varphi\, d\mathbb{Q}_n \longrightarrow \int \varphi\, d\mathbb{Q}.$$

■

The previous proposition simplifies the family of test functions, but weak convergence is still expressed through integrals against continuous functions. It is natural to ask whether one can characterize weak convergence without referring to test functions at all. The celebrated Portmanteau Theorem provides several equivalent descriptions in purely measure-theoretic terms, involving only the behavior of probability measures on open, closed, or continuity sets.

Theorem 2.3.13: Portmanteau Theorem

Let $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ be a sequence in $M_1(\mathbb{R}^d)$ and let $\mathbb{Q} \in M_1(\mathbb{R}^d)$. The following four assertions are equivalent:

1. The sequence $(\mathbb{Q}_n)$ converges weakly to $\mathbb{Q}$.
2. For every open subset G of $\mathbb{R}^d$,
$$\liminf \mathbb{Q}_n(G) \geq \mathbb{Q}(G).$$
3. For every closed subset F of $\mathbb{R}^d$,
$$\limsup \mathbb{Q}_n(F) \leq \mathbb{Q}(F).$$
4. For every Borel subset B of $\mathbb{R}^d$ such that $\mathbb{Q}(\partial B) = 0$,
$$\lim \mathbb{Q}_n(B) = \mathbb{Q}(B).$$

Proof for Theorem.

Let us start by proving that (1)⇒(2). If G is an open subset of $\mathbb{R}^d$, we can find a sequence $(\varphi_p)_{p\in\mathbb{N}}$ of bounded continuous functions such that $0 \leq \varphi_p \leq \mathbb{1}_G$ and $\varphi_p \uparrow \mathbb{1}_G$ (for instance, $\varphi_p(x) = p\,\mathrm{dist}(x, G^c) \wedge 1$). Then

$$\liminf_{n\to\infty} \mathbb{Q}_n(G) \geq \sup_p \left(\liminf_{n\to\infty} \int \varphi_p \, d\mathbb{Q}_n\right) = \sup_p \left(\int \varphi_p \, d\mathbb{Q}\right) = \mathbb{Q}(G).$$

The equivalence (2)⇔(3) is immediate since complements of open sets are closed and conversely.

Let us show that (2) and (3) imply (4). If $B \in \mathcal{B}(\mathbb{R}^d)$, write $\overline{B}$ for the closure of B and B° for the interior of B so that $\partial B = \overline{B} \setminus B^\circ$. By (2) and (3), we have :

$$\begin{cases} \limsup \mathbb{Q}_n(B) \leq & \limsup \mathbb{Q}_n(\overline{B}) \leq \mathbb{Q}(\overline{B}) \\ \liminf \mathbb{Q}_n(B) \geq & \liminf \mathbb{Q}_n(B^\circ) \geq \mathbb{Q}(B^\circ). \end{cases}$$

If $\mathbb{Q}(\partial B) = 0$ then $\mathbb{Q}(\overline{B}) = \mathbb{Q}(B^\circ) = \mathbb{Q}(B)$ and we get (4).

We still have to prove that (4)⇒(1). Let $\varphi \in C_b(\mathbb{R}^d)$. Without loss of generality, we can assume that $\varphi \geq 0$. Then, let $K \geq 0$ such that $0 \leq \varphi \leq K$. By the Fubini theorem,

$$\int \varphi(x)\mathbb{Q}(dx) = \int \left(\int_0^K \mathbb{1}_{\{t\leq\varphi(x)\}} dt\right) \mathbb{Q}(dx) = \int_0^K \mathbb{Q}(E_t^\varphi)\, dt,$$

where E_t^φ is the closed set $E_t^\varphi := \{x \in \mathbb{R}^d : \varphi(x) \geq t\}$. Similarly, for every n,

$$\int \varphi(x)\mathbb{Q}_n(dx) = \int_0^K \mathbb{Q}_n(E_t^\varphi)\, dt.$$

Now note that $\partial E_t^\varphi \subset \{x \in \mathbb{R}^d : \varphi(x) = t\}$, and that there are at most countably many values of t such that

$$\mathbb{Q}(\{x \in \mathbb{R}^d : \varphi(x) = t\}) > 0$$

(indeed, for every $k \in \mathbb{N}$, there are at most k distinct values of t such that $\mathbb{Q}(\{x \in \mathbb{R}^d : \varphi(x) = t\}) \geq \frac{1}{k}$, since the sets $\{x \in \mathbb{R}^d : \varphi(x) = t\}$ are disjoint when t varies, and $\mathbb{Q}$ is a probability measure). Hence (4) implies

$$\mathbb{Q}_n(E_t^\varphi) \xrightarrow[n\to\infty]{} \mathbb{Q}(E_t^\varphi),$$

for every $t \in [0, K]$, except possibly for a countable set of values of t, which has zero Lebesgue measure. By dominated convergence, we conclude that

$$\int \varphi(x)\mathbb{Q}_n(dx) = \int_0^K \mathbb{Q}_n(E_t^\varphi)\, dt \xrightarrow[n\to\infty]{} \int_0^K \mathbb{Q}(E_t^\varphi)\, dt = \int \varphi(x)\mathbb{Q}(dx).$$

■

Although the Portmanteau Theorem gives several equivalent formulations of weak convergence, these criteria are often difficult to verify in concrete problems. A much more tractable approach is obtained by passing to characteristic functions. Lévy's Continuity Theorem shows that weak convergence is completely characterized by pointwise convergence of Fourier transforms. Recall our notation $\hat{\mathbb{Q}}(\xi) = \int e^{i\xi\cdot x}\, \mathbb{Q}(dx)$ for the Fourier transform of $\mathbb{Q} \in M_1(\mathbb{R}^d)$, and $\Phi_X = \widehat{P_X}$ for the characteristic function of a random variable X with values in $\mathbb{R}^d$.

Theorem 2.3.14: Lévy's Theorem

A sequence $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ in $M_1(\mathbb{R}^d)$ converges weakly to $\mathbb{Q} \in M_1(\mathbb{R}^d)$ if and only if

$$\forall \xi \in \mathbb{R}^d, \qquad \hat{\mathbb{Q}}_n(\xi) \xrightarrow[n\to\infty]{} \hat{\mathbb{Q}}(\xi).$$

Equivalently, a sequence $(X_n)_{n\in\mathbb{N}}$ of random variables with values in $\mathbb{R}^d$ converges in distribution to X if and only if

$$\forall \xi \in \mathbb{R}^d, \qquad \Phi_{X_n}(\xi) \xrightarrow[n\to\infty]{} \Phi_X(\xi).$$

Proof for Theorem.

It is enough to prove the first assertion. First, if we assume that the sequence $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ converges weakly to $\mathbb{Q}$, the definition of this convergence shows that

$$\forall \xi \in \mathbb{R}^d, \qquad \hat{\mathbb{Q}}_n(\xi) = \int e^{i\xi\cdot x}\, \mathbb{Q}_n(dx) \xrightarrow[n\to\infty]{} \int e^{i\xi\cdot x}\, \mathbb{Q}(dx) = \hat{\mathbb{Q}}(\xi).$$

Conversely, assume that $\hat{\mathbb{Q}}_n(\xi) \to \hat{\mathbb{Q}}(\xi)$ for every $\xi \in \mathbb{R}^d$ and let us show that $(\mathbb{Q}_n)$ converges weakly to $\mathbb{Q}$. To simplify notation, we only treat the case $d = 1$, but the proof in the general case is exactly similar.

Let $f \in C_c(\mathbb{R})$, and, for every $\sigma > 0$, let

$$g_\sigma(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right).$$

be the density of the $\mathcal{N}(0, \sigma^2)$ distribution. We already observed in the proof of Theorem 1.2.6 that $g_\sigma * f$ converges pointwise to f as $\sigma \to 0$. Since f has compact support here, it is not hard to verify that this convergence is uniform on $\mathbb{R}$ (use Proposition 2.1.9 (i) and note that, for every $\varepsilon > 0$, we can find a compact set K such that $|g_\sigma * f| \leq \varepsilon$ for every $x \in \mathbb{R} \setminus K$ and $\sigma \in (0,1]$). Thus, if we let H be the subset of $C_b(\mathbb{R}^d)$ defined by

$$H := \{\varphi = g_\sigma * f : \sigma > 0 \text{ and } f \in C_c(\mathbb{R}^d)\}$$

the closure of H in $C_b(\mathbb{R}^d)$ contains $C_c(\mathbb{R}^d)$.

On the other hand, we also saw in the proof of Theorem 1.2.6 that, for any $\nu \in M_1(\mathbb{R})$,

$$\begin{aligned}\int g_\sigma * f \;\mathrm{d}v &= \int f(x) g_\sigma * v(x)\mathrm{d}x \\ &= \int f(x) \left((\sigma\sqrt{2\pi})^{-1} \int e^{i\xi x} g_{1/\sigma}(\xi)\widehat{v}(-\xi)\mathrm{d}\xi\right) \mathrm{d}x \qquad (1)\end{aligned}$$

We apply this formula to $\nu = \mathbb{Q}_n$ and $\nu = \mathbb{Q}$. From our assumption $\hat{\mathbb{Q}}_n(\xi) \to \hat{\mathbb{Q}}(\xi)$ for every $\xi \in \mathbb{R}$ and to the trivial bound $|\hat{\mathbb{Q}}_n(\xi)| \leq 1$, we can apply the dominated convergence theorem to get that, for every $x \in \mathbb{R}$,

$$\int e^{i\xi x} g_{1/\sigma}(\xi)\hat{\mathbb{Q}}_n(-\xi)\, d\xi \xrightarrow[n\to\infty]{} \int e^{i\xi x} g_{1/\sigma}(\xi)\hat{\mathbb{Q}}(-\xi)\, d\xi.$$

Since the quantities in the last display are bounded by 1 and $f \in C_c(\mathbb{R})$, we can use (1) and again the dominated convergence theorem to get

$$\int g_\sigma * f \, d\mathbb{Q}_n \xrightarrow[n\to\infty]{} \int g_\sigma * f \, d\mathbb{Q}.$$

Finally, we have proved that $\int \varphi \, d\mathbb{Q}_n \to \int \varphi \, d\mathbb{Q}$ for every $\varphi \in H$, and we know that the closure of H in $C_b(\mathbb{R}^d)$ contains $C_c(\mathbb{R}^d)$. By Proposition 2.3.12, this is enough to show that the sequence $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ converges weakly to $\mathbb{Q}$. ■

2.3.3 The Strong Law of Large Numbers

Let us begin with a useful implication of Kolmogorov's Zero–One Law (Theorem 2.1.20). Suppose that $(X_n)_{n\geq 1}$ is a sequence of independent random variables and that

$$Z = \lim_{n\to\infty} \frac{1}{n}(X_1 + \ldots + X_n)$$

exists almost surely. For every fixed $k \in \mathbb{N}$, the random variable Z may equivalently be expressed as a measurable function depending only on the tail variables

$$Z = \lim_{n\to\infty} \frac{1}{n}(X_k + X_{k+1} + \dots) \in \sigma(X_k, X_{k+1}, \dots)$$

then

$$Z \in \bigcap_{k=1}^{\infty} \sigma(X_k, X_{k+1}, \dots),$$

that is, Z is measurable with respect to the tail σ-algebra. Consequently, by Theorem 2.1.20, the random variable Z is almost surely constant.

Theorem 2.3.15: Strong Law of Large Numbers

Let $(X_n)_{n\geq 1}$ be a sequence of independent and identically distributed real random variables in L^1. Then

$$\frac{1}{n}(X_1 + \dots + X_n) \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[X_1].$$

Proof for Theorem.

To simplify notation, we set $S_n = X_1 + \dots + X_n$ for every $n \in \mathbb{N}$, and $S_0 = 0$. Let $a > \mathbb{E}[X_1]$, and

$$M := \sup_{n\in\mathbb{Z}_+} (S_n - na),$$

which is a random variable with values in $[0, \infty]$.
We will show that

$$M < \infty, \quad \text{a.s.} \tag{1}$$

Assume that (1) holds (for any choice of $a > \mathbb{E}[X_1]$).
Since $S_n \leq na + M$ for every $n \in \mathbb{N}$, it immediately follows from (1) that

$$\limsup_{n\to\infty} \frac{1}{n} S_n \leq a, \quad \text{a.s.}$$

By considering a sequence of values of a that decreases to $\mathbb{E}[X_1]$, we find

$$\limsup_{n\to\infty} \frac{1}{n} S_n \leq \mathbb{E}[X_1], \quad \text{a.s.}$$

Replacing X_n by $-X_n$, we get the reverse inequality

$$\liminf_{n\to\infty} \frac{1}{n} S_n \geq \mathbb{E}[X_1], \quad \text{a.s.}$$

and the convergence in the theorem follows.

It remains to prove (1). With the notation of Theorem 2.1.20, we first observe that $\{M < \infty\}$ belongs to $\mathcal{B}_\infty$. Indeed, we have, for every integer $k \in \mathbb{N}$,

$$\{M < \infty\} = \left\{ \sup_{n\in\mathbb{Z}_+} (S_n - na) < \infty \right\} = \left\{ \sup_{n\geq k}(S_n - S_k - na) < \infty \right\},$$

and the event in the right-hand side is $\mathcal{B}_{k+1}$-measurable. It will therefore be enough to prove that $\mathbb{P}(M < \infty) > 0$, or equivalently that $\mathbb{P}(M = \infty) < 1$. In the remaining part of the proof we verify that $\mathbb{P}(M = \infty) < 1$.

For every $k \in \mathbb{N}$, set

$$M_k := \sup_{0\leq n\leq k} (S_n - na), \qquad M'_k := \sup_{0\leq n\leq k} (S_{n+1} - S_1 - na).$$

Then, M_k and M'_k are nonnegative random variables with the same distribution, as a consequence of the fact that the random vectors $(X_1, \ldots, X_k)$ and $(X_2, \ldots, X_{k+1})$ have the same law. It follows that

$$M = \lim_{k\to\infty} \uparrow M_k \qquad \text{and} \qquad M' := \lim_{k\to\infty} \uparrow M'_k$$

also have the same law (just observe that $\mathbb{P}(M' \leq x) = \lim \downarrow \mathbb{P}(M'_k \leq x) = \lim \downarrow \mathbb{P}(M_k \leq x) = \mathbb{P}(M \leq x)$ for every $x \in \mathbb{R}$).

On the other hand, it follows from the definitions that, for every integer $k \geq 0$,

$$M_{k+1} = \sup\left(0, \sup_{1\leq n\leq k+1} (S_n - na)\right) = \sup(0, M'_k + X_1 - a),$$

which can also be written as

$$M_{k+1} = M'_k - \inf(a - X_1, M'_k).$$

Since M'_k has the same law as M_k (and both these variables are clearly in L^1), we get

$$\mathbb{E}[\inf(a - X_1, M'_k)] = \mathbb{E}[M'_k] - \mathbb{E}[M_{k+1}] = \mathbb{E}[M_k] - \mathbb{E}[M_{k+1}] \leq 0$$

thanks to the trivial inequality $M_k \leq M_{k+1}$. We may now apply the dominated convergence theorem to the sequence $\inf(a-X_1, M'_k)$, $k \in \mathbb{N}$, noting that these variables are bounded in absolute value by $|a - X_1|$ (recall that $M'_k \geq 0$). It follows that

$$\mathbb{E}[\inf(a - X_1, M')] = \lim_{k\to\infty} \mathbb{E}[\inf(a - X_1, M'_k)] \leq 0.$$

We now argue by contradiction to prove that $\mathbb{P}(M = \infty) < 1$. Suppose that $\mathbb{P}(M = \infty) = 1$, then we have also $\mathbb{P}(M' = \infty) = 1$, since M and M' have the same law, and it follows that $\inf(a - X_1, M') = a - X_1$ a.s. But then the inequality in the last display gives $\mathbb{E}[a - X_1] \leq 0$, which is a contradiction since we chose $a > \mathbb{E}[X_1]$. This contradiction completes the proof. ■

Remarks.

- The L^1 integrability assumption is optimal in the sense that it is needed for the limit $\mathbb{E}[X_1]$ to be defined (and finite). If we remove the L^1 assumption but assume instead that the random variables X_n are nonnegative and $\mathbb{E}[X_1] = \infty$, it is easy to obtain that
$$\frac{1}{n}(X_1 + \cdots + X_n) \xrightarrow[n\to\infty]{\text{a.s.}} +\infty,$$
by applying the theorem to the variables $X_n \wedge K$, for every $K \in \mathbb{N}$.
- One can show that the convergence of the theorem also holds in L^1. The proof will be given at the end of next chapter as an application of martingale theory. As we already mentioned, the almost sure convergence is the most interesting one from a probabilistic point of view.

2.3.4 The Central Limit Theorem

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent and identically distributed real random variables in L^1. The strong law of large numbers asserts that

$$\frac{1}{n}(X_1 + \cdots + X_n) \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[X_1].$$

We are now interested in the speed of this convergence, meaning that we want information about the typical size of

$$\frac{1}{n}(X_1 + \cdots + X_n) - \mathbb{E}[X_1]$$

when n is large.

Under the additional assumption that the variables X_n are in L^2, it is easy to guess the answer, because with a simple calculation we can conclude that

$$\mathbb{E}\left[(X_1 + \cdots + X_n - n\mathbb{E}[X_1])^2\right] = \mathrm{var}(X_1 + \cdots + X_n) = n\,\mathrm{var}(X_1).$$

This shows that the mean value of $(X_1 + \cdots + X_n - n\mathbb{E}[X_1])^2$ grows linearly with n, and thus the typical size of $X_1 + \cdots + X_n - n\mathbb{E}[X_1]$ is expected to be $\sqrt{n}$, or equivalently the typical size of
$$\frac{1}{n}(X_1 + \cdots + X_n) - \mathbb{E}[X_1]$$
should be $1/\sqrt{n}$. The central limit theorem gives a much more precise statement.

Theorem 2.3.16: Central Limit Theorem

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent and identically distributed real random variables in L^2. Set $\sigma^2 = \mathrm{var}(X_1)$ and assume that $\sigma > 0$. Then

$$\frac{1}{\sqrt{n}}(X_1 + \cdots + X_n - n\mathbb{E}[X_1]) \xrightarrow[n\to\infty]{\mathcal{D}} \mathcal{N}(0, \sigma^2).$$

where $\mathcal{N}(0, \sigma^2)$ is the Gaussian distribution with mean 0 and variance σ^2. Hence, for every $a, b \in \mathbb{R}$ with $a < b$,

$$\lim_{n\to\infty} \mathbb{P}(X_1+\cdots+X_n \in [n\mathbb{E}[X_1]+a\sqrt{n},\, n\mathbb{E}[X_1]+b\sqrt{n}]) = \frac{1}{\sigma\sqrt{2\pi}} \int_a^b \exp\left(-\frac{x^2}{2\sigma^2}\right) dx.$$

Proof for Theorem.

The second part of the statement follows from the first one, thanks to Theorem 2.3.13. To prove the first assertion, note that we can assume $\mathbb{E}[X_1] = 0$ (just replace X_n by $X_n - \mathbb{E}[X_n]$). Then set

$$Z_n = \frac{1}{\sqrt{n}}(X_1 + \cdots + X_n).$$

The characteristic function of Z_n is

$$\begin{aligned}
\Phi_{Z_n}(\xi) &= \mathbb{E}\left[\exp\left(i\xi\frac{X_1 + \cdots + X_n}{\sqrt{n}}\right)\right] \\
&= \mathbb{E}\left[\exp\left(i\frac{\xi}{\sqrt{n}}X_1\right)\right]^n \\
&= \Phi_{X_1}\left(\frac{\xi}{\sqrt{n}}\right)^n.
\end{aligned}$$

where, in the second equality, we used the fact that the random variables X_i are independent and identically distributed. On the other hand, by Proposition 1.2.7, we have

$$\Phi_{X_1}(\xi) = 1 + i\xi\mathbb{E}[X_1] - \frac{1}{2}\xi^2\mathbb{E}[X_1^2] + o(\xi^2) = 1 - \frac{\sigma^2\xi^2}{2} + o(\xi^2)$$

as $\xi \to 0$. For every fixed $\xi \in \mathbb{R}$, we have thus

$$\Phi_{X_1}\left(\frac{\xi}{\sqrt{n}}\right) = 1 - \frac{\sigma^2\xi^2}{2n} + o\left(\frac{1}{n}\right)$$

as $n \to \infty$. Thanks to proposition 2.3.8 and the last display, we have, for every $\xi \in \mathbb{R}$,

$$\lim_{n\to\infty} \Phi_{Z_n}(\xi) = \lim_{n\to\infty} \left(1 - \frac{\sigma^2\xi^2}{2n} + o\left(\frac{1}{n}\right)\right)^n = \exp\left(-\frac{\sigma^2\xi^2}{2}\right),$$

which is the characteristic function of the Gaussian $\mathcal{N}(0, \sigma^2)$ distribution as described in appendix 6. An application of Lévy's theorem 2.3.14 now completes the proof. ■

Remark When $\sigma = 0$ there is not much to say since the variables X_n are all equal a.s. to the constant $\mathbb{E}[X_1]$.

2.3.5 The Multidimensional Central Limit Theorem

We now assume that $(X_n)_{n\in\mathbb{N}}$ is a sequence of independent and identically distributed random variables with values in $\mathbb{R}^d$, such that $\mathbb{E}[\|X_1\|] < \infty$. We can apply the strong law of large numbers to each component of X_n and we have

$$\frac{1}{n}(X_1 + \cdots + X_n) \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[X_1],$$

as in the real case. If we assume furthermore that $\mathbb{E}[\|X_1\|^2] < \infty$, then we expect to have also an analog of the central limit theorem. However, in contrast with the law of large numbers, this is not immediate because the convergence in distribution of a sequence of random vectors does not follow from the convergence in distribution of each component sequence.

To extend the central limit theorem to the case of random variables with values in $\mathbb{R}^d$, we recall the main 3 results of multidimensional Gaussian distributions :

1. A random variable $\mathbf{X} = (X_1, \ldots, X_p)$ has a multidimensional Gaussian distribution (ie Gaussian vector) iff for all $a_1, \ldots, a_p \in \mathbb{R}$ the real random variable $a_1X_1 + \ldots + a_pX_p$ has a Gaussian distribution.
2. The characteristic function of Gaussian vector $\mathcal{N}(m_{\mathbf{X}}, K_{\mathbf{X}})$ is provided by :
$$\Phi_{\mathbf{X}}(\xi) = \mathbb{E}\left[e^{\xi\cdot\mathbf{X}}\right] = e^{i\xi\cdot m_{\mathbf{X}} - \frac{1}{2}{}^t\xi K_{\mathbf{X}}\xi}$$
3. If Γ be a symmetric nonnegative definite $d \times d$ matrix. Then there exists a Gaussian vector $\mathbf{X} \sim \mathcal{N}_d(0, \Gamma)$.

Theorem 2.3.17: Multidimensional Central Limit Theorem

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent and identically distributed random variables with values in $\mathbb{R}^d$, such that $\mathbb{E}[\|X_1\|^2] < \infty$. Then,

$$\frac{1}{\sqrt{n}}(X_1 + \cdots + X_n - n\mathbb{E}[X_1]) \xrightarrow[n\to\infty]{\mathcal{D}} \mathcal{N}_d(0, K_{X_1}).$$

Proof for Theorem.

Replacing X_n by $X_n - \mathbb{E}[X_n]$ allows us to assume that $\mathbb{E}[X_1] = 0$. For every $\xi \in \mathbb{R}^d$, the central limit theorem in the real case (Theorem 2.3.16) shows that

$$\frac{1}{\sqrt{n}}(\xi\cdot X_1 + \cdots + \xi\cdot X_n) \xrightarrow[n\to\infty]{\mathcal{D}} \mathcal{N}(0, \sigma^2),$$

where $\sigma^2 = \mathbb{E}[(\xi\cdot X_1)^2] = {}^t\xi K_{X_1}\xi$. This implies that

$$\mathbb{E}\left[\exp\left(i\xi\cdot\frac{X_1 + \cdots + X_n}{\sqrt{n}}\right)\right] \xrightarrow[n\to\infty]{} \exp\left(-\frac{1}{2}{}^t\xi K_{X_1}\xi\right)$$

and Lévy's theorem (Theorem 2.3.14) gives the desired result. ■

CHAPITRE 3

MARTINGALE AND CONTINUOUS SEMIMARTINGALES THEORY

3.1 Filtrations and Processes

3.1.1 Random Processes

Throughout this section, let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let (E, d) be a metric space endowed with its Borel σ-field, and let the index set T be either $\mathbb{N}$ (discrete time) or an interval of $\mathbb{R}$ (continuous time). A stochastic process is a family of E-valued random variables

$$X = (X_t)_{t \in T}.$$

Definition 3.1.1: Stochastic process

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let (E, d) be a metric space endowed with its Borel σ-field, and let T be an index set, typically $T = \mathbb{N}$ (discrete time) or $T \subseteq \mathbb{R}$ (continuous time). An *E-valued stochastic process* indexed by T is a family of random variables

$$X = (X_t)_{t \in T},$$

where each $X_t : \Omega \to E$ is measurable.

Definition 3.1.2: Sample paths

Let $X = (X_t)_{t \in T}$ be an E-valued stochastic process. For every $\omega \in \Omega$, the mapping

$$T \ni t \longmapsto X_t(\omega) \in E$$

is called the *sample path* (or *trajectory*) of X associated with ω.

Definition 3.1.3: Graph of a stochastic process

The *graph* of a sample path of a stochastic process $X = (X_t)_{t\in T}$ corresponding to $\omega \in \Omega$ is the subset of $T \times E$ defined by

$$\mathrm{Graph}(X(\omega)) = \{(t, X_t(\omega)) : t \in T\}.$$

When $E = \mathbb{R}$, this is the usual graph of the function $t \mapsto X_t(\omega)$.

Definition 3.1.4: Modification

Let $X = (X_t)_{t\in T}$ and $\widetilde{X} = (\widetilde{X}_t)_{t\in T}$ be two E-valued stochastic processes defined on the same probability space and indexed by the same set T. We say that $\widetilde{X}$ is a *modification* of X if

$$\forall t \in T, \qquad \mathbb{P}(\widetilde{X}_t = X_t) = 1.$$

A modification preserves all finite-dimensional distributions. Consequently, any probabilistic property depending only on finite-dimensional marginals is shared by both processes. In contrast, pathwise properties may differ substantially. For example, when $T = \mathbb{R}_+$ and $E = \mathbb{R}$, a process may admit a modification with continuous sample paths even though its original sample paths are almost surely discontinuous.

Definition 3.1.5: Indistinguishability

The processes X and $\widetilde{X}$ are said to be *indistinguishable* if there exists a measurable set $N \subseteq \Omega$ with $\mathbb{P}(N) = 0$ such that

$$\forall \omega \in \Omega \setminus N,\ \forall t \in T, \qquad X_t(\omega) = \widetilde{X}_t(\omega).$$

Equivalently,

$$\mathbb{P}\Big(\forall t \in T,\ X_t = \widetilde{X}_t\Big) = 1,$$

keeping in mind that, for an arbitrary index set T, the event above need not be measurable.

Every pair of indistinguishable processes are modifications of one another, but the converse is generally false. Indistinguishable processes coincide almost surely as functions of time, whereas modifications agree only at each fixed time. Throughout these notes, processes are identified up to indistinguishability. Thus, uniqueness statements should always be understood in this sense.

When $T = I \subseteq \mathbb{R}$ is an interval, the distinction disappears under mild regularity assumptions. If both X and $\widetilde{X}$ have continuous sample paths (or merely right-continuous or left-continuous sample paths) outside a $\mathbb{P}$-null set, then

$$X \text{ is a modification of } \widetilde{X} \quad \Longleftrightarrow \quad X \text{ and } \widetilde{X} \text{ are indistinguishable.}$$

Indeed, equality holds almost surely at every rational time by countability, and continuity extends this equality to every $t \in I$.

Theorem 3.1.6: Kolmogorov's Continuity Theorem

Let $I \subseteq \mathbb{R}$ be a bounded interval, and let $X = (X_t)_{t\in I}$ be an E-valued stochastic process, where (E, d) is complete. Suppose there exist constants $q, \varepsilon, C > 0$ such that

$$\mathbb{E}[d(X_s, X_t)^q] \leq C|t-s|^{1+\varepsilon}, \qquad s,t \in I.$$

Then X admits a modification $\widetilde{X}$ whose sample paths are Hölder continuous of every order $\alpha \in (0, \varepsilon/q)$. More precisely, for every $\alpha \in (0, \varepsilon/q)$ and every $\omega \in \Omega$, there exists a finite constant $C_\alpha(\omega)$ such that

$$d(\widetilde{X}_s(\omega), \widetilde{X}_t(\omega)) \leq C_\alpha(\omega)|t-s|^\alpha, \qquad s,t \in I.$$

In particular, X admits a continuous modification, which is unique up to indistinguishability.

Remarks.

(i) If I is unbounded, for example $I = \mathbb{R}_+$, the theorem can be applied successively on bounded intervals $[0,1], [1,2], \ldots$. This yields a modification whose sample paths are locally Hölder continuous of every order $\alpha \in (0, \varepsilon/q)$.

(ii) It is sufficient to prove the theorem for one fixed exponent $\alpha \in (0, \varepsilon/q)$. Applying the result to a sequence $\alpha_n \uparrow \varepsilon/q$ gives modifications that are necessarily indistinguishable, and therefore coincide almost surely.

Lemma 3.1.7: Dyadic Hölder Lemma

Let $D = \{i2^{-n} : n \geq 1,\ 0 \leq i < 2^n\} \subset [0,1)$, and let $f : D \to E$, where (E,d) is a metric space. Assume that there exist constants $\alpha > 0$ and $K < \infty$ such that

$$d\Big(f((i-1)2^{-n}), f(i2^{-n})\Big) \leq K2^{-n\alpha}$$

for every $n \geq 1$ and every $i = 1, \ldots, 2^n - 1$. Then

$$d(f(s), f(t)) \leq \frac{2K}{1-2^{-\alpha}}|t-s|^\alpha, \qquad s,t \in D.$$

3.1.2 Filtration

Definition 3.1.8: Filtration

A filtration on $(\Omega, \mathcal{A}, \mathbb{P})$ is an increasing family of sub-σ-fields of $\mathcal{A}$. The family in the most applications of this book is indexed by time, which is either discrete ($\mathbb{N}$) or continuous ($\mathbb{R}_+$).

- **The discrete-time case:** A discrete-time filtration is a family $(\mathcal{F}_n)_{n\in\mathbb{N}}$ of σ-fields such that :
$$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots \subset \mathcal{A}$$
- **The continuous-time case:** A continuous-time filtration is a family $(\mathcal{F}_t)_{t\in\mathbb{R}_+}$ of σ-fields such that :
$$\mathcal{F}_0 \subset \mathcal{F}_s \subset \mathcal{F}_t \subset \mathcal{F}$$

We also say that $(\Omega, \mathcal{A}, (\mathcal{F}_n)_{n\in\mathbb{N}}, \mathbb{P})$ and $(\Omega, \mathcal{A}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ are filtered probability spaces.

The σ-field $\mathcal{F}_n$ (resp. $\mathcal{F}_t$) gathers the information available at time n (resp. t), events that are $\mathcal{F}_n$-measurable (resp. $\mathcal{F}_t$-measurable) are interpreted as those that depend only on what has happened up to time n (resp. t). We will write :

$$\mathcal{F}_\infty = \begin{cases} \bigvee_{n\in\mathbb{N}} \mathcal{F}_n := \sigma\left(\bigcup_{n\in\mathbb{N}} \mathcal{F}_n\right) & \text{in the discrete-time case} \\ \bigvee_{t\geq 0} \mathcal{F}_t := \sigma\left(\bigcup_{t\geq 0} \mathcal{F}_t\right) & \text{in the continuous-time case} \end{cases}$$

If $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is a random process, then, for every $n \in \mathbb{N}$ (resp. $t \geq 0$), we may define $\mathcal{F}_n^X$ (resp. $\mathcal{F}_t^X$) as the smallest σ-field on Ω for which $X_0, X_1, \ldots, X_n$ (resp. $X_s \quad s \leq t$) are measurables. In other words, we have :

$$\begin{cases} \mathcal{F}_n^X = \sigma(X_0, X_1, \ldots, X_n) & \text{in the discrete-time case} \\ \mathcal{F}_t^X = \sigma(X_s : s \leq t) & \text{in the continuous-time case} \end{cases}$$

Then $(\mathcal{F}_n^X)_{n\in\mathbb{N}}$ (resp. $(\mathcal{F}_t^X)_{t\geq 0}$) is a filtration called *the canonical filtration* of $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$).

Remark: Let $(\mathcal{F}_t)_{0\leq t\leq\infty}$ be a filtration on $(\Omega, \mathcal{F}, P)$. We set, for every $t \geq 0$

$$\mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s,$$

and $\mathcal{F}_{\infty+} = \mathcal{F}_\infty$. Note that $\mathcal{F}_t \subset \mathcal{F}_{t+}$ for every $t \in [0, \infty]$. The collection $(\mathcal{F}_{t+})_{0\leq t\leq\infty}$ is also a filtration. We say that the filtration $(\mathcal{F}_t)$ is right-continuous if

$$\mathcal{F}_{t+} = \mathcal{F}_t, \qquad \forall t \geq 0.$$

Let $(\mathcal{F}_t)$ be a filtration and let $\mathcal{N}$ be the class of all $(\mathcal{F}_\infty, P)$-negligible sets (i.e. $A \in \mathcal{N}$ if there exists an $A' \in \mathcal{F}_\infty$ such $A \subset A'$ and $P(A') = 0$). The filtration is said to be complete if $\mathcal{N} \subset \mathcal{F}_0$ (and thus $\mathcal{N} \subset \mathcal{F}_t$ for every t).

If $(\mathcal{F}_t)$ is not complete, it can be completed by setting $\mathcal{F}'_t = \mathcal{F}_t \vee \sigma(\mathcal{N})$, for every $t \in [0, \infty]$, using the notation $\mathcal{F}_t \vee \sigma(\mathcal{N})$ for the smallest σ-field that contains both $\mathcal{F}_t$ and $\sigma(\mathcal{N})$ (recall that $\sigma(\mathcal{N})$ is the σ-field generated by $\mathcal{N}$). We will often apply this completion procedure to the canonical filtration of a random process $(X_t)_{t\geq 0}$, and call the resulting filtration the *completed canonical filtration* of X.

Assumption: *In this book we will consider only filtrations that are right-continuous and complete.*

3.1.3 Measurability of processes

For the discrete-time case, we have only one definition for measurablity with respect to a filtration, but for the continuous-time case, we have 3 types/definitions of measurability. We start with the discrete-time case.

Definition 3.1.9: Adapted Process [discrete]

A random process $(X_n)_{n\in\mathbb{N}}$ is said to be adapted to the filtration $(\mathcal{F}_n)_{n\in\mathbb{N}}$ if, for every $n \in \mathbb{N}$, X_n is $\mathcal{F}_n$-measurable.

In the continuous-time case, we have the following definitions.

Definition 3.1.10: Measurable Process [continuous]

A process $X = (X_t)_{t\geq 0}$ with values in a measurable space $(E, \mathcal{E})$ is said to be measurable if the mapping

$$(\omega, t) \mapsto X_t(\omega)$$

defined on $\Omega \times \mathbb{R}_+$ equipped with the product σ-field $\mathcal{F} \otimes \mathcal{B}(\mathbb{R}_+)$ is measurable. (We recall that $\mathcal{B}(\mathbb{R}_+)$ stands for the Borel σ-field of $\mathbb{R}$.)

This is stronger than saying that, for every $t \geq 0$, X_t is $\mathcal{F}$-measurable. On the other hand, considering for instance the case where $E = \mathbb{R}$, it is easy to see that if the sample paths of X are continuous, or only right-continuous, the fact that X_t is $\mathcal{F}$-measurable for every t implies that the process is measurable in the previous sense – see the argument in the proof of Proposition 3.1.12 below.

Definition 3.1.11: Adapted and Progressive Process

A random process $(X_t)_{t\geq 0}$ with values in a measurable space $(E, \mathcal{E})$ is called **adapted** if, for every $t \geq 0$, X_t is $\mathcal{F}_t$-measurable. This process is said to be **progressive** if, for every $t \geq 0$, the mapping

$$(\omega, s) \mapsto X_s(\omega)$$

defined on $\Omega \times [0, t]$ is measurable for the σ-field $\mathcal{F}_t \otimes \mathcal{B}([0, t])$.

Proposition 3.1.12

Let $(X_t)_{t\geq 0}$ be a random process with values in a metric space (E, d) (equipped with its Borel σ-field). Suppose that X is adapted and that the sample paths of X are right-continuous (i.e. for every $\omega \in \Omega$, $t \mapsto X_t(\omega)$ is right-continuous). Then X is progressive. The same conclusion holds if one replaces right-continuous by left-continuous.

Proof for Proposition.

We treat only the case of right-continuous sample paths, as the other case is similar. Fix $t > 0$. For every $n \geq 1$ and $s \in [0, t]$, define a random variable X_s^n by setting

$$X_s^n = X_{kt/n} \quad \text{if } s \in \left[(k-1)\frac{t}{n}, k\frac{t}{n}\right), k \in \{1, \dots, n\},$$

and $X_t^n = X_t$. The right-continuity of sample paths ensures that, for every $s \in [0, t]$ and $\omega \in \Omega$,

$$X_s(\omega) = \lim_{n\to\infty} X_s^n(\omega).$$

On the other hand, for every Borel subset A of E,

$$\begin{aligned}\{(\omega, s) \in \Omega \times [0, t] : X_s^n(\omega) \in A\} = (&\{X_t \in A\} \times \{t\}) \\ &\cup \left(\bigcup_{k=1}^{n} \left(\{X_{kt/n} \in A\} \times \left[\frac{(k-1)t}{n}, \frac{kt}{n}\right)\right)\right)\end{aligned}$$

which belongs to the σ-field $\mathcal{F}_t \otimes \mathcal{B}([0, t])$. Hence, for every $n \geq 1$, the mapping $(\omega, s) \mapsto X_s^n(\omega)$, defined on $\Omega \times [0, t]$, is measurable for $\mathcal{F}_t \otimes \mathcal{B}([0, t])$. Since a pointwise limit of measurable functions is also measurable, the same measurability property holds for the mapping $(\omega, s) \mapsto X_s(\omega)$ defined on $\Omega \times [0, t]$. It follows that the process X is progressive. ■

The progressive σ-field. The collection $\mathcal{P}$ of all sets $A \in \mathcal{F} \otimes \mathcal{B}(\mathbb{R}_+)$ such that the process

$$X_t(\omega) = \mathbf{1}_A(\omega, t)$$

is progressive forms a σ-field on $\Omega \times \mathbb{R}_+$, which is called the progressive σ-field. A subset A of $\Omega \times \mathbb{R}_+$ belongs to $\mathcal{P}$ if and only if, for every $t \geq 0$,

$$A \cap (\Omega \times [0, t]) \in \mathcal{F}_t \otimes \mathcal{B}([0, t]).$$

One then verifies that a process X is progressive if and only if the mapping

$$(\omega, t) \longmapsto X_t(\omega)$$

is measurable on $\Omega \times \mathbb{R}_+$ equipped with the σ-field $\mathcal{P}$.

3.2 Martingales and stopping times

3.2.1 Martingales

Definition 3.2.1: Martingale

An adapted real-valued process $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) such that $X_n \in L^1$ for every $n \in \mathbb{N}$ (resp. $X_t \in L^1$ for every $t \geq 0$) is called

- a **martingale** if :

$$\begin{cases} \forall n \in \mathbb{N} \quad X_n = \mathbb{E}[X_{n+1}|F_n] & \text{For the discret time} \\ \forall s \geq t \geq 0 \quad X_t = \mathbb{E}[X_s|F_t] & \text{For the continuous time} \end{cases}$$

- a **supermartingale** if :

$$\begin{cases} \forall n \in \mathbb{N} \quad X_n \geq \mathbb{E}[X_{n+1}|F_n] & \text{For the discret time} \\ \forall s \geq t \geq 0 \quad X_t \geq \mathbb{E}[X_s|F_t] & \text{For the continuous time} \end{cases}$$

- a **submartingale** if :

$$\begin{cases} \forall n \in \mathbb{N} \quad X_n \leq \mathbb{E}[X_{n+1}|F_n] & \text{For the discret time} \\ \forall s \geq t \geq 0 \quad X_t \leq \mathbb{E}[X_s|F_t] & \text{For the continuous time} \end{cases}$$

If $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is a submartingale, $(-X_n)_{n\in\mathbb{N}}$ (resp. $(-X_t)_{t\geq 0}$) is a supermartingale. For this reason, some of the results below are stated for supermartingales only, but the analogous results for submartingales immediately follow.

Proposition 3.2.2

Let $\varphi : \mathbb{R} \longrightarrow \mathbb{R}_+$ be a convex function, and let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be an adapted process, such that $\mathbb{E}[\varphi(X_n)] < \infty$ for every $n \in \mathbb{N}$ (resp. $\mathbb{E}[\varphi(X_t)] < \infty$ for every $t \geq 0$).

- If $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is a martingale, then $(\varphi(X_n))_{n\in\mathbb{N}}$ (resp. $(\varphi(X_t))_{t\geq 0}$) is a submartingale.
- If $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is a submartingale, and if we assume in addition that φ is increasing, $(\varphi(X_n))_{n\in\mathbb{N}}$ (resp. $(\varphi(X_t))_{t\geq 0}$) is a submartingale.

In particular, if X_n (resp. X_t) is a martingale, $|X_n|$ (resp. $|X_t|$) is a submartingale (and so is X_n^2, provided that $\mathbb{E}[X_n^2] < \infty$ for every n) and, if X_n (resp. X_t) is a submartingale, $(X_n)^+$ (resp. $(X_t)^+$) is also a submartingale.

Proposition 3.2.3

Let $(M_n)_{n\in\mathbb{N}}$ (resp. $(M_t)_{t\geq 0}$) be a square integrable martingale ($\in L^2$). Then :

- **Discrete-time case:** If $n \geq m \geq 1$. Then,

$$\mathbb{E}\left[\sum_{k=m}^{n-1}(M_{k+1}-M_k)^2 \mid \mathcal{F}_m\right] = \mathbb{E}[M_n^2 - M_m^2 \mid \mathcal{F}_m] = \mathbb{E}[(M_n - M_m)^2 \mid \mathcal{F}_m].$$

  In particular,

$$\mathbb{E}\left[\sum_{i=m}^{n-1}(M_{k+1}-M_{t_k})^2\right] = \mathbb{E}[M_n^2 - M_m^2] = \mathbb{E}[(M_n - M_m)^2].$$

- **Continuous-time case:** If $0 \leq s < t$ and let $s = t_0 < t_1 < \cdots < t_p = t$ be a subdivision of the interval $[s,t]$. Then,

$$\mathbb{E}\left[\sum_{i=1}^{p}(M_{t_i}-M_{t_{i-1}})^2 \mid \mathcal{F}_s\right] = \mathbb{E}[M_t^2 - M_s^2 \mid \mathcal{F}_s] = \mathbb{E}[(M_t - M_s)^2 \mid \mathcal{F}_s].$$

  In particular,

$$\mathbb{E}\left[\sum_{i=1}^{p}(M_{t_i}-M_{t_{i-1}})^2\right] = \mathbb{E}[M_t^2 - M_s^2] = \mathbb{E}[(M_t - M_s)^2].$$

3.2.2 Stopping times

Definition 3.2.4: Stopping time

A random variable $T : \Omega \longrightarrow \mathbb{N} \cup \{+\infty\}$ (resp. $T : \Omega \longrightarrow [0,\infty]$) is a **stopping time** of the filtration $(\mathcal{F}_n)$ (resp. $(\mathcal{F}_t)$) if $\{T \leq n\} \in \mathcal{F}_n \quad \forall n \in \mathbb{N}$ (resp. $\{T \leq t\} \in \mathcal{F}_t \quad \forall t \geq 0$) The σ-field of the past before T is then defined by

$$\begin{cases} \mathcal{F}_T = \{A \in \mathcal{F}_\infty : \forall n \in \mathbb{N}, A \cap \{T \leq n\} \in \mathcal{F}_n\}. & \text{For the discrete time} \\ \mathcal{F}_T = \{A \in \mathcal{F}_\infty : \forall t \geq 0, A \cap \{T \leq t\} \in \mathcal{F}_t\}. & \text{For the continuous time} \end{cases}$$

The reader will verify that $\mathcal{F}_T$ is indeed a σ-field.

In what follows, discrete (resp. continuous) "stopping time" will mean stopping time of the filtration $(\mathcal{F}_n)$ (resp. $(\mathcal{F}_t)$) unless otherwise specified. If T is a continuous stopping time, we also have $\{T < t\} \in \mathcal{F}_t$ for every $t > 0$, by considering $\{T < t\} = \bigcup_{n=1}^{\infty}\left\{T \leq t - \frac{1}{n}\right\}$, and moreover

$$\{T = \infty\} = \left(\bigcup_{n=1}^{\infty}\{T \leq n\}\right)^c \in \mathcal{F}_\infty.$$

A good advantage of assuming the right continuity of the filtration is that, in this case, the definition of stopping times in the continuous time case can be equivalently stated by replacing $\{T \leq t\} \in \mathcal{F}_t$ by $\{T < t\} \in \mathcal{F}_t$ for every $t \geq 0$. In fact :

$$\{T \leq t\} = \bigcap_{n=1}^{\infty} \left\{T < t + \frac{1}{n}\right\}. \in \mathcal{F}_{t+} = \mathcal{F}_t$$

Proposition 3.2.5

Let S and T be two stopping times

- If $S \leq T$. Then, $\mathcal{F}_S \subset \mathcal{F}_T$.
- We have in general, $S \wedge T$ and $S \vee T$ are also stopping times. Moreover $\mathcal{F}_{S\wedge T} = \mathcal{F}_S \cap \mathcal{F}_T$.

Proof for Proposition.

Can be easily proven by using the definition of stopping times and σ-field of stopping times and manipulation of sets. ∎

Lemma 3.2.6: Measurability w.r.t. stopped filtration

Let T be a stopping time. A function $\omega \mapsto Y(\omega)$ defined on the set $\{T < \infty\}$ and taking values in the measurable set $(E, \mathcal{E})$ is $\mathcal{F}_T$-measurable if and only if, for every $n \in \mathbb{N}$ (resp. $t \geq 0$), the restriction of Y to the set $\{T = n\}$ is $\mathcal{F}_n$ measurable (resp. $\{T \leq t\}$ $\mathcal{F}_t$-measurable).

Proof for Lemma

For the discrete-time case, first assume that, for every $n \in \mathbb{N}$, the restriction of Y to $\{T = n\}$ is $\mathcal{F}_n$-measurable. Then, for every measurable subset A of E,

$$\{Y \in A\} \cap \{T \leq n\} = \bigcup_{k=0}^{n} (\{Y \in A\} \cap \{T = k\}) \in \mathcal{F}_n.$$

Letting $n \to \infty$, we first obtain that $\{Y \in A\} \in \mathcal{F}_\infty$, and then we deduce from the previous display that $\{Y \in A\} \in \mathcal{F}_T$. Conversely, if Y is $\mathcal{F}_T$-measurable, $\{Y \in A\} \in \mathcal{F}_T$ and thus $\{Y \in A\} \cap \{T = n\} = \{Y \in A\} \cap \{T \leq n\} \setminus \{Y \in A\} \cap \{T \leq n-1\} \in \mathcal{F}_n$, giving the desired result.

For the continuous-time case, first assume that, for every $t \geq 0$, the restriction of Y to $\{T \leq t\}$ is $\mathcal{F}_t$-measurable. Then, for every measurable subset A of E,

$$\{Y \in A\} \cap \{T \leq t\} \in \mathcal{F}_t.$$

Letting $t \to \infty$, we first obtain that $\{Y \in A\} \in \mathcal{F}_\infty$, and then we deduce from the previous display that $\{Y \in A\} \in \mathcal{F}_T$.

Conversely, if Y is $\mathcal{F}_T$-measurable, $\{Y \in A\} \in \mathcal{F}_T$ and thus $\{Y \in A\} \cap \{T \leq t\} \in \mathcal{F}_t$, giving the desired result. ■

Theorem 3.2.7: Stopped process Measurability

Let $(Y_n)_{n\in\mathbb{N}}$ (resp. $(Y_t)_{t\geq 0}$) be an adapted (resp. a progressive) process, and let T be a stopping time. Then the random variable Y_T, which is defined on the event $\{T<\infty\}$ by $Y_T(\omega)=Y_{T(\omega)}(\omega)$, is $\mathcal{F}_T$-measurable.

Proof for Theorem.

Let B be a Borel subset of $\mathbb{R}$. Let's begin with the dicrete-time case, consider the restriction of Y_T to the set $\{T=n\}$, which is given by Y_n on this set. Since Y_n is $\mathcal{F}_n$-measurable, the restriction of Y_T to $\{T=n\}$ is also $\mathcal{F}_n$-measurable. Hence, by Lemma 3.2.6, Y_T is $\mathcal{F}_T$-measurable. For the continuous-time case. Let $t\geq 0$. The restriction to $\{T\leq t\}$ of the function $\omega\mapsto X_T(\omega)$ is the composition of the two mappings

$$\begin{array}{c} \{T\leq t\}\ni\omega\mapsto(\omega,T(\omega)\wedge t) \\ \mathcal{F}_t \quad \mathcal{F}_t\otimes\mathcal{B}([0,t]) \end{array}$$

and

$$\begin{array}{c} \Omega\times[0,t]\ni(\omega,s)\mapsto X_s(\omega) \\ \mathcal{F}_t\otimes\mathcal{B}([0,t]) \quad \mathcal{E} \end{array}$$

which are both measurable (the first one since $T\wedge t$ is $\mathcal{F}_t$-measurable, by Proposition 3.2.5, and the second one by the definition of a progressive process). It follows that the restriction to $\{T\leq t\}$ of the function $\omega\mapsto X_T(\omega)$ is $\mathcal{F}_t$-measurable, which gives the desired result by the lemma 3.2.6. ■

If T is a stopping time, then, for every $n\in\mathbb{N}$ (resp. $t\geq 0$), $n\wedge T$ (resp. $t\wedge T$) is also a stopping time (Proposition 3.2.5) and it follows from the preceding theorem that $Y_{n\wedge T}$ (resp. $Y_{t\wedge T}$) is $\mathcal{F}_{n\wedge T}$-measurable (resp. $\mathcal{F}_{t\wedge T}$-measurable), hence also $\mathcal{F}_n$-measurable (resp. $\mathcal{F}_t$-measurable) by Proposition 3.2.5.

Proposition 3.2.8

If (S_n) is a sequence of stopping times then :

- if (S_n) is monotone increasing, then $S=\lim\uparrow S_n$ is a stopping time.
- If (S_n) is monotone decreasing, then $S=\lim\downarrow S_n$ is a stopping time and

$$\mathcal{F}_S=\bigcap_n\mathcal{F}_{S_n}$$

Proof for Proposition.

- If (S_n) is monotone increasing, then for every $t\geq 0$, $\{S\leq t\}=\bigcap_{n=1}^{\infty}\{S_n\leq t\}$, which belongs to the σ-field $\mathcal{F}_t$ since S_n is a stopping time for every n. Hence, S is a stopping time. same argument for the discrete time case.
- If (S_n) is monotone decreasing, then for every $t\geq 0$, $\{S<t\}=\bigcup_{n=1}^{\infty}\{S_n<t\}$,

which belongs to the σ-field $\mathcal{F}_t$ since S_n is a stopping time for every n. Hence, S is a stopping time by the fact that $\mathcal{F}_t$ is right-continuous [check remarks after the definition 3.2.4].

- For the last assertion we first verify that $\mathcal{F}_S \subset \bigcap_n \mathcal{F}_{S_n}$ thanks to last proposition 3.2.5. Conversely, if $A \in \bigcap_n \mathcal{F}_{S_n}$, then

$$A \cap \{S < t\} = \bigcup_{n=1}^{\infty} A \cap \{S_n < t\} \in \mathcal{F}_t,$$

which gives $A \in \mathcal{F}_S$ again this is allowed thanks to the right-continuity of the filtration.

■

Proposition 3.2.9

Let T be a stopping time and let S be an $\mathcal{F}_T$-measurable random variable with values in $[0, \infty]$, such that $S \geq T$. Then S is also a stopping time. In particular, if T is a continuous stopping time,

$$T_n = \frac{\lceil 2^n T \rceil}{2^n} \cdot \mathbb{1}_{\{T<\infty\}} + \infty \cdot \mathbb{1}_{\{T=\infty\}}, \quad n = 0, 1, 2, \ldots$$

defines a sequence of stopping times that decreases to T.

Proof for Proposition.

For the first assertion, we write, for every $n \in \mathbb{N}$,

$$\{S \leq n\} = \{S \leq n\} \cap \{T \leq n\} \in \mathcal{F}_n$$

since $\{S \leq n\}$ is $\mathcal{F}_T$-measurable. And similar argument holds for the continuous time case. Further more, the second assertion follows since $T_n \geq T$, and T_n is a function of T, hence $\mathcal{F}_T$-measurable, and $T_n \downarrow T$ as $n \uparrow \infty$ by construction.

■

Finally lets defining hitting time.

Definition 3.2.10: Hitting time

Let $(X_t)_{t\geq 0}$ be an adapted process with values in a metric space (E, d).

1. Assume that the sample paths of X are right-continuous, and let O be an open subset of E. Then

$$T_O = \inf\{t \geq 0 : X_t \in O\}$$

is a stopping time of the filtration $(\mathcal{F}_t)$.

2. Assume that the sample paths of X are continuous, and let F be a closed

subset of E. Then

$$T_F = \inf\{t \geq 0 : X_t \in F\}$$

is a stopping time (of the filtration $(\mathcal{F}_t)$).

In fact the stopping time property can be proven by considering the two sets :

$$\begin{cases} \{T_O < t\} & = \bigcup_{s \in [0,t) \cap \mathbb{Q}} \{X_s \in O\} \in \mathcal{F}_t, \\ \{T_F \leq t\} & = \left\{ \inf_{0 \leq s \leq t} d(X_s, F) = 0 \right\} = \left\{ \inf_{s \in [0,t] \cap \mathbb{Q}} d(X_s, F) = 0 \right\} \in \mathcal{F}_t. \end{cases}$$

3.2.3 Two Important Lemmas

Lemma 3.2.11: Optional stopping theorem, first version

Suppose that $(X_n)_{n \in \mathbb{N}}$ is a martingale (resp. a supermartingale) and let T be a stopping time. Then $(X_{n \wedge T})_{n \in \mathbb{N}}$ is also a martingale (resp. a supermartingale). In particular, if the stopping time T is bounded, one has $X_T \in L^1$, and

$$\mathbb{E}[X_T] = \mathbb{E}[X_0] \quad \text{(resp. } \mathbb{E}[X_T] \leq \mathbb{E}[X_0]).$$

Proof for Lemma

Fix $n \in \mathbb{N}$. We wish to show that

$$\mathbb{E}[X_{(n+1) \wedge T} \mid \mathcal{F}_n] = X_{n \wedge T}.$$

Observe the pointwise decomposition

$$X_{(n+1) \wedge T} - X_{n \wedge T} = (X_{n+1} - X_n) \mathbb{1}_{T > n},$$

Therefore it suffices to show

$$\mathbb{E}\Big[(X_{n+1} - X_n), \mathbb{1}_{T > n} \mid \mathcal{F}_n\Big] = 0.$$

Since T is a stopping time, the event $\{T > n\} = \{T \leq n\}^c$ belongs to $\mathcal{F}_n$, so the indicator $\mathbb{1}_{\{T > n\}}$ is $\mathcal{F}_n$-measurable. Taking it outside the conditional expectation and using the martingale property of (X_n):

$$\begin{aligned} \mathbb{E}\Big[(X_{n+1} - X_n), \mathbb{1}_{\{T > n\}} \mid \mathcal{F}_n\Big] &= \mathbb{1}_{\{T > n\}}, \mathbb{E}[X_{n+1} - X_n \mid \mathcal{F}_n] \\ &= \mathbb{1}_{\{T > n\}} \cdot 0 \\ &= 0. \end{aligned}$$

Hence $(X_{n \wedge T})_{n \in \mathbb{N}}$ is a martingale.

If the stopping time T is bounded above by the constant N, $X_T = X_{N \wedge T}$ is in L^1, and $\mathbb{E}[X_T] = \mathbb{E}[X_{N\wedge T}] = \mathbb{E}[X_0]$. For the supermartingale case just replace equalities with inequalities. ■

Lemma 3.2.12

Let $(X_n)_{n\in\mathbb{N}}$ be a supermartingale, and let S and T be two bounded stopping times such that $S \leq T$. Then
$$\mathbb{E}[X_T] \leq \mathbb{E}[X_S].$$

Proof for Lemma

We began with the discrete-time case. Let $(X_n, \mathcal{F}_n)_{n\geq 0}$ be a supermartingale and let $S \leq T \leq N$ be bounded stopping times. Define the stopped process increments:

$$X_T - X_S = \sum_{k=1}^{N} \mathbb{1}_{S<k\leq T}(X_k - X_{k-1}).$$

Take expectation:

$$\mathbb{E}[X_T - X_S] = \sum_{k=1}^{N} \mathbb{E}\left[\mathbb{1}_{S<k\leq T}(X_k - X_{k-1})\right].$$

Since S, T are stopping times, the event $\{S < k \leq T\} \in \mathcal{F}_{k-1}$. Hence the indicator is $\mathcal{F}_{k-1}$-measurable. Using the supermartingale property, $\mathbb{E}[X_k - X_{k-1} \mid \mathcal{F}_{k-1}] \leq 0$, we get
$$\mathbb{E}!\left[\mathbb{1}_{S<k\leq T}(X_k - X_{k-1})\right] \leq 0.$$
Summing over k, and using the fact that $X_T, X_S \in L^1$ (by Lemma 3.2.11),

$$\mathbb{E}[X_T - X_S] \leq 0, \quad \Rightarrow \quad \mathbb{E}[X_T] \leq \mathbb{E}[X_S].$$

■

3.3 Martingale convergence theory

3.3.1 Upcrossing numbers

Discret time

Consider first a deterministic real sequence $y = (y_n)_{n\in\mathbb{N}}$, and $a < b$, the **upcrossing number** of this sequence along $[a, b]$ before time n, denoted by $M^y_{ab}(n)$, is the largest integer k such that there exists a strictly increasing finite sequence

$$m_1 < n_1 < m_2 < n_2 < \cdots < m_k < n_k$$

of nonnegative integers smaller than or equal to n with the properties $y_{m_i} \leq a$ and $y_{n_i} \geq b$, for every $i \in \{1, \ldots, k\}$. In what follows we consider a sequence $Y = (Y_n)_{n\in\mathbb{N}}$ of real random variables, and the associated upcrossing number $M^Y_{ab}(n)$ is then an integer-valued random variable. where $M^y_{ab}(0) = 0$ if such a sequence does not exist. We will use the following simple analytic lemma, whose proof is left to the reader.

Lemma 3.3.1

The sequence $(y_n)_{n\in\mathbb{N}}$ converges in $\overline{\mathbb{R}}$ if and only if, for every choice of the rationals a and b with $a < b$, we have $M^y_{ab}(n) < \infty$.

Lemma 3.3.2: Doob's Upcrossing Inequality

Let $X = (X_n)_{n\in\mathbb{N}}$ be a supermartingale. Then, for every reals $a < b$ and every $n \in \mathbb{N}$,

$$E[M^Y_{ab}(n)] \leq \frac{1}{b-a} E[(Y_n - a)^-]$$

Continuous time

Let $f : I \longrightarrow \mathbb{R}$ be a function defined on a subset I of $\mathbb{R}_+$. If $a < b$, the upcrossing number of f along $[a, b]$, denoted by $M^f_{ab}(I)$, is the maximal integer $k \geq 1$ such that there exists a finite increasing sequence $s_1 < t_1 < \cdots < s_k < t_k$ of elements of I such that $f(s_i) \leq a$ and $f(t_i) \geq b$ for every $i \in \{1, \ldots, k\}$ (if, even for $k = 1$, there is no such subsequence, we take $M^f_{ab}(I) = 0$, and if such a subsequence exists for every $k \geq 1$, we take $M^f_{ab}(I) = \infty$). Upcrossing numbers are a convenient tool to study the regularity of functions.

In the next lemma, the notation

$$\lim_{s\downarrow\downarrow t} f(s) \quad (\text{resp. } \lim_{s\uparrow\uparrow t} f(s))$$

means

$$\lim_{s\downarrow t, s>t} f(s) \quad (\text{resp. } \lim_{s\uparrow t, s<t} f(s)).$$

We say that $g : \mathbb{R}_+ \to \mathbb{R}$ is **càdlàg** (for the French "continue à droite avec des limites à gauche") if g is right-continuous and has left-limits at every $t > 0$.

Lemma 3.3.3

Let D be a countable dense subset of $\mathbb{R}_+$ and let f be a real function defined on D. We assume that, for every $T \in D$,

- the function f is bounded on $D \cap [0, T]$;
- for all rationals a and b such that $a < b$,

$$M^f_{ab}(D \cap [0, T]) < \infty.$$

Then, the right-limit

$$f(t+) := \lim_{s\downarrow\downarrow t, s\in D} f(s)$$

exists for every real $t \geq 0$, and similarly the left-limit

$$f(t-) := \lim_{s \uparrow\uparrow t, s \in D} f(s)$$

exists for every real $t > 0$. Furthermore, the function $g : \mathbb{R}_+ \to \mathbb{R}$ defined by $g(t) = f(t+)$ is càdlàg.

We omit the proof of this analytic lemma. It is important to note that the right and left-limits $f(t+)$ and $f(t-)$ are defined for every $t \geq 0$ ($t > 0$ in the case of $f(t-)$) and not only for $t \in D$.

Lemma 3.3.4: Doob's Upcrossing Inequality

Let D be a countable dense subset of $\mathbb{R}_+$ and let $X = (X_t)_{t\geq 0}$ be a supermartingale. Then, for every reals $a < b$ and every $t \geq 0$,

$$\mathbb{E}[M_{ab}^X(D \cap [0,T])] \leq \frac{1}{b-a}\mathbb{E}[(X_T - a)^-]$$

Can be obtained by monotone convergence applied on the discret version Lemma 3.3.2 by considering a sequence of finite subsets of D.

3.3.2 Lp Convergence and Doob's inequalities

Theorem 3.3.5: Doob's supermartingale inequalities

1. (Maximal inequality for submartingales) Let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be a submartingale (with right-continuous sample paths in the continuous case). Then, for every $n \in \mathbb{N}$ (resp. $t > 0$) and every $\lambda > 0$,

$$\begin{cases} \lambda\mathbb{P}\left(\sup\limits_{0\leq m\leq n} |X_m| > \lambda\right) \leq \mathbb{E}[X_n^+] & \text{For the discrete case} \\ \lambda\mathbb{P}\left(\sup\limits_{0\leq s\leq t} |X_s| > \lambda\right) \leq \mathbb{E}[X_t^+] & \text{For the continuous case} \end{cases}$$

2. (Maximal inequality for supermartingales) Let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be a supermartingale (with right-continuous sample paths in the continuous case). Then, for every $n \in \mathbb{N}$ (resp. $t > 0$) and every $\lambda > 0$,

$$\begin{cases} \lambda\mathbb{P}\left(\sup\limits_{0\leq m\leq n} |X_m| > \lambda\right) \leq \mathbb{E}[X_0] + \mathbb{E}[X_n^-] & \text{For the discrete case} \\ \lambda\mathbb{P}\left(\sup\limits_{0\leq s\leq t} |X_s| > \lambda\right) \leq \mathbb{E}[X_0] + \mathbb{E}[X_t^-] & \text{For the continuous case} \end{cases}$$

3. (Doob's inequality in L^p for submartingales) Let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be

a nonnegative submartingale (with right-continuous sample paths in the continuous case). Then, for every $n \in \mathbb{N}$ (resp. $t > 0$) and every $p > 1$,

$$\begin{cases} \mathbb{E}\left[\sup\limits_{0\le m\le n} X_m^p\right] \le \left(\dfrac{p}{p-1}\right)^p \mathbb{E}[|X_n|^p] & \text{For the discrete case} \\ \mathbb{E}\left[\sup\limits_{0\le s\le t} X_s^p\right] \le \left(\dfrac{p}{p-1}\right)^p \mathbb{E}[|X_t|^p] & \text{For the continuous case} \end{cases}$$

4. (Doob's inequality in L^p for martingales) Let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\ge 0}$) be a martingale (with right-continuous sample paths in the continuous case). Then, for every $n \in \mathbb{N}$ (resp. $t > 0$) and every $p > 1$,

$$\begin{cases} \mathbb{E}\left[\sup\limits_{0\le m\le n} |X_m|^p\right] \le \left(\dfrac{p}{p-1}\right)^p \mathbb{E}[|X_n|^p] & \text{For the discrete case} \\ \mathbb{E}\left[\sup\limits_{0\le s\le t} |X_s|^p\right] \le \left(\dfrac{p}{p-1}\right)^p \mathbb{E}[|X_t|^p] & \text{For the continuous case} \end{cases}$$

Note that part (3) and (4) of the inequality is useful only if $\mathbb{E}[|X_n|^p] < \infty$(resp. $\mathbb{E}[|X_t|^p] < \infty$).

Proof for Theorem.

- Let us prove the first assertion. Consider the stopping time

$$T = \inf\{k \ge 0 : X_k \ge a\}.$$

Then, if

$$A = \left\{\sup_{0\le k\le n} X_k \ge a\right\},$$

we have $A = \{T \le n\}$. On the other hand, by applying the preceding lemma 3.2.12 to the stopping times $T \wedge n$ and n, we have

$$\mathbb{E}[X_{T\wedge n}] \le \mathbb{E}[X_n].$$

Furthermore,

$$X_{T\wedge n} \ge a\mathbb{1}_A + X_n\mathbb{1}_{A^c}.$$

By combining these two inequalities, we get

$$\mathbb{E}[X_n] \ge a\mathbb{P}(A) + \mathbb{E}[X_n\mathbb{1}_{A^c}]$$

and it follows that $a\mathbb{P}(A) \le \mathbb{E}[X_n\mathbb{1}_A]$, which gives the first assertion of the theorem in the discrete case.

For the continuous case, Fix $t > 0$ and consider a countable dense subset D of $\mathbb{R}_+$ such that $0 \in D$ and $t \in D$. Then $D \cap [0, t]$ is the increasing union of a sequence $(D_m)_{m\ge 1}$ of finite subsets $[0, t]$ of the form $D_m = \{t_0^m, t_1^m, \ldots, t_m^m\}$ where

$0 = t_0^m < t_1^m < \cdots < t_m^m = t$. For every fixed m, we can apply the discrete time case to the sequence $Y_n = X_{t^m_{n\wedge m}}$, which is a discrete submartingale with respect to the filtration $\mathcal{G}_n = \mathcal{F}_{t^m_{n\wedge m}}$. We get

$$\lambda \mathbb{P}\left(\sup_{s\in D_m} |X_s| > \lambda\right) \leq \mathbb{E}[X_t^+].$$

Then, we observe that

$$P\left(\sup_{s\in D_m} |X_s| > \lambda\right) \uparrow \mathbb{P}\left(\sup_{s\in D\cap[0,t]} |X_s| > \lambda\right)$$

when $m \uparrow \infty$. We have thus

$$\lambda \mathbb{P}\left(\sup_{s\in D\cap[0,t]} |X_s| > \lambda\right) \leq \mathbb{E}[X_t^+].$$

Finally, the right-continuity of sample paths (and the fact that $t \in D$) ensures that

$$\sup_{s\in D\cap[0,t]} |X_s| = \sup_{s\in[0,t]} |X_s|.$$

- The proof of the (2) is very similar to the proof of the (1). We begin with the discrete case and introduce the stopping time

  $$R = \inf\{k \geq 0 : X_k \geq a\},$$

  and the event $B = \{R \leq n\}$. We have then $\mathbb{E}[X_{R\wedge n}] \leq \mathbb{E}[X_0]$, and $\mathbb{E}[X_{R\wedge n}] \geq a\mathbb{P}(B) + \mathbb{E}[X_n \mathbb{1}_{B^c}]$. It follows that $a\mathbb{P}(B) \leq \mathbb{E}[X_0] - \mathbb{E}[X_n \mathbb{1}_{B^c}]$, which leads to the desired result. Then for the continuous case, we can use the same argument as in assertion (1) by considering a countable dense subset D of $\mathbb{R}_+$ such that $0 \in D$ and $t \in D$, and then applying the discrete case to the sequence $Y_n = X_{t^m_{n\wedge m}}$ for every fixed m.

- In order to prove the assertion (3) in the discrete case, we may assume that $\mathbb{E}[(X_n)^p] < \infty$. Then, by Jensen's inequality for conditional expectations, we have, for every $0 \leq k \leq n$,

  $$\mathbb{E}[(X_k)^p] \leq \mathbb{E}[\mathbb{E}[X_n|\mathcal{F}_k]^p] \leq \mathbb{E}[\mathbb{E}[(X_n)^p|\mathcal{F}_n]] = \mathbb{E}[(X_n)^p].$$

  Denote $\widetilde{X}_n = \sup_{0\leq m\leq n} X_m$. Since $\widetilde{X}_n \leq X_0 + \cdots + X_n$, it follows that we have also $\mathbb{E}[(\widetilde{X}_n)^p] < \infty$. By assertion (1) of the theorem, for every $a > 0$,

  $$a\,\mathbb{P}(\widetilde{X}_n \geq a) \leq \mathbb{E}[X_n \mathbb{1}_{\{\widetilde{X}_n \geq a\}}].$$

  We multiply each side of this inequality by a^{p-2} and integrate with respect to

Lebesgue measure on $(0, \infty)$. For the left side, we get

$$\int_0^\infty a^{p-1}\mathbb{P}(\widetilde{X}_n \geq a)\, da = \mathbb{E}\left[\int_0^{\widetilde{X}_n} a^{p-1} da\right] = \frac{1}{p}\mathbb{E}[(\widetilde{X}_n)^p]$$

using the Fubini theorem. Similarly, we get for the right side

$$\begin{aligned}\int_0^\infty a^{p-2}\mathbb{E}[X_n \mathbb{1}_{\{\widetilde{X}_n \geq a\}}]\, da &= \mathbb{E}\left[X_n \int_0^{\widetilde{X}_n} a^{p-2} da\right] \\ &= \frac{1}{p-1}\mathbb{E}[X_n(\widetilde{X}_n)^{p-1}] \\ &\leq \frac{1}{p-1}\mathbb{E}[(X_n)^p]^{\frac{1}{p}}\mathbb{E}[(\widetilde{X}_n)^p]^{\frac{p-1}{p}}\end{aligned}$$

by the Hölder inequality. We thus get

$$\frac{1}{p}\mathbb{E}[(\widetilde{X}_n)^p] \leq \frac{1}{p-1}\mathbb{E}[(X_n)^p]^{\frac{1}{p}}\mathbb{E}[(\widetilde{X}_n)^p]^{\frac{p-1}{p}}$$

giving the assertion (1) of the theorem (we use the fact that $\mathbb{E}[(\widetilde{X}_n)^p] < \infty$). Then for the continuous case, we can use the same argument as in assertion (1) by considering a countable dense subset D of $\mathbb{R}_+$ such that $0 \in D$ and $t \in D$, and then applying the discrete case to the sequence $Y_n = X_{t^m_{n\wedge m}}$ for every fixed m.

- The assertion (4) of the theorem follows from the assertion (3) applied to the submartingale $X_n = |Y_n|$.

■

In what follows, you will find that most of the results stated in the continuous time case requiring the right-continuity of the sample paths of the process, which is not that hard to verify in most of the cases thanks to the following theorem, which gives a sufficient condition to find a modification with the desired regularity of the sample paths.

Theorem 3.3.6: Sufficient condition for càdlàg modification

Assume that the filtration $(\mathcal{F}_t)$ is right-continuous and complete. Let $X = (X_t)_{t\geq 0}$ be a supermartingale, such that the function $t \mapsto \mathbb{E}[X_t]$ is right-continuous. Then X has a modification with càdlàg [a] sample paths, which is also an $(\mathcal{F}_t)$-supermartingale.

a. refers to the french sentence "continue à droite et limite à gauche" which means "right-continuous and have left-limits"

Proof for Theorem.

The proof of this theorem could be found in [2, thm. 3.18] ■

Theorem 3.3.7: Doob's Lp convergence theorem

Let $(X_n)_{n\in\mathbb{N}}$ be a martingale and $p \in (1,\infty)$. Suppose that

$$\sup_{n\in\mathbb{N}} \mathbb{E}[|X_n|^p] < \infty,$$

Then, X_n converges a.s. and in L^p to a random variable X_∞ such that

$$\mathbb{E}[|X_\infty|^p] = \sup_{n\in\mathbb{N}} \mathbb{E}[|X_n|^p]$$

and we have

$$\mathbb{E}[(\sup_{n\in\mathbb{N}} |X_n|^p)] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|X_\infty|^p].$$

Proof for Theorem.

Since the martingale (X_n) is bounded in L^p, it is also bounded in L^1, and we know from Theorem 3.3.9 that X_n converges a.s. to a limiting random variable X_∞. Moreover, by setting $X_n^* = \sup_{1\leq k\leq n} |X_k|$, the theorem 3.3.5 shows that, for every $n \in \mathbb{Z}_+$,

$$\mathbb{E}[(X_n^*)^p] \leq \left(\frac{p}{p-1}\right)^p \sup_{k\in\mathbb{Z}_+} \mathbb{E}[|X_k|^p].$$

Since $X_n^* \uparrow X_\infty^*$ when $n \uparrow \infty$, the monotone convergence theorem gives

$$\mathbb{E}[(X_\infty^*)^p] \leq \left(\frac{p}{p-1}\right)^p \sup_{k\in\mathbb{Z}_+} \mathbb{E}[|X_k|^p] < \infty$$

and thus $X_\infty^* \in L^p$. Since all random variables $|X_n|^p$ are dominated by $|X_\infty^*|^p$, the dominated convergence theorem ensures that X_n converges to X_∞ in L^p. Finally, we know from proof of (3) in theorem 3.3.5 that the sequence $\mathbb{E}[|X_n|^p]$ is increasing, and therefore

$$\mathbb{E}[|X_\infty|^p] = \lim_{n\to\infty} \mathbb{E}[|X_n|^p] = \sup_{n\in\mathbb{Z}_+} \mathbb{E}[|X_n|^p].$$

■

Theorem 3.3.8: Kolmogorov's criterion for a.s. convergence

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent real random variables in L^2. Assume that $\mathbb{E}[X_n] = 0$ for every $n \in \mathbb{N}$. Then the following two conditions are equivalent:

1. $\sum\limits_{n=0}^{\infty} \mathbb{E}[(X_n)^2] < \infty$.

2. The series $\sum_{n=0}^{\infty} X_n$ converges a.s. and in L^2.

Proof for Theorem.

For every integer $n \geq 0$, set $\mathcal{F}_n = \sigma(X_0, X_1, \ldots, X_n)$ and

$$M_n = \sum_{k=0}^{n} X_k.$$

Since

$$\mathbb{E}[M_{n+1}|\mathcal{F}_n] = \sum_{k=0}^{n} X_k + \mathbb{E}[X_{n+1}] = \sum_{k=0}^{n} X_k,$$

$(M_n)_{n\in\mathbb{Z}_+}$ is a martingale with respect to the filtration $(\mathcal{F}_n)_{n\in\mathbb{Z}_+}$. Furthermore, since $\mathbb{E}[X_j X_k] = 0$ if $j \neq k$, one has

$$\mathbb{E}[(M_n)^2] = \sum_{k=0}^{n} \mathbb{E}[(X_k)^2].$$

Hence, if M_n converges in L^2 as $n \to \infty$, condition (1) of the theorem must hold (we have just recovered a special case of a classical result of Hilbert space theory, see Riesz Fischer Theorem 5.2.4 in the Appendix). Conversely, if (1) holds, the martingale M_n is bounded in L^2, and by Theorem 3.3.7 we get that M_n converges a.s. and in L^2. ■

3.3.3 Almost sure convergence

Theorem 3.3.9: Almost convergence of supermartingales

Let X be a supermartingale (with right-continuous sample paths if time is continuous). Assume that the collection $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is bounded in L^1. Then, it converge almost surely to a random variable $X_\infty \in L^1$.

Proof for Theorem.

We will prove it in the continuous time case, the discrete time case can be proven by similar arguments and it is left exercice for the reader. Let D be a countable dense subset of $\mathbb{R}_+$. Using lemma 3.3.4, we have, for every $a < b$ and every $T \in D$,

$$\mathbb{E}\left[M_{ab}^X(D \cap [0,T])\right] \leq \frac{1}{b-a}\mathbb{E}\left[(X_T - a)^-\right]$$

By Monotone convergence theorem, we get, for every $a < b$,

$$\mathbb{E}\left[M_{ab}^X(D)\right] \leq \frac{1}{b-a}\sup_{t\geq 0}\mathbb{E}\left[(X_t - a)^-\right] < +\infty.$$

Since the collection $(X_t)_{t\geq 0}$ is bounded in L^1, the right-hand side of the previous display is finite. Hence, a.s. for all rationals $a < b$, we have $M_{ab}^X(D) < \infty$. It follows from

Lemma 3.3.3 that the limit

$$X_\infty := \lim_{D \ni t \to \infty} X_t$$

exists a.s. in $[-\infty, +\infty]$. We can in fact exclude the values $-\infty$ and $+\infty$ since, by Fatou's lemma,

$$\mathbb{E}[|X_\infty|] \leq \liminf_{D \ni t \to \infty} \mathbb{E}[|X_t|] < +\infty$$

and we get that $X_\infty \in L^1$. The right-continuity of X samples (which we have not used yet) allows us to remove the restriction to D in the limit. ■

Corollary 3.3.10

Let $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) be a nonnegative supermartingale. Then, X_n (resp. X_t) converges a.s. as $n \to \infty$ (resp. $t \to \infty$). Its limit X_∞ is in L^1 and we have $X_n \geq \mathbb{E}[X_\infty \mid \mathcal{F}_n]$ for every $n \in \mathbb{N}$. (resp. $X_t \geq \mathbb{E}[X_\infty \mid \mathcal{F}_t]$ for every $t \geq 0$).

Backward supermartingales A useful variant of the previous theorem is the following one, which is stated in a backward form in discrete time setup, we will not prove it however proof could be found in [2, thm. 12.30]. Since it will be used only in a proof step of supermartingale optional theorem in the continuous time case.

A backward filtration is a sequence $(\mathcal{F}_n)_{n \in \mathbb{Z}_-}$ of sub-σ-fields of $\mathcal{A}$, which is indexed by the set $\mathbb{Z}_- = \{0, -1, -2, \dots\}$ of all nonpositive integers, and is such that $\mathcal{F}_n \subset \mathcal{F}_m$ whenever $n, m \in \mathbb{Z}_-$ and $n \leq m$. We then set

$$\mathcal{F}_{-\infty} = \bigcap_{n \in \mathbb{Z}_-} \mathcal{F}_n$$

which is also a sub-σ-field of $\mathcal{A}$. It is important to observe that, in contrast with the "forward case" studied in the previous sections, the σ-field $\mathcal{F}_n$ becomes smaller and smaller when $n \downarrow -\infty$ (the larger $|n|$ is, the smaller the σ-field $\mathcal{F}_n$).

In what follows, we fix a backward filtration $(\mathcal{F}_n)_{n \in \mathbb{Z}_-}$. Let $X = (X_n)_{n \in \mathbb{Z}_-}$ be a sequence of real random variables indexed by $\mathbb{Z}_-$. We say that X is a backward martingale (resp. a backward supermartingale, a backward submartingale) if X_n is $\mathcal{F}_n$-measurable and $\mathbb{E}[|X_n|] < \infty$ for every $n \in \mathbb{Z}_-$, and if, for every $m, n \in \mathbb{Z}_-$ such that $n \leq m$,

$$X_n = \mathbb{E}[X_m | \mathcal{F}_n] \quad (\text{resp. } X_n \geq \mathbb{E}[X_m | \mathcal{F}_n], X_n \leq \mathbb{E}[X_m | \mathcal{F}_n]).$$

Theorem 3.3.11: Backward supermartingale convergence

Let $(X_n)_{n \in \mathbb{Z}_-}$ be a backward supermartingale. Suppose that

$$\sup_{n \in \mathbb{Z}_-} \mathbb{E}[|X_n|] < \infty.$$

Then the sequence $(X_n)_{n \in \mathbb{Z}_-}$ is uniformly integrable and converges a.s. and in L^1 to a random variable X_∞ as $n \to -\infty$. Moreover for every $n \in \mathbb{Z}_-$,

$$\mathbb{E}[X_n|\mathcal{F}_{-\infty}] \leq X_\infty,$$

and equality holds in the last display if $(X_n)_{n\in\mathbb{Z}_-}$ is a backward martingale.

3.3.4 Martingale Closure

Definition 3.3.12: Closed Martingale

A martingale $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is said to be *closed* if there exists a random variable $Z \in L^1$ such that, for every $n \in \mathbb{N}$ (resp. $t \geq 0$),

$$\begin{cases} X_n = \mathbb{E}[Z \mid \mathcal{F}_n] & \text{For the discrete time} \\ X_t = \mathbb{E}[Z \mid \mathcal{F}_t] & \text{For the continuous time} \end{cases}$$

Definition 3.3.13: Uniform integrability

A collection $(X_i)_{i\in I}$ of random variables in $L^1(\Omega, \mathcal{A}, \mathbb{P})$ is said to be *uniformly integrable* (u.i. in short) if

$$\lim_{a\to+\infty} \left(\sup_{i\in I} \mathbb{E}[|X_i| \mathbb{1}_{\{|X_i|>a\}}] \right) = 0.$$

A uniformly integrable collection $(X_i)_{i\in I}$ is bounded in L^1. To verify this, take a large enough so that

$$\sup_{i\in I} \mathbb{E}[|X_i| \mathbb{1}_{\{|X_i|>a\}}] \leq 1$$

and write $\mathbb{E}[|X_i|] \leq \mathbb{E}[|X_i| \mathbb{1}_{\{|X_i|\leq a\}}] + \mathbb{E}[|X_i| \mathbb{1}_{\{|X_i|>a\}}] \leq a + 1$. The converse is false: a collection $(X_i)_{i\in I}$ which is bounded in L^1 may not be uniformly integrable.

Proposition 3.3.14

Let $(X_i)_{i\in I}$ be a bounded subset of L^1. The following two conditions are equivalent:

1. The collection $(X_i)_{i\in I}$ is u.i.
2. For every $\varepsilon > 0$, one can find $\delta > 0$ such that, for every event $A \in \mathcal{A}$ of probability $\mathbb{P}(A) < \delta$, one has

$$\forall i \in I, \quad \mathbb{E}[|X_i| \mathbb{1}_A] < \varepsilon.$$

Proof for Proposition.

(1)⇒(2) Let $\varepsilon > 0$. We can first fix $a > 0$ such that

$$\sup_{i\in I} \mathbb{E}[|X_i| \mathbb{1}_{\{|X_i|>a\}}] < \frac{\varepsilon}{2}.$$

If we set $\delta = \varepsilon/(2a)$, then the condition $\mathbb{P}(A) < \delta$ implies, that, for every $i \in I$,

$$\mathbb{E}[|X_i| \mathbb{1}_A] \leq \mathbb{E}[|X_i| \mathbb{1}_{A\cap\{|X_i|\leq a\}}] + \mathbb{E}[|X_i| \mathbb{1}_{\{|X_i|>a\}}] \leq a\mathbb{P}(A) + \frac{\varepsilon}{2} < \varepsilon.$$

(2)⇒(1) Set $C = \sup_{i\in I} \mathbb{E}[|X_i|]$. By the Markov inequality, we have for every $a > 0$,

$$\forall i \in I, \quad \mathbb{P}(|X_i| > a) \leq \frac{C}{a}.$$

Let $\varepsilon > 0$ and choose $\delta > 0$ so that the property stated in (2) holds. Then, if a is large enough so that $C/a < \delta$, we have $\mathbb{P}(|X_i| > a) < \delta$ for every $i \in I$, and thus, by (2),

$$\forall i \in I, \quad \mathbb{E}[|X_i| \mathbb{1}_{\{|X_i|>a\}}] < \varepsilon.$$

We conclude that the collection $(X_i)_{i\in I}$ is u.i. ■

Theorem 3.3.15: Vitali convergence theorem

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables in L^1 that converges to X_∞ in probability. The following two conditions are equivalent:

1. The sequence $(X_n)_{n\in\mathbb{N}}$ converges to X_∞ in L^1.
2. The sequence $(X_n)_{n\in\mathbb{N}}$ is uniformly integrable.

Proof for Theorem.

- (1)⇒(2) If the sequence $(X_n)_{n\in\mathbb{N}}$ converges in L^1, then it is bounded in L^1. Then, let $\varepsilon > 0$. We can choose N large enough so that, for every $n \geq N$,

  $$\mathbb{E}[|X_n - X_N|] < \frac{\varepsilon}{2}.$$

  Since the finite set $\{X_1, \dots, X_N\}$ is u.i., Proposition 3.3.14 allows us to choose $\delta > 0$ such that, for every event A of probability $\mathbb{P}(A) < \delta$, we have

  $$\forall n \in \{1, \dots, N\}, \quad \mathbb{E}[|X_n| \mathbb{1}_A] < \frac{\varepsilon}{2}.$$

  Then, if $n > N$, we have also

  $$\mathbb{E}[|X_n| \mathbb{1}_A] \leq \mathbb{E}[|X_N| \mathbb{1}_A] + \mathbb{E}[|X_n - X_N|] < \varepsilon.$$

  By Proposition 3.3.14, the sequence $(X_n)_{n\in\mathbb{N}}$ is uniformly integrable.
- (2)⇒(1) Using the characterization of uniform integrability in Proposition 3.3.14(ii), it is straightforward to verify that the collection $(X_n - X_m)_{n,m\in\mathbb{N}}$

is also u.i. Hence, if $\varepsilon > 0$ is fixed, we can choose $a > 0$ large enough so that, for every $m, n \in \mathbb{N}$,

$$\mathbb{E}[|X_n - X_m|\mathbb{1}_{\{|X_n - X_m| > a\}}] < \varepsilon.$$

Then, for every $m, n \in \mathbb{N}$,

$$\begin{aligned}\mathbb{E}[|X_n - X_m|] \leq \mathbb{E}[|X_n - X_m|\mathbb{1}_{\{|X_n - X_m| \leq \varepsilon\}}] + \mathbb{E}[|X_n - X_m|\mathbb{1}_{\{\varepsilon < |X_n - X_m| \leq a\}}] \\ + \mathbb{E}[|X_n - X_m|\mathbb{1}_{\{|X_n - X_m| > a\}}] \\ \leq 2\varepsilon + a\,\mathbb{P}(|X_n - X_m| > \varepsilon).\end{aligned}$$

The convergence in probability of the sequence $(X_n)_{n\in\mathbb{N}}$ implies that

$$\mathbb{P}(|X_n - X_m| > \varepsilon) \leq \mathbb{P}\left(|X_n - X_\infty| > \frac{\varepsilon}{2}\right) + \mathbb{P}\left(|X_m - X_\infty| > \frac{\varepsilon}{2}\right) \xrightarrow[n,m\to\infty]{} 0.$$

It follows that

$$\limsup_{m,n\to\infty} \mathbb{E}[|X_n - X_m|] \leq 2\varepsilon$$

and, since ε was arbitrary, this shows that the sequence $(X_n)_{n\in\mathbb{N}}$ is Cauchy in L^1, hence converges in L^1.

■

Theorem 3.3.16: Doob's Uniform Integrability convergence theorem

Let X be a martingale (with right-continuous sample paths if time is continuous). Then the following properties are equivalent:

1. X is closed;
2. the collection $(X_n)_{n\in\mathbb{N}}$ ($(X_t)_{t\geq 0}$ if time is continuous) is uniformly integrable;
3. X_n (resp. X_t) converges a.s. and in L^1 as $n \to \infty$ (resp. $t \to \infty$).

Moreover, if these properties hold, we have $X_n = \mathbb{E}[X_\infty \mid \mathcal{F}_n]$ (resp. $X_t = \mathbb{E}[X_\infty \mid \mathcal{F}_t]$) for every $n \in \mathbb{N}$ (resp. $t \geq 0$), where $X_\infty \in L^1$ is the a.s. limit of X_n (resp. X_t) as $n \to \infty$ (resp. $t \to \infty$).

Proof for Theorem.

- (1)$\Rightarrow$(2): if $Z \in L^1$, the collection of all random variable $\mathbb{E}[Z|\mathcal{G}]$ when $\mathcal{G}$ varies among all sub-σ-fields of $\mathcal{F}_\infty$ is u.i.
- (2)$\Rightarrow$(3): If (2) holds, then $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is bounded in L^1, and thus converges a.s. to some random variable X_∞ by Theorem 3.3.9. By uniform integrability, the latter convergence also holds in L^1.
- (3)$\Rightarrow$(1): If (3) holds, for every $m \in \mathbb{N}$ (resp. $s \geq 0$), we can pass to the limit $n \to +\infty$ (resp. $t \to +\infty$) in the equality, , $X_m = \mathbb{E}(X_n \mid \mathcal{F}_m)$ (resp. $X_s = \mathbb{E}(X_t \mid \mathcal{F}_s)$) and using the theorem 2.2.4 we get $X_m = \mathbb{E}(X_\infty \mid \mathcal{F}_m)$ (resp. $X_s = \mathbb{E}(X_\infty \mid \mathcal{F}_s)$), which gives the desired result.

Corollary 3.3.17

Let $Z \in L^1(\Omega, \mathcal{A}, \mathbb{P})$. The martingale $X_n = \mathbb{E}[Z \mid \mathcal{F}_n]$ (resp. $X_t = \mathbb{E}[Z \mid \mathcal{F}_t]$) converges a.s. and in L^1 to $X_\infty = \mathbb{E}[Z \mid \mathcal{F}_\infty]$.

3.4 Optional stopping theorems

3.4.1 The theorem for martingales

Theorem 3.4.1: Optional stopping theorem – martingale

Let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be a uniformly integrable martingale (with right-continuous sample paths if time is continnuous). Let S and T be two stopping times with $S \leq T$. Then X_S and X_T are in L^1 and

$$X_S = \mathbb{E}[X_T \mid \mathcal{F}_S].$$

Proof for Theorem.

We begins with the discrete time case. From the preceding section, we know that the martingale $(X_n)_{n\in\mathbb{N}}$ is closed and $X_n = \mathbb{E}[X_\infty|\mathcal{F}_n]$ for every n. Let us verify that $X_T \in L^1$:

$$\begin{aligned}
\mathbb{E}[|X_T|] &= \sum_{n=0}^{\infty} \mathbb{E}[\mathbb{1}_{\{T=n\}}|X_n|] + \mathbb{E}[\mathbb{1}_{\{T=\infty\}}|X_\infty|] \\
&= \sum_{n=0}^{\infty} \mathbb{E}[\mathbb{1}_{\{T=n\}}|\mathbb{E}[X_\infty|\mathcal{F}_n]|] + \mathbb{E}[\mathbb{1}_{\{T=\infty\}}|X_\infty|] \\
&\leq \sum_{n=0}^{\infty} \mathbb{E}[\mathbb{1}_{\{T=n\}}\mathbb{E}[|X_\infty||\mathcal{F}_n]] + \mathbb{E}[\mathbb{1}_{\{T=\infty\}}|X_\infty|] \\
&= \sum_{n=0}^{\infty} \mathbb{E}[\mathbb{1}_{\{T=n\}}|X_\infty|] + \mathbb{E}[\mathbb{1}_{\{T=\infty\}}|X_\infty|] \\
&= \mathbb{E}[|X_\infty|] < \infty.
\end{aligned}$$

Then, if $A \in \mathcal{F}_T$,

$$
\begin{aligned}
\mathbb{E}[\mathbb{1}_A X_T] &= \sum_{n\in\mathbb{N}\cup\{\infty\}} \mathbb{E}[\mathbb{1}_{A\cap\{T=n\}} X_T] \\
&= \sum_{n\in\mathbb{N}\cup\{\infty\}} \mathbb{E}[\mathbb{1}_{A\cap\{T=n\}} X_n] \\
&= \sum_{n\in\mathbb{N}\cup\{\infty\}} \mathbb{E}[\mathbb{1}_{A\cap\{T=n\}} X_\infty] \\
&= \mathbb{E}[\mathbb{1}_A X_\infty].
\end{aligned}
$$

In the first equality, we use the fact that $X_T \in L^1$ to justify the interchange of sum and expectation via the Fubini theorem. In the third equality, we use the properties $X_n = \mathbb{E}[X_\infty|\mathcal{F}_n]$ and $A\cap\{T=n\} \in \mathcal{F}_n$ for $A \in \mathcal{F}_T$. Since X_T is $\mathcal{F}_T$-measurable and in L^1, the preceding display shows that X_T satisfies the characteristic property of the conditional expectation $\mathbb{E}[X_\infty|\mathcal{F}_T]$.

Finally, if S and T are two stopping times such that $S \le T$, the property $\mathcal{F}_S \subset \mathcal{F}_T$ (Proposition 3.2.5) and Theorem of nested σ-fields in conditional expectation 2.2.8 imply that

$$X_S = \mathbb{E}[X_\infty|\mathcal{F}_S] = \mathbb{E}[\mathbb{E}[X_\infty|\mathcal{F}_T]|\mathcal{F}_S] = \mathbb{E}[X_T|\mathcal{F}_S].$$

For the continuous time case, Set, for every integer $n \ge 0$,

$$T_n = \frac{\lceil 2^n T \rceil}{2^n} \cdot \mathbb{1}_{\{T<\infty\}} + \infty \cdot \mathbb{1}_{\{T=\infty\}}, \quad n = 0, 1, 2, \ldots$$

and similarly

$$S_n = \frac{\lceil 2^n S \rceil}{2^n} \cdot \mathbb{1}_{\{S<\infty\}} + \infty \cdot \mathbb{1}_{\{S=\infty\}}, \quad n = 0, 1, 2, \ldots$$

By Proposition 3.2.9, (T_n) and (S_n) are two sequences of stopping times that decrease respectively to T and to S. Moreover, we have $S_n \le T_n$ for every $n \ge 0$.

Now observe that, for every fixed n, $2^n S_n$ and $2^n T_n$ are stopping times of the discrete filtration $\mathcal{H}_k^{(n)} := \mathcal{F}_{k/2^n}$, and $Y_k^{(n)} := X_{k/2^n}$ is a discrete martingale with respect to this filtration. From the discrete case we proven before, we get that $Y^{(n)}_{2^n S_n}$ and $Y^{(n)}_{2^n T_n}$ are in L^1, and

$$X_{S_n} = Y^{(n)}_{2^n S_n} = \mathbb{E}[Y^{(n)}_{2^n T_n} \mid \mathcal{H}^{(n)}_{2^n S_n}] = \mathbb{E}[X_{T_n} \mid \mathcal{F}_{S_n}]$$

(here we need to verify that $\mathcal{H}^{(n)}_{2^n S_n} = \mathcal{F}_{S_n}$, but this is straightforward).

Let $A \in \mathcal{F}_S$. Since $\mathcal{F}_S \subset \mathcal{F}_{S_n}$, we have $A \in \mathcal{F}_{S_n}$ and thus

$$\mathbb{E}[\mathbb{1}_A X_{S_n}] = \mathbb{E}[\mathbb{1}_A X_{T_n}].$$

By the right-continuity of sample paths, we get a.s.

$$X_S = \lim_{n\to\infty} X_{S_n}, \quad X_T = \lim_{n\to\infty} X_{T_n}.$$

These limits also hold in L^1. Indeed, thanks again to the optional stopping theorem for uniformly integrable discrete martingales, we have $X_{S_n} = \mathbb{E}[X_\infty \mid \mathcal{F}_{S_n}]$ for every n, and thus the sequence (X_{S_n}) is uniformly integrable (and the same holds for the sequence (X_{T_n})).

The L^1-convergence implies that X_S and X_T belong to L^1, and also allows us to pass to the limit $n \to \infty$ in the equality $\mathbb{E}[\mathbb{1}_A X_{S_n}] = \mathbb{E}[\mathbb{1}_A X_{T_n}]$ in order to get

$$\mathbb{E}[\mathbb{1}_A X_S] = \mathbb{E}[\mathbb{1}_A X_T].$$

Since this holds for every $A \in \mathcal{F}_S$, and since the variable X_S is $\mathcal{F}_S$-measurable (by the remarks before the theorem), we conclude that

$$X_S = \mathbb{E}[X_T \mid \mathcal{F}_S],$$

which completes the proof. ■

In particular, for every stopping time S, we have

$$X_S = \mathbb{E}[X_\infty \mid \mathcal{F}_S],$$

and

$$\mathbb{E}[X_S] = \mathbb{E}[X_\infty] = \mathbb{E}[X_0].$$

Corollary 3.4.2

If T and S are stopping time such that $S \leq T$, and both S and T are bounded, then X_S and X_T are in L^1 and

$$X_S = \mathbb{E}[X_T \mid \mathcal{F}_S].$$

Proof for Corollary.

Let $a \geq 0$ such that $S \leq T \leq a$. We apply Optional Stopping Theorem 3.4.1 to the martingale $\{X_{n\wedge a}\}_{n\geq 0}$ (resp. $\{X_{t\wedge a}\}_{t\geq 0}$) which is closed by X_a. ■

Corollary 3.4.3

Let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be a martingale (with right-continuous sample paths if time is continuous), and let T be a stopping time.

1. The process $(X_{n\wedge T})_{n\in\mathbb{N}}$ (resp. $(X_{t\wedge T})_{t\geq 0}$) is still a martingale.
2. Suppose in addition that the martingale $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is uniformly integrable. Then the process $(X_{n\wedge T})_{n\in\mathbb{N}}$ (resp. $(X_{t\wedge T})_{t\geq 0}$) is also a uniformly integrable martingale, and more precisely we have,

$$\begin{cases} X_{n\wedge T} = \mathbb{E}[X_T \mid \mathcal{F}_{n\wedge T}], & n \in \mathbb{N} \quad \text{for the discrete case,} \\ X_{t\wedge T} = \mathbb{E}[X_T \mid \mathcal{F}_{t\wedge T}], & t \geq 0 \quad \text{for the continuous case.} \end{cases}$$

Proof for Corollary.

We will prove it for the continuous time case, the discrete time case can be proven by similar arguments and it is left exercice for the reader. We start with the proof of (2). Note that $t \wedge T$ is a stopping time by property (f) of stopping times. By Theorem 3.4.1, $X_{t\wedge T}$ and X_T are in L^1, and we also know that $X_{t\wedge T}$ is $\mathcal{F}_{t\wedge T}$-measurable, hence $\mathcal{F}_t$-measurable since $\mathcal{F}_{t\wedge T} \subset \mathcal{F}_t$. So in order to get formula in (2), it is enough to prove that, for every $A \in \mathcal{F}_t$,

$$\mathbb{E}[\mathbb{1}_A X_T] = \mathbb{E}[\mathbb{1}_A X_{t\wedge T}].$$

Let us fix $A \in \mathcal{F}_t$. First, we have trivially

$$\mathbb{E}[\mathbb{1}_{A\cap\{T\le t\}} X_T] = \mathbb{E}[\mathbb{1}_{A\cap\{T\le t\}} X_{t\wedge T}]. \tag{*}$$

On the other hand, by Theorem 3.4.1, we have

$$X_{t\wedge T} = \mathbb{E}[X_T \mid \mathcal{F}_{t\wedge T}],$$

and we notice that we have both $A\cap\{T > t\} \in \mathcal{F}_t$ and $A\cap\{T > t\} \in \mathcal{F}_T$ (the latter as a straightforward consequence of the definition of $\mathcal{F}_T$), so that $A\cap\{T > t\} \in \mathcal{F}_t \cap \mathcal{F}_T = \mathcal{F}_{t\wedge T}$. Using the preceding display, we obtain

$$\mathbb{E}[\mathbb{1}_{A\cap\{T> t\}} X_T] = \mathbb{E}[\mathbb{1}_{A\cap\{T> t\}} X_{t\wedge T}].$$

By adding this equality to (*), we get the desired result.

To prove (1), we just need to apply (2) to the (uniformly integrable) martingale $(X_{t\wedge a})_{a\ge 0}$, for any choice of $a \ge 0$. ■

3.4.2 The theorem for supermartingales

Theorem 3.4.4: Optional stopping theorem - supermartingale

Let $(Z_n)_{n\in\mathbb{N}}$ (resp. $(Z_t)_{t\ge 0}$) be a nonnegative supermartingale (with right-continuous sample paths if time is continnuous). Let U and V be two stopping times such that $U \le V$. Then Z_U and Z_V are in L^1, and

$$Z_U \ge \mathbb{E}[Z_V \mid \mathcal{F}_U].$$

Proof for Theorem.

Let's begin with the discrete time case. We first note that $(X_n)_{n\in\mathbb{N}}$ is bounded in L^1. Theorem 3.3.9 then shows that X_n converges a.s. to X_∞, so that the definition of X_T makes sense for any stopping time, even on the event $\{T = \infty\}$. If T is a bounded stopping time, we know that $\mathbb{E}[X_T] \le \mathbb{E}[X_0]$ (Lemma 3.2.11). Fatou's lemma then shows that, for any stopping time T,

$$\mathbb{E}[X_T] \leq \liminf_{k\to\infty} \mathbb{E}[X_{T\wedge k}] \leq \mathbb{E}[X_0]$$

and thus $X_T \in L^1$.

Then let S and T be two stopping times such that $S \leq T$. Let us first assume that $T \leq N$ for some fixed integer N. According to Lemma 3.2.12, we have then $\mathbb{E}[X_S] \geq \mathbb{E}[X_T]$. More generally, let $A \in \mathcal{F}_S$, and set

$$S^A(\omega) = \begin{cases} S(\omega) & \text{if } \omega \in A, \\ N & \text{if } \omega \notin A. \end{cases}$$

It is straightforward to verify that S^A is a stopping time (just note that $\{S^A \leq n\} = \Omega$ if $n \geq N$, and $\{S^A \leq n\} = A \cap \{S \leq n\} \in \mathcal{F}_n$ if $n < N$, using the definition of $\mathcal{F}_S$). We define T^A in a similar manner, and note that T^A is also a stopping time since $A \in \mathcal{F}_S \subset \mathcal{F}_T$. Since $S^A \leq T^A$, we have $\mathbb{E}[X_{S^A}] \geq \mathbb{E}[X_{T^A}]$, and it follows that

$$\mathbb{E}[X_S \mathbb{1}_A] \geq \mathbb{E}[X_T \mathbb{1}_A].$$

Let us come back to the general case where S and T satisfy $S \leq T$ but are no longer assumed to be bounded. Let $A \in \mathcal{F}_S$, and $k \in \mathbb{N}$. Then $A \cap \{S \leq k\}$ belongs both to $\mathcal{F}_k$ (by the definition of $\mathcal{F}_S$) and to $\mathcal{F}_S$ (recall that S is $\mathcal{F}_S$-measurable). From Proposition 3.2.5, $A \cap \{S \leq k\}$ belongs to $\mathcal{F}_{S\wedge k}$. Hence, by applying the bounded case to the stopping times $S \wedge k$ and $T \wedge k$, we get

$$\mathbb{E}[X_S \mathbb{1}_{A\cap\{S\leq k\}}] = \mathbb{E}[X_{S\wedge k} \mathbb{1}_{A\cap\{S\leq k\}}] \geq \mathbb{E}[X_{T\wedge k} \mathbb{1}_{A\cap\{S\leq k\}}].$$

By monotone convergence,

$$\lim_{k\to\infty} \uparrow \mathbb{E}[X_S \mathbb{1}_{A\cap\{S\leq k\}}] = \mathbb{E}[X_S \mathbb{1}_{A\cap\{S<\infty\}}].$$

On the other hand, Fatou's lemma gives

$$\liminf_{k\to\infty} \mathbb{E}[X_{T\wedge k} \mathbb{1}_{A\cap\{S\leq k\}}] \geq \mathbb{E}[X_T \mathbb{1}_{A\cap\{S<\infty\}}].$$

So we have obtained that

$$\mathbb{E}[X_S \mathbb{1}_{A\cap\{S<\infty\}}] \geq \mathbb{E}[X_T \mathbb{1}_{A\cap\{S<\infty\}}].$$

Since $X_S = X_T = X_\infty$ on the event $\{S = \infty\}$, it follows that

$$\mathbb{E}[X_S \mathbb{1}_A] \geq \mathbb{E}[X_T \mathbb{1}_A] = \mathbb{E}[\mathbb{E}[X_T \mid \mathcal{F}_S] \mathbb{1}_A].$$

Since this holds for every $A \in \mathcal{F}_S$, and both X_S and $\mathbb{E}[X_T \mid \mathcal{F}_S]$ are $\mathcal{F}_S$-measurable, it follows that $X_S \geq \mathbb{E}[X_T \mid \mathcal{F}_S]$ a.s. (apply the last display with the set $A = \{X_S < \mathbb{E}[X_T \mid \mathcal{F}_S]\}$). This completes the proof of the theorem in the discrete case.

Now the continuous time case, Firstly, we make the extra assumption that U and V

are bounded and we verify that we then have $\mathbb{E}[Z_U] \geq \mathbb{E}[Z_V]$. Let $p \geq 1$ be an integer such that $U \leq p$ and $V \leq p$. For every integer $n \geq 0$, set

$$U_n = \sum_{k=0}^{p2^n-1} \frac{k+1}{2^n} \mathbb{1}_{\{k2^{-n}<U\leq(k+1)2^{-n}\}}, \quad V_n = \sum_{k=0}^{p2^n-1} \frac{k+1}{2^n} \mathbb{1}_{\{k2^{-n}<V\leq(k+1)2^{-n}\}}$$

in such a way (by Proposition 3.2.9) that (U_n) and (V_n) are two sequences of bounded stopping times that decrease respectively to U and V, and additionally we have $U_n \leq V_n$ for every $n \geq 0$. The right-continuity of sample paths ensures that $Z_{U_n} \to Z_U$ and $Z_{V_n} \to Z_V$ a.s. as $n \to \infty$. Then, by the optional stopping theorem for discrete supermartingales in the case of bounded stopping times (which we just proved), with respect to the filtration $(\mathcal{F}_{k/2^{n+1}})_{k\geq 0}$, we have for every $n \geq 0$,

$$Z_{U_{n+1}} \geq \mathbb{E}[Z_{U_n} \mid \mathcal{F}_{U_{n+1}}].$$

Setting $Y_n = Z_{U_{-n}}$ and $\mathcal{H}_n = \mathcal{F}_{U_{-n}}$, for every integer $n \leq 0$, we get that the sequence $(Y_n)_{n\leq 0}$ is a backward supermartingale with respect to the filtration $(\mathcal{H}_n)_{n\leq 0}$. Since, for every $n \geq 0$, $\mathbb{E}[Z_{U_n}] \leq \mathbb{E}[Z_0]$ (by another application of the discrete optional stopping theorem), the sequence $(Y_n)_{n\leq 0}$ is bounded in L^1, and by the convergence theorem for backward supermartingales in Theorem 3.3.11, it converges in L^1. Hence the convergence of Z_{U_n} to Z_U also holds in L^1 and similarly the convergence of Z_{V_n} to Z_V holds in L^1. Since $U_n \leq V_n$, by yet another application of the discrete optional stopping theorem, we have $\mathbb{E}[Z_{U_n}] \geq \mathbb{E}[Z_{V_n}]$. Using the L^1-convergence of Z_{U_n} and Z_{V_n} we can pass to the limit $n \to \infty$ and obtain that $\mathbb{E}[Z_U] \geq \mathbb{E}[Z_V]$ as claimed.

Let us prove the statement of the theorem (no longer assuming that U and V are bounded). By the first step of the proof applied to the stopping times 0 and $U \wedge p$, we have $\mathbb{E}[Z_{U\wedge p}] \leq \mathbb{E}[Z_0]$ for every $p \geq 1$, and Fatou's lemma gives $\mathbb{E}[Z_U] \leq \mathbb{E}[Z_0] < \infty$ and similarly $\mathbb{E}[Z_V] < \infty$. Fix $A \in \mathcal{F}_U \subset \mathcal{F}_V$ and recall our notation U^A for the stopping time defined by $U^A(\omega) = U(\omega)$ if $\omega \in A$ and $U^A(\omega) = \infty$ otherwise (cf. property (d) of stopping times). By the first part of the proof, we have, for every $p \geq 1$,

$$\mathbb{E}[Z_{U^A\wedge p}] \geq \mathbb{E}[Z_{V^A\wedge p}].$$

By writing each of these two expectations as a sum of expectations over the sets A^c, $A \cap \{U \leq p\}$ and $A \cap \{U > p\}$, and noting that $U > p$ implies $V > p$, we get

$$\mathbb{E}[Z_U \mathbb{1}_{A\cap\{U\leq p\}}] \geq \mathbb{E}[Z_V \mathbb{1}_{A\cap\{U\leq p\}}].$$

Letting $p \to \infty$ then gives

$$\mathbb{E}[Z_U \mathbb{1}_{A\cap\{U<\infty\}}] \geq \mathbb{E}[Z_V \mathbb{1}_{A\cap\{U<\infty\}}].$$

is trivial and by adding it to the preceding display, we obtain

$$\mathbb{E}[Z_U \mathbb{1}_A] \geq \mathbb{E}[Z_V \mathbb{1}_A] = \mathbb{E}[\mathbb{E}[Z_V \mid \mathcal{F}_U]\mathbb{1}_A].$$

Since this holds for every $A \in \mathcal{F}_U$ and Z_U is $\mathcal{F}_U$-measurable, the desired result follows. ■

Remark This implies that $\mathbb{E}[Z_U] \geq \mathbb{E}[Z_V]$, and since $Z_U = Z_V = Z_\infty$ on the event $\{U = \infty\}$, it also follows that

$$\mathbb{E}[\mathbb{1}_{\{U<\infty\}} Z_U] \geq \mathbb{E}[\mathbb{1}_{\{U<\infty\}} Z_V] \geq \mathbb{E}[\mathbb{1}_{\{V<\infty\}} Z_V].$$

3.5 Finite Variation Processes

3.5.1 Finite Variation Functions

In our discussion of functions with finite variation, we restrict our attention to *continuous* functions, as this is the case of interest in the subsequent developments. Recall that a **signed measure** on a compact interval $[0, T]$ is the difference of two finite positive measures on $[0, T]$.

Definition 3.5.1: Finite Variation function

Let $T \geq 0$. A continuous function $a : [0, T] \longrightarrow \mathbb{R}$ such that $a(0) = 0$ is said to have *finite variation* if there exists a signed measure μ on $[0, T]$ such that $a(t) = \mu([0, t])$ for every $t \in [0, T]$.

The measure μ is then determined uniquely by a. Since a is continuous and $a(0) = 0$, it follows that μ has no atoms [1].

By the Jordan Decomposition Theorem, any signed measure can be split uniquely as $\mu = \mu^+ - \mu^-$, where μ^+ and μ^- are mutually singular positive measures, ensuring that the structural decomposition of such restricted systems is completely rigid and unique. The decomposition of as a difference of two finite positive measures on $[0, T]$ is not unique, but there exists a unique decomposition $\mu = \mu^+ - \mu^-$ such that μ^+ and μ^- are supported on disjoint Borel sets (D_+ and D_-). We write $|\mu|$ for the (finite) positive measure $|\mu| = \mu_+ + \mu_-$. The measure $|\mu|$ is called the *total variation* of a. We have $|\mu(A)| \leq |\mu|(A)$ for every $A \in \mathcal{B}([0, T])$. Moreover, the Radon-Nikodym derivative of μ with respect to $|\mu|$ is

$$\frac{d\mu}{d|\mu|} = \mathbb{1}_{D_+} - \mathbb{1}_{D_-}.$$

The fact that $a(t) = \mu_+([0, t]) - \mu_-([0, t])$ shows that a is the difference of two monotone nondecreasing continuous functions that vanish at 0 (since μ has no atoms, the same holds for μ_+ of μ_-). Conversely, the difference of two monotone nondecreasing continuous functions that vanish at 0 has finite variation in the sense of the previous definition. Indeed, this follows from the well-known fact that the formula $g(t) = \theta([0, t]), t \in [0, T]$ induces a bijection between monotone nondecreasing right-continuous functions $g : [0, T] \longrightarrow \mathbb{R}_+$ and finite positive measures θ on $[0, T]$.

1. A measure μ has an atom A if $\mu(A) > 0$ and for every $B \in \mathcal{A}$ with $B \subseteq A$, either $\mu(B) = 0$ or $\mu(B) = \mu(A)$.

Let $f : [0,T] \longrightarrow \mathbb{R}$ be a measurable function such that $\int_{[0,T]} |f(s)|\, |\mu|(ds) < \infty$. We set

$$\int_0^T f(s)\, da(s) = \int_{[0,T]} f(s)\, \mu(ds),$$
$$\int_0^T f(s)\, |da(s)| = \int_{[0,T]} f(s)\, |\mu|(ds).$$

Then the bound

$$\left|\int_0^T f(s)\, da(s)\right| \leq \int_0^T |f(s)|\, |da(s)|$$

holds. By restricting a to $[0,t]$ (which amounts to restricting μ, μ_+, μ_-), we can define $\int_0^t f(s)\, da(s)$ for every $t \in [0,T]$, and we observe that the function $t \mapsto \int_0^t f(s)\, da(s)$ also has finite variation on $[0,T]$ (the associated measure is just $\mu'(ds) = f(s)\mu(ds)$).

Proposition 3.5.2

For every $t \in (0,T]$,

$$\int_0^t |da(s)| = \sup\left\{\sum_{i=1}^{p} |a(t_i) - a(t_{i-1})|\right\},$$

where the supremum is over all subdivisions $0 = t_0 < t_1 < \cdots < t_p = t$ of $[0,t]$. More precisely, for any increasing sequence $0 = t_0^n < t_1^n < \cdots < t_{p_n}^n = t$ of subdivisions of $[0,t]$ whose mesh tends to 0, we have

$$\lim_{n\to\infty} \sum_{i=1}^{p_n} |a(t_i^n) - a(t_{i-1}^n)| = \int_0^t |da(s)|.$$

Proof for Proposition.

Clearly, it is enough to treat the case $t = T$. The inequality $\geq$ in the first assertion is very easy since, for any subdivision $0 = t_0 < t_1 < \cdots < t_p = T$ of $[0,T]$,

$$|a(t_i) - a(t_{i-1})| = |\mu((t_{i-1}, t_i])| \leq |\mu|((t_{i-1}, t_i]), \quad \forall i \in \{1, \ldots, p\},$$

and

$$\sum_{i=1}^{p} |\mu|((t_{i-1}, t_i]) = |\mu|([0,T]) = \int_0^T |da(s)|.$$

In order to get the reverse inequality, it suffices to prove the second assertion. So we consider an increasing sequence $0 = t_0^n < t_1^n < \cdots < t_{p_n}^n = T$ of subdivisions of $[0,T]$, whose mesh $\max\{t_i^n - t_{i-1}^n : 1 \leq i \leq p_n\}$ tends to 0. Although we are proving a "deterministic" result, we will use a martingale argument. Leaving aside the trivial case where $|\mu| = 0$, we introduce the probability space $\Omega = [0,T]$, which is equipped with the Borel σ-field $\mathcal{B} = \mathcal{B}([0,T])$ and the probability measure $P(ds) = (|\mu|([0,T]))^{-1}|\mu|(ds)$. On this probability space, we consider the discrete filtration $(\mathcal{B}_n)_{n\geq 0}$ such that, for every integer $n \geq 0$, $\mathcal{B}_n$ is the σ-field generated by the intervals $(t_{i-1}^n, t_i^n]$, $1 \leq i \leq p_n$.

We then set

$$X(s) = \mathbb{1}_{D_+}(s) - \mathbb{1}_{D_-}(s) = \frac{d\mu}{d|\mu|}(s),$$

and, for every $n \geq 0$,

$$X_n = \mathbb{E}[X \mid \mathcal{B}_n].$$

Properties of conditional expectation show that X_n is constant on every interval $(t_{i-1}^n, t_i^n]$ and takes the value

$$\frac{\mu((t_{i-1}^n, t_i^n])}{|\mu|((t_{i-1}^n, t_i^n])} = \frac{a(t_i^n) - a(t_{i-1}^n)}{|\mu|((t_{i-1}^n, t_i^n])}$$

on this interval. On the other hand, the sequence (X_n) is a closed martingale, with respect to the discrete filtration $(\mathcal{B}_n)$. Since X is measurable with respect to $\mathcal{B} = \bigvee_n \mathcal{B}_n$, this martingale converges to X in L^1, by the convergence theorem for closed discrete martingales 3.3.16. In particular,

$$\lim_{n\to\infty} \mathbb{E}[|X_n|] = \mathbb{E}[|X|] = 1,$$

where the last equality is clear since $|X(s)| = 1$, $|\mu|(ds)$ a.e. The desired result follows by noting that

$$\mathbb{E}[|X_n|] = (|\mu|([0,T]))^{-1} \sum_{i=1}^{p_n} |a(t_i^n) - a(t_{i-1}^n)|,$$

and recalling that $|\mu|([0,T]) = \int_0^T |da(s)|$. ■

Remark In the usual presentation of functions with finite variation, one starts from the property that the supremum in the first display of the proposition is finite.

We now give a useful approximation lemma for the integral of a continuous function with respect to a function with finite variation

Lemma 3.5.3: Approximation for integral w.r.t finite variation function

If $f : [0,T] \longrightarrow \mathbb{R}$ is a continuous function, and if $0 = t_0^n < t_1^n < \cdots < t_{p_n}^n = T$ is a sequence of subdivisions of $[0,T]$ whose mesh tends to 0, we have

$$\int_0^T f(s)\, da(s) = \lim_{n\to\infty} \sum_{i=1}^{p_n} f(t_{i-1}^n) \left(a(t_i^n) - a(t_{i-1}^n)\right).$$

Proof for Lemma

Let f_n be defined on $[0,T]$ by $f_n(s) = f(t_{i-1}^n)$ if $s \in (t_{i-1}^n, t_i^n]$, $1 \leq i \leq p_n$, and $f_n(0) = f(0)$. Then,

$$\sum_{i=1}^{p_n} f(t_{i-1}^n) \left(a(t_i^n) - a(t_{i-1}^n)\right) = \int_{[0,T]} f_n(s)\, \mu(ds),$$

and the desired result follows by dominated convergence since $f_n(s) \longrightarrow f(s)$ as $n \to \infty$, for every $s \in [0,T]$. ■

We say that a function $a : \mathbb{R}_+ \longrightarrow \mathbb{R}$ is a *finite variation function on* $\mathbb{R}_+$ if the restriction of a to $[0,T]$ has finite variation on $[0,T]$, for every $T > 0$. Then there is a unique σ-finite (positive) measure on $\mathbb{R}_+$ whose restriction to every interval $[0,T]$ is the total variation measure of the restriction of a to $[0,T]$, and we write

$$\int_0^\infty f(s)\,|da(s)|$$

for the integral of a nonnegative Borel function f on $\mathbb{R}_+$ with respect to this σ-finite measure. Furthermore, we can define

$$\int_0^\infty f(s)\,da(s) = \lim_{T\to\infty} \int_0^T f(s)\,da(s) \in (-\infty,\infty)$$

for any real Borel function f on $\mathbb{R}_+$ such that $\int_0^\infty |f(s)|\,|da(s)| < \infty$.

3.5.2 Finite Variation Processes

We now consider random variables and processes defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), P)$.

Definition 3.5.4: Finite Variation process

An adapted process $A = (A_t)_{t\geq 0}$ is called a *finite variation process* if all its sample paths are finite variation functions on $\mathbb{R}_+$. If in addition the sample paths are nondecreasing functions, the process A is called an *increasing process.*

Remark In particular, $A_0 = 0$ and the sample paths of A are continuous – one can define finite variation processes with càdlàg sample paths, but in this book we consider **only the case of continuous sample paths**. Our special convention that the initial value of a finite variation process is 0 will be convenient for certain uniqueness statements. If A is a finite variation process, the process

$$V_t = \int_0^t |dA_s|$$

is an increasing process. Indeed, it is clear that the sample paths of V are nondecreasing functions (as well as continuous functions that vanish at $t = 0$). The fact that V_t is an $\mathcal{F}_t$-measurable random variable can be deduced from the second part of Proposition 3.5.2. Writing $A_t = \frac{1}{2}(V_t + A_t) - \frac{1}{2}(V_t - A_t)$ shows that any finite variation process can be written as the difference of two increasing processes (the converse is obvious).

Proposition 3.5.5

Let A be a finite variation process, and let H be a progressive process such that

$$\forall t \geq 0,\ \forall \omega \in \Omega,\ \int_0^t |H_s(\omega)|\,|dA_s(\omega)| < \infty.$$

Then the process $H \cdot A = ((H \cdot A)_t)_{t\geq 0}$ defined by

$$(H \cdot A)_t = \int_0^t H_s\,dA_s$$

is also a finite variation process.

Proof for Proposition.

By the observations preceding the statement of Proposition 3.5.2, we know that the sample paths of $H \cdot A$ are finite variation functions. It remains to verify that the process $H \cdot A$ is adapted. To this end, it is enough to check that, if $t > 0$ is fixed, if $h : \Omega \times [0,t] \longrightarrow \mathbb{R}$ is measurable for the σ-field $\mathcal{F}_t \otimes \mathcal{B}([0,t])$, and if $\int_0^t |h(\omega,s)|\,|dA_s(\omega)| < \infty$ for every ω, then the variable $\int_0^t h(\omega,s)\,dA_s(\omega)$ is $\mathcal{F}_t$-measurable.

If $h(\omega,s) = \mathbb{1}_{(u,v]}(s)\mathbb{1}_\Gamma(\omega)$ with $(u,v] \subset [0,t]$ and $\Gamma \in \mathcal{F}_t$, the result is immediate since $\int_0^t h(\omega,s)\,dA_s(\omega) = \mathbb{1}_\Gamma(\omega)(A_v(\omega) - A_u(\omega))$ in that case. A monotone class argument 1.1.5 then gives the case $h = \mathbb{1}_G$, $G \in \mathcal{F}_t \otimes \mathcal{B}([0,t])$. Finally, in the general case, we observe that we can write h as a pointwise limit of a sequence of simple functions (i.e., finite linear combinations of indicator functions of measurable sets) h_n such that $|h_n| \leq |h|$ for every n, and that we then have $\int_0^t h_n(\omega,s)\,dA_s(\omega) \longrightarrow \int_0^t h(\omega,s)\,dA_s(\omega)$ by dominated convergence, for every $\omega \in \Omega$. □ ■

Remarks

(i) It happens frequently that instead of the assumption of the proposition we have the weaker assumption

$$\text{a.s. } \forall t \geq 0,\ \int_0^t |H_s(\omega)|\,|dA_s(\omega)| < \infty.$$

If the filtration is complete, we can still define $H \cdot A$ as a finite variation process under this weaker assumption. We replace H by the process H' defined by

$$H'_t(\omega) = \begin{cases} H_t(\omega) & \text{if } \int_0^n |H_s(\omega)|\,|dA_s(\omega)| < \infty,\ \forall n, \\ 0 & \text{otherwise.} \end{cases}$$

Thanks to the fact that the filtration is complete, the process H' is still progressive, which allows us to define $H \cdot A = H' \cdot A$. We will use this extension of Proposition 3.5.5 implicitly in what follows.

(ii) Under appropriate assumptions (if H and K are progressive and $\int_0^t |H_s|\,|dA_s| < \infty$,

$\int_0^t |H_s K_s| \, |dA_s| < \infty$ for every $t \geq 0$), we have the *associativity property*

$$K \cdot (H \cdot A) = (KH) \cdot A.$$

This indeed follows from the analogous deterministic result saying informally that $k(s)(h(s)\,\mu(ds)) = (k(s)h(s))\,\mu(ds)$ if $h(s)$ and $k(s)h(s)$ are integrable with respect to the signed measure μ on $[0,t]$.

An important special case of the proposition is the case where $A_t = t$. If H is a progressive process such that

$$\forall t \geq 0, \, \forall \omega \in \Omega, \int_0^t |H_s(\omega)| \, ds < \infty,$$

the process $\int_0^t H_s \, ds$ is a finite variation process.

3.6 Continuous Local Martingales

We consider again a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), P)$. If T is a stopping time, and if $X = (X_t)_{t \geq 0}$ is an adapted process with continuous sample paths, we will write X^T for process X stopped at T, defined by $X_t^T = X_{t \wedge T}$ for every $t \geq 0$. It is useful to observe that, if S is another stopping time,

$$(X^T)^S = (X^S)^T = X^{S \wedge T}.$$

3.6.1 Definition and Basic Properties of Continuous Local Martingales

Definition 3.6.1: continuous local martingale

An adapted process $M = (M_t)_{t \geq 0}$ with continuous sample paths and such that $M_0 = 0$ a.s. is called a *continuous local martingale* if there exists a nondecreasing sequence $(T_n)_{n \geq 0}$ of stopping times such that $T_n \uparrow \infty$ (i.e. $T_n(\omega) \uparrow \infty$ for every ω) and, for every n, the stopped process M^{T_n} is a uniformly integrable martingale.

More generally, when we do not assume that $M_0 = 0$ a.s., we say that M is a continuous local martingale if the process $N_t = M_t - M_0$ is a continuous local martingale. In all cases, we say that the sequence of stopping times (T_n) *reduces* M if $T_n \uparrow \infty$ and, for every n, the stopped process M^{T_n} is a uniformly integrable martingale. **Remarks**

1. We do not require in the definition of a continuous local martingale that the variables M_t are in L^1 (compare with the definition of martingales). In particular, the variable M_0 may be any $\mathcal{F}_0$-measurable random variable.
2. Any martingale with continuous sample paths is a continuous local martingale (see property (a) below) but the converse is false, and for this reason we will sometimes speak of *true martingales* to emphasize the difference with local martingales.

3. One can define a notion of local martingale with càdlàg sample paths. In this course, however, we consider only continuous local martingales.

The following properties are easily established.

Proposition 3.6.2

1. A martingale with continuous sample paths is a continuous local martingale, and the sequence $T_n = n$ reduces M.
2. In the definition of a continuous local martingale starting from 0, one can replace uniformly integrable martingale by *martingale* (indeed, one can then observe that $M^{T_n \wedge n}$ is uniformly integrable, and we still have $T_n \wedge n \uparrow \infty$).
3. If M is a continuous local martingale, then, for every stopping time T, M^T is a continuous local martingale (this follows from Corollary of Optional Stopping Theorem 3.4.1).
4. If (T_n) reduces M and if (S_n) is a sequence of stopping times such that $S_n \uparrow \infty$, then the sequence $(T_n \wedge S_n)$ also reduces M (use same Corollary again).
5. The space of all continuous local martingales is a vector space (to check stability under addition, note that if M and M' are two continuous local martingales such that $M_0 = 0$ and $M'_0 = 0$, if the sequence (T_n) reduces M and if the sequence (T'_n) reduces M', property (d) shows that the sequence $T_n \wedge T'_n$ reduces $M + M'$).

The next proposition gives three other useful properties of local martingales.

Proposition 3.6.3

1. A *nonnegative continuous local martingale* M such that $M_0 \in L^1$ is a *supermartingale*.
2. A *continuous local martingale* M such that there exists a random variable $Z \in L^1$ with $|M_t| \leq Z$ for every $t \geq 0$ (in particular a *bounded continuous local martingale*) is a *uniformly integrable martingale*.
3. If M is a *continuous local martingale* and $M_0 = 0$ (or more generally $M_0 \in L^1$), the sequence of stopping times

$$T_n = \inf\{t \geq 0 : |M_t| \geq n\}$$

reduces M.

Theorem 3.6.4: Total variation of Local Martingales

Let M be a continuous local martingale. Assume that M is also a finite variation process (in particular $M_0 = 0$). Then $M_t = 0$ for every $t \geq 0$, a.s.

Proof for Theorem.

Set

$$\tau_n = \inf\left\{t \geq 0 : \int_0^t |dM_s| \geq n\right\}$$

for every integer $n \geq 0$. By Definition 3.2.10, τ_n is a stopping time (recall that $\int_0^t |dM_s|$ is an increasing process if M is a finite variation process).

Fix $n \geq 0$ and set $N = M^{\tau_n}$. Note that, for every $t \geq 0$,

$$|N_t| = |M_{t\wedge\tau_n}| \leq \int_0^t \wedge\tau_n |dM_s| \leq n.$$

By Proposition 3.6.3, N is a (bounded) martingale. Let $t > 0$ and let $0 = t_0 < t_1 < \cdots < t_p = t$ be any subdivision of $[0, t]$. Then, from Proposition 3.2.3, we have

$$\begin{aligned} E[N_t^2] &= \sum_{i=1}^{p} E[(N_{t_i} - N_{t_{i-1}})^2] \\ &\leq E\left[\left(\sup_{1\leq i\leq p} |N_{t_i} - N_{t_{i-1}}|\right) \sum_{i=1}^{p} |N_{t_i} - N_{t_{i-1}}|\right] \\ &\leq n\, E\left[\sup_{1\leq i\leq p} |N_{t_i} - N_{t_{i-1}}|\right] \end{aligned}$$

noting that $\int_0^t |dN_s| \leq n$ by the definition of τ_n, and using Proposition 3.5.2.

We now apply the preceding bound to a sequence $0 = t_0^k < t_1^k < \cdots < t_{p_k}^k = t$ of subdivisions of $[0, t]$ whose mesh tends to 0. Using the continuity of sample paths, and the fact that N is bounded (to justify dominated convergence), we get

$$\lim_{k\to\infty} E\left[\sup_{1\leq i\leq p_k} |N_{t_i^k} - N_{t_{i-1}^k}|\right] = 0.$$

We then conclude that $E[N_t^2] = 0$, hence $M_{t\wedge\tau_n} = 0$ a.s. Letting n tend to ∞, we get that $M_t = 0$ a.s. ■

3.6.2 The Quadratic Variation of a Continuous Local Martingale

Theorem 3.6.5: Quadratic variation existence

Let $M = (M_t)_{t\geq 0}$ be a continuous local martingale. There exists an increasing process denoted by $(\langle M, M\rangle_t)_{t\geq 0}$, which is unique up to indistinguishability, such that $M_t^2 - \langle M, M\rangle_t$ is a continuous local martingale. Furthermore, for every fixed $t > 0$, if $0 = t_0^n < t_1^n < \cdots < t_{p_n}^n = t$ is an increasing sequence of subdivisions of $[0, t]$ with mesh tending to 0, we have :

$$\langle M, M\rangle_t = \lim_{n\to\infty} \sum_{i=1}^{p_n} (M_{t_i^n} - M_{t_{i-1}^n})^2$$

In probability. The process $\langle M, M\rangle$ is called the quadratic variation of M.

Proof for Theorem.

The proof can be found in [2, thm. 4.9] ∎

Remarks:

1. We observe that the process $\langle M, M\rangle$ does not depend on the initial value M_0, but only on the increments of M: if $M_t = M_0 + N_t$, we have $\langle M, M\rangle = \langle N, N\rangle$. This is obvious from the second assertion of the theorem, and this will also be clear from the proof that follows.
2. In the second assertion of the theorem, it is in fact not necessary to assume that the sequence of subdivisions is increasing.

Now we present some properties of the quadratic variation :

Proposition 3.6.6

Let M be a continuous local martingale and let T be a stopping time.

1. We have a.s. for every $t \geq 0$,

$$\langle M^T, M^T\rangle_t = \langle M, M\rangle_{t\wedge T}.$$

2. If $M_0 = 0$. Then we have $\langle M, M\rangle = 0$ if and only if $M = 0$.

Proof for Proposition.

1. This follows from the fact that $M^2_{t\wedge T} - \langle M, M\rangle_{t\wedge T}$ is a continuous local martingale (cf. property (3) of continuous local martingales in Proposition 3.6.2).
2. Suppose that $\langle M, M\rangle = 0$. Then M_t^2 is a nonnegative continuous local martingale and, by Proposition 3.6.3 (1), M_t^2 is a supermartingale, hence $E[M_t^2] \leq E[M_0^2] = 0$, so that $M_t = 0$ for every t. The converse is obvious.

∎

The next theorem shows that properties of a continuous local martingale are closely related to those of its quadratic variation. If A is an increasing process, A_∞ denotes the increasing limit of A_t as $t \to \infty$ (this limit always exists in $[0, \infty]$).

Theorem 3.6.7

Let M be a continuous local martingale with $M_0 \in L^2$.

1. *The following are equivalent:*
   (a) *M is a (true) martingale bounded in L^2.*
   (b) *$E[\langle M, M\rangle_\infty] < \infty$.*
   Furthermore, if these properties hold, the process $M_t^2 - \langle M, M\rangle_t$ is a uniformly integrable martingale, and in particular $E[M_\infty^2] = E[M_0^2] + E[\langle M, M\rangle_\infty]$.
2. *The following are equivalent:*
   (a) *M is a (true) martingale and $M_t \in L^2$ for every $t \geq 0$.*
   (b) *$E[\langle M, M\rangle_t] < \infty$ for every $t \geq 0$.*

Furthermore, if these properties hold, the process $M_t^2 - \langle M, M\rangle_t$ is a martingale.

Proof for Theorem.

Let's prove (1), Replacing M by $M - M_0$, we may assume that $M_0 = 0$ in the proof. Let us first assume that M is a martingale bounded in L^2. Doob's inequality in L^2 (Proposition 3.3.5 (4)) shows that, for every $T > 0$,

$$E\left[\sup_{0\le t\le T} M_t^2\right] \le 4E[M_T^2].$$

By letting T go to ∞, we have

$$E\left[\sup_{t\ge 0} M_t^2\right] \le 4\sup_{t\ge 0} E[M_t^2] =: C < \infty.$$

Set $S_n = \inf\{t \ge 0 : \langle M, M\rangle_t \ge n\}$. Then the continuous local martingale $M^2_{t\wedge S_n} - \langle M, M\rangle_{t\wedge S_n}$ is dominated by the variable

$$\sup_{s\ge 0} M_s^2 + n,$$

which is integrable. From Proposition 3.6.3 (2), we get that this continuous local martingale is a uniformly integrable martingale, hence

$$E[\langle M, M\rangle_{t\wedge S_n}] = E[M^2_{t\wedge S_n}] \le E\left[\sup_{s\ge 0} M_s^2\right] \le C.$$

By letting n, and then t tend to infinity, and using monotone convergence, we get $E[\langle M, M\rangle_\infty] \le C < \infty$.

Conversely, assume that $E[\langle M, M\rangle_\infty] < \infty$. Set $T_n = \inf\{t \ge 0 : |M_t| \ge n\}$. Then the continuous local martingale $M^2_{t\wedge T_n} - \langle M, M\rangle_{t\wedge T_n}$ is dominated by the variable

$$n^2 + \langle M, M\rangle_\infty,$$

which is integrable. From Proposition 3.6.3 (2) again, this continuous local martingale is a uniformly integrable martingale, hence, for every $t \ge 0$,

$$E[M^2_{t\wedge T_n}] = E[\langle M, M\rangle_{t\wedge T_n}] \le E[\langle M, M\rangle_\infty] =: C' < \infty.$$

By letting $n \to \infty$ and using Fatou's lemma, we get $E[M_t^2] \le C'$, so that the collection $(M_t)_{t\ge 0}$ is bounded in L^2. We have not yet verified that $(M_t)_{t\ge 0}$ is a martingale. However, the previous bound on $E[M^2_{t\wedge T_n}]$ shows that the sequence $(M_{t\wedge T_n})_{n\ge 1}$ is uniformly integrable, and therefore converges both a.s. and in L^1 to M_t, for every $t \ge 0$. Recalling that M^{T_n} is a martingale (Proposition 3.6.3 (3)), the L^1-convergence allows us to pass to the limit $n \to \infty$ in the martingale property $E[M_{t\wedge T_n} \mid \mathcal{F}_s] = M_{s\wedge T_n}$, for $0 \le s < t$, and to get that M is a martingale.

Finally, if properties (a) and (b) hold, the continuous local martingale $M^2 - \langle M, M\rangle$

is dominated by the integrable variable

$$\sup_{t\geq 0} M_t^2 + \langle M, M\rangle_\infty$$

and is therefore (by Proposition 3.6.3 (2)) a uniformly integrable martingale. ■

Remark:

1. In property (a) of (1) (or of (2)), it is essential to suppose that M is a martingale, and not only a continuous local martingale. Doob's inequality used in the previous proof is not valid in general for a continuous local martingale!
2. It suffices to apply (1) to $(M_{t\wedge a})_{t\geq 0}$ for every choice of $a \geq 0$, to prove (2).

Lemma 3.6.8: Quadratic-variation integral formula for continuous integrands

Let $X = (X_s)_{s\geq 0}$ be a continuous process for which Theorem 3.6.5 holds, so that the quadratic variation $\langle X, X\rangle$ exists as a continuous, nondecreasing, adapted process. Fix $t > 0$ and let $(\pi_n)_{n\geq 1}$,

$$\pi_n : \quad 0 = t_0^n < t_1^n < \cdots < t_{p_n}^n = t,$$

be a sequence of subdivisions of $[0, t]$ whose mesh $|\pi_n| := \max_i(t_{i+1}^n - t_i^n)$ tends to 0 as $n \to \infty$. Let $H = (H_s)_{0\leq s\leq t}$ be a process, jointly measurable in (s, ω), whose sample paths $s \mapsto H_s(\omega)$ are almost surely continuous. Then

$$\sum_{i=0}^{p_n-1} H_{t_i^n} \left(X_{t_{i+1}^n} - X_{t_i^n}\right)^2 \xrightarrow[n\to\infty]{\mathbb{P}} \int_0^t H_s \, d\langle X, X\rangle_s.$$

Proof for Lemma

Since convergence in probability is metrizable, it suffices to show that every subsequence of (n) has a further subsequence along which the stated convergence holds almost surely.

1. **Step 1 (Random measures).** For each n, define the (a.s. finite, purely atomic, positive) random measure on $[0, t]$

   $$\mu_n(dr) \; := \; \sum_{i=0}^{p_n-1} \left(X_{t_{i+1}^n} - X_{t_i^n}\right)^2 \delta_{t_i^n}(dr).$$

   By construction,

   $$\int_{[0,t]} H_s \, \mu_n(ds) \; = \; \sum_{i=0}^{p_n-1} H_{t_i^n} \left(X_{t_{i+1}^n} - X_{t_i^n}\right)^2. \quad [1]$$

2. **Step 2 (Convergence of cumulative masses via Proposition 3.6.5).** Fix $r \in [0, t]$ and let $k = k_n(r) := \max\{\, i : t_i^n \leq r \,\}$. Applying Proposition 3.6.5 (with

$Y = X$) to the subdivision $t_0^n < t_1^n < \cdots < t_k^n < r$ of $[0, r]$ — whose mesh also tends to 0 — gives

$$\sum_{i=0}^{k-1} \left(X_{t_{i+1}^n} - X_{t_i^n}\right)^2 + \left(X_r - X_{t_k^n}\right)^2 \xrightarrow[n\to\infty]{\mathbb{P}} \langle X, X\rangle_r.$$

On the other hand, directly from the definition of μ_n,

$$\mu_n([0,r]) \;=\; \sum_{i=0}^{k-1} \left(X_{t_{i+1}^n} - X_{t_i^n}\right)^2 + \left(X_{t_{k+1}^n} - X_{t_k^n}\right)^2.$$

The two displays differ only in their last term, and

$$\left(X_{t_{k+1}^n} - X_{t_k^n}\right)^2 - \left(X_r - X_{t_k^n}\right)^2 \xrightarrow[n\to\infty]{\text{a.s.}} 0,$$

since $t_{k+1}^n - t_k^n \to 0$ and $r - t_k^n \to 0$ as $|\pi_n| \to 0$, and X has (a.s. uniformly) continuous paths on $[0, t]$. Hence

$$\mu_n([0,r]) \xrightarrow[n\to\infty]{\mathbb{P}} \langle X, X\rangle_r \qquad \text{for every } r \in [0,t]. \quad [1]$$

3. **Step 3 (Diagonal extraction).** Let $D := \bigcup_{n\geq 1}\{t_0^n, \dots, t_{p_n}^n\}$, a countable dense subset of $[0, t]$ containing t. Enumerate $D = \{r_1, r_2, \dots\}$. By $(**)$ and a diagonal argument, there is a subsequence $(n_k)_{k\geq 1}$ along which

$$\mu_{n_k}([0, r_j]) \xrightarrow[k\to\infty]{\text{a.s.}} \langle X, X\rangle_{r_j} \qquad \text{simultaneously for every } j \geq 1.$$

4. **Step 4 (Weak convergence of μ_{n_k}).** Let Ω_0 be the a.s. event on which: (i) the convergence of Step 3 holds at every $r \in D$; (ii) $s \mapsto \langle X, X\rangle_s(\omega)$ is continuous and nondecreasing; (iii) $s \mapsto H_s(\omega)$ is continuous. Fix $\omega \in \Omega_0$. The nondecreasing functions $r \mapsto \mu_{n_k}([0, r])(\omega)$ converge on the dense set D to the continuous nondecreasing function $r \mapsto \langle X, X\rangle_r(\omega)$; a standard squeeze argument using monotonicity and density then extends this to convergence at *every* $r \in [0, t]$, including $r = t$ (total mass). Convergence of these "distribution functions" at every point, together with convergence of total mass, implies

$$\mu_{n_k}(\omega) \xrightarrow[k\to\infty]{w} \mathbf{1}_{[0,t]}(r)\, d\langle X, X\rangle_r(\omega)$$

weakly as finite positive measures on $[0, t]$.

5. **Step 5 (Passing to the limit against H).** Since $H_{\cdot}(\omega)$ is continuous, hence bounded, on the compact interval $[0, t]$, weak convergence yields

$$\int_{[0,t]} H_s(\omega)\, \mu_{n_k}(ds)(\omega) \xrightarrow[k\to\infty]{} \int_0^t H_s(\omega)\, d\langle X, X\rangle_s(\omega), \qquad \text{for every } \omega \in \Omega_0.$$

By $(*)$, this is exactly

$$\sum_{i=0}^{p_{n_k}-1} H_{t_i^{n_k}} \left(X_{t_{i+1}^{n_k}} - X_{t_i^{n_k}}\right)^2 \xrightarrow[k\to\infty]{\text{a.s.}} \int_0^t H_s \, d\langle X, X\rangle_s.$$

6. **Step 6 (Conclusion).** Every subsequence of (n) admits a further subsequence along which the convergence holds almost surely; therefore the original sequence converges in probability, which proves the lemma.

■

3.6.3 The Bracket of Two Continuous Local Martingales

Definition 3.6.9: Bracket of two continuous local martingales

If M and N are two continuous local martingales, the *bracket* $\langle M, N\rangle$ is the finite variation process defined by setting, for every $t \geq 0$,

$$\langle M, N\rangle_t = \frac{1}{2}(\langle M+N, M+N\rangle_t - \langle M, M\rangle_t - \langle N, N\rangle_t).$$

Let us state a few easy properties of the bracket.

Proposition 3.6.10

1. *$\langle M, N\rangle$ is the unique (up to indistinguishability) finite variation process such that $M_tN_t - \langle M, N\rangle_t$ is a continuous local martingale.*
2. *The mapping $(M, N) \mapsto \langle M, N\rangle$ is bilinear and symmetric.*
3. *If $0 = t_0^n < t_1^n < \cdots < t_{p_n}^n = t$ is an increasing sequence of subdivisions of $[0, t]$ with mesh tending to 0, we have*

$$\lim_{n\to\infty} \sum_{i=1}^{p_n} (M_{t_i^n} - M_{t_{i-1}^n})(N_{t_i^n} - N_{t_{i-1}^n}) = \langle M, N\rangle_t$$

   in probability.
4. *For every stopping time T, $\langle M^T, N^T\rangle_t = \langle M^T, N\rangle_t = \langle M, N\rangle_{t\wedge T}$.*
5. *If M and N are two martingales (with continuous sample paths) bounded in L^2, $M_tN_t - \langle M, N\rangle_t$ is a uniformly integrable martingale. Consequently, $\langle M, N\rangle_\infty$ is well defined as the almost sure limit of $\langle M, N\rangle_t$ as $t \to \infty$, is integrable, and satisfies*

$$\mathbb{E}[M_\infty N_\infty] = \mathbb{E}[M_0 N_0] + \mathbb{E}[\langle M, N\rangle_\infty].$$

Remark: A consequence of (iv) is the fact that $M^T(N - N^T)$ is a continuous local martingale, which is not so easy to prove directly.

Definition 3.6.11: Orthogonal continuous local martingales

Two continuous local martingales M and N are said to be orthogonal if $\langle M, N\rangle = 0$, which holds if and only if MN is a continuous local martingale.

If M and N are two orthogonal martingales bounded in L^2, we have $E[M_tN_t] = E[M_0N_0]$, and even $E[M_SN_S] = E[M_0N_0]$ for any stopping time S. This follows from Theorem 3.4.1, using property (5) of Proposition 3.6.10.

Theorem 3.6.12: Kunita-Watanabe inequality

Let M and N be two continuous local martingales and let H and K be two measurable processes. Then, a.s.,

$$\int_0^\infty |H_s||K_s||\mathrm{d}\langle M, N\rangle_s| \leq \left(\int_0^\infty H_s^2 \mathrm{d}\langle M, M\rangle_s\right)^{1/2} \left(\int_0^\infty K_s^2 \mathrm{d}\langle N, N\rangle_s\right)^{1/2}.$$

Proof for Theorem.

Only in this proof, we use the special notation $\langle M, N\rangle_s^t = \langle M, N\rangle_t - \langle M, N\rangle_s$ for $0 \leq s \leq t$. The first step of the proof is to observe that we have a.s. for every choice of the rationals $s < t$ (and also by continuity for every reals $s < t$),

$$|\langle M, N\rangle_s^t| \leq \sqrt{\langle M, M\rangle_s^t}\sqrt{\langle N, N\rangle_s^t}.$$

Indeed, this follows from the approximations of $\langle M, M\rangle$ and $\langle M, N\rangle$ given in Theorem 3.6.5 and in Proposition 3.6.10 respectively (note that these approximations are easily extended to the increments of $\langle M, M\rangle$ and $\langle M, N\rangle$), together with the Cauchy-Schwarz inequality. From now on, we fix ω such that the inequality of the last display holds for every $s < t$, and we argue with this value of ω (the remaining part of the argument is *deterministic*).

We then observe that we also have, for every $0 \leq s \leq t$,

$$\int_s^t |\mathrm{d}\langle M, N\rangle_u| \leq \sqrt{\langle M, M\rangle_s^t}\sqrt{\langle N, N\rangle_s^t}. \tag{*}$$

Indeed, we use Proposition 3.5.2, noting that, for any subdivision $s = t_0 < t_1 < \cdots < t_p = t$, we can bound

$$\begin{aligned}
\sum_{i=1}^p |\langle M, N\rangle_{t_{i-1}}^{t_i}| &\leq \sum_{i=1}^p \sqrt{\langle M, M\rangle_{t_{i-1}}^{t_i}}\sqrt{\langle N, N\rangle_{t_{i-1}}^{t_i}} \\
&\leq \left(\sum_{i=1}^p \langle M, M\rangle_{t_{i-1}}^{t_i}\right)^{1/2} \left(\sum_{i=1}^p \langle N, N\rangle_{t_{i-1}}^{t_i}\right)^{1/2} \\
&= \sqrt{\langle M, M\rangle_s^t}\sqrt{\langle N, N\rangle_s^t}.
\end{aligned}$$

We then get that, for every bounded Borel subset A of $\mathbb{R}_+$,

$$\int_A |\mathrm{d}\langle M, N\rangle_u| \le \sqrt{\int_A \mathrm{d}\langle M, M\rangle_u}\sqrt{\int_A \mathrm{d}\langle N, N\rangle_u}.$$

When $A = [s, t]$, this is the bound (*). If A is a finite union of intervals, this follows from (*) and another application of the Cauchy-Schwarz inequality. A monotone class argument shows that the inequality of the last display remains valid for any bounded Borel set A (Theorem 1.1.5).

Next let $h = \sum_{i=1}^{p} \lambda_i \mathbf{1}_{A_i}$ and $k = \sum_{i=1}^{p} \mu_i \mathbf{1}_{A_i}$ be two nonnegative simple functions on $\mathbb{R}_+$ with bounded support contained in $[0, K]$, for some $K > 0$. Here $A_1, \dots, A_p$ is a measurable partition of $[0, K]$, and $\lambda_1, \dots, \lambda_p, \mu_1, \dots, \mu_p$ are reals (we can always assume that h and k are expressed in terms of the same partition). Then,

$$\begin{aligned}
\int h(s)k(s)|\mathrm{d}\langle M, N\rangle_s| &= \sum_{i=1}^{p} \lambda_i\mu_i \int_{A_i} |\mathrm{d}\langle M, N\rangle_s| \\
&\le \sum_{i=1}^{p} \lambda_i\mu_i \sqrt{\int_{A_i} \mathrm{d}\langle M, M\rangle_s}\sqrt{\int_{A_i} \mathrm{d}\langle N, N\rangle_s} \\
&\le \left(\sum_{i=1}^{p} \lambda_i^2 \int_{A_i} \mathrm{d}\langle M, M\rangle_s\right)^{1/2} \left(\sum_{i=1}^{p} \mu_i^2 \int_{A_i} \mathrm{d}\langle N, N\rangle_s\right)^{1/2} \\
&= \left(\int h(s)^2 \mathrm{d}\langle M, M\rangle_s\right)^{1/2} \left(\int k(s)^2 \mathrm{d}\langle N, N\rangle_s\right)^{1/2},
\end{aligned}$$

which gives the desired inequality for simple functions. Since every nonnegative Borel function is a monotone increasing limit of simple functions with bounded support, an application of the monotone convergence theorem completes the proof. ■

3.6.4 The Hilbert Space $\mathbb{H}^2$

We write $\mathbb{H}^2$ for the space of all continuous martingales M which are bounded in L^2 and such that $M_0 = 0$, with the usual convention that two indistinguishable processes are identified. Equivalently, $M \in \mathbb{H}^2$ if and only if M is a continuous local martingale such that $M_0 = 0$ and

$$\mathbb{E}[\langle M, M\rangle_\infty] < \infty;$$

see Theorem 3.6.7. By Theorem 3.3.16, every $M \in \mathbb{H}^2$ admits an almost sure and L^2 limit $M_\infty \in L^2$, and

$$M_t = \mathbb{E}[M_\infty \mid \mathcal{F}_t], \qquad t \ge 0.$$

If $M, N \in \mathbb{H}^2$, Proposition 3.6.10 (v) shows that $\langle M, N\rangle_\infty$ is well defined and integrable. We may therefore define a symmetric bilinear form on $\mathbb{H}^2$ by

$$(M, N)_{\mathbb{H}^2} = \mathbb{E}[\langle M, N\rangle_\infty] = \mathbb{E}[M_\infty N_\infty].$$

The second equality follows again from Proposition 3.6.10 (v), since $M_0 = N_0 = 0$. In particular, $(M,M)_{\mathbb{H}^2} = 0$ if and only if $M = 0$. The associated norm is

$$\|M\|_{\mathbb{H}^2} = \Big((M,M)_{\mathbb{H}^2}\Big)^{1/2} = (\mathbb{E}[\langle M,M\rangle_\infty])^{1/2} = \Big(\mathbb{E}[M_\infty^2]\Big)^{1/2}.$$

Proposition 3.6.13

The space $\mathbb{H}^2$, equipped with the scalar product $(\cdot,\cdot)_{\mathbb{H}^2}$, is a Hilbert space.

Proof for Proposition.

Let $(M^n)_{n\geq 1}$ be a Cauchy sequence in $\mathbb{H}^2$. Then $(M_\infty^n)_{n\geq 1}$ is Cauchy in L^2, hence converges in L^2 to some $X \in L^2$. For each $t \geq 0$, the sequence $(M_t^n)_{n\geq 1}$ converges in L^2 to $M_t := \mathbb{E}[X \mid \mathcal{F}_t]$. Since $\mathbb{E}[M_\infty^n \mid \mathcal{F}_0] = M_0^n = 0$ for every n, we also have $M_0 = \mathbb{E}[X \mid \mathcal{F}_0] = 0$. Thus $(M_t)_{t\geq 0}$ is a martingale bounded in L^2.

Moreover, Doob's L^2 inequality gives, for every $n, m \geq 1$,

$$\mathbb{E}\left[\sup_{t\geq 0} |M_t^n - M_t^m|^2\right] \leq 4\,\mathbb{E}\left[|M_\infty^n - M_\infty^m|^2\right].$$

By extracting a subsequence, we obtain almost sure uniform convergence of M^n to M on $\mathbb{R}_+$. Since each M^n has continuous sample paths, M has a continuous modification. Hence $M \in \mathbb{H}^2$. Finally,

$$\|M^n - M\|_{\mathbb{H}^2}^2 = \mathbb{E}\left[|M_\infty^n - X|^2\right] \longrightarrow 0,$$

which proves that $\mathbb{H}^2$ is complete. ■

3.7 Continuous Semimartingales

We now introduce the class of processes for which we will develop the theory of stochastic integrals.

Definition 3.7.1: Continuous semimartingale

A process $X = (X_t)_{t\geq 0}$ is a *continuous semimartingale* if it can be written in the form

$$X_t = M_t + A_t,$$

where M is a continuous local martingale and A is a finite variation process.

The decomposition $X = M + A$ is then unique up to indistinguishability thanks to Theorem 3.6.4. We say that this is the *canonical decomposition* of X.

By construction, continuous semimartingales have continuous sample paths. It is possible to define a notion of semimartingale with càdlàg sample paths, but in this book, we will only deal with *continuous* semimartingales, and for this reason we sometimes omit the word continuous.

Definition 3.7.2: Bracket of semimartingales

Let $X = M + A$ and $Y = M' + A'$ be the canonical decompositions of two continuous semimartingales X and Y. The *bracket* $\langle X, Y\rangle$ is the finite variation process defined by

$$\langle X, Y\rangle_t = \langle M, M'\rangle_t.$$

In particular, we have $\langle X, X\rangle_t = \langle M, M\rangle_t$.

Proposition 3.7.3

Let $0 = t_0^n < t_1^n < \cdots < t_{p_n}^n = t$ be an increasing sequence of subdivisions of $[0, t]$ whose mesh tends to 0. Then,

$$\lim_{n\to\infty} \sum_{i=1}^{p_n} (X_{t_i^n} - X_{t_{i-1}^n})(Y_{t_i^n} - Y_{t_{i-1}^n}) = \langle X, Y\rangle_t$$

in probability.

Proof for Proposition.

We treat the case where $X = Y$ and leave the general case to the reader. We have

$$\sum_{i=1}^{p_n}(X_{t_i^n} - X_{t_{i-1}^n})^2 = \sum_{i=1}^{p_n}(M_{t_i^n} - M_{t_{i-1}^n})^2 + \sum_{i=1}^{p_n}(A_{t_i^n} - A_{t_{i-1}^n})^2 + 2\sum_{i=1}^{p_n}(M_{t_i^n} - M_{t_{i-1}^n})(A_{t_i^n} - A_{t_{i-1}^n}).$$

By Theorem 3.6.5,

$$\lim_{n\to\infty} \sum_{i=1}^{p_n}(M_{t_i^n} - M_{t_{i-1}^n})^2 = \langle M, M\rangle_t = \langle X, X\rangle_t,$$

in probability. On the other hand,

$$\begin{aligned}\sum_{i=1}^{p_n}(A_{t_i^n} - A_{t_{i-1}^n})^2 &\le \left(\sup_{1\le i\le p_n} |A_{t_i^n} - A_{t_{i-1}^n}|\right)\sum_{i=1}^{p_n}|A_{t_i^n} - A_{t_{i-1}^n}| \\ &\le \left(\int_0^t |dA_s|\right)\sup_{1\le i\le p_n}|A_{t_i^n} - A_{t_{i-1}^n}|,\end{aligned}$$

which tends to 0 a.s. when $n \to \infty$ by the continuity of sample paths of A. The same argument shows that

$$\left|\sum_{i=1}^{p_n}(A_{t_i^n} - A_{t_{i-1}^n})(M_{t_i^n} - M_{t_{i-1}^n})\right| \le \left(\int_0^t |dA_s|\right)\sup_{1\le i\le p_n}|M_{t_i^n} - M_{t_{i-1}^n}|$$

tends to 0 a.s. ∎

CHAPITRE 4

STOCHASTIC INTEGRATION

This chapter is at the core of the present book. We start by defining the stochastic integral with respect to a square-integrable continuous martingale, considering first the integral of elementary processes (which play a role analogous to step functions in the theory of the Riemann integral) and then using an isometry between Hilbert spaces to deal with the general case.

It is easy to extend the definition of stochastic integrals to continuous local martingales and semimartingales. We then derive the celebrated Itô's formula, which shows that the image of one or several continuous semimartingales under a smooth function is still a continuous semimartingale, whose canonical decomposition is given in terms of stochastic integrals.

Itô's formula is the main technical tool of stochastic calculus, and we discuss several important applications of this formula, including Lévy's theorem characterizing Brownian motion as a continuous local martingale with quadratic variation process equal to t, the Burkholder-Davis-Gundy inequalities, and the representation of martingales as stochastic integrals in a Brownian filtration.

The end of the chapter is devoted to Girsanov's theorem, which deals with the stability of the notions of a martingale and a semimartingale under an absolutely continuous change of probability measure. As an application of Girsanov's theorem, we establish the famous Cameron-Martin formula giving the image of the Wiener measure under a translation by a deterministic function.

4.1 The Construction of Stochastic Integrals

Throughout this chapter, we argue on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$, and we assume that the filtration $(\mathcal{F}_t)$ is complete. Unless otherwise specified, all processes in this chapter are indexed by $\mathbb{R}_+$ and take real values. We often say *continuous martingale* instead of *martingale with continuous sample paths.*

We recall that $\mathbb{H}^2$ denotes the space of continuous martingales starting from 0 and bounded in L^2. As established in Proposition 3.6.13, endowed with its usual scalar product, $\mathbb{H}^2$ is a Hilbert space. We also denote by $\mathcal{P}$ the progressive σ-field on $\Omega \times \mathbb{R}_+$; see the end of Section 3.1.2.

4.1.1 Stochastic Integrals for Martingales Bounded in L^2

Let $M \in \mathbb{H}^2$, we let $L^2(M)$ be the set of all progressive processes H such that

$$\mathbb{E}\left[\int_0^\infty H_s^2 \, d\langle M, M\rangle_s\right] < \infty,$$

with the convention that two progressive processes H and H' satisfying this integrability condition are identified if $H_s = H'_s$, $d\langle M, M\rangle_s$-a.e., a.s. We can view $L^2(M)$ as an ordinary L^2-space, namely

$$L^2(M) = L^2\Big(\Omega \times \mathbb{R}_+, \mathcal{P}, \, d\mathbb{P} \, d\langle M, M\rangle_s\Big),$$

where $d\mathbb{P} \, d\langle M, M\rangle_s$ refers to the finite measure on $(\Omega \times \mathbb{R}_+, \mathcal{P})$ that assigns the mass

$$\mathbb{E}\left[\int_0^\infty \mathbb{1}_A(\omega, s) \, d\langle M, M\rangle_s\right]$$

to a set $A \in \mathcal{P}$ (the total mass of this measure is $\mathbb{E}[\langle M, M\rangle_\infty] = \|M\|^2_{\mathbb{H}^2}$). Just like any L^2-space, the space $L^2(M)$ is a Hilbert space for the scalar product

$$(H, K)_{L^2(M)} = \mathbb{E}\left[\int_0^\infty H_s K_s \, d\langle M, M\rangle_s\right],$$

and the associated norm is

$$\|H\|_{L^2(M)} = \left(\mathbb{E}\left[\int_0^\infty H_s^2 \, d\langle M, M\rangle_s\right]\right)^{1/2}.$$

Definition 4.1.1: elementary process

An *elementary process* is a progressive process of the form

$$H_s(\omega) = \sum_{i=0}^{p-1} H_{(i)}(\omega) \, \mathbb{1}_{(t_i, t_{i+1}]}(s),$$

where $0 = t_0 < t_1 < t_2 < \cdots < t_p$ and, for every $i \in \{0, 1, \ldots, p-1\}$, $H_{(i)}$ is a bounded $\mathcal{F}_{t_i}$-measurable random variable.

The set $\mathcal{E}$ of all elementary processes forms a linear subspace of $L^2(M)$. To be precise, we should here say "equivalence classes of elementary processes" (recall that H and H' are identified in $L^2(M)$ if $\|H - H'\|_{L^2(M)} = 0$).

Proposition 4.1.2

For every $M \in \mathbb{H}^2$, $\mathcal{E}$ is dense in $L^2(M)$.

Proof for Proposition.

By elementary Hilbert space theory [Proposition **??** in Appendix 5], it is enough to verify that, if $K \in L^2(M)$ is orthogonal to $\mathcal{E}$, then $K = 0$. Assume that $K \in L^2(M)$ is orthogonal to $\mathcal{E}$, and set, for every $t \geq 0$,

$$X_t = \int_0^t K_u \, d(M,M)_u.$$

To see that the integral in the right-hand side makes sense, and defines a finite variation process $(X_t)_{t\geq 0}$, we use the Cauchy-Schwarz inequality to observe that

$$\mathbb{E}\Big[\int_0^t |K_u| \, d(M,M)_u\Big] \leq \Big(\mathbb{E}\Big[\int_0^t (K_u)^2 \, d(M,M)_u\Big]\Big)^{1/2} \times (\mathbb{E}[(M,M)_\infty])^{1/2}.$$

The right-hand side is finite since $M \in \mathcal{H}^2$ and $K \in L^2(M)$, and thus we have in particular

$$\text{a.s.} \qquad \forall t \geq 0, \qquad \int_0^t |K_u| \, d(M,M)_u < \infty.$$

By Proposition 3.5.5 (and Remark (i) following this proposition), $(X_t)_{t\geq 0}$ is well defined as a finite variation process. The preceding bound also shows that $X_t \in L^1$ for every $t \geq 0$. Let $0 \leq s < t$, and let $H \in \mathcal{E}$ be an elementary process in the form $H_r(\omega) = F(\omega)\mathbb{1}_{(s,t]}(r)$. Writing the orthogonality $H \perp K$ in $L^2(M)$, we have:

$$0 = (H,K)_{L^2(M)} = \mathbb{E}\left[F \int_s^t K_r \, d\langle M,M\rangle_r\right]$$

It follows that

$$\mathbb{E}[F(X_t - X_s)] = 0$$

for every $s < t$ and every bounded $\mathcal{F}_s$-measurable variable F. Since the process X is adapted and we know that $X_r \in L^1$ for every $r \geq 0$, this implies that X is a (continuous) martingale. On the other hand, X is also a finite variation process and, by Theorem 3.6.4, this is only possible if $X = 0$. We have thus proved that

$$\int_0^t K_u \, d\langle M,M\rangle_u = 0 \qquad \forall t \geq 0, \quad \text{a.s.}$$

which implies that, a.s., the signed measure having density K_u with respect to $d\langle M,M\rangle_u$ is the zero measure, which is only possible if

$$K_u = 0, \qquad d\langle M,M\rangle_u\text{-a.e.,} \quad \text{a.s.}$$

or equivalently $K = 0$ in $L^2(M)$. ■

Recall our notation X^T for the process X stopped at the stopping time T: $X_t^T = X_{t\wedge T}$. If $M \in \mathbb{H}^2$, the fact that $\langle M^T, M^T\rangle_\infty = \langle M,M\rangle_T$ immediately implies that M^T also belongs to $\mathbb{H}^2$. Furthermore, if $H \in L^2(M)$, the process $\mathbb{1}_{[0,T]}H$ defined by $(\mathbb{1}_{[0,T]}H)_s(\omega) = \mathbb{1}_{\{0\leq s\leq T(\omega)\}}H_s(\omega)$ also belongs to $L^2(M)$ (note that $\mathbb{1}_{[0,T]}$ is progressive since it is adapted with left-continuous sample paths).

The quantity $H \cdot \langle M,N\rangle$ in the right-hand side of (4.1.1 in the Theorem bellow) is an integral with respect to the finite variation process $\langle M,N\rangle$, as defined in Section 3.5. This integral is well defined because $\langle M,N\rangle$ is continuous and of finite variation, and H

is progressive. The notation $H \cdot A$ for an integral with respect to a finite variation process A, and $H \cdot M$ for an integral with respect to a martingale M, creates no ambiguity since these two classes of integrators are essentially disjoint: a nontrivial continuous martingale cannot be a finite variation process, and a nonconstant finite variation process cannot be a martingale.

Theorem 4.1.3: Stochastic integral wrt a martingale in $\mathbb{H}^2$

Let $M \in \mathbb{H}^2$. For every elementary process

$$H_s(\omega) = \sum_{i=0}^{p-1} H_{(i)}(\omega)\, \mathbb{1}_{(t_i,t_{i+1}]}(s),$$

the formula

$$(H \cdot M)_t = \sum_{i=0}^{p-1} H_{(i)}\Big(M_{t_{i+1}\wedge t} - M_{t_i \wedge t}\Big)$$

defines a process $H \cdot M \in \mathbb{H}^2$. The mapping $H \mapsto H \cdot M$ extends uniquely to an isometry from $L^2(M)$ into $\mathbb{H}^2$. Furthermore, $H \cdot M$ is the unique martingale in $\mathbb{H}^2$ such that

$$\langle H \cdot M, N\rangle = H \cdot \langle M, N\rangle, \qquad \forall N \in \mathbb{H}^2. \tag{4.1.1}$$

If T is a stopping time, then

$$(\mathbb{1}_{[0,T]}H) \cdot M = (H \cdot M)^T = H \cdot M^T. \tag{4.1.2}$$

We often use the notation

$$(H \cdot M)_t = \int_0^t H_s\, dM_s$$

and call $H \cdot M$ the *stochastic integral* of H with respect to M.

Proof for Theorem.

We divide this proof into 3 steps :

1. *Construction and Isometry extension* which is done via four substeps :
   - We can verify that the mapping $H \mapsto H \cdot M$ does not depend on the decomposition chosen for H, which implies that the mapping is linear and is an isometry from $\mathcal{E}$ (viewed as a subspace of $L^2(M)$) into $\mathbb{H}^2$.
   - We study each terms $M^i = H_{(i)}\left(M_{t_{i+1}\wedge t} - M_{t_i \wedge t}\right)$ in the sum

     $$H \cdot M = \sum_{i=0}^{p-1} H_{(i)}\left(M_{t_{i+1}\wedge t} - M_{t_i \wedge t}\right) = \sum_{i=0}^{p-1} M^i$$

     It's left to reader to prove that M^i are continuous martingales in $\mathbb{H}^2$ and that M^i are orthogonal and their quadratic variations are given by

$$\langle M^i, M^i\rangle_t = H_{(i)}^2 \left(\langle M, M\rangle_{t_{i+1}\wedge t} - \langle M, M\rangle_{t_i\wedge t}\right)$$

- We conclude that the mapping $H \mapsto H \cdot M$ is an isometry from $\mathcal{E}$ (equipped with the norm of $L^2(M)$) into $\mathbb{H}^2$. For instance by using the approximations of $\langle M, N\rangle$. We conclude that

$$\langle H \cdot M, H \cdot M\rangle_t = \sum_{i=0}^{p-1} H_{(i)}^2 \left(\langle M, M\rangle_{t_{i+1}\wedge t} - \langle M, M\rangle_{t_i\wedge t}\right) = \int_0^t H_s^2 \, d\langle M, M\rangle_s.$$

Consequently,

$$\|H \cdot M\|_{\mathbb{H}^2}^2 = E\left[\langle H \cdot M, H \cdot M\rangle_\infty\right] = E\left[\int_0^\infty H_s^2 \, d\langle M, M\rangle_s\right] = \|H\|_{L^2(M)}^2.$$

By linearity, this implies that $H \cdot M = H' \cdot M$ if H' is another elementary process that is identified with H in $L^2(M)$. Therefore the mapping $H \mapsto H \cdot M$ makes sense from $\mathcal{E}$ viewed as a subspace of $L^2(M)$ into $\mathbb{H}^2$. The latter mapping is linear, and, since it preserves the norm, it is an isometry from $\mathcal{E}$ (equipped with the norm of $L^2(M)$) into $\mathbb{H}^2$.
- Finally, since $\mathcal{E}$ is dense in $L^2(M)$ (Proposition 4.1.2) and $\mathbb{H}^2$ is a Hilbert space (Proposition **??**), this mapping can be extended in a unique way to an isometry from $L^2(M)$ into $\mathbb{H}^2$.

2. *The Bracket identity :* We fix $N \in \mathbb{H}^2$. We can prove a simpler formula which is :

$$\langle H \cdot M, N\rangle_\infty = H \cdot \langle M, N\rangle_\infty$$

And then replace N by the stopped martingale N^t to get the desired formula. First note that, if $H \in L^2(M)$, the Kunita–Watanabe inequality (Proposition 3.6.12) shows that

$$E\left[\int_0^\infty |H_s| \, |d\langle M, N\rangle_s|\right] \leq \|H\|_{L^2(M)} \|N\|_{\mathbb{H}^2} < \infty$$

and thus the variable $\int_0^\infty H_s \, d\langle M, N\rangle_s = (H \cdot \langle M, N\rangle)_\infty$ is well defined and in L^1. We firstly consider the case of elementary process then generalize:

- If H is an elementary process then by simple calculation we have

$$\begin{aligned}\langle H \cdot M, N\rangle_\infty &= \sum_{i=0}^{p-1} \langle M^i, N\rangle_\infty \\ &= \sum_{i=0}^{p-1} H_{(i)} \left(\langle M, N\rangle_{t_{i+1}} - \langle M, N\rangle_{t_i}\right) \\ &= \int_0^\infty H_s \, \mathrm{d}\langle M, N\rangle_s\end{aligned}$$

- Now we need to generalize for $H \in L^2(M)$. Let $(H^n)_{n\geq 1}$ be a sequence of elementary processes such that $H^n \to H$ in $L^2(M)$. To pass the limit a clever idea is to prove that the linear mapping $X \mapsto \langle X, N\rangle_\infty$ is continuous. Indeed, by the Kunita–Watanabe inequality,

$$E\left[|\langle X, N\rangle_\infty|\right] \le E\left[\langle X, X\rangle_\infty\right]^{1/2} E\left[\langle N, N\rangle_\infty\right]^{1/2} = \|N\|_{\mathbb{H}^2}\|X\|_{\mathbb{H}^2}.$$

If $(H^n)_{n\ge 1}$ is a sequence in $\mathcal{E}$, such that $H^n \to H$ in $L^2(M)$, we have therefore

$$\begin{aligned}\langle H\cdot M, N\rangle_\infty &= \lim_{n\to\infty}\langle H^n\cdot M, N\rangle_\infty \\ &= \lim_{n\to\infty}(H^n\cdot\langle M, N\rangle)_\infty \\ &= (H\cdot\langle M, N\rangle)_\infty,\end{aligned}$$

where the convergences hold in L^1, and the last equality again follows from the Kunita–Watanabe inequality by writing

$$E\left[\left|\int_0^\infty (H_s^n - H_s)\, d\langle M, N\rangle_s\right|\right] \le E\left[\langle N, N\rangle_\infty\right]^{1/2}\|H^n - H\|_{L^2(M)}.$$

- It is easy to see that (4.1.1) characterizes $H\cdot M$ among the martingales of $\mathbb{H}^2$. Indeed, if X is another martingale of $\mathbb{H}^2$ that satisfies the same identity, we get, for every $N \in \mathbb{H}^2$,

$$\langle H\cdot M - X, N\rangle = 0.$$

Taking $N = H\cdot M - X$ and using Theorem 3.6.4 we obtain that $X = H\cdot M$.

3. *The Bracket identity for stopping time :* It remains to verify (4.1.2). Using the properties of the bracket of two continuous local martingales, we observe that, if $N \in \mathbb{H}^2$,

$$\langle (H\cdot M)^T, N\rangle_t = \langle H\cdot M, N\rangle_{t\wedge T} = (H\cdot\langle M, N\rangle)_{t\wedge T} = (1_{[0,T]}H\cdot\langle M, N\rangle)_t$$

which shows that the stopped martingale $(H\cdot M)^T$ satisfies the characteristic property of the stochastic integral $(1_{[0,T]}H)\cdot M$. The first equality in (4.1.2) follows. The second one is proved analogously, writing

$$\langle H\cdot M^T, N\rangle = H\cdot\langle M^T, N\rangle = H\cdot\langle M, N\rangle^T = 1_{[0,T]}H\cdot\langle M, N\rangle.$$

This completes the proof of the theorem.

■

Remark We could have used the relation (4.1.1) to **define** the stochastic integral $H\cdot M$, observing that the mapping $N \mapsto \mathbb{E}[(H\cdot\langle M, N\rangle)_\infty]$ yields a continuous linear form on $\mathbb{H}^2$, and thus there exists a unique martingale $H\cdot M$ in $\mathbb{H}^2$ such that

$$\mathbb{E}[(H\cdot\langle M, N\rangle)_\infty] = (H\cdot M, N)_{\mathbb{H}^2} = E[\langle H\cdot M, N\rangle_\infty].$$

Using the notation introduced at the end of Theorem 4.1.3, we can rewrite (4.1.1) in the form

$$\left\langle \int_0^\cdot H_s\, dM_s, N \right\rangle_t = \int_0^t H_s\, d\langle M, N\rangle_s.$$

We interpret this by saying that the stochastic integral *commutes* with the bracket. Let us immediately mention a very important consequence. If $M \in \mathbb{H}^2$, and $H \in L^2(M)$, two successive applications of (4.1.1) give

$$\langle H \cdot M, H \cdot M\rangle = H \cdot (H \cdot \langle M, M\rangle) = H^2 \cdot \langle M, M\rangle,$$

using the *associativity property* (4.1.3) of integrals with respect to finite variation processes. Put differently, the quadratic variation of the continuous martingale $H \cdot M$ is

$$\left\langle \int_0^\cdot H_s\, dM_s, \int_0^\cdot H_s\, dM_s \right\rangle_t = \int_0^t H_s^2\, d\langle M, M\rangle_s.$$

More generally, if N is another martingale of $\mathbb{H}^2$ and $K \in L^2(N)$, the same argument gives

$$\left\langle \int_0^\cdot H_s\, dM_s, \int_0^\cdot K_s\, dN_s \right\rangle_t = \int_0^t H_s K_s\, d\langle M, N\rangle_s. \tag{4.1.3}$$

The following *associativity* property of stochastic integrals, which is analogous to property (4.1.3) for integrals with respect to finite variation processes, is very useful.

Proposition 4.1.4

Let $H \in L^2(M)$ and let K be a progressive process. Then $KH \in L^2(M)$ if and only if $K \in L^2(H \cdot M)$. If these equivalent properties hold, then

$$(KH) \cdot M = K \cdot (H \cdot M).$$

Proof for Proposition.

Using property (4.1.3), we have

$$\mathbb{E}\left[\int_0^\infty K_s^2 H_s^2\, d\langle M, M\rangle_s\right] = \mathbb{E}\left[\int_0^\infty K_s^2\, d\langle H \cdot M, H \cdot M\rangle_s\right],$$

which gives the first assertion.

For the second one, we write for $N \in \mathbb{H}^2$,

$$\langle (KH) \cdot M, N\rangle = KH \cdot \langle M, N\rangle = K \cdot \langle H \cdot M, N\rangle,$$

and by the uniqueness statement in (4.1.1), this implies that

$$(KH) \cdot M = K \cdot (H \cdot M).$$

■

Moments of stochastic integrals. Let $M, N \in \mathbb{H}^2$, $H \in L^2(M)$, and $K \in L^2(N)$.

For every $t \in [0, \infty]$, we have

$$\begin{aligned} \mathbb{E}\left[\int_0^t H_s\, dM_s\right] &= 0, \\ \mathbb{E}\left[\left(\int_0^t H_s\, dM_s\right)\left(\int_0^t K_s\, dN_s\right)\right] &= \mathbb{E}\left[\int_0^t H_s K_s\, d\langle M, N\rangle_s\right], \\ \mathbb{E}\left[\left(\int_0^t H_s\, dM_s\right)^2\right] &= \mathbb{E}\left[\int_0^t H_s^2\, d\langle M, M\rangle_s\right]. \end{aligned} \tag{4.1.4}$$

using (5) in Proposition 3.6.10 and equation 4.1.3. Furthermore, since $H\dot{M}$ is a (true) martingale, we also have for every $0 \leq s \leq t \leq \infty$,

$$\mathbb{E}\left[\int_0^t H_r\, dM_r \,\middle|\, \mathcal{F}_s\right] = \int_0^s H_r\, dM_r, \tag{4.1.5}$$

or, equivalently,

$$\mathbb{E}\left[\int_s^t H_r\, dM_r \,\middle|\, \mathcal{F}_s\right] = 0.$$

with an obvious notation for $\int_s^t H_r\, dM_r$. It is important to observe that these formulas (and particularly (1) and (3) in equation 4.1.4) may no longer hold for the extensions of stochastic integrals that we will now describe.

4.1.2 Stochastic Integrals for Local Martingales

We will now use the identities (4.1.2) above to extend the definition of $H \cdot M$ to an arbitrary continuous local martingale. If M is a continuous local martingale, we write $L^2_{\text{loc}}(M)$ (respectively $L^2(M)$) for the set of all progressive processes H such that

$$\int_0^t H_s^2\, d\langle M, M\rangle_s < \infty, \qquad \forall t \geq 0, \quad \text{a.s.}$$

respectively, such that

$$\mathbb{E}\Big[\int_0^\infty H_s^2\, d\langle M, M\rangle_s\Big] < \infty.$$

For future reference, we note that $L^2(M)$ (with the same identifications as in the case where $M \in \mathbb{H}^2$) can again be viewed as an *ordinary* L^2-space and thus has a Hilbert space structure.

Theorem 4.1.5: Stochastic integral wrt a continuous local martingale

Let M be a continuous local martingale. For every $H \in L^2_{\text{loc}}(M)$, there exists a unique continuous local martingale, starting from 0, denoted by $H \cdot M$, such that, for every continuous local martingale N,

$$\langle H \cdot M, N\rangle = H \cdot \langle M, N\rangle. \tag{4.1.6}$$

If T is a stopping time, then

$$(\mathbb{1}_{[0,T]}H) \cdot M = (H \cdot M)^T = H \cdot M^T. \tag{4.1.7}$$

If $H \in L^2_{loc}(M)$ and K is a progressive process, then $K \in L^2_{\text{loc}}(H \cdot M)$ if and only if $HK \in L^2_{\text{loc}}(M)$, and in this case

$$K \cdot (H \cdot M) = HK \cdot M.$$

Finally, if $M \in \mathbb{H}^2$ and $H \in L^2(M)$, this definition is consistent with that of Theorem 4.1.3.

Proof for Theorem.

We may assume $M_0 = 0$ (in general write $M = M_0 + M'$ and set $H \cdot M = H \cdot M'$, noting that $\langle M, N\rangle = \langle M', N\rangle$ for every continuous local martingale N). Also we may assume the property $\int_0^t H_s^2 \, d\langle M, M\rangle_s < \infty$ for every $t \geq 0$ for every $\omega \in \Omega$ (on the negligible set where this fails replace H by 0).

For every $n \geq 1$, set

$$T_n = \inf\Big\{t \geq 0 : \int_0^t (1 + H_s^2) \, d\langle M, M\rangle_s \geq n\Big\},$$

so that (T_n) is an increasing sequence of stopping times with $T_n \uparrow \infty$. Since $\langle M^{T_n}, M^{T_n}\rangle = \langle M, M\rangle_{\cdot\wedge T_n}$, the stopped martingale M^{T_n} is in $\mathbb{H}^2$ by Theorem 3.6.7. Moreover,

$$\int_0^\infty H_s^2 \, d\langle M^{T_n}, M^{T_n}\rangle_s = \int_0^{T_n} H_s^2 \, d\langle M, M\rangle_s \leq n.$$

Hence $H \in L^2(M^{T_n})$, and the definition of $H \cdot M^{T_n}$ makes sense by Theorem 4.1.3. Moreover, by property (4.1.2), we have, if $m > n$,

$$H \cdot M^{T_n} = (H \cdot M^{T_m})^{T_n}.$$

It follows that there exists a unique process denoted by $H \cdot M$ such that, for every n,

$$(H \cdot M)^{T_n} = H \cdot M^{T_n}.$$

Clearly $H \cdot M$ has continuous sample paths and is adapted since $(H \cdot M)_t = \lim_n (H \cdot M)_t^{T_n}$. Since the processes $(H \cdot M)^{T_n}$ are martingales in $\mathbb{H}^2$, we get that $H \cdot M$ is a continuous local martingale.

To verify (4.1.6), assume that N is a continuous local martingale with $N_0 = 0$. For every $n \geq 1$, set $T'_n = \inf\{t \geq 0 : |N_t| \geq n\}$ and $S_n = T_n \wedge T'_n$. Then, noting that

$N^{T'_n} \in \mathbb{H}^2$, we have

$$\begin{aligned}\langle H \cdot M, N\rangle^{S_n} &= \langle (H \cdot M)^{T_n}, N^{T'_n}\rangle \\ &= \langle H \cdot M^{T_n}, N^{T'_n}\rangle \\ &= H \cdot \langle M^{T_n}, N^{T'_n}\rangle \\ &= H \cdot \langle M, N\rangle^{S_n}.\end{aligned}$$

Letting $n \to \infty$ and using $S_n \uparrow \infty$, this gives $\langle H \cdot M, N\rangle = H \cdot \langle M, N\rangle$.

The fact that this equality, written for every continuous local martingale N, characterizes $H \cdot M$ among continuous local martingales with initial value 0 is derived from Proposition **??** in the proof of Theorem 4.1.3. The property (4.1.7) is obtained by the same arguments as for property (4.1.2) in Theorem 4.1.3 (these arguments only depend on the characteristic property (4.1.1) which we have just extended in (4.1.6)). Similarly, the proof of the associativity assertion is analogous to the proof of Proposition 4.1.4. Finally, if $M \in \mathbb{H}^2$ and $H \in L^2(M)$, the equality $\langle H \cdot M, H \cdot M\rangle = H^2 \cdot \langle M, M\rangle$ follows from (4.1.6), and implies $H \cdot M \in \mathbb{H}^2$. Then the characteristic property (4.1.1) shows that the definitions of Theorems 4.1.3 and 4.1.5 are consistent. ■

In the setting of Theorem 4.1.5, we again write

$$(H \cdot M)_t = \int_0^t H_s \, dM_s.$$

It is worth pointing out that formula (4.1.3) **remain valid** when M and N are continuous local martingales and $H \in L^2_{\text{loc}}(M)$, $K \in L^2_{\text{loc}}(N)$. Indeed, these formulas immediately follow from (4.1.6).

Connection with the Wiener integral Suppose that B is an $(\mathcal{F}_t)$-Brownian motion, and $h \in L^2(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+), dt)$ is a deterministic square integrable function. We can then define the Wiener integral $\int_0^t h(s)\, dB_s = G(f\, \mathbb{1}_{[0,t]})$, where G is the Gaussian white noise associated with B (see the end of Section 7). It is easy to verify that this integral coincides with the stochastic integral $(h \cdot B)_t$, which makes sense by viewing h as a (deterministic) progressive process. This is immediate when h is a simple function, and the general case follows from a density argument.

Let us now discuss the extension of the moment formulas that we stated above in the setting of Theorem 4.1.3. Let M be a continuous local martingale, $H \in L^2_{\text{loc}}(M)$ and $t \in [0, \infty]$. Then, *under the condition*

$$E\left[\int_0^t H_s^2 \, d\langle M, M\rangle_s\right] < \infty, \tag{4.1.8}$$

we can apply Theorem 3.6.7 to $(H \cdot M)^t$, and get that $(H \cdot M)^t$ is a martingale of $\mathbb{H}^2$. It follows that properties (4.1.4) still hold:

$$E\left[\int_0^t H_s \, dM_s\right] = 0, \quad E\left[\left(\int_0^t H_s \, dM_s\right)^2\right] = E\left[\int_0^t H_s^2 \, d\langle M, M\rangle_s\right],$$

and similarly (4.1.5) is valid for $0 \le s \le t$. In particular (case $t = \infty$), if $H \in L^2(M)$, the continuous local martingale $H \cdot M$ is in $\mathbb{H}^2$ and its terminal value satisfies

$$E\left[\left(\int_0^\infty H_s\,dM_s\right)^2\right] = E\left[\int_0^\infty H_s^2\,d\langle M, M\rangle_s\right].$$

If the condition (4.1.8) does not hold, the previous formulas may fail. However, we always have the bound

$$E\left[\left(\int_0^t H_s\,dM_s\right)^2\right] \le E\left[\int_0^t H_s^2\,d\langle M, M\rangle_s\right]. \tag{4.1.9}$$

Indeed, if the right-hand side is finite, this is an equality by the preceding observations. If the right-hand side is infinite, the bound is trivial.

4.1.3 Stochastic Integrals for Semimartingales

We finally extend the definition of stochastic integrals to continuous semimartingales. We say that a progressive process H is *locally bounded* if

$$\forall t \ge 0, \qquad \sup_{s \le t} |H_s| < \infty, \qquad \text{a.s.}$$

In particular, any adapted process with continuous sample paths is a locally bounded progressive process. If H is progressive and locally bounded, then for every finite-variation process V, we have

$$\forall t \ge 0, \qquad \int_0^t |H_s|\,d|V|_s < \infty, \qquad \text{a.s.}$$

and similarly $H \in L^2_{\text{loc}}(M)$ for every continuous local martingale M.

Definition 4.1.6: Stochastic integral wrt a continuous semimartingale

Let X be a continuous semimartingale with canonical decomposition $X = M + V$. If H is a locally bounded progressive process, the *stochastic integral* $H \cdot X$ is the continuous semimartingale with canonical decomposition

$$H \cdot X = H \cdot M + H \cdot V.$$

We write

$$(H \cdot X)_t = \int_0^t H_s\,dX_s.$$

It has the following properties:

1. The mapping $(H, X) \mapsto H \cdot X$ is bilinear.
2. If H and K are progressive and locally bounded, then

$$H \cdot (K \cdot X) = (HK) \cdot X.$$

3. For every stopping time T,

$$(H \cdot X)^T = (H\mathbb{1}_{[0,T]}) \cdot X = H \cdot X^T.$$

4. If X is a continuous local martingale, respectively a finite variation process, then $H \cdot X$ has the same property.
5. If

$$H_s(\omega) = \sum_{i=0}^{p-1} H_{(i)}(\omega)\mathbb{1}_{(t_i,t_{i+1}]}(s),$$

where $0 = t_0 < t_1 < \cdots < t_p$ and each $H_{(i)}$ is $\mathcal{F}_{t_i}$-measurable, then

$$(H \cdot X)_t = \sum_{i=0}^{p-1} H_{(i)}\Big(X_{t_{i+1}\wedge t} - X_{t_i\wedge t}\Big).$$

We can restate the *associativity* property (ii) by saying that, if $Y_t = \int_0^t K_s\,dX_s$ then

$$\int_0^t H_s\,dY_s = \int_0^t H_s K_s\,dX_s.$$

Properties (i)–(iv) easily follow from the results obtained when X is a continuous local martingale, resp. a finite variation process. As for property (v), we first note that it is enough to consider the case where $X = M$ is a continuous local martingale with $M_0 = 0$, and by stopping M at suitable stopping times (and using (**??**)), we can even assume that M is in $\mathbb{H}^2$. There is a minor difficulty coming from the fact that the variables $H_{(i)}$ are not assumed to be bounded (and therefore we cannot directly use the construction of the integral of elementary processes). To circumvent this difficulty, we set, for every $n \geq 1$,

$$T_n = \inf\{t \geq 0 : |H_t| \geq n\} = \inf\{t_i : |H_{(i)}| \geq n\} \quad (\text{where } \inf\varnothing = \infty).$$

It is easy to verify that T_n is a stopping time, and we have $T_n \uparrow \infty$ as $n \to \infty$. Furthermore, we have for every n,

$$H_s\,\mathbb{1}_{[0,T_n]}(s) = \sum_{i=0}^{p-1} H^n_{(i)}\,\mathbb{1}_{(t_i,t_{i+1}]}(s)$$

where the random variables $H^n_{(i)} = H_{(i)}\mathbb{1}_{\{T_n > t_i\}}$ satisfy the same properties as the $H_{(i)}$'s and additionally are bounded by n. Hence $H\,\mathbb{1}_{[0,T_n]}$ is an elementary process, and by the very definition of the stochastic integral with respect to a martingale of $\mathbb{H}^2$, we have

$$(H \cdot M)_{t\wedge T_n} = (H\,\mathbb{1}_{[0,T_n]} \cdot M)_t = \sum_{i=0}^{p-1} H^n_{(i)}(M_{t_{i+1}\wedge t} - M_{t_i\wedge t}).$$

The desired result now follows by letting n tend to infinity.

4.1.4 Convergence of Stochastic Integrals

We start by giving a "dominated convergence theorem" for stochastic integrals.

Proposition 4.1.7

Let $X = M + V$ be the canonical decomposition of a continuous semimartingale X, and let $t > 0$. Let $(H^n)_{n\geq 1}$ and H be locally bounded progressive processes, and let K be a nonnegative progressive process. Assume that the following properties hold a.s.:

1. $H_s^n \longrightarrow H_s$ as $n \to \infty$, for every $s \in [0, t]$;
2. $|H_s^n| \leq K_s$, for every $n \geq 1$ and $s \in [0, t]$;
3. $\int_0^t (K_s)^2 \, d\langle M, M\rangle_s < \infty$ and $\int_0^t K_s \, |dV_s| < \infty$.

Then,

$$\int_0^t H_s^n \, dX_s \xrightarrow[n\to\infty]{} \int_0^t H_s \, dX_s$$

in probability.

Proof for Proposition.

The a.s. convergence

$$\int_0^t H_s^n \, dV_s \longrightarrow \int_0^t H_s \, dV_s$$

follows from the usual dominated convergence theorem. We just have to verify that $\int_0^t H_s^n \, dM_s$ converges in probability to $\int_0^t H_s \, dM_s$. For every integer $p \geq 1$, consider the stopping time

$$T_p := \inf\{r \in [0, t] : \int_0^r (K_s)^2 \, d\langle M, M\rangle_s \geq p\} \wedge t,$$

and observe that $T_p = t$ for all large enough p, a.s., by assumption (3). Then, the bound (4.1.9) gives

$$\mathbb{E}\left[\left(\int_0^{T_p} H_s^n \, dM_s - \int_0^{T_p} H_s \, dM_s\right)^2\right] \leq \mathbb{E}\left[\int_0^{T_p} (H_s^n - H_s)^2 \, d\langle M, M\rangle_s\right],$$

which tends to 0 as $n \to \infty$ by dominated convergence, using (1)-(2) and the fact that $\int_0^{T_p} (K_s)^2 \, d\langle M, M\rangle_s \leq p$. Since $P(T_p = t) \to 1$ as $p \to \infty$, the desired result follows. ■

Remarks

1. Assertion (3) holds automatically if K is locally bounded.
2. Instead of assuming that (1) and (2) hold for every $s \in [0, t]$ (a.s.), it is enough to assume that these conditions hold for $d\langle M, M\rangle_s$-a.e. $s \in [0, t]$ and for $|dV_s|$-a.e. $s \in [0, t]$, a.s. This will be clear from the proof.

We apply the preceding proposition to an approximation result in the case of continuous integrands, which will be useful in the next section.

Proposition 4.1.8

Let X be a continuous semimartingale, and let H be an adapted process with continuous sample paths. Then, for every $t > 0$, for every sequence $0 = t_0^n < \cdots < t_{p_n}^n = t$ of subdivisions of $[0, t]$ whose mesh tends to 0, we have

$$\lim_{n\to\infty} \sum_{i=0}^{p_n-1} H_{t_i^n}\Big(X_{t_{i+1}^n} - X_{t_i^n}\Big) = \int_0^t H_s\, dX_s,$$

in probability.

Proof for Proposition.

For every $n \geq 1$ define the process H^n by

$$H_s^n = \begin{cases} H_{t_i^n}, & \text{if } t_i^n < s \leq t_{i+1}^n, \quad i \in \{0, 1, \ldots, p_n - 1\}, \\ H_0, & \text{if } s = 0, \\ 0, & \text{if } s > t. \end{cases}$$

Note that H^n is progressive. Set

$$K_s = \max_{0\leq r\leq s} |H_r|,$$

which is a locally bounded progressive process. Hence all assumptions of Proposition 4.1.7 hold with this choice of K, and we conclude that

$$\int_0^t H_s^n\, dX_s \xrightarrow[n\to\infty]{} \int_0^t H_s\, dX_s$$

in probability. By property (v) following Definition 4.1.6 we have, for every n,

$$\int_0^t H_s^n\, dX_s = \sum_{i=0}^{p_n-1} H_{t_i^n}\Big(X_{t_{i+1}^n} - X_{t_i^n}\Big),$$

and the proof is complete. ■

Remark The preceding proposition can be viewed as a generalization of Lemma 3.5.3 to stochastic integrals. However, in contrast with that lemma, it is essential in Proposition 4.1.7 to evaluate H at the left end of the interval $(t_i^n, t_{i+1}^n]$: The result will fail if we replace $H_{t_i^n}$ by $H_{t_{i+1}^n}$. Let us give a simple counterexample. We take $H_t = X_t$ and we assume that the sequence of subdivisions $(t_i^n)_{0\leq i\leq p_n}$ is increasing. By the proposition, we have

$$\lim_{n\to\infty} \sum_{i=0}^{p_n-1} X_{t_i^n}\Big(X_{t_{i+1}^n} - X_{t_i^n}\Big) = \int_0^t X_s\, dX_s,$$

in probability. On the other hand, writing

$$\sum_{i=0}^{p_n-1} X_{t^n_{i+1}}\left(X_{t^n_{i+1}} - X_{t^n_i}\right) = \sum_{i=0}^{p_n-1} X_{t^n_i}\left(X_{t^n_{i+1}} - X_{t^n_i}\right) + \sum_{i=0}^{p_n-1}\left(X_{t^n_{i+1}} - X_{t^n_i}\right)^2,$$

and using Proposition 3.7.3, we get

$$\lim_{n\to\infty} \sum_{i=0}^{p_n-1} X_{t^n_{i+1}}\left(X_{t^n_{i+1}} - X_{t^n_i}\right) = \int_0^t X_s\, dX_s + \langle X, X\rangle_t,$$

in probability. The resulting limit is different from $\int_0^t X_s\, dX_s$ unless the martingale part of X is degenerate. Note that, if we add the previous two convergences, we arrive at the formula

$$(X_t)^2 - (X_0)^2 = 2\int_0^t X_s\, dX_s + \langle X, X\rangle_t,$$

which is a special case of Itô's formula of the next section.

4.2 Itô's Formula

Itô's formula is the cornerstone of stochastic calculus. It provides the explicit canonical decomposition of a twice continuously differentiable function of continuous semimartingales.

Theorem 4.2.1: Itô's formula

Let $X^1, \dots, X^p$ be p continuous semimartingales, and let F be a twice continuously differentiable real function on $\mathbb{R}^p$. Then, for every $t \geq 0$,

$$\begin{aligned} F(X^1_t, \dots, X^p_t) = F(X^1_0, \dots, X^p_0) &+ \sum_{i=1}^{p} \int_0^t \frac{\partial F}{\partial x^i}(X^1_s, \dots, X^p_s)\, dX^i_s \\ &+ \frac{1}{2}\sum_{i,j=1}^{p} \int_0^t \frac{\partial^2 F}{\partial x^i \partial x^j}(X^1_s, \dots, X^p_s)\, d\langle X^i, X^j\rangle_s. \end{aligned}$$

Proof for Theorem.

We first deal with the case $p = 1$ and we write $X = X^1$ for simplicity. Fix $t > 0$ and consider an increasing sequence $0 = t^n_0 < \cdots < t^n_{p_n} = t$ of subdivisions of $[0, t]$ whose mesh tends to 0. Then, for every n,

$$F(X_t) = F(X_0) + \sum_{i=0}^{p_n-1} (F(X_{t^n_{i+1}}) - F(X_{t^n_i})).$$

For every $i \in \{0, 1, \ldots, p_n - 1\}$, we apply the Taylor–Lagrange formula to the function $[0, 1] \ni \theta \mapsto F(X_{t_i^n} + \theta(X_{t_{i+1}^n} - X_{t_i^n}))$, between $\theta = 0$ and $\theta = 1$, and we get that

$$F(X_{t_{i+1}^n}) - F(X_{t_i^n}) = F'(X_{t_i^n})(X_{t_{i+1}^n} - X_{t_i^n}) + \frac{1}{2} f_{n,i}(X_{t_{i+1}^n} - X_{t_i^n})^2,$$

where the quantity $f_{n,i}$ can be written as $F''(X_{t_i^n} + c(X_{t_{i+1}^n} - X_{t_i^n}))$ for some $c \in [0, 1]$. By Proposition 4.1.8 with $H_s = F'(X_s)$, we have

$$\lim_{n\to\infty} \sum_{i=0}^{p_n-1} F'(X_{t_i^n})(X_{t_{i+1}^n} - X_{t_i^n}) = \int_0^t F'(X_s)\,\mathrm{d}X_s,$$

in probability. To complete the proof of the case $p = 1$ of the theorem, it is therefore enough to verify that

$$\lim_{n\to\infty} \sum_{i=0}^{p_n-1} f_{n,i}(X_{t_{i+1}^n} - X_{t_i^n})^2 = \int_0^t F''(X_s)\,\mathrm{d}\langle X, X\rangle_s, \tag{4.2.1}$$

in probability. We observe that

$$\sup_{0\le i\le p_n-1} |f_{n,i} - F''(X_{t_i^n})| \le \sup_{0\le i\le p_n-1} \left(\sup_{x\in[X_{t_i^n}\wedge X_{t_{i+1}^n}, X_{t_i^n}\vee X_{t_{i+1}^n}]} |F''(x) - F''(X_{t_i^n})| \right).$$

The right-hand side of the preceding display tends to 0 a.s. as $n \to \infty$, as a simple consequence of the uniform continuity of F'' (and of the sample paths of X) over a compact interval.

Since $\sum_{i=0}^{p_n-1}(X_{t_{i+1}^n} - X_{t_i^n})^2$ converges in probability (Theorem 3.6.5), it follows from the last display that

$$\left| \sum_{i=0}^{p_n-1} f_{n,i}(X_{t_{i+1}^n} - X_{t_i^n})^2 - \sum_{i=0}^{p_n-1} F''(X_{t_i^n})(X_{t_{i+1}^n} - X_{t_i^n})^2 \right| \xrightarrow[n\to\infty]{} 0$$

in probability. So the convergence (4.2.1) will follow if we can verify that

$$\lim_{n\to\infty} \sum_{i=0}^{p_n-1} F''(X_{t_i^n})(X_{t_{i+1}^n} - X_{t_i^n})^2 = \int_0^t F''(X_s)\,\mathrm{d}\langle X, X\rangle_s,$$

in probability. In fact, this can be achieved by Lemma 3.6.8 which completes the proof of the case $p = 1$. In the general case, use same arguments as in the case $p = 1$ in the Taylor–Lagrange formula,

$$\begin{aligned} F(X^1_{t_{i+1}^n}, \ldots, X^p_{t_{i+1}^n}) - F(X^1_{t_i^n}, \ldots, X^p_{t_i^n}) &= \sum_{k=1}^{p} \frac{\partial F}{\partial x_k}(X^1_{t_i^n}, \ldots, X^p_{t_i^n})(X^k_{t_{i+1}^n} - X^k_{t_i^n}) \\ &\quad + \sum_{k,l=1}^{p} \frac{f^{k,l}_{n,i}}{2}(X^k_{t_{i+1}^n} - X^k_{t_i^n})(X^l_{t_{i+1}^n} - X^l_{t_i^n}) \end{aligned}$$

An important special case of Itô's formula is the formula of integration by parts, which is obtained by taking $p = 2$ and $F(x, y) = xy$: if X and Y are two continuous semimartingales, we have

$$X_tY_t = X_0Y_0 + \int_0^t X_s\,dY_s + \int_0^t Y_s\,dX_s + \langle X, Y\rangle_t.$$

In particular, if $Y = X$,

$$X_t^2 = X_0^2 + 2\int_0^t X_s\,dX_s + \langle X, X\rangle_t.$$

When $X = M$ is a continuous local martingale, we know from the definition of the quadratic variation that $M^2 - \langle M, M\rangle$ is a continuous local martingale. The previous formula shows that this continuous local martingale is

$$M_0^2 + 2\int_0^t M_s\,dM_s.$$

We could have seen this directly from the construction of $\langle M, M\rangle$ in Theorem 3.6.5 (this construction involved approximations of the stochastic integral $\int_0^t M_s\,dM_s$).

Let B be an $(\mathcal{F}_t)$-real Brownian motion (recall from Definition that this means that B is a Brownian motion, which is adapted to the filtration $(\mathcal{F}_t)$ and such that, for every $0 \le s < t$, the variable $B_t - B_s$ is independent of the σ-field $\mathcal{F}_s$). An $(\mathcal{F}_t)$-Brownian motion is a continuous local martingale (a martingale if $B_0 \in L^1$) and we already noticed that its quadratic variation is $\langle B, B\rangle_t = t$.

In this particular case, Itô's formula reads

$$F(B_t) = F(B_0) + \int_0^t F'(B_s)\,dB_s + \frac{1}{2}\int_0^t F''(B_s)\,ds.$$

Taking $X_t^1 = t$, $X_t^2 = B_t$, we also get for every twice continuously differentiable function $F(t, x)$ on $\mathbb{R}_+ \times \mathbb{R}$,

$$F(t, B_t) = F(0, B_0) + \int_0^t \frac{\partial F}{\partial x}(s, B_s)\,dB_s + \int_0^t \left(\frac{\partial F}{\partial t} + \frac{1}{2}\frac{\partial^2 F}{\partial x^2}\right)(s, B_s)\,ds.$$

Let $B_t = (B_t^1, \ldots, B_t^d)$ be a d-dimensional $(\mathcal{F}_t)$-Brownian motion. Note that the components B^i are $(\mathcal{F}_t)$-Brownian motions. By Proposition 4.16, $\langle B^i, B^j\rangle = 0$ when $i \neq j$ (by subtracting the initial value, which does not change the bracket $\langle B^i, B^j\rangle$, we are reduced to the case where $B_0^1, \ldots, B_0^d$ are independent). Itô's formula then shows that, for every twice continuously differentiable function F on $\mathbb{R}^d$,

$$\begin{aligned} F(B_t^1, \ldots, B_t^d) = F(B_0^1, \ldots, B_0^d) &+ \sum_{i=1}^d \int_0^t \frac{\partial F}{\partial x^i}(B_s^1, \ldots, B_s^d)\,dB_s^i \\ &+ \frac{1}{2}\int_0^t \Delta F(B_s^1, \ldots, B_s^d)\,ds. \end{aligned}$$

The latter formula is often written in the shorter form

$$F(B_t) = F(B_0) + \int_0^t \nabla F(B_s) \cdot dB_s + \frac{1}{2} \int_0^t \Delta F(B_s)\, ds,$$

where ∇F stands for the vector of first partial derivatives of F. There is again an analogous formula for $F(t, B_t)$.

Important remark It frequently occurs that one needs to apply Itô's formula to a function F which is only defined (and twice continuously differentiable) on an open subset U of $\mathbb{R}^p$. In that case, we can argue in the following way. Suppose that there exists another open set V, such that $(X_t^1, \dots, X_t^p) \in V$ a.s. and $V \subset U$ (here $\overline{V}$ denotes the closure of V). Typically V will be the set of all points whose distance from U^c is strictly greater than ε, for some $\varepsilon > 0$. Set

$$T_V := \inf\{t \geq 0 : (X_t^1, \dots, X_t^p) \notin V\},$$

which is a stopping time by Proposition **??**. Simple analytic arguments allow us to find a function G which is twice continuously differentiable on $\mathbb{R}^p$ and coincides with F on $\overline{V}$. We can now apply Itô's formula to obtain the canonical decomposition of the semimartingale

$$G(X_{t\wedge T_V}^1, \dots, X_{t\wedge T_V}^p) = F(X_{t\wedge T_V}^1, \dots, X_{t\wedge T_V}^p),$$

and this decomposition only involves the first and second derivatives of F on V. If in addition we know that the process $(X_t^1, \dots, X_t^p)$ a.s. does not exit U, we can let the open set V increase to U, and we get that Itô's formula for $F(X_t^1, \dots, X_t^p)$ remains valid exactly in the same form as in Theorem 4.2.1. These considerations can be applied, for instance, to the function $F(x) = \log x$ and to a semimartingale X taking strictly positive values: see the proof of Proposition 4.4.2 below.

We now use Itô's formula to exhibit a remarkable class of (local) martingales, which extends the exponential martingales associated with processes with independent increments. A random process with values in the complex plane $\mathbb{C}$ is called a complex continuous local martingale if both its real part and its imaginary part are continuous local martingales.

Proposition 4.2.2

Let M be a continuous local martingale and, for every $\lambda \in \mathbb{C}$, let

$$\mathcal{E}(\lambda M)_t = \exp\left(\lambda M_t - \frac{\lambda^2}{2}\langle M, M\rangle_t\right).$$

The process $\mathcal{E}(\lambda M)$ is a complex continuous local martingale, which can be written

$$\mathcal{E}(\lambda M)_t = e^{\lambda M_0} + \lambda \int_0^t \mathcal{E}(\lambda M)_s\, dM_s.$$

Proof for Proposition.

If $F(r,x)$ is twice continuously differentiable on $\mathbb{R}^2$, Itô's formula gives

$$F\Big(\langle M,M\rangle_t, M_t\Big) = F(0,M_0) + \int_0^t \frac{\partial F}{\partial x}\Big(\langle M,M\rangle_s, M_s\Big)\,dM_s + \int_0^t \left(\frac{\partial F}{\partial r} + \tfrac{1}{2}\frac{\partial^2 F}{\partial x^2}\right)\Big(\langle M,M\rangle_s, M_s\Big)\,d\langle M,M\rangle_s.$$

Hence $F(\langle M,M\rangle_t, M_t)$ is a continuous local martingale as soon as F satisfies the PDE

$$\frac{\partial F}{\partial r} + \tfrac{1}{2}\frac{\partial^2 F}{\partial x^2} = 0.$$

This equation holds for $F(r,x) = \exp\Big(\lambda x - \frac{\lambda^2}{2} r\Big)$ (for both the real and imaginary parts), and for this choice of F we have $\partial F/\partial x = \lambda F$. Therefore

$$\mathcal{E}(\lambda M)_t = \exp\Big(\lambda M_t - \tfrac{\lambda^2}{2}\langle M,M\rangle_t\Big)$$

is a continuous local martingale and, differentiating the exponential, one obtains

$$\mathcal{E}(\lambda M)_t = \mathcal{E}(\lambda M)_0 + \lambda \int_0^t \mathcal{E}(\lambda M)_s\,dM_s,$$

which is the stated formula. ∎

Remark The stochastic integral in the right-hand side of the last display is defined by dealing separately with the real and the imaginary part.

4.3 The Representation of Martingales as Stochastic Integrals

In the special setting where the filtration on Ω is the completed canonical filtration of a Brownian motion, we will now show that all martingales can be represented as stochastic integrals with respect to that Brownian motion. For the sake of simplicity, we first consider a one-dimensional Brownian motion, but we will discuss the extension to Brownian motion in higher dimensions at the end of this section.

Note: The following lemma is stated before the theorem and is used exclusively to prove this theorem.

Lemma 4.3.1: Useful density Lemma

Under the assumptions of the theorem below, the vector space generated by the random variables

$$\exp\left(\mathrm{i}\sum_{j=1}^{n} \lambda_j (B_{t_j} - B_{t_{j-1}})\right),$$

for any choice of $0 = t_0 < t_1 < \cdots < t_n$ and $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$, is dense in the space $L^2_{\mathbb{C}}(\Omega, \mathcal{F}_\infty, \mathbb{P})$ of all square-integrable complex-valued $\mathcal{F}_\infty$-measurable random variables.

Proof for Lemma

It is enough to prove that, if $Z \in L^2_{\mathbb{C}}(\Omega, \mathcal{F}_\infty, \mathbb{P})$ is such that

$$E\left[Z \exp\left(\mathrm{i}\sum_{j=1}^{n} \lambda_j (B_{t_j} - B_{t_{j-1}})\right)\right] = 0$$

for any choice of $0 = t_0 < t_1 < \cdots < t_n$ and $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$, then $Z = 0$.

Fix $0 = t_0 < t_1 < \cdots < t_n$, and consider the complex measure μ on $\mathbb{R}^n$ defined by

$$\mu(F) = E\left[Z \mathbb{1}_F (B_{t_1}, B_{t_2} - B_{t_1}, \ldots, B_{t_n} - B_{t_{n-1}})\right]$$

for any Borel subset F of $\mathbb{R}^n$. Then the equation above exactly shows that the Fourier transform of μ is identically zero. By the injectivity of the Fourier transform on complex measures on $\mathbb{R}^d$, it follows that $\mu = 0$. We have thus $E[Z\mathbb{1}_A] = 0$ for every $A \in \sigma(B_{t_1}, \ldots, B_{t_n})$.

A monotone class argument then shows that the identity $E[Z\mathbb{1}_A] = 0$ remains valid for any $A \in \sigma(B_t, t \geq 0)$, and then by completion for any $A \in \mathcal{F}_\infty$. It follows that $Z = 0$. ■

Theorem 4.3.2: Martingal Representation Theorem

Assume that the filtration $(\mathcal{F}_t)$ on Ω is the completed canonical filtration of a real Brownian motion B started from 0. Then, for every random variable $Z \in L^2(\Omega, \mathcal{F}_\infty, \mathbb{P})$, there exists a unique progressive process $h \in L^2(B)$ (i.e., $E[\int_0^\infty h_s^2 \, \mathrm{d}s] < \infty$) such that

$$Z = E[Z] + \int_0^\infty h_s \, \mathrm{d}B_s.$$

Proof for Theorem.

Uniqueness is immediate: if $Z = E[Z] + \displaystyle\int_0^\infty h_s \, dB_s = E[Z] + \int_0^\infty h'_s \, dB_s$ with $h, h' \in L^2(B)$, then

$$E\left[\int_0^\infty (h_s - h'_s)^2 \, ds\right] = E\left[\left(\int_0^\infty (h_s - h'_s) \, dB_s\right)^2\right] = 0,$$

hence $h = h'$ in $L^2(B)$.

For existence let $\mathcal{H}$ be the vector space of all $Z \in L^2(\Omega, \mathcal{F}_\infty, P)$ admitting a representation $Z = E[Z] + \int_0^\infty h_s \, dB_s$ with $h \in L^2(B)$. If $Z \in \mathcal{H}$ with associated h, then

$$E[Z^2] = (E[Z])^2 + E\left[\int_0^\infty h_s^2 \, ds\right],$$

so $\mathcal{H}$ is closed in $L^2(\Omega, \mathcal{F}_\infty, P)$. Indeed, if $Z^{(n)} \to Z$ in L^2 and $h^{(n)}$ are the associated processes in $L^2(B)$, then by the above identity the sequence $h^{(n)}$ is Cauchy in $L^2(B)$ and converges to some $h \in L^2(B)$; passing to the limit yields the representation for Z.

It remains to show that $\mathcal{H}$ is dense in $L^2(\Omega, \mathcal{F}_\infty, P)$. Fix $0 = t_0 < t_1 < \cdots < t_n$ and $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$, and set $f(s) = \sum_{j=1}^n \lambda_j \mathbb{1}_{(t_{j-1}, t_j]}(s)$. Write $\mathcal{E}^f$ for the exponential martingale $\mathcal{E}(f) = \mathcal{E}\Big(\mathrm{i} \int_0^\cdot f(s)\, dB_s\Big)$. By Proposition 4.2.2 we have

$$\exp\left(\mathrm{i}\sum_{j=1}^n \lambda_j (B_{t_j} - B_{t_{j-1}}) + \tfrac{1}{2}\sum_{j=1}^n \lambda_j^2 (t_j - t_{j-1})\right) = \mathcal{E}^f_\infty = 1 + \mathrm{i}\int_0^\infty \mathcal{E}^f_s f(s)\, dB_s.$$

Thus both the real and imaginary parts of $\exp\Big(\mathrm{i}\sum_{j=1}^n \lambda_j (B_{t_j} - B_{t_{j-1}})\Big)$ belong to $\mathcal{H}$. By Lemma 4.3.1 the linear span of such variables is dense in $L^2(\Omega, \mathcal{F}_\infty, P)$, hence $\mathcal{H} = L^2(\Omega, \mathcal{F}_\infty, P)$. This proves the first assertion of the theorem.

For the second assertion, let M be a (real) martingale bounded in L^2. Then $M_\infty \in L^2(\Omega, \mathcal{F}_\infty, P)$ and so $M_\infty = E[M_\infty] + \int_0^\infty h_s\, dB_s$ for some $h \in L^2(B)$. Using (4.1.5) we get, for every t,

$$M_t = E[M_\infty \mid \mathcal{F}_t] = E[M_\infty] + \int_0^t h_s\, dB_s,$$

and uniqueness of h follows from the first part.

Finally, if M is a continuous local martingale, set $T_n := \inf\{t \geq 0 : |M_t| \geq n\}$. Each stopped martingale M^{T_n} is bounded in L^2, so there exists $h^{(n)} \in L^2(B)$ with $M_t^{T_n} = M_0^{T_n} + \int_0^t h_s^{(n)}\, dB_s$. By uniqueness the processes $h^{(n)}$ are consistent on increasing time intervals, hence they patch together to define $h \in L^2_{\mathrm{loc}}(B)$ with $M_t = M_0 + \int_0^t h_s\, dB_s$. Uniqueness of h is immediate from the uniqueness in the bounded case. ■

Consequently, for every martingale M that is bounded in L^2 (respectively, for every continuous local martingale M), there exists a unique process $h \in L^2(B)$ (resp. $h \in L^2_{\mathrm{loc}}(B)$) and a constant $C \in \mathbb{R}$ such that

$$M_t = C + \int_0^t h_s\, \mathrm{d}B_s.$$

Remark: As the proof will show, the second part of the statement applies to a martingale M that is bounded in L^2, **without any assumption** on the continuity of sample paths of M. This observation will be useful later when we discuss consequences of the representation theorem. Note that continuous local martingales have continuous sample paths by definition.

4.4 Girsanov's Theorem and its applications

Throughout this section, we assume that the filtration $(\mathcal{F}_t)$ is both complete and right-continuous. Our goal is to study how the notions of a martingale and of a semimartingale are affected when the underlying probability measure P is replaced by another probability measure Q. Most of the time we will assume that P and Q are mutually absolutely continuous, and then the fact that the filtration $(\mathcal{F}_t)$ is complete with respect to P

implies that it is complete with respect to Q. When there is a risk of confusion, we will write E_P for the expectation under the probability measure P, and similarly E_Q for the expectation under Q. Unless otherwise specified, the notions of a (local) martingale or of a semimartingale refer to the underlying probability measure P (when we consider these notions under Q we will say so explicitly). Note that, in contrast with the notion of a martingale, the notion of a finite variation process does not depend on the underlying probability measure.

Proposition 4.4.1

Assume that Q is a probability measure on $(\Omega, \mathcal{F})$, which is absolutely continuous with respect to P on the σ-field $\mathcal{F}_\infty$. For every $t \in [0, \infty]$, let

$$D_t = \frac{\mathrm{d}Q}{\mathrm{d}P}\bigg|_{\mathcal{F}_t}$$

be the Radon–Nikodym derivative of Q with respect to P on the σ-field $\mathcal{F}_t$. The process $(D_t)_{t\geq 0}$ is a uniformly integrable martingale. Consequently $(D_t)_{t\geq 0}$ has a càdlàg modification. Keeping the same notation $(D_t)_{t\geq 0}$ for this modification, we have, for every stopping time T,

$$D_T = \frac{\mathrm{d}Q}{\mathrm{d}P}\bigg|_{\mathcal{F}_T}.$$

Finally, if we assume furthermore that P and Q are mutually absolutely continuous on $\mathcal{F}_\infty$, we have

$$\inf_{t\geq 0} D_t > 0 \quad P\text{-a.s.}$$

Proof for Proposition.

If $A \in \mathcal{F}_t$, we have

$$Q(A) = E_Q[\,\mathbb{1}_A\,] = E_P[\,\mathbb{1}_A D_\infty\,] = E_P[\mathbb{1}_A E_P[D_\infty \mid \mathcal{F}_t]]\,,$$

and by the uniqueness of the Radon–Nikodym derivative on $\mathcal{F}_t$, it follows that

$$D_t = E_P[D_\infty \mid \mathcal{F}_t], \qquad \text{a.s.}$$

Hence D is a uniformly integrable martingale, which is closed by D_∞. Theorem 3.3.6 (using the fact that $(\mathcal{F}_t)$ is both complete and right-continuous) then allows us to find a càdlàg modification of $(D_t)_{t\geq 0}$, which we consider from now on.

Then, if T is a stopping time, the optional stopping theorem (Theorem 3.4.1) gives for every $A \in \mathcal{F}_T$,

$$Q(A) = E_Q[\,\mathbb{1}_A\,] = E_P[\,\mathbb{1}_A D_\infty\,] = E_P[\mathbb{1}_A E_P[D_\infty \mid \mathcal{F}_T]] = E_P[\,\mathbb{1}_A D_T\,],$$

and, since D_T is $\mathcal{F}_T$-measurable, it follows that

$$D_T = \left.\frac{dQ}{dP}\right|_{\mathcal{F}_T}.$$

Let us prove the last assertion. For every $\varepsilon > 0$, set

$$T_\varepsilon = \inf\{t \geq 0 : D_t < \varepsilon\}.$$

Then T_ε is a stopping time as the first hitting time of an open set by a càdlàg process (recall Definition 3.2.10 and the fact that the filtration is right-continuous). Note that $\{T_\varepsilon < \infty\}$ is $\mathcal{F}_{T_\varepsilon}$-measurable, and hence

$$Q(T_\varepsilon < \infty) = E_P[\mathbb{1}_{\{T_\varepsilon<\infty\}} D_{T_\varepsilon}] \leq \varepsilon,$$

since $D_{T_\varepsilon} \leq \varepsilon$ on $\{T_\varepsilon < \infty\}$ by right-continuity of the sample paths. It immediately follows that

$$Q\left(\bigcap_{n=1}^{\infty} \{T_{1/n} < \infty\}\right) = 0,$$

and since P is absolutely continuous with respect to Q we also have

$$P\left(\bigcap_{n=1}^{\infty} \{T_{1/n} < \infty\}\right) = 0.$$

But this exactly means that, P-a.s., there exists an integer $n \geq 1$ such that $T_{1/n} = \infty$, which gives the last assertion of the proposition. ■

Proposition 4.4.2

Let D be a continuous local martingale taking (strictly) positive values. There exists a unique continuous local martingale L such that

$$D_t = \exp\left(L_t - \frac{1}{2}\langle L, L\rangle_t\right) = \mathcal{E}(L)_t.$$

Moreover, L is given by the formula

$$L_t = \log D_0 + \int_0^t D_s^{-1}\, \mathrm{d}D_s.$$

Proof for Proposition.

Uniqueness is an easy consequence of Theorem 3.6.4. Then, since D takes positive values, we can apply Itô's formula to $\log D_t$ (see the remark before Proposition 4.2.2), and we get

$$\log D_t = \log D_0 + \int_0^t \frac{\mathrm{d}D_s}{D_s} - \frac{1}{2}\int_0^t \frac{\mathrm{d}\langle D, D\rangle_s}{D_s^2} = L_t - \frac{1}{2}\langle L, L\rangle_t,$$

where L is as in the statement. ■

Theorem 4.4.3: Girsanov Theorem

Assume that the probability measures P and Q are mutually absolutely continuous on $\mathcal{F}_\infty$. Let $(D_t)_{t\geq 0}$ be the martingale with càdlàg sample paths such that, for every $t \geq 0$,

$$D_t = \left.\frac{\mathrm{d}Q}{\mathrm{d}P}\right|_{\mathcal{F}_t}.$$

Assume that D has continuous sample paths, and let L be the unique continuous local martingale such that $D_t = \mathcal{E}(L)_t$. Then, if M is a continuous local martingale under P, the process

$$\tilde{M} = M - \langle M, L\rangle$$

is a continuous local martingale under Q.

Proof for Theorem.

The fact that D_t can be written in the form $D_t = \mathcal{E}(L)_t$ follows from Proposition 4.4.2 (we are assuming that D has continuous sample paths, and we also know from Proposition 4.4.1 that D takes positive values). Then, let T be a stopping time and let X be an adapted process with continuous sample paths. We claim that, if $(XD)^T$ is a martingale under P, then X^T is a martingale under Q.

By Proposition 4.4.1, $D_{T\wedge t} = \left.\frac{dQ}{dP}\right|_{\mathcal{F}_{T\wedge t}}$ and $D_{T\wedge s} = \left.\frac{dQ}{dP}\right|_{\mathcal{F}_{T\wedge s}}$, and thus, since $A\cap\{T > s\} \in \mathcal{F}_{T\wedge s} \subset \mathcal{F}_{T\wedge t}$, it follows that

$$E_Q\left[\mathbb{1}_{A\cap\{T>s\}}X_{T\wedge t}\right] = E_P\left[\mathbb{1}_{A\cap\{T>s\}}X_{T\wedge t}D_{T\wedge t}\right] = E_P\left[\mathbb{1}_{A\cap\{T>s\}}X_{T\wedge s}D_{T\wedge s}\right].$$

On the other hand, it is immediate that

$$E_Q\left[\mathbb{1}_{A\cap\{T\leq s\}}X_{T\wedge t}\right] = E_Q\left[\mathbb{1}_{A\cap\{T\leq s\}}X_{T\wedge s}\right].$$

Combining the two displays, we obtain

$$E_Q[\mathbb{1}_A X_{T\wedge t}] = E_Q[\mathbb{1}_A X_{T\wedge s}],$$

which proves the claim. As a consequence, if XD is a continuous local martingale under P, then X is a continuous local martingale under Q.

Next let M be a continuous local martingale under P, and let $\tilde{M}$ be as in the statement of the theorem. We apply the preceding observation to $X = \tilde{M}$. By Itô's formula,

$$\tilde{M}_t D_t = M_0 D_0 + \int_0^t \tilde{M}_s\, dD_s + \int_0^t D_s\, dM_s - \int_0^t D_s\, d\langle M, L\rangle_s + \langle M, D\rangle_t.$$

Since $d\langle M, L\rangle_s = D_s^{-1}\, d\langle M, D\rangle_s$ by Proposition 4.4.2, the last two terms cancel, and we get

$$\tilde{M}_t D_t = M_0 D_0 + \int_0^t \tilde{M}_s\, dD_s + \int_0^t D_s\, dM_s.$$

Therefore $\tilde{M}D$ is a continuous local martingale under P, and the preceding claim yields that $\tilde{M}$ is a continuous local martingale under Q. ■

Remark By consequences of the martingale representation theorem explained at the end of the previous section, the continuity assumption for the sample paths of D always holds when $(\mathcal{F}_t)$ is the (completed) canonical filtration of a Brownian motion. In applications of Theorem 4.4.3, one often starts from the martingale (D_t) to define the probability measure Q, so that the continuity assumption is satisfied by construction (see the examples in the next section).

Consequences

1. A process M which is a continuous local martingale under P remains a semimartingale under Q, and its canonical decomposition under Q is $M = \tilde{M} + \langle M, L\rangle$ (recall that the notion of a finite variation process does not depend on the underlying probability measure). It follows that the class of semimartingales under P is contained in the class of semimartingales under Q.

   In fact these two classes are equal. Indeed, under the assumptions of Theorem 4.4.3, P and Q play symmetric roles, since the Radon–Nikodym derivative of P with respect to Q on the σ-field $\mathcal{F}_t$ is D_t^{-1}, which has continuous sample paths if D does. We may furthermore notice that

   $$D_t^{-1} = \exp\left(-L_t + \langle L, L\rangle_t - \frac{1}{2}\langle L, L\rangle_t\right) = \exp\left(-\tilde{L}_t - \frac{1}{2}\langle \tilde{L}, \tilde{L}\rangle_t\right) = \mathcal{E}(-\tilde{L})_t,$$

   where $\tilde{L} = L - \langle L, L\rangle$ is a continuous local martingale under Q, and $\langle \tilde{L}, \tilde{L}\rangle = \langle L, L\rangle$. So, under the assumptions of Theorem 4.4.3, the roles of P and Q can be interchanged provided D is replaced by D^{-1} and L is replaced by $-\tilde{L}$.

2. Let X and Y be two semimartingales (under P or under Q). The bracket $\langle X, Y\rangle$ is the same under P and under Q. In fact this bracket is given in both cases by the approximation of Proposition 3.7.3 (this observation was used implicitly in (a) above).

   Similarly, if H is a locally bounded progressive process, the stochastic integral $H \cdot X$ is the same under P and under Q. To see this it is enough to consider the case when $X = M$ is a continuous local martingale (under P). Write $(H \cdot M)_P$ for the stochastic integral under P and $(H \cdot M)_Q$ for the one under Q. By linearity,

   $$(H \cdot \tilde{M})_P = (H \cdot M)_P - H \cdot \langle M, L\rangle = (H \cdot M)_P - \langle (H \cdot M)_P, L\rangle,$$

   and Theorem 4.4.3 shows that $(H \cdot \tilde{M})_P$ is a continuous local martingale under Q. Furthermore the bracket of this continuous local martingale with any continuous local martingale N under Q is equal to $H \cdot \langle M, N\rangle = H \cdot \langle \tilde{M}, N\rangle$, and it follows from Theorem 4.1.5 that $(H \cdot \tilde{M})_P = (H \cdot \tilde{M})_Q$ hence also $(H \cdot M)_P = (H \cdot M)_Q$. With the notation of Theorem 4.4.3, set $\tilde{M} = \mathcal{G}_Q^P(M)$. Then $\mathcal{G}_Q^P$ maps the set of all P-continuous local martingales onto the set of all Q-continuous local martingales. One easily verifies, using the remarks in (a) above, that $\mathcal{G}_P^Q \circ \mathcal{G}_Q^P = \mathrm{Id}$. Furthermore, the mapping $\mathcal{G}_Q^P$ commutes with the stochastic integral: if H is a locally bounded progressive process, $H \cdot \mathcal{G}_Q^P(M) = \mathcal{G}_Q^P(H \cdot M)$.

3. Suppose that $M = B$ is an $(\mathcal{F}_t)$-Brownian motion under P, then $\tilde{B} = B - \langle B, L\rangle$ is a continuous local martingale under Q, with quadratic variation $\langle \tilde{B}, \tilde{B}\rangle_t = \langle B, B\rangle_t = t$. By Theorem 5.12, it follows that $\tilde{B}$ is an $(\mathcal{F}_t)$-Brownian motion under Q.

In most applications of Girsanov's theorem, one constructs the probability measure Q in the following way. Start from a continuous local martingale L such that $L_0 = 0$ and $\langle L, L\rangle_\infty < \infty$ a.s. The latter condition implies that the limit $L_\infty := \lim_{t\to\infty} L_t$ exists a.s. . Then $\mathcal{E}(L)_t$ is a nonnegative continuous local martingale hence a supermartingale (Proposition 3.6.3), which converges a.s. to $\mathcal{E}(L)_\infty = \exp(L_\infty - \frac{1}{2}\langle L, L\rangle_\infty)$, and $E[\mathcal{E}(L)_\infty] \leq 1$ by Fatou's lemma. If the property

$$E[\mathcal{E}(L)_\infty] = 1 \tag{1}$$

holds, then $\mathcal{E}(L)$ is a uniformly integrable martingale (by Fatou's lemma again, one has $\mathcal{E}(L)_t \geq E[\mathcal{E}(L)_\infty \mid \mathcal{F}_t]$, but (1) implies that $E[\mathcal{E}(L)_\infty] = E[\mathcal{E}(L)_0] = E[\mathcal{E}(L)_t]$ for every $t \geq 0$). If we let Q be the probability measure with density $\mathcal{E}(L)_\infty$ with respect to P, we are in the setting of Theorem 4.4.3, with $D_t = \mathcal{E}(L)_t$. It is therefore very important to give conditions that ensure that (1) holds.

Theorem 4.4.4: Novikov-Kazamaki criterions

Let L be a continuous local martingale such that $L_0 = 0$. Consider the following properties:

1. $E[\exp \frac{1}{2}\langle L, L\rangle_\infty] < \infty$ (Novikov's criterion);
2. L is a uniformly integrable martingale, and $E[\exp \frac{1}{2} L_\infty] < \infty$ (Kazamaki's criterion);
3. $\mathcal{E}(L)$ is a uniformly integrable martingale.

Then, (1) $\implies$ (2) $\implies$ (3).

Proof for Theorem.

(i) $\Rightarrow$ (ii) Property (i) implies that $E[\langle L, L\rangle_\infty] < \infty$, hence also that L is a continuous martingale bounded in L^2 (Theorem 3.6.7). Then,

$$\exp\left(\frac{1}{2} L_\infty\right) = (\mathcal{E}(L)_\infty)^{1/2} \exp\left(\frac{1}{2}\langle L, L\rangle_\infty\right)^{1/2},$$

so that, by the Cauchy–Schwarz inequality,

$$E\Big[\exp\Big(\frac{1}{2}L_\infty\Big)\Big] \leq E[\mathcal{E}(L)_\infty]^{1/2}\, E\Big[\exp\Big(\frac{1}{2}\langle L,L\rangle_\infty\Big)\Big]^{1/2} \leq E\Big[\exp\Big(\frac{1}{2}\langle L,L\rangle_\infty\Big)\Big]^{1/2} < \infty.$$

(ii) $\Rightarrow$ (iii) Since L is a uniformly integrable martingale, Theorem 3.4.1 shows that, for any stopping time T, we have $L_T = E[L_\infty \mid \mathcal{F}_T]$. Jensen's inequality then gives

$$\exp\Big(\frac{1}{2}L_T\Big) \leq E\Big[\exp\Big(\frac{1}{2}L_\infty\Big)\;\Big|\;\mathcal{F}_T\Big].$$

By assumption $E[\exp(\frac{1}{2}L_\infty)] < \infty$, which implies that the collection of all variables of the form $E[\exp(\frac{1}{2}L_\infty) \mid \mathcal{F}_T]$, for any stopping time T, is uniformly integrable. The preceding bound then shows that the collection of all variables $\exp(\frac{1}{2}L_T)$, for any stopping time T, is also uniformly integrable.

For $0 < a < 1$, set

$$Z_t^{(a)} = \exp\Big(\frac{a}{1+a}L_t\Big).$$

Then one easily verifies that

$$\mathcal{E}(aL)_t = (\mathcal{E}(L)_t)^a (Z_t^{(a)})^{1-a}.$$

If $\Gamma \in \mathcal{F}$ and T is a stopping time, Hölder's inequality gives

$$E[\mathbb{1}_\Gamma \mathcal{E}(aL)_T] \leq E[\mathcal{E}(L)_T]^a E\Big[\mathbb{1}_\Gamma Z_T^{(a)}\Big]^{1-a} \leq E\Big[\mathbb{1}_\Gamma \exp\Big(\frac{1}{2}L_T\Big)\Big]^{2a(1-a)}.$$

In the second inequality we used the property $E[\mathcal{E}(L)_T] \leq 1$, which holds by Theorem 3.4.4 because $\mathcal{E}(L)$ is a nonnegative supermartingale and $\mathcal{E}(L)_0 = 1$. In the third inequality we used Jensen's inequality, noting that $\frac{1+a}{2a} > 1$. Since the collection of all variables of the form $\exp(\frac{1}{2}L_T)$, for any stopping time T, is uniformly integrable, the preceding bound shows that so is the collection of all variables $\mathcal{E}(aL)_T$ for any stopping time T.

By the definition of a continuous local martingale, there is an increasing sequence $T_n \uparrow \infty$ of stopping times such that, for every n, $\mathcal{E}(aL)_{\cdot\wedge T_n}$ is a martingale. If $0 \leq s \leq t$, we can use uniform integrability to pass to the limit $n \to \infty$ in the equality $E[\mathcal{E}(aL)_{t\wedge T_n} \mid \mathcal{F}_s] = \mathcal{E}(aL)_{s\wedge T_n}$, and we get that $\mathcal{E}(aL)$ is a uniformly integrable martingale. It follows that

$$1 = E[\mathcal{E}(aL)_\infty] \leq E[\mathcal{E}(aL)_\infty]^a\, E[Z_\infty^{(a)}]^{1-a} \leq E[\mathcal{E}(L)_\infty]^a\, E\Big[\exp\Big(\frac{1}{2}L_\infty\Big)\Big]^{2a(1-a)},$$

using again Jensen's inequality as above. When $a \to 1$, this gives $E[\mathcal{E}(L)_\infty] \geq 1$, hence $E[\mathcal{E}(L)_\infty] = 1$. ∎

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CHAPITRE 5

A FEW FACTS FROM FUNCTIONAL ANALYSIS

In this appendix, we recall several basic results from functional analysis that are used throughout the text. Proofs may be found in [4] or in most standard textbooks on functional analysis.

5.1 Normed Linear Spaces and Banach Spaces

Let E be a linear space over the field $\mathbb{R}$. A *norm* on E is a mapping

$$x \mapsto \|x\|$$

from E into $\mathbb{R}_+$ satisfying the following properties:

- For every $x, y \in E$,
$$\|x + y\| \leq \|x\| + \|y\|.$$
- For every $x \in E$ and $a \in \mathbb{R}$,
$$\|ax\| = |a|\,\|x\|.$$
- For every $x \in E$,
$$\|x\| = 0 \iff x = 0.$$

The pair $(E, \|\cdot\|)$ is called a *normed linear space*. The norm induces a distance d on E defined by

$$d(x, y) = \|x - y\|.$$

A normed linear space $(E, \|\cdot\|)$ is called a *Banach space* if it is complete with respect to the distance induced by the norm. In other words, if $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in E, namely

$$\lim_{m,n\to\infty} \|x_m - x_n\| = 0,$$

then there exists $x \in E$ such that $x_n \to x$.

A *linear form* on E is a linear mapping from E into $\mathbb{R}$. A linear form Φ is continuous if and only if

$$\|\Phi\| := \sup_{\|x\|\leq 1} |\Phi(x)| < \infty.$$

In that case,

$$|\Phi(x)| \leq \|\Phi\| \, \|x\|, \qquad x \in E.$$

The space of all continuous linear forms on E is denoted by E'. Equipped with the norm $\|\cdot\|$, the space $(E', \|\cdot\|)$ is a normed linear space, called the *topological dual* of E.

The dual space $(E', \|\cdot\|)$ is always a Banach space, even when $(E, \|\cdot\|)$ itself is not complete.

The following special case of the Hahn–Banach theorem is used in this book only to construct certain counterexamples.

Theorem 5.1.1: Hahn–Banach Theorem

Let $(E, \|\cdot\|)$ be a normed linear space, and let F be a linear subspace of E. Let ϕ be a linear form on F such that

$$|\phi(y)| \leq \|y\|, \qquad y \in F.$$

Then there exists a continuous linear form Φ on E such that

$$\Phi(y) = \phi(y), \qquad y \in F,$$

and

$$\|\Phi\| \leq 1.$$

5.2 Hilbert Spaces

Let H be a linear space over the field $\mathbb{R}$. A *scalar product* on H is a mapping

$$(x, y) \mapsto \langle x, y\rangle$$

from $H \times H$ into $\mathbb{R}$ satisfying the following properties:

- For every $x, y \in H$,
$$\langle x, y\rangle = \langle y, x\rangle.$$
- For every $x, y, z \in H$ and $a \in \mathbb{R}$,
$$\langle x, y + z\rangle = \langle x, y\rangle + \langle x, z\rangle, \qquad \langle ax, y\rangle = a\langle x, y\rangle.$$
- For every $x \in H$,
$$\langle x, x\rangle \geq 0, \qquad \langle x, x\rangle = 0 \Longleftrightarrow x = 0.$$

The scalar product induces a norm on H defined by

$$\|x\| := \sqrt{\langle x, x\rangle}.$$

The triangle inequality follows from the *Cauchy–Schwarz inequality*:

$$|\langle x, y\rangle| \leq \|x\|\,\|y\|, \qquad x, y \in H.$$

If $(H, \|\cdot\|)$ is complete, then $(H, \langle\cdot,\cdot\rangle)$ is called a *Hilbert space*.
An **Isometry** is a map $T : H \to H$ such that $\|T(x) - T(y)\| = \|x - y\|$ for all $x, y \in H$.
An **Isomorphism** is a bijective linear map $T : H \to K$ that preserves structure (e.g., inner product or norm), with a linear inverse.

In the sequel, $(H, \langle\cdot,\cdot\rangle)$ denotes a Hilbert space.

The following theorem, applied with $H = L^2(E, \mathcal{A}, \mu)$, plays a key role in the proof of the Radon–Nikodym theorem 1.3.11.

Theorem 5.2.1: Riesz Representation Theorem

For every $y \in H$, define a continuous linear form Φ_y on H by

$$\Phi_y(x) := \langle x, y\rangle, \qquad x \in H.$$

Then the mapping

$$y \mapsto \Phi_y$$

is a linear isometric isomorphism from H onto H'. In particular,

$$\|\Phi_y\| = \|y\|, \qquad y \in H.$$

We now state the orthogonal projection theorem, which is used in Theorem **??** to interpret conditional expectation.

Theorem 5.2.2: Orthogonal Projection Theorem

Let F be a closed linear subspace of H, and let $x \in H$. Then there exists a unique element $p_F(x) \in F$ such that

$$\|x - p_F(x)\| = \min\{\|x - y\| : y \in F\}.$$

Moreover, $p_F(x)$ is characterized by the properties

- $p_F(x) \in F$,
- $$\langle x - p_F(x), y\rangle = 0, \qquad y \in F.$$

The element $p_F(x)$ is called the *orthogonal projection* of x onto F.

Two elements $x, y \in H$ are said to be *orthogonal* if

$$\langle x, y \rangle = 0.$$

Theorem 5.2.3: Density Criterion

Let H be a Hilbert space and let E be a linear subspace of H. Then the following are equivalent:

1. The linear subspace E is dense in H, that is,
$$\overline{E} = H.$$
2. The only vector orthogonal to every element of E is the zero vector, namely,
$$E^{\perp} := \{x \in H : \langle x, y \rangle = 0,\ \forall\, y \in E\} = \{0\}.$$

Theorem 5.2.4: Riesz–Fischer Theorem

Let $(x_n)_{n\in\mathbb{N}}$ be a sequence of pairwise orthogonal elements of H. Then the limit

$$\lim_{k\to\infty} \sum_{n=1}^{k} x_n$$

exists in H if and only if

$$\sum_{n=1}^{\infty} \|x_n\|^2 < \infty.$$

In this case, the limit is denoted by

$$\sum_{n=1}^{\infty} x_n,$$

and satisfies

$$\left\| \sum_{n=1}^{\infty} x_n \right\|^2 = \sum_{n=1}^{\infty} \|x_n\|^2.$$

We finally record two results used in the construction of Brownian motion in Chapter 14.

A sequence $(e_n)_{n\in\mathbb{N}}$ in H is called an *orthonormal system* if

- $$\langle e_m, e_n \rangle = 0, \qquad m \neq n,$$
- $$\|e_n\| = 1, \qquad n \in \mathbb{N}.$$

An orthonormal system $(e_n)_{n\in\mathbb{N}}$ is called an *orthonormal basis* of H if the linear span of $\{e_n : n \in \mathbb{N}\}$ is dense in H.

Theorem 5.2.5: Parseval's Identity

Let $(e_n)_{n\in\mathbb{N}}$ be an orthonormal basis of H. Then, for every $x \in H$,

$$\sum_{n=1}^{\infty} |\langle x, e_n\rangle|^2 = \|x\|^2,$$

and

$$x = \sum_{n=1}^{\infty} \langle x, e_n\rangle e_n.$$

The series in the second identity converges in the sense of Theorem **??**.

The following result shows that an isometric mapping between Hilbert spaces can be constructed by mapping an orthonormal basis of one space onto an orthonormal system in another.

Proposition 5.2.6

Let H and K be Hilbert spaces. We use the same notation $\|\cdot\|$ and $\langle\cdot,\cdot\rangle$ for the norm and scalar product on both spaces.
Suppose that $(e_n)_{n\in\mathbb{N}}$ is an orthonormal basis of H and $(f_n)_{n\in\mathbb{N}}$ is an orthonormal system in K.
Then there exists a unique linear isometry

$$\Psi : H \to K$$

such that

$$\Psi(e_n) = f_n, \qquad n \in \mathbb{N}.$$

Moreover, for every $x \in H$,

$$\Psi(x) = \sum_{n=1}^{\infty} \langle x, e_n\rangle f_n.$$

Theorem 5.2.7: Bounded Linear Operators are Continuous

Let X and Y be normed vector spaces, and let $T : X \to Y$ be a linear mapping. The following assertions are equivalent:

1. T is continuous.
2. T is continuous at 0.
3. There exists $C \geq 0$ such that

$$\|Tx\|_Y \leq C\|x\|_X, \qquad x \in X.$$

In particular, every bounded linear operator is continuous.

Theorem 5.2.8: Extension of an Isometry

Let X be a normed vector space, let Y be a Banach space, and let $E \subset X$ be a dense linear subspace. Suppose that $T : E \to Y$ is a linear isometry, namely

$$\|Tx\|_Y = \|x\|_X, \qquad x \in E.$$

Then there exists a unique linear isometry

$$\widetilde{T} : X \to Y$$

such that

$$\widetilde{T}|_E = T.$$

CHAPITRE 6

GAUSSIAN VARIABLES AND GAUSSIAN PROCESSES

Gaussian random processes play an important role both in theoretical probability and in various applied models. This appendix begins by recalling basic facts about Gaussian random variables and Gaussian vectors. We then discuss Gaussian spaces and Gaussian processes, and we establish the fundamental properties concerning independence and conditioning in the Gaussian setting. We finally introduce the notion of a Gaussian white noise, which will be used to give a simple construction of Brownian motion in the next appendix.

6.1 Gaussian Random Variables

Throughout this appendix, we deal with random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For some of the existence statements that follow, this probability space should be chosen in an appropriate way.
A real random variable X is said to be a *standard Gaussian (or normal) variable* if its law has density

$$p_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$$

with respect to Lebesgue measure on $\mathbb{R}$.

The complex Laplace transform of X is then given by

$$\mathbb{E}[e^{zX}] = e^{z^2/2}, \qquad \forall z \in \mathbb{C}.$$

By taking $z = i\lambda$, $\lambda \in \mathbb{R}$, we get the characteristic function of X:

$$\mathbb{E}[e^{i\lambda X}] = e^{-\lambda^2/2}.$$

From the expansion

$$\mathbb{E}[e^{i\lambda X}] = 1 + i\lambda \mathbb{E}[X] + \cdots + \frac{(i\lambda)^n}{n!}\mathbb{E}[X^n] + O(|\lambda|^{n+1}),$$

as $\lambda \to 0$ (this expansion holds for every $n \geq 1$ when X belongs to all spaces L^p, $1 \leq p < \infty$, which is the case here), we get

$$\mathbb{E}[X] = 0, \qquad \mathbb{E}[X^2] = 1,$$

and more generally, for every integer $n \geq 0$,

$$\mathbb{E}[X^{2n}] = \frac{(2n)!}{2^n n!}, \qquad \mathbb{E}[X^{2n+1}] = 0.$$

We now extend this definition. Let $\sigma > 0$ and $m \in \mathbb{R}$, we say that a real random variable Y is Gaussian with $\mathcal{N}(m, \sigma^2)$-distribution if Y satisfies any of the three equivalent properties:

1. $Y = \sigma X + m$, where X is a standard Gaussian variable (i.e. X follows the $\mathcal{N}(0,1)$-distribution);
2. the law of Y has density
$$p_Y(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right);$$
3. the characteristic function of Y is
$$\mathbb{E}[e^{i\lambda Y}] = \exp\left(i\lambda m - \frac{\lambda^2}{2}\sigma^2\right).$$

We have then $\mathbb{E}[Y] = m$, $\mathrm{var}(Y) = \sigma^2$. By extension, we say that Y is Gaussian with $\mathcal{N}(m, 0)$-distribution if $Y = m$ a.s. (property (3) still holds in that case).

We now interest in the sum and limit of such gaussian variables :

1. **Sum of independent Gaussian variables:** Suppose that Y follows the $\mathcal{N}(m, \sigma^2)$-distribution, Y' follows the $\mathcal{N}(m', \sigma'^2)$-distribution, and Y and Y' are independent. Then $Y + Y'$ follows the $\mathcal{N}(m + m', \sigma^2 + \sigma'^2)$-distribution.
2. **Limit of Gaussian variables sequence:** Let $(X_n)_{n\geq 1}$ be a sequence of real random variables such that, for every $n \geq 1$, X_n follows the $\mathcal{N}(m_n, \sigma_n^2)$-distribution. Suppose that X_n converges in L^2 to X. Then:
   (a) The random variable X follows the $\mathcal{N}(m, \sigma^2)$-distribution, where $m = \lim m_n$ and $\sigma = \lim \sigma_n$.
   (b) The convergence also holds in all L^p spaces, $1 \leq p < \infty$.

   Remark. The assumption that X_n converges in L^2 to X can be weakened to convergence in probability (and in fact the convergence in distribution of the sequence $(X_n)_{n\geq 1}$ suffices to get part (1)).

6.2 Gaussian Vectors

Let E be a d-dimensional Euclidean space (E is isomorphic to $\mathbb{R}^d$ and we may take $E = \mathbb{R}^d$, with the usual inner product, but it will be more convenient to work with an

abstract space). We write $\langle u, v\rangle$ for the inner product in $\mathcal{E}$.
A random variable $\mathbf{X}$ with values in E is called a *Gaussian vector* if, for every $u \in E$, $\langle u, X\rangle$ is a (real) Gaussian variable. (For instance, if $E = \mathbb{R}^d$, and if $X_1, \dots, X_d$ are independent Gaussian variables, the property of sums of independent Gaussian variables shows that the random vector $\mathbf{X} = (X_1, \dots, X_d)$ is a Gaussian vector.) Let $\mathbf{X}$ be a

Gaussian vector with values in E. Then there exist $m_{\mathbf{X}} \in E$ and a nonnegative quadratic form $q_{\mathbf{X}}$ on E such that, for every $u \in E$,

$$\mathbb{E}[\langle u, \mathbf{X}\rangle] = \langle u, m_{\mathbf{X}}\rangle, \qquad \mathrm{var}(\langle u, \mathbf{X}\rangle) = q_{\mathbf{X}}(u).$$

Indeed, let $(e_1, \dots, e_d)$ be an orthonormal basis on E, and write $\mathbf{X} = \sum_{j=1}^d X_j e_j$ and $u = \sum_{j=1}^d u_j e_j$ in this basis. Then :

$$\begin{aligned} m_{\mathbf{X}} &= \textstyle\sum_{j=1}^d \mathbb{E}[X_j] e_j \overset{\text{not.}}{=} \mathbb{E}[\mathbf{X}] \\ q_{\mathbf{X}}(u) &= \textstyle\sum_{j,k=1}^d u_j u_k \,\mathrm{cov}(X_j, X_k). \end{aligned}$$

Since $\langle u, \mathbf{X}\rangle$ follows the $\mathcal{N}(\langle u, m_{\mathbf{X}}\rangle, q_{\mathbf{X}}(u))$-distribution, we get the characteristic function of the random vector $\mathbf{X}$,

$$\mathbb{E}[\exp(i\langle u, \mathbf{X}\rangle)] = \exp\Big(i\langle u, m_{\mathbf{X}}\rangle - \frac{1}{2} q_{\mathbf{X}}(u)\Big).$$

Furthermore, the random variables $X_1, \dots, X_d$ are independent if and only if the covariance matrix $(\mathrm{cov}(X_j, X_k))_{1\le j,k\le d}$ is diagonal, or equivalently if and only if $q_{\mathbf{X}}$ is of diagonal form in the basis $(e_1, \dots, e_d)$.
With the quadratic form $q_{\mathbf{X}}$, we associate the unique symmetric endomorphism $\gamma_{\mathbf{X}}$ of E such that

$$q_{\mathbf{X}}(u) = \langle u, \gamma_{\mathbf{X}}(u)\rangle$$

(the matrix of $\gamma_{\mathbf{X}}$ in the basis $(e_1, \dots, e_d)$ is $(\mathrm{cov}(X_j, X_k))_{1\le j,k\le d}$, but of course the definition of $\gamma_{\mathbf{X}}$ does not depend on the choice of a basis). Note that $\gamma_{\mathbf{X}}$ is nonnegative in the sense that its eigenvalues are all nonnegative. From now on, to simplify the statements, we restrict our attention to centered Gaussian vectors, i.e. such that $m_{\mathbf{X}} = 0$, but the following result are easily adapted to the non-centered case.

Theorem 6.2.1: Existence and diagonalization of Gaussian vectors

1. Let γ be a nonnegative symmetric endomorphism of E. Then there exists a Gaussian vector $\mathbf{X}$ such that $\gamma_{\mathbf{X}} = \gamma$.
2. Let $\mathbf{X}$ be a centered Gaussian vector. Let $(\varepsilon_1, \dots, \varepsilon_d)$ be a basis of E in which $\gamma_{\mathbf{X}}$ is diagonal, $\gamma_{\mathbf{X}}\varepsilon_j = \lambda_j \varepsilon_j$ for every $1 \le j \le d$, where

$$\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_r > 0 = \lambda_{r+1} = \cdots = \lambda_d,$$

so that r is the rank of $\gamma_{\mathbf{X}}$. Then,

$$\mathbf{X} = \sum_{j=1}^{r} Y_j \varepsilon_j,$$

where Y_j, $1 \leq j \leq r$, are independent (centered) Gaussian variables and the variance of Y_j is λ_j. Consequently, if $\mathbb{P}_{\mathbf{X}}$ denotes the distribution of $\mathbf{X}$, the topological support of $\mathbb{P}_{\mathbf{X}}$ is the vector space spanned by $\varepsilon_1, \ldots, \varepsilon_r$. Furthermore, $P_{\mathbf{X}}$ is absolutely continuous with respect to Lebesgue measure on E if and only if $r = d$, and in that case the density of $\mathbf{X}$ is

$$p_{\mathbf{X}}(x) = \frac{1}{(2\pi)^{d/2}\sqrt{\det \gamma_{\mathbf{X}}}} \exp\left(-\frac{1}{2}\langle x, \gamma_{\mathbf{X}}^{-1}(x)\rangle\right).$$

6.3 Gaussian Processes and Gaussian Spaces

From now on until the end of this appendix, we consider only centered Gaussian variables, and we frequently omit the word *centered*.
A (centered) *Gaussian space* is a closed linear subspace of $L^2(\Omega, \mathcal{F}, P)$ which contains only centered Gaussian variables.
A simple example is the vector space spanned by $\{X_1, \ldots, X_d\}$ is a Gaussian space, where $X = (X_1, \ldots, X_d)$ is a centered Gaussian vector in $\mathbb{R}^d$.
A (real-valued) random process $(X_t)_{t\geq 0}$ [1] is called a (centered) *Gaussian process* if any finite linear combination of the variables X_t, $t \geq 0$, is centered Gaussian.
Furthermore, the closed linear subspace of L^2 spanned by the variables X_t, $t \geq 0$, is a Gaussian space, which is called the Gaussian space generated by the process X.

6.3.1 Independence properties

We now turn to independence properties in a Gaussian space. The next theorem shows that, in some sense, independence is equivalent to orthogonality in a Gaussian space. This is a very particular property of the Gaussian distribution.

Theorem 6.3.1: Independence vs orthogonality in a Gaussian space

Let H be a centered Gaussian space and let $(H_i)_{i\in I}$ be a collection of linear subspaces of H. Then the subspaces H_i, $i \in I$, are (pairwise) orthogonal in L^2 if and only if the σ-fields $\sigma(H_i)$, $i \in I$, are independent.

As an application of the previous theorem, we now discuss conditional expectations in the Gaussian framework. The fact that these conditional expectations can be computed as

1. Here we consider continuous time for simplicity, however the resuts are valable for any indexing set including $\mathbb{N}$

orthogonal projections (as shown in the next corollary) is very particular to the Gaussian setting.

Corollary 6.3.2

Let H be a (centered) Gaussian space and let K be a closed linear subspace of H. Let p_K denote the orthogonal projection onto K in the Hilbert space L^2, and let $X \in H$.

1. We have
$$\mathbb{E}[X \mid \sigma(K)] = p_K(X).$$

2. Let $\sigma^2 = \mathbb{E}[(X - p_K(X))^2]$. Then, for every Borel subset A of $\mathbb{R}$, the random variable $\mathbb{P}[X \in A \mid \sigma(K)]$ is given by
$$\mathbb{P}[X \in A \mid \sigma(K)](\omega) = \mathbb{Q}(\omega, A),$$
where $\mathbb{Q}(\omega, \cdot)$ denotes the $\mathcal{N}(p_K(X)(\omega), \sigma^2)$-distribution:
$$\mathbb{Q}(\omega, A) = \frac{1}{\sigma\sqrt{2\pi}} \int_A dy \, \exp\left(-\frac{(y - p_K(X)(\omega))^2}{2\sigma^2}\right)$$
(and by convention $\mathbb{Q}(\omega, A) = \mathbf{1}_A(p_K(X))$ if $\sigma = 0$).

Remarks.

1. Part (2) of the statement can be interpreted by saying that the conditional distribution of X knowing $\sigma(K)$ is $\mathcal{N}(p_K(X), \sigma^2)$.
2. For a general random variable X in L^2, one has (see Theorem 2.2.6)
$$\mathbb{E}[X \mid \sigma(K)] = p_{L^2(\Omega,\sigma(K),P)}(X).$$
Assertion (1) shows that, in our Gaussian framework, this orthogonal projection coincides with the orthogonal projection onto the space K, which is *much smaller* than $L^2(\Omega, \sigma(K), P)$.
3. Assertion (1) also gives the principle of linear regression. For instance, if (X_1, X_2, X_3) is a (centered) Gaussian vector in $\mathbb{R}^3$, the best approximation in L^2 of X_3 as a (not necessarily linear) function of X_1 and X_2 can be written $\lambda_1 X_1 + \lambda_2 X_2$ where λ_1 and λ_2 are computed by saying that $X_3 - (\lambda_1 X_1 + \lambda_2 X_2)$ is orthogonal to the vector space spanned by X_1 and X_2.

Let $(X_t)_{t\geq 0}$ be a (centered) Gaussian process. The *covariance function* of X is the function $\Gamma : \mathbb{R}^+ \times \mathbb{R}^+ \to \mathbb{R}$ defined by $\Gamma(s,t) = \mathrm{cov}(X_s, X_t) = \mathbb{E}[X_s X_t]$.
This function characterizes the collection of *finite-dimensional marginal distributions* of the process X, that is, the collection consisting, for every choice of distinct indices $t_1, \ldots, t_p$ in $\mathbb{R}^+$, of the law of the vector $(X_{t_1}, \ldots, X_{t_p})$. Indeed this vector is a centered Gaussian vector in $\mathbb{R}^p$ with covariance matrix $(\Gamma(t_i, t_j))_{1\leq i,j\leq p}$.

6.3.2 Existence of Gaussian Processes

Given a function Γ on $\mathbb{R}^+ \times \mathbb{R}^+$, one may ask whether there exists a Gaussian process X whose covariance function is Γ. The function Γ must be symmetric ($\Gamma(s,t) = \Gamma(t,s)$) and *of positive type* in the following sense: if c is a real function on $\mathbb{R}^+$ with finite support T, then

$$\sum_{T\times T} c(s)c(t)\Gamma(s,t) \geq 0.$$

Indeed, if Γ is the covariance function of the process X, we have immediately

$$\sum_{T\times T} c(s)c(t)\Gamma(s,t) = \mathrm{var}\Big(\sum_{T} c(s)X_s\Big) \geq 0.$$

The next theorem solves the existence problem in the general case. This theorem is a direct consequence of the *Kolmogorov extension theorem* is not stated yet.

Theorem 6.3.3: Existence of Gaussian processes with prescribed covariance

Let Γ be a symmetric function of positive type on $\mathbb{R}^+ \times \mathbb{R}^+$. There exists, on an appropriate probability space $(\Omega, \mathcal{F}, P)$, a centered Gaussian process whose covariance function is Γ.

Example. Let μ be a finite measure on $\mathbb{R}$, which is also symmetric (i.e. $\mu(-A) = \mu(A)$). Then set, for every $s,t \in \mathbb{R}$,

$$\Gamma(s,t) = \int e^{i\xi(t-s)}\,\mu(d\xi).$$

It is easy to verify that Γ has the required properties. In particular, if c is a real function on $\mathbb{R}$ with finite support T,

$$\sum_{T\times T} c(s)c(t)\Gamma(s,t) = \int \Big|\sum_{T} c(s)e^{i\xi s}\Big|^2 \mu(d\xi) \geq 0.$$

The function Γ enjoys the additional property that $\Gamma(s,t)$ only depends on $t-s$. It immediately follows that any (centered) Gaussian process $(X_t)_{t\geq 0}$ with covariance function Γ is stationary (in a strong sense), meaning that

$$(X_{t_1+t}, X_{t_2+t}, \ldots, X_{t_n+t}) \overset{(d)}{=} (X_{t_1}, X_{t_2}, \ldots, X_{t_n})$$

for any choice of $t_1, \ldots, t_n, t \in \mathbb{R}$. Conversely, any stationary Gaussian process X indexed by $\mathbb{R}^+$ has a covariance of the preceding type (this is Bochner's theorem, which we will not use in this book), and the measure μ is called the *spectral measure* of X.

6.4 Gaussian White Noise

Let $(E, \mathcal{E})$ be a measurable space, and let μ be a σ-finite measure on $(E, \mathcal{E})$. A *Gaussian white noise* with intensity μ is an isometry G from $L^2(E, \mathcal{E}, \mu)$ into a (centered) Gaussian space.

Hence, if $f \in L^2(E, \mathcal{E}, \mu)$, $G(f)$ is centered Gaussian with variance

$$\mathbb{E}[G(f)^2] = \|G(f)\|^2_{L^2(\Omega,\mathcal{F},P)} = \|f\|^2_{L^2(E,\mathcal{E},\mu)} = \int f^2 \, d\mu.$$

If $f, g \in L^2(E, \mathcal{E}, \mu)$, the covariance of $G(f)$ and $G(g)$ is

$$\mathbb{E}[G(f)G(g)] = \langle f, g \rangle_{L^2(E,\mathcal{E},\mu)} = \int fg \, d\mu.$$

In particular, if $f = \mathbf{1}_A$ with $\mu(A) < \infty$, $G(\mathbf{1}_A)$ is $\mathcal{N}(0, \mu(A))$-distributed. To simplify notation, we will write $G(A) = G(\mathbf{1}_A)$.

Let $A_1, \ldots, A_n \in \mathcal{E}$ be disjoint and such that $\mu(A_j) < \infty$ for every j. Then the vector

$$(G(A_1), \ldots, G(A_n))$$

is a Gaussian vector in $\mathbb{R}^n$ and its covariance matrix is diagonal since, if $i \neq j$,

$$\mathbb{E}[G(A_i)G(A_j)] = \langle \mathbf{1}_{A_i}, \mathbf{1}_{A_j} \rangle_{L^2(E,\mathcal{E},\mu)} = 0.$$

Then we get that the variables $G(A_1), \ldots, G(A_n)$ are independent.

Suppose that $A \in \mathcal{E}$ is such that $\mu(A) < \infty$ and that A is the disjoint union of a countable collection $A_1, A_2, \ldots$ of measurable subsets of E. Then, $\mathbf{1}_A = \sum_{j=1}^{\infty} \mathbf{1}_{A_j}$, where the series converges in $L^2(E, \mathcal{E}, \mu)$, and by the isometry property this implies that

$$G(A) = \sum_{j=1}^{\infty} G(A_j),$$

where the series converges in $L^2(\Omega, \mathcal{F}, P)$ (since the random variables $G(A_j)$ are independent, an easy application of the convergence theorem for discrete martingales also shows that the series converges a.s.).

Properties of the mapping $A \mapsto G(A)$ are therefore very similar to those of a measure depending on the parameter $\omega \in \Omega$. However, one can show that, if ω is fixed, the mapping $A \mapsto G(A)(\omega)$ does not (in general) define a measure. We will come back to this point later.

Now we discuss the existence of such isometry given a measurable space and a given intensity.

Proposition 6.4.1

Let $(E, \mathcal{E})$ be a measurable space, and let μ be a σ-finite measure on $(E, \mathcal{E})$. There exists, on an appropriate probability space $(\Omega, \mathcal{F}, P)$, a Gaussian white noise with intensity μ.

Remark. In what follows, we will only consider the case when $L^2(E, \mathcal{E}, \mu)$ is separable[2]. For instance, if $(E, \mathcal{E}) = (\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))$ and μ is Lebesgue measure (which would be the case in most of this book), the construction of G only requires a sequence $(\xi_n)_{n\geq 0}$ of independent $\mathcal{N}(0,1)$ random variables, and the choice of an orthonormal basis $(\varphi_n)_{n\geq 0}$ of $L^2(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+), dt)$: we get G by setting

$$G(f) = \sum_{n\geq 0} \langle f, \varphi_n \rangle \, \xi_n.$$

Our last proposition gives a way of recovering the intensity $\mu(A)$ of a measurable set A from the values of G on the atoms of finer and finer partitions of A.

Proposition 6.4.2

Let G be a Gaussian white noise on $(E, \mathcal{E})$ with intensity μ. Let $A \in \mathcal{E}$ be such that $\mu(A) < \infty$. Assume that there exists a sequence of partitions of A,

$$A = A_1^n \cup \cdots \cup A_{k_n}^n,$$

whose "mesh" tends to 0, in the sense that

$$\lim_{n\to\infty} \left(\sup_{1\leq j\leq k_n} \mu(A_j^n) \right) = 0.$$

Then,

$$\lim_{n\to\infty} \sum_{j=1}^{k_n} G(A_j^n)^2 = \mu(A)$$

in L^2.

2. A metric space (or Hilbert space) is separable if it contains a countable dense subset.

CHAPITRE 7

BROWNIAN MOTION

In this appendix, we construct Brownian motion and investigate some of its properties. We start by introducing the "pre-Brownian motion" (this is not a canonical terminology), which is easily defined from a Gaussian white noise on $\mathbb{R}_+$ whose intensity is Lebesgue measure. Going from pre-Brownian motion to Brownian motion requires the additional property of continuity of sample paths, which is derived here via the classical Kolmogorov lemma. The end of the chapter discusses several properties of Brownian sample paths, and establishes the strong Markov property, with its classical application to the reflection principle.

7.1 Pre-Brownian Motion

Throughout this appendix, we argue on a probability space $(\Omega, \mathcal{F}, P)$. Most of the time, but not always, random processes will be indexed by $\mathbb{R}_+$ and take values in $\mathbb{R}$.
Let G be a Gaussian white noise on $\mathbb{R}_+$ whose intensity is Lebesgue measure. The random process $(B_t)_{t\in\mathbb{R}_+}$ defined by

$$B_t = G(\mathbf{1}_{[0,t]})$$

is called *pre-Brownian motion*. We have :

$$K(s,t) = \min\{s,t\} \overset{\text{not.}}{=} s \wedge t.$$

The next proposition gives different ways of characterizing pre-Brownian motion.

Proposition 7.1.1

Let $(X_t)_{t\geq 0}$ be a (real-valued) random process. The following properties are equivalent:

1. $(X_t)_{t\geq 0}$ is a pre-Brownian motion;
2. $(X_t)_{t\geq 0}$ is a centered Gaussian process with covariance $K(s,t) = s \wedge t$;
3. $X_0 = 0$ a.s., and, for every $0 \leq s < t$, the random variable $X_t - X_s$ is independent of $\sigma(X_r, r \leq s)$ and distributed according to $\mathcal{N}(0, t-s)$;

4. $X_0 = 0$ a.s., and, for every choice of $0 = t_0 < t_1 < \cdots < t_p$, the variables $X_{t_i} - X_{t_{i-1}}$, $1 \le i \le p$, are independent, and, for every $1 \le i \le p$, the variable $X_{t_i} - X_{t_{i-1}}$ is distributed according to $\mathcal{N}(0, t_i - t_{i-1})$.

Let $(B_t)_{t\ge 0}$ be a pre-Brownian motion. Then, for every choice of $0 = t_0 < t_1 < \cdots < t_n$, the law of the vector $(B_{t_1}, B_{t_2}, \ldots, B_{t_n})$ has density

$$p(x_1, \ldots, x_n) = \frac{1}{(2\pi)^{n/2}\sqrt{t_1(t_2 - t_1)\cdots(t_n - t_{n-1})}} \exp\left(-\sum_{i=1}^{n} \frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})}\right),$$

where by convention $x_0 = 0$.
We have the property of invariance of pre-Brownian motion :

1. $-B$ is also a pre-Brownian motion (symmetry property);
2. for every $\lambda > 0$, the process $B_t^\lambda = \frac{1}{\lambda} B_{\lambda^2 t}$ is also a pre-Brownian motion (invariance under scaling);
3. for every $s \ge 0$, the process $B_t^{(s)} = B_{s+t} - B_s$ is a pre-Brownian motion and is independent of $\sigma(B_r, r \le s)$ (simple Markov property).

Let B be a pre-Brownian motion and let G be the associated Gaussian white noise. Note that G is determined by B: if f is a step function there is an explicit formula for $G(f)$ in terms of B, and one then uses a density argument. One often writes for $f \in L^2(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+), dt)$,

$$G(f) = \int_0^\infty f(s)\, dB_s$$

and similarly

$$G(f\mathbf{1}_{[0,t]}) = \int_0^t f(s)\, dB_s\ , \qquad G(f\mathbf{1}_{(s,t]}) = \int_s^t f(r)\, dB_r.$$

This notation is justified by the fact that, if $u < v$,

$$\int_u^v dB_s = G((u, v]) = G([0, v]) - G([0, u]) = B_v - B_u.$$

The mapping $f \mapsto \int_0^\infty f(s)\, dB_s$ (that is, the Gaussian white noise G) is then called the *Wiener integral* with respect to B. Recall that $\int_0^\infty f(s)\, dB_s$ is distributed according to $\mathcal{N}(0, \int_0^\infty f(s)^2\, ds)$.

Since a Gaussian white noise is not a *real* measure depending on ω, $\int_0^\infty f(s)\, dB_s$ is not a *real* integral depending on ω. Much of what follows in this book is devoted to extending the definition of $\int_0^\infty f(s)\, dB_s$ to functions f that may depend on ω.

7.2 The Continuity of Sample Paths

Definition 7.2.1: Brownnian Motion

A process $(B_t)_{t\geq 0}$ is a *Brownian motion* if:

(i) $(B_t)_{t\geq 0}$ is a pre-Brownian motion.

(ii) All sample paths of B are continuous.

This is in fact the definition of a *real* (or *linear*) Brownian motion *started from* 0. Extensions to arbitrary starting points and to higher dimensions will be discussed later. The existence of Brownian motion in the sense of the preceding definition follows from Theorem **??**. Indeed, starting from a pre-Brownian motion, this theorem provides a modification with continuous sample paths (and even locally Hôlder continuous with exponent $\frac{1}{2} - \delta$ for every $\delta \in (0, \frac{1}{2})$.), which is still a pre-Brownian motion. In what follows we no longer consider pre-Brownian motion, as we will be interested only in Brownian motion.
It is important to note that the preceding pre-Brownian motion properties hold without change if pre-Brownian motion is replaced everywhere by Brownian motion.

Definition 7.2.2: $(\mathcal{F}_t)$-Brownian Motion

A real-valued process $B = (B_t)_{t\geq 0}$ is called an $(\mathcal{F}_t)$-*Brownian motion* if B is a Brownian motion, B is adapted to the filtration $(\mathcal{F}_t)_{t\geq 0}$, and B has independent increments with respect to $(\mathcal{F}_t)$.
Similarly, an $\mathbb{R}^d$-valued process $B = (B_t)_{t\geq 0}$ is called a *d-dimensional* $(\mathcal{F}_t)$-*Brownian motion* if B is a d-dimensional Brownian motion, B is adapted to $(\mathcal{F}_t)_{t\geq 0}$, and has independent increments with respect to $(\mathcal{F}_t)$.
If B is a (possibly d-dimensional) Brownian motion and $(\mathcal{F}_t^B)_{t\geq 0}$ denotes its (possibly completed) canonical filtration, then B is an $(\mathcal{F}_t^B)$-Brownian motion.

The Wiener measure. Let $C(\mathbb{R}_+, \mathbb{R})$ be the space of all continuous functions from $\mathbb{R}_+$ into $\mathbb{R}$. We equip $C(\mathbb{R}_+, \mathbb{R})$ with the σ-field $\mathcal{C}$ defined as the smallest σ-field on $C(\mathbb{R}_+, \mathbb{R})$ for which the coordinate mappings $w \mapsto w(t)$ are measurable for every $t \geq 0$ (alternatively, one checks that $\mathcal{C}$ coincides with the Borel σ-field on $C(\mathbb{R}_+, \mathbb{R})$ associated with the topology of uniform convergence on every compact set). Given a Brownian motion B, we can consider the mapping

$$\Omega \longrightarrow C(\mathbb{R}_+, \mathbb{R}), \qquad \omega \mapsto (t \mapsto B_t(\omega))$$

and one verifies that this mapping is measurable (if we take its composition with a coordinate map $w \mapsto w(t)$ we get the random variable B_t, and a simple argument shows that this suffices for the desired measurability).

The *Wiener measure* (or law of Brownian motion) is by definition the image of the probability measure $P(d\omega)$ under this mapping. The Wiener measure, which we denote by $W(dw)$, is thus a probability measure on $C(\mathbb{R}_+, \mathbb{R})$, and, for every measurable subset A of $C(\mathbb{R}_+, \mathbb{R})$, we have

$$W(A) = P(B_{\cdot} \in A),$$

where in the right-hand side $B_{\cdot}$ stands for the random continuous function $t \mapsto B_t(\omega)$.

We can specialize the last equality to a "cylinder set" of the form

$$A = \{w \in C(\mathbb{R}_+, \mathbb{R}) : w(t_0) \in A_0,\ w(t_1) \in A_1, \ldots, w(t_n) \in A_n\},$$

where $0 = t_0 < t_1 < \cdots < t_n$, and $A_0, A_1, \ldots, A_n \in \mathcal{B}(\mathbb{R})$ (recall that $\mathcal{B}(\mathbb{R})$ stands for the Borel σ-field on $\mathbb{R}$). Corollary **??** then gives

$$W(\{w; w(t_0) \in A_0, w(t_1) \in A_1, \ldots, w(t_n) \in A_n\})$$

$$= P(B_{t_0} \in A_0, B_{t_1} \in A_1, \ldots, B_{t_n} \in A_n)$$

$$= \mathbf{1}_{A_0}(0) \int_{A_1 \times \cdots \times A_n} \frac{dx_1 \cdots dx_n}{(2\pi)^{n/2}\sqrt{t_1(t_2 - t_1)\cdots(t_n - t_{n-1})}} \exp\left(-\sum_{i=1}^{n} \frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})}\right),$$

where $x_0 = 0$ by convention.

This formula for the W-measure of cylinder sets characterizes the probability measure W. Indeed, the class of cylinder sets is stable under finite intersections and generates the σ-field $\mathcal{C}$, which by a standard monotone class argument (see Appendix A1) implies that a probability measure on $\mathcal{C}$ is characterized by its values on this class. A consequence of the preceding formula for the W-measure of cylinder sets is the (fortunate) fact that the definition of the Wiener measure does not depend on the choice of the Brownian motion B: the law of Brownian motion is uniquely defined!

Suppose that B' is another Brownian motion. Then, for every $A \in \mathcal{C}$,

$$P(B'_{\cdot} \in A) = W(A) = P(B_{\cdot} \in A).$$

This means that the probability that a given property (corresponding to a measurable subset A of $C(\mathbb{R}_+, \mathbb{R})$) holds is the same for the sample paths of B and for the sample paths of B'. We will use this observation many times in what follows (see in particular the second part of the proof of Proposition 7.3.2 below).

Consider now the special choice of a probability space,

$$\Omega = C(\mathbb{R}_+, \mathbb{R}), \qquad \mathcal{F} = \mathcal{C}, \qquad P(dw) = W(dw).$$

Then on this probability space, the so-called *canonical process*

$$X_t(w) = w(t)$$

is a Brownian motion (the continuity of sample paths is obvious, and the fact that X has the right finite-dimensional marginals follows from the preceding formula for the W-measure of cylinder sets). This is the *canonical construction* of Brownian motion.

7.3 Properties of Brownian Sample Paths

In this section, we investigate properties of sample paths of Brownian motion. We fix a Brownian motion $(B_t)_{t\geq 0}$. For every $t \geq 0$, we set

$$\mathcal{F}_t = \sigma(B_s, s \leq t).$$

Note that $\mathcal{F}_s \subset \mathcal{F}_t$ if $s \leq t$. We also set

$$\mathcal{F}_{0+} = \bigcap_{s>0} \mathcal{F}_s.$$

We start by stating a useful 0–1 law.

Theorem 7.3.1: Blumenthal's zero-one law

The σ-field $\mathcal{F}_{0+}$ is trivial, in the sense that $P(A) = 0$ or 1 for every $A \in \mathcal{F}_{0+}$.

Proposition 7.3.2

1. We have, a.s. for every $\varepsilon > 0$,

$$\sup_{0\leq s\leq\varepsilon} B_s > 0, \qquad \inf_{0\leq s\leq\varepsilon} B_s < 0.$$

2. For every $a \in \mathbb{R}$, let $T_a = \inf\{t \geq 0 : B_t = a\}$ (with the convention $\inf \emptyset = \infty$). Then,

$$\text{a.s.}, \quad \forall a \in \mathbb{R}, \quad T_a < \infty.$$

   Consequently, we have a.s.

$$\limsup_{t\to\infty} B_t = +\infty, \qquad \liminf_{t\to\infty} B_t = -\infty.$$

3. Almost surely, the function $t \mapsto B_t$ is not monotone on any non-trivial interval.

Remark. It is not a priori obvious that $\sup_{0\leq s\leq\varepsilon} B_s$ is measurable, since this is an uncountable supremum of random variables. However, we can take advantage of the continuity of sample paths to restrict the supremum to *rational* values of $s \in [0, \varepsilon]$, so that we have a supremum of a countable collection of random variables. We will implicitly use such remarks in what follows.

Proposition 7.3.3

- Let $0 = t_0^n < t_1^n < \cdots < t_{p_n}^n = t$ be a sequence of subdivisions of $[0, t]$ whose mesh tends to 0 (i.e. $\sup_{1\le i\le p_n}(t_i^n - t_{i-1}^n) \to 0$ as $n\to\infty$). Then,

$$\lim_{n\to\infty}\sum_{i=1}^{p_n}(B_{t_i^n} - B_{t_{i-1}^n})^2 = t,$$

in L^2.

- If $a < b$ and f is a real function defined on $[a,b]$, the function f is said to have *infinite variation* if the supremum of the quantities $\sum_{i=1}^{p}|f(t_i) - f(t_{i-1})|$, over all subdivisions $a = t_0 < t_1 < \cdots < t_p = b$, is infinite. Almost surely, the function $t \mapsto B_t$ has infinite variation on any non-trivial interval.

The part (1) is in fact defining and calculating the *quadratic variation* of Brownian motion, this concept is studied in more details in Section 3.6.2 and use the notation $\langle B, B\rangle = t$. The part (2) shows that it is not possible to define the integral $\int_0^t f(s)\,dB_s$ as a special case of the usual (Stieltjes) integral with respect to functions of finite variation (see Section 3.5 for a brief presentation of the integral with respect to functions of finite variation, and also the comments on *Weirner measure* in the previous section).

We recall that the bracket of two brownian motion $B^{(1)}$ and $B^{(2)}$ is defined as process

$$\langle B^{(1)}, B^{(2)}\rangle_t = \frac{1}{2}\left((B^{(1)} + B^{(2)})_t - \langle B^{(1)}, B^{(1)}\rangle_t - \langle B^{(2)}, B^{(2)}\rangle_t\right).$$

Ând we have the following result.

Proposition 7.3.4

Let $B = (B_t)_{t\ge 0}$ and $B' = (B'_t)_{t\ge 0}$ be two $(\mathcal{F}_t)$-Brownian motions. Assume that B and B' are *independent*, that is,

$$\sigma(B_t : t \ge 0) \quad \text{and} \quad \sigma(B'_t : t \ge 0)$$

are independent σ-algebras. Then,

$$\langle B, B'\rangle_t = 0, \qquad \forall\, t \ge 0.$$

7.4 The Strong Markov Property of Brownian Motion

Our goal is to extend the simple Markov property to the case where the deterministic time s is replaced by a stopping time T (see Section 3.2). As in the previous section, we fix a Brownian motion $(B_t)_{t\ge 0}$. We keep the notation $\mathcal{F}_t$ introduced before Theorem **??** and we also set $\mathcal{F}_\infty = \sigma(B_s, s \ge 0)$.

Theorem 7.4.1: strong Markov property

Let T be a stopping time. We assume that $P(T < \infty) > 0$ and we set, for every $t \geq 0$,

$$B_t^{(T)} = \mathbf{1}_{\{T<\infty\}}(B_{T+t} - B_T).$$

Then under the probability measure $P(\cdot \mid T < \infty)$, the process $(B_t^{(T)})_{t\geq 0}$ is a Brownian motion independent of $\mathcal{F}_T$.

An important application of the strong Markov property is the “reflection principle” that leads to the following theorem.

Theorem 7.4.2: reflection principle

For every $t > 0$, set $S_t = \sup_{s\leq t} B_s$. Then, if $a \geq 0$ and $b \in (-\infty, a]$, we have

$$P(S_t \geq a,\ B_t \leq b) = P(B_t \geq 2a - b).$$

In particular, S_t has the same distribution as $|B_t|$.

It follows from the previous theorem that the law of the pair (S_t, B_t) has density

$$g(a, b) = \frac{2(2a-b)}{\sqrt{2\pi t^3}} \exp\left(-\frac{(2a-b)^2}{2t}\right) \mathbf{1}_{\{a>0,\, b<a\}}.$$

It follows also that : for every $a > 0$, T_a has the same distribution as $\dfrac{a^2}{B_1^2}$ and has density

$$f(t) = \frac{a}{\sqrt{2\pi t^3}} \exp\left(-\frac{a^2}{2t}\right) \mathbf{1}_{\{t>0\}}.$$

Remark. From the form of the density of T_a, we immediately get that $\mathbb{E}[T_a] = \infty$.

7.5 Extension for Brownian Motion

We finally extend the definition of Brownian motion to the case of an arbitrary (possibly random) initial value and to any dimension.

7.5.1 real Brownian motion started from Z

If Z is a real random variable, a process $(X_t)_{t\geq 0}$ is a *real Brownian motion started from* Z if we can write $X_t = Z + B_t$ where B is a real Brownian motion started from 0 and is independent of Z.

7.5.2 d-dimensional Brownian motion

A random process $B_t = (B_t^1, \dots, B_t^d)$ with values in $\mathbb{R}^d$ is a *d-dimensional Brownian motion started from* 0 if its components $B^1, \dots, B^d$ are independent real Brownian motions started from 0. If Z is a random variable with values in $\mathbb{R}^d$ and $X_t = (X_t^1, \dots, X_t^d)$ is a process with values in $\mathbb{R}^d$, we say that X is a *d-dimensional Brownian motion started from Z* if we can write $X_t = Z + B_t$ where B is a d-dimensional Brownian motion started from 0 and is independent of Z.

Note that, if X is a d-dimensional Brownian motion and the initial value of X is random, the components of X may not be independent because the initial value may introduce some dependence (this does not occur if the initial value is deterministic).

In a way similar to the end of first section, the Wiener measure in dimension d is defined as the probability measure on $C(\mathbb{R}_+, \mathbb{R}^d)$ which is the law of a d-dimensional Brownian motion started from 0. The canonical construction of Sect. 2.2 also applies to d-dimensional Brownian motion.

7.5.3 Extended results

Many of the preceding results can be extended to d-dimensional Brownian motion with an arbitrary starting point.

Furthermore, the symetry property can be extended as follows. If X is a d-dimensional Brownian motion and Φ is an isometry of $\mathbb{R}^d$, the process $(\Phi(X_t))_{t\geq 0}$ is still a d-dimensional Brownian motion.

7.6 Levy's Characterisation

Theorem 7.6.1: Levy's Characterisation

Let $X = (X^1, \dots, X^d)$ be an adapted process with continuous sample paths. The following are equivalent:

1. X is a d-dimensional $(\mathcal{F}_t)$-Brownian motion.
2. The processes $X^1, \dots, X^d$ are continuous local martingales, and

$$\langle X^i, X^j \rangle_t = \delta_{ij} t, \qquad \forall\, t \geq 0,\ i, j \in \{1, \dots, d\},$$

where δ_{ij} denotes the Kronecker delta.

In particular, a continuous local martingale M is an $(\mathcal{F}_t)$-Brownian motion if and only if

$$\langle M, M \rangle_t = t, \qquad \forall\, t \geq 0,$$

or equivalently, if and only if

$$M_t^2 - t$$

is a continuous local martingale.